In [2]:
```
import numpy as np
import pandas as pd
import difflib
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
```

In [4]:
```
#Loading Data from the csv file
movies_data= pd.read_csv(r"E:\ANN-project\movies_updated.csv")
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[4], line 2
      1 #Loading Data from the csv file
----> 2 movies_data= pd.read_csv(r"E:\ANN-project\movies_updated.csv")

NameError: name 'pd' is not defined
```

In [2]:
```
#printing first 5 rows of dataset
movies_data.head()
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[2], line 2
      1 #printing first 5 rows of dataset
----> 2 movies_data.head()

NameError: name 'movies_data' is not defined
```

In [ ]:

In [5]:
```
#No. of Rows & Columns
movies_data.shape
```

Out[5]:
```
(4803, 24)
```

In [6]:
```
#selecting the relevant features for recommendation
selected_features = ['genres','keywords','tagline','cast','director']
print(selected_features)
```

```
['genres', 'keywords', 'tagline', 'cast', 'director']
```

In [7]:
```
# replacing the null valuess with null string
for feature in selected_features:
  movies_data[feature] = movies_data[feature].fillna('')
```

In [8]:
```
movies_data.isnull().sum()
```

Out[8]:
```
index                     0
budget                    0
genres                    0
homepage               3091
id                        0
keywords                  0
original_language         0
original_title            0
overview                  3
popularity                0
production_companies      0
production_countries      0
release_date              1
revenue                   0
runtime                   2
spoken_languages          0
status                    0
tagline                   0
title                     0
vote_average              0
vote_count                0
cast                      0
crew                      0
director                  0
dtype: int64
```

In [9]:
```
# combining all the 5 selected features
combined_features = movies_data['genres']+' '+movies_data['keywords']+' '+movies_da
```

In [10]:
```
print(combined_features)
```

```
0       Action Adventure Fantasy Science Fiction cultu...
1       Adventure Fantasy Action ocean drug abuse exot...
2       Action Adventure Crime spy based on novel secr...
3       Action Crime Drama Thriller dc comics crime fi...
4       Action Adventure Science Fiction based on nove...
                                ...
4798    Action Crime Thriller united states\u2013mexic...
4799    Comedy Romance  A newlywed couple's honeymoon ...
4800    Comedy Drama Romance TV Movie date love at fir...
4801      A New Yorker in Shanghai Daniel Henney Eliza...
4802    Documentary obsession camcorder crush dream gi...
Length: 4803, dtype: object
```

In [11]:
```
# converting the text data to feature vectors

vectorizer = TfidfVectorizer()
```

In [12]:
```
feature_vectors = vectorizer.fit_transform(combined_features)
```

In [13]:
```
print(feature_vectors)
```

```
(0, 2432)     0.17272411194153
(0, 7755)     0.1128035714854756
(0, 13024)    0.1942362060108871
(0, 10229)    0.16058685400095302
(0, 8756)     0.22709015857011816
(0, 14608)    0.15150672398763912
(0, 16668)    0.19843263965100372
(0, 14064)    0.20596090415084142
(0, 13319)    0.2177470539412484
(0, 17290)    0.20197912553916567
(0, 17007)    0.23643326319898797
(0, 13349)    0.15021264094167086
(0, 11503)    0.27211310056983656
(0, 11192)    0.09049319826481456
(0, 16998)    0.1282126322850579
(0, 15261)    0.07095833561276566
(0, 4945)     0.24025852494110758
(0, 14271)    0.21392179219912877
(0, 3225)     0.24960162956997736
(0, 16587)    0.12549432354918996
(0, 14378)    0.33962752210959823
(0, 5836)     0.1646750903586285
(0, 3065)     0.22208377802661425
(0, 3678)     0.21392179219912877
(0, 5437)     0.1036413987316636
:     :
(4801, 17266) 0.2886098184932947
(4801, 4835)  0.24713765026963996
(4801, 403)   0.17727585190343226
(4801, 6935)  0.2886098184932947
(4801, 11663) 0.21557500762727902
(4801, 1672)  0.1564793427630879
(4801, 10929) 0.13504166990041588
(4801, 7474)  0.11307961713172225
(4801, 3796)  0.3342808988877418
(4802, 6996)  0.5700048226105303
(4802, 5367)  0.22969114490410403
(4802, 3654)  0.262512960498006
(4802, 2425)  0.24002350969074696
(4802, 4608)  0.24002350969074696
(4802, 6417)  0.21753405888348784
(4802, 4371)  0.1538239182675544
(4802, 12989) 0.1696476532191718
(4802, 1316)  0.1960747079005741
(4802, 4528)  0.19504460807622875
(4802, 3436)  0.21753405888348784
(4802, 6155)  0.18056463596934083
(4802, 4980)  0.16078053641367315
(4802, 2129)  0.3099656128577656
(4802, 4518)  0.16784466610624255
(4802, 11161) 0.17867407682173203
```

In [15]:

```
#getting the similarity scores using cosine similarity

similarity = cosine_similarity(feature_vectors)
```

In [16]:
```
print(similarity)
```

```
[[1.         0.07219487 0.037733   ... 0.         0.         0.        ]
 [0.07219487 1.         0.03281499 ... 0.03575545 0.         0.        ]
 [0.037733   0.03281499 1.         ... 0.         0.05389661 0.        ]
 ...
 [0.         0.03575545 0.         ... 1.         0.         0.02651502]
 [0.         0.         0.05389661 ... 0.         1.         0.        ]
 [0.         0.         0.         ... 0.02651502 0.         1.        ]]
```

In [17]:
```
similarity.shape
```

Out[17]:
```
(4803, 4803)
```

In [18]:
```
movie_name = input(' Enter your favourite movie name : ')
```

In [19]:
```
#creating a list with all the movie names given in the dataset

list_of_all_titles = movies_data['title'].tolist()
print(list_of_all_titles)
```

```
['Avatar', "Pirates of the Caribbean: At World's End", 'Spectre', 'The Dark Knight R
ises', 'John Carter', 'Spider-Man 3', 'Tangled', 'Avengers: Age of Ultron', 'Harry P
otter and the Half-Blood Prince', 'Batman v Superman: Dawn of Justice', 'Superman Re
turns', 'Quantum of Solace', "Pirates of the Caribbean: Dead Man's Chest", 'The Lone
Ranger', 'Man of Steel', 'The Chronicles of Narnia: Prince Caspian', 'The Avengers',
'Pirates of the Caribbean: On Stranger Tides', 'Men in Black 3', 'The Hobbit: The Ba
ttle of the Five Armies', 'The Amazing Spider-Man', 'Robin Hood', 'The Hobbit: The D
esolation of Smaug', 'The Golden Compass', 'King Kong', 'Titanic', 'Captain America:
Civil War', 'Battleship', 'Jurassic World', 'Skyfall', 'Spider-Man 2', 'Iron Man 3',
'Alice in Wonderland', 'X-Men: The Last Stand', 'Monsters University', 'Transformer
s: Revenge of the Fallen', 'Transformers: Age of Extinction', 'Oz: The Great and Pow
erful', 'The Amazing Spider-Man 2', 'TRON: Legacy', 'Cars 2', 'Green Lantern', 'Toy
Story 3', 'Terminator Salvation', 'Furious 7', 'World War Z', 'X-Men: Days of Future
Past', 'Star Trek Into Darkness', 'Jack the Giant Slayer', 'The Great Gatsby', 'Prin
ce of Persia: The Sands of Time', 'Pacific Rim', 'Transformers: Dark of the Moon',
'Indiana Jones and the Kingdom of the Crystal Skull', 'The Good Dinosaur', 'Brave',
'Star Trek Beyond', 'WALL·E', 'Rush Hour 3', '2012', 'A Christmas Carol', 'Jupiter A
scending', 'The Legend of Tarzan', 'The Chronicles of Narnia: The Lion, the Witch an
d the Wardrobe', 'X-Men: Apocalypse', 'The Dark Knight', 'Up', 'Monsters vs Aliens',
'Iron Man', 'Hugo', 'Wild Wild West', 'The Mummy: Tomb of the Dragon Emperor', 'Suic
ide Squad', 'Evan Almighty', 'Edge of Tomorrow', 'Waterworld', 'G.I. Joe: The Rise o
f Cobra', 'Inside Out', 'The Jungle Book', 'Iron Man 2', 'Snow White and the Huntsma
n', 'Maleficent', 'Dawn of the Planet of the Apes', 'The Lovers', '47 Ronin', 'Capta
in America: The Winter Soldier', 'Shrek Forever After', 'Tomorrowland', 'Big Hero
6', 'Wreck-It Ralph', 'The Polar Express', 'Independence Day: Resurgence', 'How to T
rain Your Dragon', 'Terminator 3: Rise of the Machines', 'Guardians of the Galaxy',
'Interstellar', 'Inception', 'Shin Godzilla', 'The Hobbit: An Unexpected Journey',
'The Fast and the Furious', 'The Curious Case of Benjamin Button', 'X-Men: First Cla
ss', 'The Hunger Games: Mockingjay - Part 2', "The Sorcerer's Apprentice", 'Poseido
n', 'Alice Through the Looking Glass', 'Shrek the Third', 'Warcraft', 'Terminator Ge
nisys', 'The Chronicles of Narnia: The Voyage of the Dawn Treader', 'Pearl Harbor',
'Transformers', 'Alexander', 'Harry Potter and the Order of the Phoenix', 'Harry Pot
ter and the Goblet of Fire', 'Hancock', 'I Am Legend', 'Charlie and the Chocolate Fa
ctory', 'Ratatouille', 'Batman Begins', 'Madagascar: Escape 2 Africa', 'Night at the
Museum: Battle of the Smithsonian', 'X-Men Origins: Wolverine', 'The Matrix Revoluti
ons', 'Frozen', 'The Matrix Reloaded', 'Thor: The Dark World', 'Mad Max: Fury Road',
'Angels & Demons', 'Thor', 'Bolt', 'G-Force', 'Wrath of the Titans', 'Dark Shadows',
'Mission: Impossible - Rogue Nation', 'The Wolfman', 'Bee Movie', 'Kung Fu Panda 2',
'The Last Airbender', 'Mission: Impossible III', 'White House Down', 'Mars Needs Mom
s', 'Flushed Away', 'Pan', 'Mr. Peabody & Sherman', 'Troy', "Madagascar 3: Europe's
Most Wanted", 'Die Another Day', 'Ghostbusters', 'Armageddon', 'Men in Black II', 'B
eowulf', 'Kung Fu Panda 3', 'Mission: Impossible - Ghost Protocol', 'Rise of the Gua
rdians', 'Fun with Dick and Jane', 'The Last Samurai', 'Exodus: Gods and Kings', 'St
ar Trek', 'Spider-Man', 'How to Train Your Dragon 2', 'Gods of Egypt', 'Stealth', 'W
atchmen', 'Lethal Weapon 4', 'Hulk', 'G.I. Joe: Retaliation', 'Sahara', 'Final Fanta
sy: The Spirits Within', 'Captain America: The First Avenger', 'The World Is Not Eno
ugh', 'Master and Commander: The Far Side of the World', 'The Twilight Saga: Breakin
g Dawn - Part 2', 'Happy Feet Two', 'The Incredible Hulk', 'The BFG', 'The Revenan
t', 'Turbo', 'Rango', 'Penguins of Madagascar', 'The Bourne Ultimatum', 'Kung Fu Pan
da', 'Ant-Man', 'The Hunger Games: Catching Fire', 'Home', 'War of the Worlds', 'Bad
Boys II', 'Puss in Boots', 'Salt', 'Noah', 'The Adventures of Tintin', 'Harry Potter
and the Prisoner of Azkaban', 'Australia', 'After Earth', 'Dinosaur', 'Night at the
Museum: Secret of the Tomb', 'Megamind', "Harry Potter and the Philosopher's Stone",
'R.I.P.D.', 'Pirates of the Caribbean: The Curse of the Black Pearl', 'The Hunger Ga
mes: Mockingjay - Part 1', 'The Da Vinci Code', 'Rio 2', 'X2', 'Fast Five', 'Sherloc
k Holmes: A Game of Shadows', 'Clash of the Titans', 'Total Recall', 'The 13th Warri
```

```
or', 'The Bourne Legacy', 'Batman & Robin', 'How the Grinch Stole Christmas', 'The D
ay After Tomorrow', 'Mission: Impossible II', 'The Perfect Storm', 'Fantastic 4: Ris
e of the Silver Surfer', 'Life of Pi', 'Ghost Rider', 'Jason Bourne', "Charlie's Ang
els: Full Throttle", 'Prometheus', 'Stuart Little 2', 'Elysium', 'The Chronicles of
Riddick', 'RoboCop', 'Speed Racer', 'How Do You Know', 'Knight and Day', 'Oblivion',
'Star Wars: Episode III - Revenge of the Sith', 'Star Wars: Episode II - Attack of t
he Clones', 'Monsters, Inc.', 'The Wolverine', 'Star Wars: Episode I - The Phantom M
enace', 'The Croods', 'Asterix at the Olympic Games', 'Windtalkers', "The Huntsman:
Winter's War", 'Teenage Mutant Ninja Turtles', 'Gravity', "Dante's Peak", 'Teenage M
utant Ninja Turtles: Out of the Shadows', 'Fantastic Four', 'Night at the Museum',
'San Andreas', 'Tomorrow Never Dies', 'The Patriot', "Ocean's Twelve", 'Mr. & Mrs. S
mith', 'Insurgent', 'The Aviator', "Gulliver's Travels", 'The Green Hornet', '300: R
ise of an Empire', 'The Smurfs', 'Home on the Range', 'Allegiant', 'Real Steel', 'Th
e Smurfs 2', 'Speed 2: Cruise Control', "Ender's Game", 'Live Free or Die Hard', 'Th
e Lord of the Rings: The Fellowship of the Ring', 'Around the World in 80 Days', 'Al
i', 'The Cat in the Hat', 'I, Robot', 'Kingdom of Heaven', 'Stuart Little', 'The Pri
ncess and the Frog', 'The Martian', 'The Island', 'Town & Country', 'Gone in Sixty S
econds', 'Gladiator', 'Minority Report', 'Harry Potter and the Chamber of Secrets',
'Casino Royale', 'Planet of the Apes', 'Terminator 2: Judgment Day', 'Public Enemie
s', 'American Gangster', 'True Lies', 'The Taking of Pelham 1 2 3', 'Little Focker
s', 'The Other Guys', 'Eraser', 'Django Unchained', 'The Hunchback of Notre Dame',
"The Emperor's New Groove", 'The Expendables 2', 'National Treasure', 'Eragon', 'Whe
re the Wild Things Are', 'Epic', 'The Tourist', 'End of Days', 'Blood Diamond', 'The
Wolf of Wall Street', 'Batman Forever', 'Starship Troopers', 'Cloud Atlas', "Legend
of the Guardians: The Owls of Ga'Hoole", 'Catwoman', 'Hercules', 'Treasure Planet',
'Land of the Lost', 'The Expendables 3', 'Point Break', 'Son of the Mask', 'In the H
eart of the Sea', 'The Adventures of Pluto Nash', 'Green Zone', 'The Peanuts Movie',
'The Spanish Prisoner', 'The Mummy Returns', 'Gangs of New York', 'The Flowers of Wa
r', "Surf's Up", 'The Stepford Wives', 'Black Hawk Down', 'The Campaign', 'The Fifth
Element', 'Sex and the City 2', 'The Road to El Dorado', 'Ice Age: Continental Drif
t', 'Cinderella', 'The Lovely Bones', 'Finding Nemo', 'The Lord of the Rings: The Re
turn of the King', 'The Lord of the Rings: The Two Towers', 'Seventh Son', 'Lara Cro
ft: Tomb Raider', 'Transcendence', 'Jurassic Park III', 'Rise of the Planet of the A
pes', 'The Spiderwick Chronicles', 'A Good Day to Die Hard', 'The Alamo', 'The Incre
dibles', 'Cutthroat Island', 'Percy Jackson & the Olympians: The Lightning Thief',
'Men in Black', 'Toy Story 2', 'Unstoppable', 'Rush Hour 2', 'What Lies Beneath', 'C
loudy with a Chance of Meatballs', 'Ice Age: Dawn of the Dinosaurs', 'The Secret Lif
e of Walter Mitty', "Charlie's Angels", 'The Departed', 'Mulan', 'Tropic Thunder',
'The Girl with the Dragon Tattoo', 'Die Hard: With a Vengeance', 'Sherlock Holmes',
'Ben-Hur', 'Atlantis: The Lost Empire', 'Alvin and the Chipmunks: The Road Chip', 'V
alkyrie', "You Don't Mess with the Zohan", 'Pixels', 'A.I. Artificial Intelligence',
'The Haunted Mansion', 'Contact', 'Hollow Man', 'The Interpreter', 'Percy Jackson: S
ea of Monsters', 'Lara Croft Tomb Raider: The Cradle of Life', 'Now You See Me 2',
'The Saint', 'Spy Game', 'Mission to Mars', 'Rio', 'Bicentennial Man', 'Volcano', "T
he Devil's Own", 'K-19: The Widowmaker', 'Conan the Barbarian', 'Cinderella Man', 'T
he Nutcracker: The Untold Story', 'Seabiscuit', 'Twister', 'Cast Away', 'Happy Fee
t', 'The Bourne Supremacy', 'Air Force One', "Ocean's Eleven", 'The Three Musketeer
s', 'Hotel Transylvania', 'Enchanted', 'Safe House', '102 Dalmatians', 'Tower Heis
t', 'The Holiday', 'Enemy of the State', "It's Complicated", "Ocean's Thirteen", 'Op
en Season', 'Divergent', 'Enemy at the Gates', 'The Rundown', 'Last Action Hero', 'M
emoirs of a Geisha', 'The Fast and the Furious: Tokyo Drift', 'Arthur Christmas', 'M
eet Joe Black', 'Collateral Damage', 'All That Jazz', 'Mirror Mirror', 'Scott Pilgri
m vs. the World', 'The Core', 'Nutty Professor II: The Klumps', 'Scooby-Doo', 'Dred
d', 'Click', 'Creepshow', 'Cats & Dogs 2 : The Revenge of Kitty Galore', 'Jumper',
'Hellboy II: The Golden Army', 'Zodiac', 'The 6th Day', 'Bruce Almighty', 'The Expen
dables', 'Mission: Impossible', 'The Hunger Games', 'The Hangover Part II', 'Batman
```

Returns', 'Over the Hedge', 'Lilo & Stitch', "Charlotte's Web", 'Deep Impact', 'RED 2', 'The Longest Yard', 'Alvin and the Chipmunks: Chipwrecked', 'Grown Ups 2', 'Get Smart', "Something's Gotta Give", 'Shutter Island', 'Four Christmases', 'Robots', 'Face/Off', 'Bedtime Stories', 'Road to Perdition', 'Just Go with It', 'Con Air', 'Eagle Eye', 'Cold Mountain', 'The Book of Eli', 'Flubber', 'The Haunting', 'Space Jam', 'The Pink Panther', 'The Day the Earth Stood Still', 'Conspiracy Theory', 'Fury', 'Six Days Seven Nights', 'Yogi Bear', 'Spirit: Stallion of the Cimarron', 'Zookeeper', 'Lost in Space', 'The Manchurian Candidate', 'Déjà Vu', 'Hotel Transylvania 2', 'Fantasia 2000', 'The Time Machine', 'Mighty Joe Young', 'Swordfish', 'The Legend of Zorro', 'What Dreams May Come', 'Little Nicky', 'The Brothers Grimm', 'Mars Attacks!', 'Evolution', 'The Edge', 'Surrogates', 'Thirteen Days', 'Daylight', 'Walking With Dinosaurs', 'Battlefield Earth', 'Looney Tunes: Back in Action', 'Nine', 'Timeline', 'The Postman', 'Babe: Pig in the City', 'The Last Witch Hunter', 'Red Planet', 'Arthur and the Invisibles', 'Oceans', 'A Sound of Thunder', 'Pompeii', 'Top Cat Begins', 'A Beautiful Mind', 'The Lion King', 'Journey 2: The Mysterious Island', 'Cloudy with a Chance of Meatballs 2', 'Red Dragon', 'Hidalgo', 'Jack and Jill', '2 Fast 2 Furious', 'The Little Prince', 'The Invasion', 'The Adventures of Rocky & Bullwinkle', 'The Secret Life of Pets', 'The League of Extraordinary Gentlemen', 'Despicable Me 2', 'Independence Day', 'The Lost World: Jurassic Park', 'Madagascar', 'Children of Men', 'X-Men', 'Wanted', 'The Rock', 'Ice Age: The Meltdown', '50 First Dates', 'Hairspray', 'Exorcist: The Beginning', 'Inspector Gadget', 'Now You See Me', 'Grown Ups', 'The Terminal', 'Hotel for Dogs', 'Vertical Limit', "Charlie Wilson's War", 'Shark Tale', 'Dreamgirls', 'Be Cool', 'Munich', 'Tears of the Sun', 'Killers', 'The Man from U.N.C.L.E.', 'Spanglish', 'Monster House', 'Bandits', 'First Knight', 'Anna and the King', 'Immortals', 'Hostage', 'Titan A.E.', 'Hollywood Homicide', 'Soldier', 'Carriers', 'Monkeybone', 'Flight of the Phoenix', 'Unbreakable', 'Minions', 'Sucker Punch', 'Snake Eyes', 'Sphere', 'The Angry Birds Movie', "Fool's Gold", 'Funny People', 'The Kingdom', 'Talladega Nights: The Ballad of Ricky Bobby', 'Dr. Dolittle 2', 'Braveheart', 'Jarhead', 'The Simpsons Movie', 'The Majestic', 'Driven', 'Two Brothers', 'The Village', 'Doctor Dolittle', 'Signs', 'Shrek 2', 'Cars', 'Runaway Bride', 'xXx', 'The SpongeBob Movie: Sponge Out of Water', 'Ransom', 'Inglourious Basterds', 'Hook', 'Die Hard 2', 'S.W.A.T.', 'Vanilla Sky', 'Lady in the Water', 'AVP: Alien vs. Predator', 'Alvin and the Chipmunks: The Squeakquel', 'We Were Soldiers', 'Olympus Has Fallen', 'Star Trek: Insurrection', 'Battle: Los Angeles', 'Big Fish', 'Wolf', 'War Horse', 'The Monuments Men', 'The Abyss', 'Wall Street: Money Never Sleeps', 'Dracula Untold', 'The Siege', 'Stardust', 'Seven Years in Tibet', 'The Dilemma', 'Bad Company', 'Doom', 'I Spy', 'Underworld: Awakening', 'Rock of Ages', "Hart's War", 'Killer Elite', 'Rollerball', 'Ballistic: Ecks vs. Sever', 'Hard Rain', 'Osmosis Jones', "Legends of Oz: Dorothy's Return", 'Blackhat', 'Sky Captain and the World of Tomorrow', 'Basic Instinct 2', 'Escape Plan', 'The Legend of Hercules', 'The Sum of All Fears', 'The Twilight Saga: Eclipse', 'The Score', 'Despicable Me', 'Money Train', 'Ted 2', 'Agora', 'Mystery Men', 'Hall Pass', 'The Insider', 'The Finest Hours', 'Body of Lies', 'Dinner for Schmucks', 'Abraham Lincoln: Vampire Hunter', 'Entrapment', 'The X Files', 'The Last Legion', 'Saving Private Ryan', 'Need for Speed', 'What Women Want', 'Ice Age', 'Dreamcatcher', 'Lincoln', 'The Matrix', 'Apollo 13', 'The Santa Clause 2', 'Les Misérables', "You've Got Mail", 'Step Brothers', 'The Mask of Zorro', 'Due Date', 'Unbroken', 'Space Cowboys', 'Cliffhanger', 'Broken Arrow', 'The Kid', 'World Trade Center', 'Mona Lisa Smile', 'The Dictator', 'Eyes Wide Shut', 'Annie', 'Focus', 'This Means War', 'Blade: Trinity', 'Red Dawn', 'Primary Colors', 'Resident Evil: Retribution', 'Death Race', 'The Long Kiss Goodnight', 'Proof of Life', 'Zathura: A Space Adventure', 'Fight Club', 'We Are Marshall', 'Hudson Hawk', 'Lucky Numbers', 'I, Frankenstein', 'Oliver Twist', 'Elektra', 'Sin City: A Dame to Kill For', 'Random Hearts', 'Everest', 'Perfume: The Story of a Murderer', 'Austin Powers in Goldmember', 'Astro Boy', 'Jurassic Park', 'Wyatt Earp', 'Clear and Present Danger', 'Dragon Blade', 'Little Man', 'U-571', 'The American President', 'The Love Guru', '3000 Miles to Graceland', 'The Hateful Eight', 'Blades of Glory', 'Hop', '300',

'Meet the Fockers', 'Marley & Me', 'The Green Mile', 'Wild Hogs', 'Chicken Little', 'Gone Girl', 'The Bourne Identity', 'GoldenEye', "The General's Daughter", 'The Truman Show', 'The Prince of Egypt', 'Daddy Day Care', '2 Guns', 'Cats & Dogs', 'The Italian Job', 'Two Weeks Notice', 'Antz', 'Couples Retreat', 'Days of Thunder', 'Cheaper by the Dozen 2', 'Maze Runner: The Scorch Trials', 'Eat Pray Love', 'The Family Man', 'RED', 'Any Given Sunday', 'The Horse Whisperer', 'Collateral', 'The Scorpion King', 'Ladder 49', 'Jack Reacher', 'Deep Blue Sea', 'This Is It', 'Contagion', 'Kangaroo Jack', 'Coraline', 'The Happening', 'Man on Fire', 'The Shaggy Dog', 'Starsky & Hutch', 'Jingle All the Way', 'Hellboy', 'A Civil Action', 'ParaNorman', 'The Jackal', 'Paycheck', 'Up Close & Personal', 'The Tale of Despereaux', 'The Tuxedo', 'Under Siege 2: Dark Territory', 'Jack Ryan: Shadow Recruit', 'Joy', 'London Has Fallen', 'Alien: Resurrection', 'Shooter', 'The Boxtrolls', 'Practical Magic', 'The Lego Movie', 'Miss Congeniality 2: Armed and Fabulous', 'Reign of Fire', 'Gangster Squad', 'Year One', 'Invictus', 'State of Play', 'Duplicity', 'My Favorite Martian', 'The Sentinel', 'Planet 51', 'Star Trek: Nemesis', 'Intolerable Cruelty', 'Trouble with the Curve', 'Edge of Darkness', 'The Relic', 'Analyze That', 'Righteous Kill', 'Mercury Rising', 'The Soloist', 'The Legend of Bagger Vance', 'Almost Famous', 'Garfield: A Tail of Two Kitties', 'xXx: State of the Union', 'Priest', 'Sinbad: Legend of the Seven Seas', 'Event Horizon', 'Dragonfly', 'The Black Dahlia', 'Flyboys', 'The Last Castle', 'Supernova', "Winter's Tale", 'The Mortal Instruments: City of Bones', 'Meet Dave', 'Dark Water', 'Edtv', 'Inkheart', 'The Spirit', 'Mortdecai', 'In the Name of the King: A Dungeon Siege Tale', 'Beyond Borders', 'The Monkey King 2', 'The Great Raid', 'Deadpool', 'Holy Man', 'American Sniper', 'Goosebumps', 'Just Like Heaven', 'The Flintstones in Viva Rock Vegas', 'Rambo III', 'Leatherheads', 'The Ridiculous 6', 'Did You Hear About the Morgans?', 'The Internship', 'Resident Evil: Afterlife', 'Red Tails', "The Devil's Advocate", "That's My Boy", 'DragonHeart', 'After the Sunset', 'Ghost Rider: Spirit of Vengeance', "Captain Corelli's Mandolin", 'The Pacifier', 'Walking Tall', 'Forrest Gump', 'Alvin and the Chipmunks', 'Meet the Parents', 'Pocahontas', 'Superman', 'The Nutty Professor', 'Hitch', 'George of the Jungle', 'American Wedding', 'Captain Phillips', 'Date Night', 'Casper', 'The Equalizer', 'Maid in Manhattan', 'Crimson Tide', 'The Pursuit of Happyness', 'Flightplan', 'Disclosure', 'City of Angels', 'Kill Bill: Vol. 1', 'Bowfinger', 'Kill Bill: Vol. 2', 'Tango & Cash', 'Death Becomes Her', 'Shanghai Noon', 'Executive Decision', "Mr. Popper's Penguins", 'The Forbidden Kingdom', 'Free Birds', 'Alien³', 'Evita', 'Ronin', 'The Ghost and the Darkness', 'Paddington', 'The Watch', 'The Hunted', 'Instinct', 'Stuck on You', 'Semi-Pro', 'The Pirates! In an Adventure with Scientists!', 'Changeling', 'Chain Reaction', 'The Fan', 'The Phantom of the Opera', 'Elizabeth: The Golden Age', 'Æon Flux', 'Gods and Generals', 'Turbulence', 'Imagine That', 'Muppets Most Wanted', 'Thunderbirds', 'Burlesque', 'A Very Long Engagement', 'Lolita', 'D-Tox', 'Blade II', 'Seven Pounds', 'Bullet to the Head', 'The Godfather: Part III', 'Elizabethtown', 'You, Me and Dupree', 'Superman II', 'Gigli', "All the King's Men", 'Shaft', 'Anastasia', 'Moulin Rouge!', 'Domestic Disturbance', 'Black Mass', 'Flags of Our Fathers', 'Law Abiding Citizen', 'Grindhouse', 'Beloved', 'Lucky You', 'Catch Me If You Can', 'Zero Dark Thirty', 'The Break-Up', 'Mamma Mia!', "Valentine's Day", 'The Dukes of Hazzard', 'The Thin Red Line', 'The Change-Up', 'Man on the Moon', 'Casino', 'From Paris with Love', 'Bulletproof Monk', 'Me, Myself & Irene', 'Barnyard', 'Deck the Halls', 'The Twilight Saga: New Moon', 'Shrek', 'The Adjustment Bureau', 'Robin Hood: Prince of Thieves', 'Jerry Maguire', 'Ted', 'As Good as It Gets', 'Patch Adams', 'Anchorman 2: The Legend Continues', 'Mr. Deeds', 'Super 8', 'Erin Brockovich', 'How to Lose a Guy in 10 Days', '22 Jump Street', 'Interview with the Vampire', 'Yes Man', 'Central Intelligence', 'Stepmom', "Daddy's Home", 'Into the Woods', 'Inside Man', 'Payback', 'Congo', 'We Bought a Zoo', 'Knowing', 'Failure to Launch', 'The Ring Two', 'Crazy, Stupid, Love.', 'Garfield', 'Christmas with the Kranks', 'Moneyball', 'Outbreak', 'Non-Stop', 'Race to Witch Mountain', 'V for Vendetta', 'Shanghai Knights', 'Curious George', 'Herbie Fully Loaded', "Don't Say a Word", 'Hansel & Gretel: Witch Hunters', 'Unfaithful', 'I Am Number Four', 'Syriana', '13 Hours: The Secre

t Soldiers of Benghazi', 'The Book of Life', 'Firewall', 'Absolute Power', 'G.I. Jan
e', 'The Game', 'Silent Hill', 'The Replacements', 'American Reunion', 'The Negotiat
or', 'Into the Storm', 'Beverly Hills Cop III', 'Gremlins 2: The New Batch', 'The Ju
dge', 'The Peacemaker', 'Resident Evil: Apocalypse', 'Bridget Jones: The Edge of Rea
son', 'Out of Time', 'On Deadly Ground', 'The Adventures of Sharkboy and Lavagirl',
'The Beach', 'Raising Helen', 'Ninja Assassin', 'For Love of the Game', 'Stripteas
e', 'Marmaduke', 'Hereafter', 'Murder by Numbers', 'Assassins', 'Hannibal Rising',
'The Story of Us', 'The Host', 'Basic', 'Blood Work', 'The International', 'Escape f
rom L.A.', 'The Iron Giant', 'The Life Aquatic with Steve Zissou', 'Free State of Jo
nes', 'The Life of David Gale', 'Man of the House', 'Run All Night', 'Eastern Promis
es', 'Into the Blue', 'The Messenger: The Story of Joan of Arc', 'Your Highness', 'D
ream House', 'Mad City', "Baby's Day Out", 'The Scarlet Letter', 'Fair Game', 'Domin
o', 'Jade', 'Gamer', 'Beautiful Creatures', 'Death to Smoochy', 'Zoolander 2', 'The
Big Bounce', 'What Planet Are You From?', 'Drive Angry', 'Street Fighter: The Legend
of Chun-Li', 'The One', 'The Adventures of Ford Fairlane', 'The Boat That Rocked',
'Traffic', 'Indiana Jones and the Last Crusade', 'Anna Karenina', 'Chappie', 'The Bo
ne Collector', 'Panic Room', 'The Tooth Fairy', 'Three Kings', 'Child 44', 'Rat Rac
e', 'K-PAX', 'Kate & Leopold', 'Bedazzled', 'The Cotton Club', '3:10 to Yuma', 'Take
n 3', 'Out of Sight', 'The Cable Guy', 'Earth', 'Dick Tracy', 'The Thomas Crown Affa
ir', 'Riding in Cars with Boys', 'First Blood', 'Solaris', "Happily N'Ever After",
'Mary Reilly', "My Best Friend's Wedding", "America's Sweethearts", 'Insomnia', 'Sta
r Trek: First Contact', 'Jonah Hex', 'Courage Under Fire', 'Liar Liar', 'The Infiltr
ator', 'Inchon', 'The Flintstones', 'Taken 2', 'Scary Movie 3', 'Miss Congeniality',
'Journey to the Center of the Earth', 'The Princess Diaries 2: Royal Engagement', 'T
he Pelican Brief', 'The Client', 'The Bucket List', 'Patriot Games', 'Monster-in-La
w', 'Prisoners', 'Training Day', 'Galaxy Quest', 'Scary Movie 2', 'The Muppets', 'Bl
ade', 'Coach Carter', 'Changing Lanes', 'Anaconda', 'Coyote Ugly', 'Love Actually',
"A Bug's Life", 'From Hell', 'The Specialist', 'Tin Cup', 'Yours, Mine and Ours', 'K
icking & Screaming', "The Hitchhiker's Guide to the Galaxy", 'Fat Albert', 'Resident
Evil: Extinction', 'Blended', 'Last Holiday', 'The River Wild', 'The Indian in the C
upboard', 'Savages', 'Cellular', 'Johnny English', 'The Ant Bully', 'Dune', 'Across
the Universe', 'Revolutionary Road', '16 Blocks', 'Babylon A.D.', 'The Glimmer Man',
'Multiplicity', 'Aliens in the Attic', 'The Pledge', 'The Producers', 'The Phantom',
'All the Pretty Horses', 'Nixon', 'The Ghost Writer', 'Deep Rising', 'Miracle at St.
Anna', 'Curse of the Golden Flower', 'Bangkok Dangerous', 'Big Trouble', 'Love in th
e Time of Cholera', 'Shadow Conspiracy', 'Johnny English Reborn', 'Foodfight!', 'Arg
o', 'The Fugitive', 'The Bounty Hunter', 'Sleepers', 'Rambo: First Blood Part II',
'The Juror', 'Pinocchio', "Heaven's Gate", 'Underworld: Evolution', 'Victor Frankens
tein', 'Finding Forrester', '28 Days', 'Unleashed', 'The Sweetest Thing', 'The Fir
m', 'Charlie St. Cloud', 'The Mechanic', '21 Jump Street', 'Notting Hill', 'Chicken
Run', 'Along Came Polly', 'Boomerang', 'The Heat', 'Cleopatra', 'Here Comes the Boo
m', 'High Crimes', 'The Mirror Has Two Faces', 'The Mothman Prophecies', 'Brüno', 'L
icence to Kill', 'Red Riding Hood', '15 Minutes', 'Super Mario Bros.', 'Lord of Wa
r', 'Hero', 'One for the Money', 'The Interview', "The Warrior's Way", 'Micmacs', '8
Mile', 'Why I Did (Not) Eat My Father', "A Knight's Tale", 'The Medallion', 'The Six
th Sense', 'Man on a Ledge', 'The Big Year', 'The Karate Kid', 'American Hustle', 'T
he Proposal', 'Double Jeopardy', 'Back to the Future Part II', 'Lucy', 'Fifty Shades
of Grey', 'Spy Kids 3-D: Game Over', 'A Time to Kill', 'Cheaper by the Dozen', 'Lone
Survivor', 'A League of Their Own', 'The Conjuring 2', 'The Social Network', "He's J
ust Not That Into You", 'Scary Movie 4', 'Scream 3', 'Back to the Future Part III',
'Get Hard', 'Dracula', 'Julie & Julia', '42', 'The Talented Mr. Ripley', 'Dumb and D
umber To', 'Eight Below', 'The Intern', 'Ride Along 2', 'The Last of the Mohicans',
'Ray', 'Sin City', 'Vantage Point', 'I Love You, Man', 'Shallow Hal', 'JFK', "Big Mo
mma's House 2", 'The Mexican', '17 Again', 'The Other Woman', 'The Final Destinatio
n', 'Bridge of Spies', 'Behind Enemy Lines', 'Get Him to the Greek', 'Shall We Danc
e?', 'Small Soldiers', 'Spawn', 'The Count of Monte Cristo', 'The Lincoln Lawyer',

'Unknown', 'The Prestige', 'Horrible Bosses 2', 'Escape from Planet Earth', 'Apocaly
pto', 'The Living Daylights', 'Predators', 'Legal Eagles', 'Secret Window', 'The Lak
e House', 'The Skeleton Key', 'The Odd Life of Timothy Green', 'Made of Honor', 'Jer
sey Boys', 'The Rainmaker', 'Gothika', 'Amistad', 'Medicine Man', 'Aliens vs Predato
r: Requiem', 'Ri¢hie Ri¢h', 'Autumn in New York', 'Music and Lyrics', 'Paul', 'The G
uilt Trip', 'Scream 4', '8MM', 'The Doors', 'Sex Tape', 'Hanging Up', 'Final Destina
tion 5', 'Mickey Blue Eyes', 'Pay It Forward', 'Fever Pitch', 'Drillbit Taylor', 'A
Million Ways to Die in the West', 'The Shadow', 'Extremely Loud & Incredibly Close',
'Morning Glory', "Get Rich or Die Tryin'", 'The Art of War', 'Rent', 'Bless the Chil
d', 'The Out-of-Towners', 'The Island of Dr. Moreau', 'The Musketeer', 'The Other Bo
leyn Girl', 'Sweet November', 'The Reaping', 'Mean Streets', 'Renaissance Man', 'Col
ombiana', 'Quest for Camelot', 'City By The Sea', 'At First Sight', 'Torque', 'City
Hall', 'Showgirls', 'Marie Antoinette', 'Kiss of Death', 'Get Carter', 'The Impossib
le', 'Ishtar', 'Fantastic Mr. Fox', 'Life or Something Like It', 'Memoirs of an Invi
sible Man', 'Amélie', 'New York Minute', 'Alfie', 'Big Miracle', 'The Deep End of th
e Ocean', 'FearDotCom', "Cirque du Freak: The Vampire's Assistant", 'Duplex', 'Soul
Men', 'Raise the Titanic', 'Universal Soldier: The Return', 'Pandorum', 'Impostor',
'Extreme Ops', 'Just Visiting', 'Sunshine', 'A Thousand Words', 'Delgo', 'The Gunma
n', 'Stormbreaker', 'Disturbia', 'Hackers', 'The Hunting Party', 'The Hudsucker Prox
y', 'The Warlords', 'Nomad: The Warrior', 'Snowpiercer', 'A Monster in Paris', 'The
Last Shot', 'The Crow', 'Baahubali: The Beginning', "The Time Traveler's Wife", 'Bec
ause I Said So', 'Frankenweenie', 'Serenity', 'Against the Ropes', 'Superman III',
'Grudge Match', 'Red Cliff', 'Sweet Home Alabama', 'The Ugly Truth', 'Sgt. Bilko',
'Spy Kids 2: The Island of Lost Dreams', 'Star Trek: Generations', 'The Grandmaste
r', 'Water for Elephants', "Dragon Nest: Warriors' Dawn", 'The Hurricane', 'Enough',
'Heartbreakers', 'Paul Blart: Mall Cop 2', 'Angel Eyes', 'Joe Somebody', 'The Ninth
Gate', 'Extreme Measures', 'Rock Star', 'Precious', 'White Squall', 'The Thing', 'Ri
ddick', 'Switchback', 'Texas Rangers', 'City of Ember', 'The Master', 'Virgin Territ
ory', 'The Express', 'The 5th Wave', 'Creed', 'The Town', "What to Expect When You'r
e Expecting", 'Burn After Reading', "Nim's Island", 'Rush', 'Magnolia', 'Cop Out',
'How to Be Single', 'Dolphin Tale', 'Twilight', 'John Q', 'Blue Streak', "We're the
Millers", 'The Inhabited Island', 'Breakdown', 'Never Say Never Again', 'Hot Tub Tim
e Machine', 'Dolphin Tale 2', 'Reindeer Games', 'A Man Apart', 'Aloha', 'Ghosts of M
ississippi', 'Snow Falling on Cedars', 'The Rite', 'Gattaca', "Isn't She Great", 'Sp
ace Chimps', 'Head of State', 'The Hangover', 'Ip Man 3', 'Austin Powers: The Spy Wh
o Shagged Me', 'Batman', 'There Be Dragons', 'Lethal Weapon 3', 'The Blind Side', 'S
py Kids', 'Horrible Bosses', 'True Grit', 'The Devil Wears Prada', 'Star Trek: The M
otion Picture', 'Identity Thief', 'Cape Fear', '21', 'Trainwreck', 'Guess Who', 'The
English Patient', 'L.A. Confidential', 'Sky High', 'In & Out', 'Species', 'A Nightma
re on Elm Street', 'The Cell', 'The Man in the Iron Mask', 'Secretariat', 'TMNT', 'R
adio', 'Friends with Benefits', 'Neighbors 2: Sorority Rising', 'Saving Mr. Banks',
'Malcolm X', 'This Is 40', 'Old Dogs', 'Underworld: Rise of the Lycans', 'License to
Wed', 'The Benchwarmers', 'Must Love Dogs', 'Donnie Brasco', 'Resident Evil', 'Polte
rgeist', 'The Ladykillers', 'Max Payne', 'In Time', 'The Back-Up Plan', 'Something B
orrowed', 'Black Knight', 'The Bad News Bears', 'Street Fighter', 'The Pianist', 'Th
e Nativity Story', 'House of Wax', 'Closer', 'J. Edgar', 'Mirrors', 'Queen of the Da
mned', 'Predator 2', 'Untraceable', 'Blast from the Past', 'Flash Gordon', 'Jersey G
irl', 'Alex Cross', 'Midnight in the Garden of Good and Evil', 'Heist', 'Nanny McPhe
e and the Big Bang', 'Hoffa', 'The X Files: I Want to Believe', 'Ella Enchanted', 'C
oncussion', 'Abduction', 'Valiant', 'Wonder Boys', 'Superhero Movie', 'Broken City',
'Cursed', 'Premium Rush', 'Hot Pursuit', 'The Four Feathers', 'Parker', 'Wimbledon',
'Furry Vengeance', 'Bait', 'Krull', 'Lions for Lambs', 'Flight of the Intruder', 'Wa
lk Hard: The Dewey Cox Story', 'The Shipping News', 'American Outlaws', 'The Young V
ictoria', 'Whiteout', 'The Tree of Life', 'Knock Off', 'Sabotage', 'The Order', 'Pun
isher: War Zone', 'Zoom', 'The Walk', 'Warriors of Virtue', 'A Good Year', 'Radio Fl
yer', 'Bound by Honor', "Smilla's Sense of Snow", 'Femme Fatale', 'Lion of the Deser

t', 'The Horseman on the Roof', 'Ride with the Devil', 'Biutiful', 'Bandidas', 'Blac
k Water Transit', 'The Maze Runner', 'Unfinished Business', 'The Age of Innocence',
'The Fountain', 'Chill Factor', 'Stolen', 'Ponyo', 'The Longest Ride', "The Astronau
t's Wife", 'I Dreamed of Africa', 'Playing for Keeps', 'Mandela: Long Walk to Freedo
m', 'Reds', 'A Few Good Men', 'Exit Wounds', "Big Momma's House", 'Thunder and the H
ouse of Magic', 'The Darkest Hour', 'Step Up Revolution', 'Snakes on a Plane', 'The
Watcher', 'The Punisher', 'Goal!: The Dream Begins', 'Safe', 'Pushing Tin', 'Return
of the Jedi', 'Doomsday', 'The Reader', 'Wanderlust', 'Elf', 'Phenomenon', 'Snow Dog
s', 'Scrooged', 'Nacho Libre', 'Bridesmaids', 'This Is the End', 'Stigmata', 'Men of
Honor', 'Takers', 'The Big Wedding', 'Big Mommas: Like Father, Like Son', 'Source Co
de', 'Alive', 'The Number 23', 'The Young and Prodigious T.S. Spivet', '1941', 'Drea
mer: Inspired By a True Story', 'A History of Violence', 'Transporter 2', 'The Quick
and the Dead', 'Laws of Attraction', 'Bringing Out the Dead', 'Repo Men', 'Dragon Wa
rs: D-War', 'Bogus', 'The Incredible Burt Wonderstone', "Cats Don't Dance", 'Cradle
Will Rock', 'The Good German', 'George and the Dragon', 'Apocalypse Now', 'Going the
Distance', "Mr. Holland's Opus", 'Criminal', 'Out of Africa', 'Flight', 'Moonraker',
'The Grand Budapest Hotel', 'Hearts in Atlantis', 'Arachnophobia', 'Frequency', 'Vac
ation', 'Get Shorty', 'Chicago', 'Big Daddy', 'American Pie 2', 'Toy Story', 'Spee
d', 'The Vow', 'Extraordinary Measures', 'Remember the Titans', 'The Hunt for Red Oc
tober', 'The Butler', 'DodgeBall: A True Underdog Story', 'The Addams Family', 'Ace
Ventura: When Nature Calls', 'The Princess Diaries', 'The First Wives Club', 'Se7e
n', 'District 9', 'The SpongeBob SquarePants Movie', 'Mystic River', 'Million Dollar
Baby', 'Analyze This', 'The Notebook', '27 Dresses', 'Hannah Montana: The Movie', 'R
ugrats in Paris: The Movie', 'The Prince of Tides', 'Legends of the Fall', 'Up in th
e Air', 'About Schmidt', 'Warm Bodies', 'Looper', 'Down to Earth', 'Babe', 'Hope Spr
ings', 'Forgetting Sarah Marshall', 'Four Brothers', 'Baby Mama', 'Hope Floats', 'Br
ide Wars', 'Without a Paddle', '13 Going on 30', 'Midnight in Paris', 'The Nut Job',
'Blow', 'Message in a Bottle', 'Star Trek V: The Final Frontier', 'Like Mike', 'The
Naked Gun 33⅓: The Final Insult', 'A View to a Kill', 'The Curse of the Were-Rabbi
t', 'P.S. I Love You', 'Racing Stripes', 'Atonement', 'Letters to Juliet', 'Black Ra
in', 'The Three Stooges', 'Corpse Bride', 'Glory Road', 'Sicario', 'Southpaw', 'Drag
Me to Hell', 'The Age of Adaline', 'Secondhand Lions', 'Step Up 3D', 'Blue Crush',
'Stranger Than Fiction', '30 Days of Night', 'The Cabin in the Woods', 'Meet the Spa
rtans', 'Midnight Run', 'The Running Man', 'Little Shop of Horrors', 'Hanna', 'Morta
l Kombat: Annihilation', 'Larry Crowne', 'Carrie', 'Take the Lead', 'Gridiron Gang',
"What's the Worst That Could Happen?", '9', 'Side Effects', 'The Prince & Me', 'Winn
ie the Pooh', 'Dumb and Dumberer: When Harry Met Lloyd', 'Bulworth', 'Get on Up', 'O
ne True Thing', 'Virtuosity', 'My Super Ex-Girlfriend', 'Deliver Us from Evil', 'San
ctum', 'Little Black Book', 'The Five-Year Engagement', 'Mr. 3000', 'The Next Three
Days', 'Ultraviolet', 'Assault on Precinct 13', 'The Replacement Killers', 'Fled',
'Eight Legged Freaks', 'Love & Other Drugs', '88 Minutes', 'North Country', 'The Who
le Ten Yards', 'Splice', 'Howard the Duck', 'Pride and Glory', 'The Cave', 'Alex & E
mma', 'Wicker Park', 'Fright Night', 'The New World', 'Wing Commander', 'In Dreams',
'Dragonball Evolution', 'The Last Stand', 'Godsend', 'Chasing Liberty', 'Hoodwinked
Too! Hood VS. Evil', 'An Unfinished Life', 'The Imaginarium of Doctor Parnassus', "B
arney's Version", 'Runner Runner', 'Antitrust', 'Glory', 'Once Upon a Time in Americ
a', 'Dead Man Down', 'The Merchant of Venice', 'The Good Thief', 'Supercross', 'Miss
Potter', 'The Promise', 'DOA: Dead or Alive', 'The Assassination of Jesse James by t
he Coward Robert Ford', '1911', 'Little Nicholas', 'Wild Card', 'Machine Gun Preache
r', 'Animals United', 'Goodbye Bafana', 'United Passions', 'Grace of Monaco', 'Savv
a. Heart of the Warrior', "Ripley's Game", 'Sausage Party', 'Pitch Perfect 2', 'Walk
the Line', 'Keeping the Faith', 'The Borrowers', 'Frost/Nixon', 'Confessions of a Da
ngerous Mind', 'Serving Sara', 'The Boss', 'Cry Freedom', 'Mumford', 'Seed of Chuck
y', 'The Jacket', 'Aladdin', 'Straight Outta Compton', 'Indiana Jones and the Temple
of Doom', 'The Rugrats Movie', 'Along Came a Spider', 'Florence Foster Jenkins', 'On
ce Upon a Time in Mexico', 'Die Hard', 'Role Models', 'The Big Short', 'Taking Woods

```
tock', 'Miracle', 'Dawn of the Dead', 'The Wedding Planner', 'Space Pirate Captain H
arlock', 'The Royal Tenenbaums', 'Identity', 'Last Vegas', 'For Your Eyes Only', 'Se
rendipity', 'Timecop', 'Zoolander', 'Safe Haven', 'Hocus Pocus', 'No Reservations',
'Kick-Ass', '30 Minutes or Less', 'Dracula 2000', 'Alexander and the Terrible, Horri
ble, No Good, Very Bad Day', 'Pride & Prejudice', 'Blade Runner', 'Rob Roy', '3 Days
to Kill', 'We Own the Night', 'Lost Souls', 'Winged Migration', 'Just My Luck', 'Mys
tery, Alaska', 'The Spy Next Door', 'A Simple Wish', 'Ghosts of Mars', 'Our Brand Is
Crisis', 'Pride and Prejudice and Zombies', 'Kundun', 'How to Lose Friends & Alienat
e People', 'Kick-Ass 2', 'Alatriste', 'Brick Mansions', 'Octopussy', 'Knocked Up',
"My Sister's Keeper", 'Welcome Home Roscoe Jenkins', 'A Passage to India', 'Notes on
a Scandal', 'Rendition', 'Star Trek VI: The Undiscovered Country', 'Divine Secrets o
f the Ya-Ya Sisterhood', 'Kiss the Girls', 'The Blues Brothers', 'The Sisterhood of
the Traveling Pants 2', 'Joyful Noise', 'About a Boy', 'Lake Placid', 'Lucky Number
Slevin', 'The Right Stuff', 'Anonymous', 'The NeverEnding Story', 'Dark City', 'The
Duchess', 'Return to Oz', 'The Newton Boys', 'Case 39', 'Suspect Zero', 'Martian Chi
ld', 'Spy Kids: All the Time in the World', 'Money Monster', 'The 51st State', 'Flaw
less', 'Mindhunters', 'What Just Happened', 'The Statement', 'The Magic Flute', 'Pau
l Blart: Mall Cop', 'Freaky Friday', 'The 40 Year Old Virgin', 'Shakespeare in Lov
e', 'A Walk Among the Tombstones', 'Kindergarten Cop', 'Pineapple Express', 'Ever Af
ter: A Cinderella Story', 'Open Range', 'Flatliners', 'A Bridge Too Far', 'Red Eye',
'Final Destination 2', 'O Brother, Where Art Thou?', 'Legion', 'Pain & Gain', 'In Go
od Company', 'Clockstoppers', 'Silverado', 'Brothers', 'Agent Cody Banks 2: Destinat
ion London', "New Year's Eve", 'Original Sin', 'The Raven', 'Welcome to Mooseport',
'Highlander: The Final Dimension', 'Blood and Wine', 'Snow White: A Tale of Terror',
'The Curse of the Jade Scorpion', 'Accidental Love', 'Flipper', 'Self/less', 'The Co
nstant Gardener', 'The Passion of the Christ', 'Mrs. Doubtfire', 'Rain Man', 'Gran T
orino', 'W.', 'Taken', 'The Best of Me', 'The Bodyguard', "Schindler's List", 'The H
elp', 'The Fifth Estate', 'Scooby-Doo 2: Monsters Unleashed', 'Forbidden Kingdom',
'Freddy vs. Jason', 'The Face of an Angel', 'Jimmy Neutron: Boy Genius', 'Cloverfiel
d', 'Teenage Mutant Ninja Turtles II: The Secret of the Ooze', 'The Untouchables',
'No Country for Old Men', 'Ride Along', "Bridget Jones's Diary", 'Chocolat', 'Legall
y Blonde 2: Red, White & Blonde', 'Parental Guidance', 'No Strings Attached', 'Tombs
tone', 'Romeo Must Die', 'The Omen', 'Final Destination 3', 'The Lucky One', 'Bridge
to Terabithia', 'Finding Neverland', 'A Madea Christmas', 'The Grey', 'Hide and See
k', 'Anchorman: The Legend of Ron Burgundy', 'GoodFellas', 'Agent Cody Banks', 'Nann
y McPhee', 'Scarface', 'Nothing to Lose', 'The Last Emperor', 'Contraband', 'Money T
alks', 'There Will Be Blood', 'The Wild Thornberrys Movie', 'Rugrats Go Wild', 'Unde
rcover Brother', 'The Sisterhood of the Traveling Pants', 'Kiss of the Dragon', 'The
House Bunny', 'Beauty Shop', 'Million Dollar Arm', 'The Giver', 'What a Girl Wants',
'Jeepers Creepers 2', 'Good Luck Chuck', 'Cradle 2 the Grave', 'The Hours', "She's t
he Man", "Mr. Bean's Holiday", 'Anacondas: The Hunt for the Blood Orchid', 'Blood Ti
es', 'August Rush', 'Elizabeth', 'Bride of Chucky', 'Tora! Tora! Tora!', 'Spice Worl
d', 'The Sitter', 'Dance Flick', 'The Shawshank Redemption', 'Crocodile Dundee in Lo
s Angeles', 'Kingpin', 'The Gambler', 'August: Osage County', 'Ice Princess', 'A Lot
Like Love', 'Eddie the Eagle', 'He Got Game', 'Don Juan DeMarco', 'Dear John', 'The
Losers', "Don't Be Afraid of the Dark", 'War', 'Punch-Drunk Love', 'EuroTrip', 'Half
Past Dead', 'Unaccompanied Minors', 'Bright Lights, Big City', 'The Adventures of Pi
nocchio', 'The Greatest Game Ever Played', 'The Box', 'The Ruins', 'The Next Best Th
ing', 'My Soul to Take', 'The Girl Next Door', 'Maximum Risk', 'Stealing Harvard',
'Legend', 'Hot Rod', 'Shark Night', "Angela's Ashes", 'Draft Day', 'Lifeforce', 'The
Conspirator', 'Lords of Dogtown', 'The 33', 'Big Trouble in Little China', 'Fly Me t
o the Moon', 'Warrior', 'Michael Collins', 'Gettysburg', 'Stop-Loss', 'Abandon', 'Br
okedown Palace', 'The Possession', 'Mrs. Winterbourne', 'Straw Dogs', 'The Hoax', 'S
tone Cold', 'The Road', 'Sheena', 'Underclassman', "Say It Isn't So", "The World's F
astest Indian", 'Tank Girl', "King's Ransom", 'Blindness', 'BloodRayne', 'Carnage',
'Where the Truth Lies', 'Cirque du Soleil: Worlds Away', 'Without Limits', 'Me and O
```

rson Welles', 'The Best Offer', 'The Bad Lieutenant: Port of Call - New Orleans', "A Turtle's Tale: Sammy's Adventures", 'Little White Lies', 'Love Ranch', "The True Story of Puss 'n Boots", 'Space Dogs', 'The Counselor', 'Ironclad', 'Waterloo', 'Kung Fu Jungle', 'Red Sky', 'Dangerous Liaisons', 'On the Road', 'Star Trek IV: The Voyage Home', 'Rocky Balboa', 'Scream 2', 'Jane Got a Gun', 'Think Like a Man Too', 'The Whole Nine Yards', 'Footloose', 'Old School', 'The Fisher King', 'I Still Know What You Did Last Summer', 'Return to Me', 'Zack and Miri Make a Porno', 'Nurse Betty', 'The Men Who Stare at Goats', 'Double Take', 'Girl, Interrupted', 'Win a Date with Tad Hamilton!', 'Muppets from Space', 'The Wiz', 'Ready to Rumble', 'Play It to the Bone', "I Don't Know How She Does It", 'Piranha 3D', 'Beyond the Sea', 'Meet the Deedles', 'The Thief and the Cobbler', 'The Bridge of San Luis Rey', 'Faster', "Howl's Moving Castle", 'Zombieland', 'The Waterboy', 'The Empire Strikes Back', 'Bad Boys', 'The Naked Gun 2½: The Smell of Fear', 'Final Destination', 'The Ides of March', 'Pitch Black', 'Someone Like You...', 'Her', 'Joy Ride', 'The Adventurer: The Curse of the Midas Box', 'Anywhere But Here', 'The Crew', 'Haywire', 'Jaws: The Revenge', "Marvin's Room", 'The Longshots', 'The End of the Affair', 'Harley Davidson and the Marlboro Man', 'In the Valley of Elah', 'Coco Before Chanel', 'Forsaken', 'Cheri', 'Vanity Fair', 'Bodyguards and Assassins', '1408', 'Spaceballs', 'The Water Diviner', 'Ghost', "There's Something About Mary", 'The Santa Clause', 'The Rookie', 'The Game Plan', 'The Bridges of Madison County', 'The Animal', 'Gandhi', 'The Hundred-Foot Journey', 'The Net', 'I Am Sam', 'Son of God', 'Underworld', 'Derailed', 'The Informant!', 'Shadowlands', 'Deuce Bigalow: European Gigolo', 'Delivery Man', 'Our Kind of Traitor', 'Saving Silverman', 'Diary of a Wimpy Kid: Dog Days', 'Summer of Sam', 'Jay and Silent Bob Strike Back', 'The Glass House', 'Hail, Caesar!', 'Josie and the Pussycats', 'Homefront', 'The Little Vampire', 'I Heart Huckabees', 'RoboCop 3', 'Megiddo: The Omega Code 2', 'Darling Lili', 'Dudley Do-Right', 'The Transporter Refueled', 'The Libertine', 'Black Book', 'Joyeux Noël', 'Hit & Run', 'Mad Money', 'Before I Go to Sleep', 'Sorcerer', 'Stone', 'Moliere', 'Out of the Furnace', 'Michael Clayton', 'My Fellow Americans', 'Arlington Road', 'Underdogs', 'To Rome with Love', 'Firefox', 'South Park: Bigger, Longer & Uncut', 'Death at a Funeral', 'Teenage Mutant Ninja Turtles III', 'Hardball', 'Silver Linings Playbook', 'Freedom Writers', 'For Colored Girls', 'The Transporter', 'Never Back Down', 'The Rage: Carrie 2', 'Away We Go', 'Swing Vote', 'Moonlight Mile', 'Tinker Tailor Soldier Spy', 'Molly', 'The Beaver', 'The Best Little Whorehouse in Texas', 'eXistenZ', 'Raiders of the Lost Ark', 'Home Alone 2: Lost in New York', 'Close Encounters of the Third Kind', 'Pulse', 'Beverly Hills Cop II', 'Bringing Down the House', 'The Silence of the Lambs', "Wayne's World", 'Jackass 3D', 'Jaws 2', 'Beverly Hills Chihuahua', 'The Conjuring', 'Are We There Yet?', 'Tammy', 'School of Rock', 'Mortal Kombat', 'White Chicks', 'The Descendants', 'Holes', 'The Last Song', '12 Years a Slave', 'Drumline', 'Why Did I Get Married Too?', 'Edward Scissorhands', 'Me Before You', "Madea's Witness Protection", 'The French Connection', 'Bad Moms', 'Date Movie', 'Return to Never Land', 'Selma', 'The Jungle Book 2', 'Boogeyman', 'Premonition', 'The Tigger Movie', 'Orphan', 'Max', 'Epic Movie', 'Spotlight', 'Lakeview Terrace', 'The Grudge 2', 'How Stella Got Her Groove Back', "Bill & Ted's Bogus Journey", 'Man of the Year', 'The Black Hole', 'The American', 'Selena', 'Vampires Suck', 'Babel', 'This Is Where I Leave You', 'Doubt', 'Team America: World Police', 'Texas Chainsaw 3D', 'Copycat', 'Scary Movie 5', 'Paint Your Wagon', 'Milk', 'Risen', 'Ghost Ship', 'A Very Harold & Kumar Christmas', 'Wild Things', 'The Stepfather', 'The Debt', 'High Fidelity', 'One Missed Call', 'Eye for an Eye', 'The Bank Job', 'Eternal Sunshine of the Spotless Mind', 'You Again', 'Street Kings', "The World's End", 'Nancy Drew', 'Daybreakers', "She's Out of My League", 'Monte Carlo', 'Stay Alive', 'Quigley Down Under', 'Alpha and Omega', 'The Covenant', 'Stick It', 'Shorts', 'To Die For', 'Nerve', 'Appaloosa', 'Vampires', 'Psycho', "My Best Friend's Girl", 'Endless Love', 'Georgia Rule', 'Under the Rainbow', 'Ladyhawke', 'Simon Birch', 'Reign Over Me', 'Into the Wild', 'School for Scoundrels', 'Silent Hill: Revelation 3D', 'From Dusk Till Dawn', "Pooh's Heffalump Movie", 'Home for the Holidays', 'Kung Fu Hustle', 'The Country Bears', 'The Kite Runner', '21 Gram

```
s', 'Paparazzi', 'A Guy Thing', 'Loser', 'Capitalism: A Love Story', 'The Greatest S
tory Ever Told', 'Secret in Their Eyes', 'Disaster Movie', 'Armored', 'The Man Who K
new Too Little', "What's Your Number?", 'Lockout', 'Envy', 'Crank: High Voltage', 'B
ullets Over Broadway', 'One Night with the King', 'The Quiet American', 'The Weather
Man', 'Undisputed', 'Ghost Town', '12 Rounds', 'Let Me In', '3 Ninjas Kick Back', 'B
e Kind Rewind', 'Mrs Henderson Presents', 'Triple 9', 'Deconstructing Harry', 'Three
to Tango', 'Burnt', "We're No Angels", 'Everyone Says I Love You', 'Death Sentence',
"Everybody's Fine", 'Superbabies: Baby Geniuses 2', 'The Man', 'Code Name: The Clean
er', 'Connie and Carla', 'Sweet Charity', 'Inherent Vice', 'Doogal', 'Battle of the
Year', 'An American Carol', 'Machete Kills', 'Willard', 'Strange Wilderness', 'Topsy
-Turvy', 'Little Boy', 'A Dangerous Method', 'A Scanner Darkly', 'Chasing Maverick
s', 'Alone in the Dark', 'Bandslam', 'Birth', 'A Most Violent Year', 'Passchendael
e', 'Flash of Genius', "I'm Not There.", 'The Cold Light of Day', 'The Brothers Bloo
m', 'Synecdoche, New York', 'Princess Mononoke', 'Bon voyage', "Can't Stop the Musi
c", 'The Proposition', 'My All American', 'Marci X', 'Equilibrium', 'The Children of
Huang Shi', 'The Yards', 'The Oogieloves in the Big Balloon Adventure', 'By the Se
a', 'Steamboy', 'The Game of Their Lives', 'All Good Things', 'Rapa Nui', 'CJ7', 'Th
e Visitors II: The Corridors of Time', 'Dylan Dog: Dead of Night', 'People I Know',
'The Tempest', 'Regression', 'Three Kingdoms: Resurrection of the Dragon', 'Butterfl
y on a Wheel', 'Zambezia', 'Ramanujan', 'Dwegons', 'Hands of Stone', 'Survivor', 'Th
e Frozen Ground', 'The Painted Veil', 'The Baader Meinhof Complex', 'Dances with Wol
ves', 'Bad Teacher', 'Sea of Love', 'A Cinderella Story', 'Scream', 'Thir13en Ghost
s', 'The Shining', 'Back to the Future', 'House on Haunted Hill', 'I Can Do Bad All
By Myself', 'Fight Valley', 'The Switch', 'Just Married', "The Devil's Double", 'Tho
mas and the Magic Railroad', 'The Crazies', 'Spirited Away', 'Firestorm', 'The Bount
y', 'The Book Thief', 'Sex Drive', 'Leap Year', 'The Fall of the Roman Empire', 'Tak
e Me Home Tonight', "Won't Back Down", 'The Nutcracker', 'Kansas City', 'Indignatio
n', 'The Amityville Horror', 'Adaptation.', 'Land of the Dead', 'Out of Inferno', 'F
ear and Loathing in Las Vegas', 'The Invention of Lying', 'Neighbors', 'The Mask',
'Big', 'Borat: Cultural Learnings of America for Make Benefit Glorious Nation of Kaz
akhstan', 'Legally Blonde', 'Star Trek III: The Search for Spock', 'The Exorcism of
Emily Rose', 'Deuce Bigalow: Male Gigolo', 'Left Behind', 'The Family Stone', 'Barbe
rshop 2:  Back in Business', 'Bad Santa', 'Austin Powers: International Man of Myste
ry', 'My Big Fat Greek Wedding 2', 'Diary of a Wimpy Kid: Rodrick Rules', 'Predato
r', 'Amadeus', 'Prom Night', 'Mean Girls', 'Under the Tuscan Sun', 'Gosford Park',
'Peggy Sue Got Married', 'Birdman', 'Blue Jasmine', 'United 93', 'Honey', 'Spy Har
d', 'The Fog', 'Soul Surfer', 'Catch-22', 'Observe and Report', 'Conan the Destroye
r', 'Raging Bull', 'Love Happens', 'Young Sherlock Holmes', 'Fame', '127 Hours', 'Sm
all Time Crooks', 'Center Stage', 'Love the Coopers', 'Catch That Kid', 'Life as a H
ouse', 'Steve Jobs', 'I Love You, Beth Cooper', 'Youth in Revolt', 'The Legend of th
e Lone Ranger', 'The Tailor of Panama', 'Blow Out', 'Getaway', 'The Ice Storm', 'And
So It Goes', 'Troop Beverly Hills', 'Being Julia', 'Nine 1/2 Weeks', 'Dragonslayer',
'The Last Station', 'Ed Wood', 'Labor Day', 'Mongol: The Rise of Genghis Khan', 'Roc
kNRolla', 'Megaforce', 'Hamlet', "Mao's Last Dancer", 'Midnight Special', 'Anything
Else', 'The Railway Man', 'The White Ribbon', 'Restoration', 'The Wraith', 'Salton S
ea', 'Metallica: Through the Never', 'The Informers', 'Carlos', 'I Come with the Rai
n', "One Man's Hero", 'Day of the Dead', 'I Am Wrath', 'Renaissance', 'Red Sonja',
'Red Lights', 'Superbad', 'Madea Goes to Jail', 'Wolves', 'Step Up 2: The Streets',
'Hoodwinked!', 'Hotel Rwanda', 'Hitman', 'Black Nativity', 'The Prince', 'City of Gh
osts', 'The Others', 'Aliens', 'My Fair Lady', 'I Know What You Did Last Summer', "L
et's Be Cops", 'Sideways', 'Beerfest', 'Halloween', 'Good Boy!', 'The Best Man Holid
ay', "Smokin' Aces", 'Saw: The Final Chapter', '40 Days and 40 Nights', 'A Night at
the Roxbury', 'Beastly', 'The Hills Have Eyes', 'Dickie Roberts: Former Child Star',
'McFarland, USA', 'Lottery Ticket', 'ATL', 'Pitch Perfect', 'Summer Catch', 'A Simpl
e Plan', 'They', 'Larry the Cable Guy: Health Inspector', 'The Adventures of Elmo in
Grouchland', "Brooklyn's Finest", '55 Days at Peking', 'Evil Dead', 'My Life in Ruin
```

s', 'American Dreamz', 'Superman IV: The Quest for Peace', 'How She Move', 'Running
Scared', 'Bobby Jones: Stroke of Genius', 'Shanghai Surprise', 'The Illusionist', 'R
oar', 'Veronica Guerin', 'Escobar: Paradise Lost', 'Southland Tales', 'Dragon Hunter
s', 'Damnation Alley', 'The Apparition', 'My Girl', 'Fur: An Imaginary Portrait of D
iane Arbus', 'Wall Street', 'Sense and Sensibility', 'Becoming Jane', 'Sydney Whit
e', 'House of Sand and Fog', 'Dead Poets Society', 'Dumb and Dumber', 'When Harry Me
t Sally...', 'The Verdict', 'Road Trip', 'Varsity Blues', 'The Artist', 'The Unbor
n', 'Moonrise Kingdom', 'The Texas Chainsaw Massacre: The Beginning', 'The Young Mes
siah', 'The Master of Disguise', "Pan's Labyrinth", 'See Spot Run', 'Baby Boy', 'The
Roommate', 'Joe Dirt', 'Double Impact', 'Hot Fuzz', 'The Women', 'Vicky Cristina Bar
celona', 'Arn: The Knight Templar', 'Boys and Girls', 'White Oleander', "Jennifer's
Body", 'Drowning Mona', 'Radio Days', 'Remember Me', 'How to Deal', 'My Stepmother i
s an Alien', 'Philadelphia', 'The Thirteenth Floor', 'The Cookout', 'Meteor', 'Duet
s', 'Hollywood Ending', 'Detroit Rock City', 'Highlander', 'Things We Lost in the Fi
re', 'Steel', 'The Immigrant', 'The White Countess', 'Trance', 'Soul Plane', 'Welcom
e to the Sticks', 'Good', 'Enter the Void', 'Vamps', "Hachi: A Dog's Tale", 'Zulu',
'The Homesman', 'Juwanna Mann', 'Ararat', 'Madison', 'Slow Burn', 'Wasabi', 'Slithe
r', 'Beverly Hills Cop', 'Home Alone', 'Three Men and a Baby', 'Tootsie', 'Top Gun',
'Crouching Tiger, Hidden Dragon', 'American Beauty', "The King's Speech", 'Twins',
'The Yellow Handkerchief', 'The Color Purple', 'Tidal Wave', 'The Imitation Game',
'Private Benjamin', "Coal Miner's Daughter", 'Diary of a Wimpy Kid', 'Mama', "Nation
al Lampoon's Vacation", 'Bad Grandpa', 'The Queen', 'Beetlejuice', 'Why Did I Get Ma
rried?', 'Little Women', 'The Woman in Black', 'When a Stranger Calls', 'Big Fat Lia
r', 'The Deer Hunter', 'Wag the Dog', 'The Lizzie McGuire Movie', 'Snitch', 'Krampu
s', 'The Faculty', "What's Love Got to Do with It", 'Cop Land', 'Not Another Teen Mo
vie', 'End of Watch', 'The Skulls', 'The Theory of Everything', "Malibu's Most Wante
d", 'Where the Heart Is', 'Lawrence of Arabia', 'Halloween II', 'Wild', 'The Last Ho
use on the Left', 'The Wedding Date', 'Halloween: Resurrection', 'The Princess Brid
e', 'The Great Debaters', 'Drive', 'Confessions of a Teenage Drama Queen', 'The Obje
ct of My Affection', '28 Weeks Later', 'When the Game Stands Tall', 'Because of Winn
-Dixie', 'Love & Basketball', 'Grosse Pointe Blank', 'All About Steve', 'Book of Sha
dows: Blair Witch 2', 'The Craft', 'Match Point', 'Ramona and Beezus', 'The Remains
of the Day', 'Boogie Nights', 'Nowhere to Run', 'Flicka', 'The Hills Have Eyes 2',
'Urban Legends: Final Cut', 'Tuck Everlasting', 'The Marine', 'Keanu', 'Country Stro
ng', 'Disturbing Behavior', 'The Place Beyond the Pines', 'The November Man', 'Eye o
f the Beholder', 'The Hurt Locker', 'Firestarter', 'Killing Them Softly', 'A Most Wa
nted Man', 'Freddy Got Fingered', "VeggieTales: The Pirates Who Don't Do Anything",
'U2 3D', 'Highlander: Endgame', 'Idlewild', 'One Day', 'Whip It', 'Knockaround Guy
s', 'Confidence', 'The Muse', 'De-Lovely', 'New York Stories', "Barney's Great Adven
ture", 'The Man with the Iron Fists', 'Home Fries', 'Here On Earth', 'Brazil', 'Rais
e Your Voice', 'The Big Lebowski', 'Black Snake Moan', 'Dark Blue', 'A Mighty Hear
t', 'Whatever It Takes', 'Boat Trip', 'The Importance of Being Earnest', 'The Love L
etter', 'Hoot', 'In Bruges', 'Peeples', 'The Rocker', 'Post Grad', 'Promised Land',
'Whatever Works', 'The In Crowd', 'The Three Burials of Melquiades Estrada', 'Jakob
the Liar', 'Kiss Kiss Bang Bang', 'Idle Hands', 'Mulholland Drive', 'Blood and Choco
late', 'You Will Meet a Tall Dark Stranger', 'Never Let Me Go', 'The Company', 'Tran
ssiberian', 'The Clan of the Cave Bear', 'Crazy in Alabama', 'Funny Games', 'Listeni
ng', "Felicia's Journey", 'Metropolis', 'District B13', "Things to Do in Denver When
You're Dead", 'The Assassin', 'Buffalo Soldiers', 'The Return', 'Ong Bak 2', 'Centur
ion', 'Silent Trigger', 'The Midnight Meat Train', 'Winnie Mandela', 'The Son of No
One', "All The Queen's Men", 'The Good Night', 'Bathory: Countess of Blood', 'Khumb
a', 'Automata', 'Dungeons & Dragons: Wrath of the Dragon God', 'Chiamatemi Francesco
- Il Papa della gente', 'Shinjuku Incident', 'Pandaemonium', 'Groundhog Day', 'Magic
Mike XXL', 'Romeo + Juliet', "Sarah's Key", 'Freedom', 'Unforgiven', 'Manderlay', 'S
lumdog Millionaire', 'Fatal Attraction', 'Pretty Woman', 'Crocodile Dundee II', 'Bro
ken Horses', 'Born on the Fourth of July', 'Cool Runnings', 'My Bloody Valentine',

'Stomp the Yard', 'The Spy Who Loved Me', 'Urban Legend', 'Good Deeds', 'White Fan
g', 'Superstar', 'The Iron Lady', 'Jonah: A VeggieTales Movie', 'Poetic Justice', 'A
ll About the Benjamins', 'Vampire in Brooklyn', 'Exorcist II: The Heretic', 'An Amer
ican Haunting', "My Boss's Daughter", 'A Perfect Getaway', 'Our Family Wedding', 'De
ad Man on Campus', 'Tea with Mussolini', 'Thinner', 'New York, New York', 'Crookly
n', 'I Think I Love My Wife', 'Jason X', 'Bobby', 'Head Over Heels', 'Fun Size', 'Th
e Diving Bell and the Butterfly', 'Little Children', 'Gossip', 'A Walk on the Moon',
'Catch a Fire', 'Soul Survivors', 'Jefferson in Paris', 'Easy Virtue', 'Caravans',
'Mr. Turner', 'Wild Grass', 'Amen.', 'Reign of Assassins', 'The Lucky Ones', 'Margar
et', 'Stan Helsing', 'Flipped', 'Brokeback Mountain', 'Clueless', 'Far from Heaven',
'Hot Tub Time Machine 2', 'Quills', 'Seven Psychopaths', "The Caveman's Valentine",
'Downfall', 'The Sea Inside', 'Under the Skin', 'Good Morning, Vietnam', 'The Last G
odfather', 'Justin Bieber: Never Say Never', 'Black Swan', 'The Godfather: Part II',
'Save the Last Dance', 'A Nightmare on Elm Street 4: The Dream Master', 'Miracles fr
om Heaven', 'Dude, Where’s My Car?', 'Young Guns', 'St. Vincent', 'About Last Nigh
t', '10 Things I Hate About You', 'The New Guy', "National Lampoon's Loaded Weapon
1", 'The Shallows', 'The Butterfly Effect', 'Snow Day', 'This Christmas', 'Baby Geni
uses', 'The Big Hit', 'Harriet the Spy', "Child's Play 2", 'No Good Deed', 'The Mis
t', 'Ex Machina', 'Being John Malkovich', 'Two Can Play That Game', 'Earth to Echo',
'Crazy/Beautiful', 'Letters from Iwo Jima', 'The Astronaut Farmer', 'Woo', 'Room',
'Dirty Work', 'Serial Mom', 'Dick', 'Light It Up', '54', 'Bubble Boy', 'Birthday Gir
l', '21 & Over', "Paris, je t'aime", 'Resurrecting the Champ', 'Admission', 'The Wid
ow of Saint-Pierre', 'Chloe', 'Faithful', 'Find Me Guilty', 'The Perks of Being a Wa
llflower', 'Excessive Force', 'Infamous', 'The Claim', 'The Vatican Tapes', 'Attack
the Block', 'In the Land of Blood and Honey', 'The Call', 'Operation Chromite', 'The
Crocodile Hunter: Collision Course', 'I Love You Phillip Morris', 'Quest for Fire',
'Antwone Fisher', "The Emperor's Club", 'True Romance', 'Womb', 'Glengarry Glen Ros
s', 'The Killer Inside Me', 'Cat People', 'Sorority Row', 'The Prisoner of Zenda',
'Lars and the Real Girl', 'The Boy in the Striped Pyjamas', 'Dancer in the Dark', 'O
scar and Lucinda', 'The Funeral', 'Solitary Man', 'Machete', 'Casino Jack', 'The Lan
d Before Time', 'Tae Guk Gi: The Brotherhood of War', 'The Perfect Game', 'The Exorc
ist', 'Jaws', 'American Pie', 'Ernest & Celestine', 'The Golden Child', 'Think Like
a Man', 'Barbershop', 'Star Trek II: The Wrath of Khan', 'Ace Ventura: Pet Detectiv
e', 'WarGames', 'Witness', 'Act of Valor', 'Step Up', 'Beavis and Butt-Head Do Ameri
ca', 'Jackie Brown', 'Harold & Kumar Escape from Guantanamo Bay', 'Chronicle', 'Yent
l', 'Time Bandits', 'Crossroads', 'Project X', 'Patton', 'One Hour Photo', 'Quaranti
ne', 'The Eye', 'Johnson Family Vacation', 'How High', 'The Muppet Christmas Carol',
'Frida', 'Katy Perry: Part of Me', 'The Fault in Our Stars', 'Rounders', 'Top Five',
'Prophecy', 'Stir of Echoes', 'Philomena', 'The Upside of Anger', 'The Boys from Bra
zil', 'Aquamarine', 'Paper Towns', "My Baby's Daddy", 'Nebraska', 'Tales from the Cr
ypt: Demon Knight', "Max Keeble's Big Move", 'Young Adult', 'Crank', "Def Jam's How
to Be a Player", 'Living Out Loud', 'Just Wright', 'Rachel Getting Married', 'The Po
stman Always Rings Twice', 'Girl with a Pearl Earring', 'Das Boot', 'Sorority Boys',
'About Time', 'House of Flying Daggers', 'Arbitrage', 'Project Almanac', 'Cadillac R
ecords', 'Screwed', 'Fortress', 'For Your Consideration', 'Celebrity', 'Running with
Scissors', 'From Justin to Kelly', 'Girl 6', 'In the Cut', 'Two Lovers', 'Last Order
s', 'The Host', 'The Pursuit of D.B. Cooper', 'Ravenous', 'Charlie Bartlett', 'The G
reat Beauty', 'The Dangerous Lives of Altar Boys', 'Stoker', '2046', 'Married Life',
'Duma', 'Ondine', 'Brother', 'Welcome to Collinwood', 'Critical Care', 'The Life Bef
ore Her Eyes', 'Darling Companion', 'Trade', 'Fateless', 'Breakfast of Champions',
'A Woman, a Gun and a Noodle Shop', 'Cypher', 'City of Life and Death', 'Legend of a
Rabbit', 'Space Battleship Yamato', '5 Days of War', 'Triangle', '10 Days in a Madho
use', 'Heaven is for Real', 'Snatch', "Dancin' It's On", 'Pet Sematary', 'Madadayo',
'The Cry of the Owl', 'A Tale of Three Cities', 'Gremlins', 'Star Wars', 'Dirty Gran
dpa', 'Doctor Zhivago', 'Trash', 'High School Musical 3: Senior Year', 'The Fighte
r', 'Jackass Number Two', 'My Cousin Vinny', 'If I Stay', 'Drive Hard', 'Major Leagu

e', "St. Trinian's", 'Phone Booth', 'A Walk to Remember', 'Dead Man Walking', 'Cruel Intentions', 'Saw VI', 'History of the World: Part I', 'The Secret Life of Bees', 'Corky Romano', 'Raising Cain', 'F.I.S.T.', 'Invaders from Mars', 'Brooklyn', 'Barry Lyndon', 'Out Cold', 'The Ladies Man', 'Quartet', 'Tomcats', 'Frailty', 'Woman in Gold', 'Kinsey', 'Army of Darkness', 'Slackers', "What's Eating Gilbert Grape", 'The Visual Bible: The Gospel of John', 'Vera Drake', 'The Guru', 'The Perez Family', 'Inside Llewyn Davis', 'O', 'Return to the Blue Lagoon', 'The Molly Maguires', 'Romance & Cigarettes', 'Copying Beethoven', 'Brighton Rock', 'Saw V', 'Machine Gun McCain', 'LOL', 'Jindabyne', 'Kabhi Alvida Naa Kehna', 'An Ideal Husband', 'The Last Days on Mars', 'Darkness', '2001: A Space Odyssey', 'E.T. the Extra-Terrestrial', 'In the Land of Women', 'The Blue Butterfly', 'There Goes My Baby', 'Housefull', 'September Dawn', 'For Greater Glory - The True Story of Cristiada', 'The Bélier Family', 'Good Will Hunting', 'Misconduct', 'Saw III', 'Stripes', 'Bring It On', 'The Purge: Election Year', "She's All That", 'Saw IV', 'White Noise', "Madea's Family Reunion", 'The Color of Money', 'The Longest Day', 'The Mighty Ducks', 'The Grudge', 'Happy Gilmore', 'Jeepers Creepers', "Bill & Ted's Excellent Adventure", 'Oliver!', 'The Best Exotic Marigold Hotel', "Recess: School's Out", 'Mad Max Beyond Thunderdome', 'Commando', 'The Boy', 'Devil', 'Friday After Next', 'Insidious: Chapter 3', 'The Last Dragon', 'The Lawnmower Man', "Nick and Norah's Infinite Playlist", 'Dogma', 'The Banger Sisters', 'Twilight Zone: The Movie', 'Road House', 'A Low Down Dirty Shame', 'Swimfan', 'Employee of the Month', "Can't Hardly Wait", 'The Outsiders', "Pete's Dragon", 'The Dead Zone', 'Sinister 2', 'Sparkle', 'Valentine', 'The Fourth Kind', 'A Prairie Home Companion', 'Sugar Hill', 'Invasion U.S.A.', 'Roll Bounce', 'Rushmore', 'Skyline', 'The Second Best Exotic Marigold Hotel', 'Kit Kittredge: An American Girl', 'The Perfect Man', "Mo' Better Blues", 'Kung Pow: Enter the Fist', 'Tremors', 'Wrong Turn', 'The Long Riders', 'The Corruptor', 'Mud', 'Reno 911!: Miami', 'One Direction: This Is Us', 'The Goods: Live Hard, Sell Hard', 'Hey Arnold! The Movie', 'My Week with Marilyn', 'The Matador', 'Love Jones', 'The Gift', 'End of the Spear', 'Get Over It', 'Office Space', 'Drop Dead Gorgeous', 'Big Eyes', 'Very Bad Things', 'Sleepover', 'Body Double', 'MacGruber', 'Dirty Pretty Things', 'Movie 43', 'Over Her Dead Body', 'Seeking a Friend for the End of the World', 'Cedar Rapids', 'American History X', 'The Collection', "Teacher's Pet", 'The Red Violin', 'The Straight Story', 'Deuces Wild', 'Bad Words', 'Run, Fatboy, Run', 'Heartbeeps', 'Black or White', 'On the Line', 'Rescue Dawn', 'Danny Collins', 'Jeff, Who Lives at Home', 'I Am Love', 'Atlas Shrugged Part II', 'Romeo Is Bleeding', 'The Limey', 'Crash', 'The House of Mirth', 'Malone', 'Peaceful Warrior', 'Bucky Larson: Born to Be a Star', 'Bamboozled', 'The Forest', 'Sphinx', "While We're Young", 'A Better Life', 'Spider', 'Gun Shy', 'Nicholas Nickleby', 'The Iceman', 'Krrish', 'Cecil B. Demented', 'Killer Joe', 'The Jones es', 'Owning Mahowny', 'The Brothers Solomon', 'My Blueberry Nights', 'Illuminata', 'Swept Away', 'War, Inc.', 'Shaolin Soccer', 'The Brown Bunny', 'The Swindle', 'Rosewater', 'The Chambermaid on the Titanic', 'Coriolanus', 'Imaginary Heroes', 'High Heels and Low Lifes', "World's Greatest Dad", 'Severance', 'Edmond', 'Welcome to the Rileys', 'Police Academy: Mission to Moscow', 'Blood Done Sign My Name', 'Cinco de Mayo: La Batalla', 'Elsa & Fred', 'An Alan Smithee Film: Burn, Hollywood, Burn', 'The Open Road', 'The Good Guy', 'Motherhood', 'Free Style', 'Strangerland', 'Janky Promoters', 'Blonde Ambition', 'The Oxford Murders', 'The Reef', 'Eulogy', 'White Noise 2: The Light', 'Beat the World', 'Fifty Dead Men Walking', 'Jungle Shuffle', 'Adam Resurrected', 'Of Horses and Men', "It's a Wonderful Afterlife", "The Devil's Tomb", 'Partition', 'Good Intentions', 'The Good, The Bad, The Weird', 'Nurse 3-D', 'Gunless', 'Adventureland', 'The Lost City', 'Next Friday', 'American Heist', 'You Only Live Twice', 'Plastic', 'Amour', 'Poltergeist III', 'Re-Kill', "It's a Mad, Mad, Mad, Mad World", 'Volver', 'Heavy Metal', 'Gentlemen Broncos', 'Richard III', 'Into the Grizzly Maze', 'Kites', 'Melancholia', 'Red Dog', 'Jab Tak Hai Jaan', 'Alien', 'The Texas Chain Saw Massacre', 'The Runaways', 'Fiddler on the Roof', 'Thunderball', 'Detention', 'Loose Cannons', 'Set It Off', 'The Best Man', "Child's Play", 'Sicko', 'The Purge: Anarchy', 'Down to You', 'Harold & Kumar Go to White Castle', 'The Contende

```
r', 'Boiler Room', 'Trading Places', 'Black Christmas', "Breakin' All the Rules", 'H
enry V', 'The Savages', 'Chasing Papi', 'The Way of the Gun', 'Igby Goes Down', 'PC
U', 'The Ultimate Gift', 'The Ice Pirates', 'Gracie', 'Trust the Man', 'Hamlet 2',
'Velvet Goldmine', 'The Wailing', 'Glee: The Concert Movie', 'The Legend of Suriyoth
ai', 'Two Evil Eyes', 'Barbecue', 'All or Nothing', 'Princess Kaiulani', 'Opal Drea
m', 'Flame & Citron', 'Undiscovered', 'Red Riding: In the Year of Our Lord 1974', 'T
he Girl on the Train', 'Veronika Decides to Die', 'Crocodile Dundee', 'Ultramarines:
A Warhammer 40,000 Movie', 'The I Inside', 'Beneath Hill 60', 'Polisse', 'Awake', 'S
tar Wars: Clone Wars: Volume 1', 'Skin Trade', 'The Lost Boys', 'Crazy Heart', 'The
Rose', 'Baggage Claim', 'Barbarella', 'Shipwrecked', 'Election', 'The Namesake', 'Th
e DUFF', 'Glitter', 'The Haunting in Connecticut 2: Ghosts of Georgia', 'Silmido',
'Bright Star', 'My Name Is Khan', 'All Is Lost', 'Limbo', 'Namastey London', 'The Wi
nd That Shakes the Barley', 'Yeh Jawaani Hai Deewani', 'Quo Vadis', 'Repo! The Genet
ic Opera', 'Valley of the Wolves: Iraq', 'Pulp Fiction', 'The Muppet Movie', 'Nightc
rawler', 'Club Dread', 'The Sound of Music', 'Splash', 'Little Miss Sunshine', 'Stan
d by Me', '28 Days Later', 'You Got Served', 'Escape from Alcatraz', 'Brown Sugar',
'A Thin Line Between Love and Hate', '50/50', 'Shutter', 'That Awkward Moment', 'Mod
ern Problems', 'Kicks', 'Much Ado About Nothing', "On Her Majesty's Secret Service",
'New Nightmare', 'Drive Me Crazy', 'Akeelah and the Bee', 'Half Baked', 'New in Tow
n', 'American Psycho', 'The Good Girl', 'Bon Cop Bad Cop', 'The Boondock Saints II:
All Saints Day', 'The City of Your Final Destination', 'Enough Said', 'Easy A', 'The
Inkwell', 'Shadow of the Vampire', 'Prom', 'The Pallbearer', 'Held Up', 'Woman on To
p', 'Howards End', 'Anomalisa', 'Another Year', '8 Women', 'Showdown in Little Toky
o', 'Clay Pigeons', "It's Kind of a Funny Story", 'Made in Dagenham', 'When Did You
Last See Your Father?', 'Prefontaine', 'The Wicked Lady', 'The Secret of Kells', 'Be
gin Again', 'Down in the Valley', 'Brooklyn Rules', 'Restless', 'The Singing Detecti
ve', 'The Land Girls', 'Fido', 'The Wendell Baker Story', 'Wild Target', 'Patholog
y', 'Wuthering Heights', '10th & Wolf', 'Dear Wendy', 'Aloft', 'Akira', 'The Death a
nd Life of Bobby Z', 'The Rocket: The Legend of Rocket Richard', 'Swelter', 'My Luck
y Star', 'Imagine Me & You', 'Mr. Church', 'Swimming Pool', 'Green Street Hooligans:
Underground', 'The Blood of Heroes', 'Code of Honor', 'Driving Miss Daisy', 'Soul Fo
od', 'Rumble in the Bronx', 'Far from Men', 'Thank You for Smoking', 'Hostel: Part I
I', 'An Education', 'Shopgirl', 'The Hotel New Hampshire', 'Narc', 'Men with Broom
s', 'Witless Protection', 'The Work and the Glory', 'Extract', 'Masked and Anonymou
s', 'Betty Fisher and Other Stories', 'Code 46', 'Outside Bet', 'Albert Nobbs', 'Bla
ck November', 'Ta Ra Rum Pum', 'Persepolis', 'The Hole', 'The Wave', 'The Neon Demo
n', 'Harry Brown', 'The Omega Code', 'Juno', 'Pound of Flesh', 'Diamonds Are Foreve
r', 'The Godfather', 'Flashdance', '(500) Days of Summer', 'The Piano', 'Magic Mik
e', 'Darkness Falls', 'Live and Let Die', 'My Dog Skip', 'Definitely, Maybe', 'Jumpi
ng the Broom', 'Good Night, and Good Luck.', 'Capote', 'Desperado', "Logan's Run",
'The Man with the Golden Gun', 'Action Jackson', 'The Descent', 'Michael Jordan to t
he Max', "Devil's Due", 'Flirting with Disaster', "The Devil's Rejects", 'Dope', 'In
Too Deep', 'House of 1000 Corpses', 'Alien Zone', 'A Serious Man', 'Get Low', 'Warlo
ck', 'Beyond the Lights', 'A Single Man', 'The Last Temptation of Christ', 'Outside
Providence', 'Bride & Prejudice', 'Rabbit-Proof Fence', "Who's Your Caddy?", 'Split
Second', 'The Other Side of Heaven', 'Veer-Zaara', 'Redbelt', 'Cyrus', 'A Dog Of Fla
nders', 'Auto Focus', 'Factory Girl', 'We Need to Talk About Kevin', 'The Christmas
Candle', 'The Mighty Macs', "Losin' It", 'Mother and Child', 'March or Die', 'The Vi
sitors', 'Somewhere', 'I Hope They Serve Beer in Hell', 'Chairman of the Board', 'He
sher', 'Dom Hemingway', 'Gerry', 'The Heart of Me', 'Freeheld', 'The Extra Man', 'Ha
rd to Be a God', 'Ca$h', 'Wah-Wah', 'The Boondock Saints', 'Z Storm', 'Twixt', 'The
Snow Queen', 'Alpha and Omega: The Legend of the Saw Tooth Cave', 'Pale Rider', 'Sta
rgate: The Ark of Truth', 'Dazed and Confused', 'High School Musical 2', 'Two Lovers
and a Bear', 'Criminal Activities', 'Aimee & Jaguar', 'The Chumscrubber', 'Shade',
'House at the End of the Street', 'Incendies', 'Remember Me, My Love', 'Perrier’s Bo
unty', 'Elite Squad', 'Annabelle', 'Bran Nue Dae', 'Boyz n the Hood', 'La Bamba', 'T
```

he Four Seasons', 'Dressed to Kill', 'The Adventures of Huck Finn', 'Go', 'Friends w
ith Money', 'The Andromeda Strain', 'Bats', 'Nowhere in Africa', 'Shame', 'Layer Cak
e', 'The Work and the Glory II: American Zion', 'The East', 'A Home at the End of th
e World', 'Aberdeen', 'The Messenger', 'Tracker', 'Control', 'The Terminator', 'Good
bye, Lenin!', 'The Damned United', 'The Return of the Living Dead', 'Mallrats', 'Gre
ase', 'Platoon', 'Fahrenheit 9/11', 'Butch Cassidy and the Sundance Kid', 'Mary Popp
ins', 'Ordinary People', 'West Side Story', 'Caddyshack', 'The Brothers', 'The Woo
d', 'The Usual Suspects', 'A Nightmare on Elm Street 5: The Dream Child', 'National
Lampoon’s Van Wilder', 'The Wrestler', 'Duel in the Sun', 'Best in Show', 'Escape fr
om New York', 'School Daze', 'Daddy Day Camp', 'Mr. Nice Guy', 'A Mighty Wind', 'Mys
tic Pizza', 'Sliding Doors', 'Tales from the Hood', 'The Last King of Scotland', 'Ha
lloween 5: The Revenge of Michael Myers', 'Bernie', 'Dolphins and Whales: Tribes of
the Ocean', 'Pollock', '200 Cigarettes', 'The Words', 'Casa De Mi Padre', 'City Isla
nd', 'The Guard', 'College', 'The Virgin Suicides', 'Little Voice', 'Miss March', 'W
ish I Was Here', 'Simply Irresistible', 'Hedwig and the Angry Inch', 'Only the Stron
g', 'Goddess of Love', 'Shattered Glass', 'Novocaine', 'The Business of Strangers',
'The Wild Bunch', 'The Wackness', 'The First Great Train Robbery', 'Morvern Callar',
'Beastmaster 2: Through the Portal of Time', 'The 5th Quarter', 'The Flower of Evi
l', 'The Greatest', 'Snow Flower and the Secret Fan', 'Come Early Morning', 'Lucky B
reak', 'Julia', 'Surfer, Dude', 'Lake of Fire', '11:14', 'Men of War', 'Don McKay',
'Deadfall', 'A Shine of Rainbows', 'The Hit List', 'Emma', 'Videodrome', 'The Spanis
h Apartment', 'Song One', 'Winter in Wartime', 'Freaky Deaky', 'The Train', 'Trade O
f Innocents', 'The Protector', 'Stiff Upper Lips', 'Bend It Like Beckham', 'Sunshine
State', 'Crossover', 'Khiladi 786', '[REC]²', 'Standing Ovation', 'The Sting', 'Char
iots of Fire', 'Diary of a Mad Black Woman', 'Shine', 'Don Jon', 'High Plains Drifte
r', 'Ghost World', 'Iris', 'Galaxina', 'The Chorus', 'Mambo Italiano', 'Wonderland',
'Do the Right Thing', 'Harvard Man', 'Le Havre', 'Irreversible', 'R100', 'Rang De Ba
santi', 'Animals', 'Salvation Boulevard', 'The Ten', 'A Room for Romeo Brass', 'Head
hunters', 'Grabbers', 'Saint Ralph', 'Miss Julie', 'Somewhere in Time', 'Dum Maaro D
um', 'Insidious: Chapter 2', 'Saw II', '10 Cloverfield Lane', 'Jackass: The Movie',
'Lights Out', 'Paranormal Activity 3', 'Ouija', 'A Nightmare on Elm Street 3: Dream
Warriors', 'Instructions Not Included', 'Paranormal Activity 4', 'The Robe', 'The Re
turn of the Pink Panther', "Freddy's Dead: The Final Nightmare", 'Monster', '20,000
Leagues Under the Sea', 'Paranormal Activity: The Marked Ones', 'The Elephant Man',
'Dallas Buyers Club', 'The Lazarus Effect', 'Memento', 'Oculus', 'Clerks II', 'Billy
Elliot', 'The Way Way Back', 'House Party 2', 'The Man from Snowy River', "Doug's 1s
t Movie", 'The Apostle', 'Mommie Dearest', 'Our Idiot Brother', 'Race', 'The Players
Club', 'As Above, So Below', 'Addicted', "Eve's Bayou", 'Still Alice', 'The Egyptia
n', 'Nighthawks', 'Friday the 13th Part VIII: Jason Takes Manhattan', 'My Big Fat Gr
eek Wedding', 'Spring Breakers', 'Halloween: The Curse of Michael Myers', 'Y Tu Mamá
También', 'Shaun of the Dead', 'The Haunting of Molly Hartley', 'Lone Star', 'Hallow
een 4: The Return of Michael Myers', "April Fool's Day", 'Diner', 'Lone Wolf McQuad
e', 'Apollo 18', 'Sunshine Cleaning', 'No Escape', 'The Beastmaster', 'Solomon and S
heba', 'Fifty Shades of Black', 'Not Easily Broken', 'A Farewell to Arms', 'The Perf
ect Match', 'Digimon: The Movie', 'Saved!', 'The Barbarian Invasions', 'Robin and Ma
rian', 'The Forsaken', 'Force 10 from Navarone', 'UHF', "Grandma's Boy", 'Slums of B
everly Hills', 'Once Upon a Time in the West', 'Made', 'Moon', 'Keeping Up with the
Steins', 'Sea Rex 3D: Journey to a Prehistoric World', 'The Sweet Hereafter', 'Of Go
ds and Men', 'Bottle Shock', 'Jekyll and Hyde ... Together Again', 'Heavenly Creatur
es', '90 Minutes in Heaven', 'Everything Must Go', 'Zero Effect', 'The Machinist',
'Light Sleeper', 'Kill the Messenger', 'Rabbit Hole', 'Party Monster', 'Green Room',
'The Oh in Ohio', 'Atlas Shrugged Part III: Who is John Galt?', 'Bottle Rocket', 'Al
bino Alligator', 'Gandhi, My Father', 'Standard Operating Procedure', 'Out of the Bl
ue', 'Tucker and Dale vs Evil', 'Lovely, Still', 'Tycoon', 'Desert Blue', 'Decoys',
'The Visit', 'Redacted', 'Fascination', 'Area 51', 'Sleep Tight', 'The Cottage', 'De
ad Like Me: Life After Death', 'Farce of the Penguins', 'Flying By', 'Rudderless',

'Henry & Me', 'Christmas Eve', 'We Have Your Husband', 'Dying of the Light', 'Born O
f War', 'Capricorn One', "Should've Been Romeo", 'Running Forever', 'Yoga Hosers',
'Navy Seals vs. Zombies', 'I Served the King of England', 'Soul Kitchen', 'Sling Bla
de', 'The Awakening', 'Hostel', 'A Cock and Bull Story', 'Take Shelter', 'Lady in Wh
ite', 'Driving Lessons', "Let's Kill Ward's Wife", 'The Texas Chainsaw Massacre 2',
'Pat Garrett & Billy the Kid', 'Only God Forgives', 'Camping Sauvage', 'Without Me
n', 'Dear Frankie', 'All Hat', 'The Names of Love', 'Treading Water', 'Savage Grac
e', 'Out of the Blue', 'Police Academy', 'The Blue Lagoon', 'Four Weddings and a Fun
eral', 'Fast Times at Ridgemont High', 'Moby Dick', '25th Hour', 'Bound', 'Requiem f
or a Dream', 'State Fair', 'Tango', 'Salvador', "Moms' Night Out", 'Donnie Darko',
'Saving Private Perez', 'Character', 'Spun', 'Life During Wartime', 'Sympathy for La
dy Vengeance', "Mozart's Sister", 'Mean Machine', 'Exiled', 'Blackthorn', 'Lilya 4-e
ver', 'After.Life', 'Fugly', "One Flew Over the Cuckoo's Nest", "R.L. Stine's Monste
rville: The Cabinet of Souls", 'Silent Movie', 'Airlift', 'Anne of Green Gables', 'F
alcon Rising', 'The Sweeney', 'Sexy Beast', 'Easy Money', 'Whale Rider', 'Paa', 'Car
go', 'High School Musical', 'Love and Death on Long Island', 'Night Watch', 'The Cry
ing Game', "Porky's", 'Survival of the Dead', 'Night of the Living Dead', 'Lost in T
ranslation', 'Annie Hall', 'The Greatest Show on Earth', "Monster's Ball", 'Maggie',
'Leaving Las Vegas', 'Hansel and Gretel Get Baked', 'The Front Page', 'The Boy Next
Door', 'Trapeze', 'The Kids Are All Right', 'They Live', 'The Great Escape', 'What t
he #$*! Do We (K)now!?', 'The Last Exorcism Part II', 'Boyhood', 'Scoop', 'The Was
h', '3 Strikes', 'The Cooler', 'The Misfits', 'The Night Listener', 'The Jerky Boy
s', 'The Orphanage', 'A Haunted House 2', 'The Rules of Attraction', 'Topaz', "Let's
Go to Prison", 'Four Rooms', 'Secretary', 'The Real Cancun', 'Talk Radio', 'Waiting
for Guffman', 'Love Stinks', 'You Kill Me', 'Thumbsucker', 'Red State', 'Mirrormas
k', 'Samsara', 'The Barbarians', 'The Art of Getting By', 'Zipper', 'Poolhall Junkie
s', 'The Loss of Sexual Innocence', 'Holy Motors', 'Joe', 'Shooting Fish', 'Prison',
'Psycho Beach Party', 'The Big Tease', 'Guten Tag, Ramón', 'Trust', 'An Everlasting
Piece', 'Among Giants', 'Adore', 'The Velocity of Gary', 'Mondays in the Sun', 'Stak
e Land', 'The Last Time I Committed Suicide', 'Futuro Beach', 'Another Happy Day',
'A Lonely Place to Die', 'Nothing', 'The Geographer Drank His Globe Away', '1776',
'Inescapable', "Hell's Angels", 'Purple Violets', 'The Veil', 'The Loved Ones', 'The
Helpers', 'How to Fall in Love', 'The Perfect Wave', 'A Man for All Seasons', 'Netwo
rk', 'Gone with the Wind', 'Desert Dancer', 'Major Dundee', 'Annie Get Your Gun', 'F
our Lions', 'The House of Sand', 'Defendor', 'The Pirate', 'The Good Heart', 'The Hi
story Boys', 'Midnight Cowboy', 'The Full Monty', 'Airplane!', 'Chain of Command',
'Friday', 'Menace II Society', 'Creepshow 2', 'The Ballad of Cable Hogue', 'In Cold
Blood', "The Nun's Story", 'Harper', 'Frenzy', 'The Witch', 'I Got the Hook Up', "Sh
e's the One", 'Gods and Monsters', 'The Secret in Their Eyes', 'Train', 'Evil Dead I
I', 'Pootie Tang', 'Sharknado', 'The Other Conquest', 'Troll Hunter', 'Ira & Abby',
'Winter Passing', 'D.E.B.S.', 'The Masked Saint', 'The Betrayed', 'Taxman', 'The Sec
ret', '2:13', 'Batman: The Dark Knight Returns, Part 2', 'Time to Choose', 'In the N
ame of the King III', 'Wicked Blood', 'Stranded', 'Lords of London', 'High Anxiety',
'March of the Penguins', 'Margin Call', 'August', 'Choke', 'Whiplash', 'City of Go
d', 'Human Traffic', 'To Write Love on Her Arms', 'The Dead Girl', 'The Hunt', 'A Ch
ristmas Story', 'Bella', 'Class of 1984', 'The Opposite Sex', 'Dreaming of Joseph Le
es', 'The Class', "Rosemary's Baby", 'The Man Who Shot Liberty Valance', 'Adam', 'Ma
ria Full of Grace', 'Beginners', 'Feast', 'Animal House', 'Goldfinger', 'Antiviral',
"It's a Wonderful Life", 'Trainspotting', 'The Original Kings of Comedy', 'Paranorma
l Activity 2', 'Waking Ned', 'Bowling for Columbine', 'Coming Home', "A Nightmare on
Elm Street Part 2: Freddy's Revenge", 'A Room with a View', 'The Purge', 'Sinister',
'Martin Lawrence Live: Runteldat', 'Cat on a Hot Tin Roof', 'Beneath the Planet of t
he Apes', 'Air Bud', 'Pokémon: Spell of the Unknown', 'Friday the 13th Part VI: Jaso
n Lives', 'The Bridge on the River Kwai', 'Spaced Invaders', 'Family Plot', 'The Apa
rtment', 'Jason Goes to Hell: The Final Friday', 'Torn Curtain', "Dave Chappelle's B
lock Party", 'Slow West', 'Krush Groove', 'Next Day Air', 'Elmer Gantry', 'Judgment

at Nuremberg', "Trippin'", 'Red River', 'Phat Girlz', 'Before Midnight', 'Teen Wolf Too', 'Phantasm II', 'Woman Thou Art Loosed', 'Real Women Have Curves', 'Water', 'East Is East', 'Whipped', 'Kama Sutra - A Tale of Love', 'Please Give', 'Willy Wonka & the Chocolate Factory', 'Warlock: The Armageddon', '8 Heads in a Duffel Bag', 'Days of Heaven', 'Thirteen Conversations About One Thing', 'Jawbreaker', 'Basquiat', 'Frances Ha', 'Tsotsi', 'Happiness', 'DysFunktional Family', 'Tusk', 'Oldboy', 'Letters to God', 'Hobo with a Shotgun', 'Compadres', 'Freeway', "Love's Abiding Joy", 'Fish Tank', 'Damsels in Distress', 'Creature', 'Bachelorette', 'Brave New Girl', "Tim and Eric's Billion Dollar Movie", 'Summer Storm', 'Fort McCoy', 'Chain Letter', 'Just Looking', 'The Divide', 'The Eclipse', 'Demonic', 'My Big Fat Independent Movie', 'The Deported', 'Tanner Hall', 'Open Road', 'They Came Together', '30 Nights of Paranormal Activity With the Devil Inside the Girl With the Dragon Tattoo', 'Never Back Down 2: The Beatdown', 'Point Blank', 'Four Single Fathers', 'Enter the Dangerous Mind', 'Something Wicked', 'AWOL-72', 'Iguana', 'Chicago Overcoat', 'Barry Munday', 'Central Station', 'Pocketful of Miracles', 'Close Range', 'Boynton Beach Club', 'Amnesiac', 'Freakonomics', 'High Tension', 'Griff the Invisible', 'Unnatural', 'Hustle & Flow', 'Some Like It Hot', 'Friday the 13th Part VII: The New Blood', 'The Wizard of Oz', 'Young Frankenstein', 'Diary of the Dead', 'Lage Raho Munna Bhai', "Ulee's Gold", 'The Black Stallion', 'Sardaarji', 'Journey to Saturn', "Donovan's Reef", 'The Dress', 'A Guy Named Joe', 'Blazing Saddles', 'Friday the 13th: The Final Chapter', 'Ida', 'Maurice', 'Beer League', 'Riding Giants', 'Timecrimes', 'Silver Medalist', 'Timber Falls', "Singin' in the Rain", 'Fat, Sick & Nearly Dead', 'A Haunted House', "2016: Obama's America", 'That Thing You Do!', 'Halloween III: Season of the Witch', 'Escape from the Planet of the Apes', 'Hud', 'Kevin Hart: Let Me Explain', 'My Own Private Idaho', 'Garden State', 'Before Sunrise', 'Evil Words', "Jesus' Son", 'Saving Face', 'Brick Lane', 'Robot & Frank', 'My Life Without Me', 'The Spectacular Now', 'Religulous', 'Fuel', "Valley of the Heart's Delight", 'Eye of the Dolphin', '8: The Mormon Proposition', 'The Other End of the Line', 'Anatomy', 'Sleep Dealer', 'Super', 'Christmas Mail', 'Stung', 'Antibirth', 'Get on the Bus', 'Thr3e', 'Idiocracy', 'The Rise of the Krays', 'This Is England', 'U.F.O.', 'Bathing Beauty', 'Go for It!', 'Dancer, Texas Pop. 81', 'Show Boat', 'Redemption Road', 'The Calling', 'The Brave Little Toaster', 'Fantasia', '8 Days', 'Friday the 13th Part III', 'Friday the 13th: A New Beginning', 'The Last Sin Eater', 'Do You Believe?', 'Impact Point', 'The Valley of Decision', 'Eden', 'Chicken Tikka Masala', "There's Always Woodstock", 'Jack Brooks: Monster Slayer', 'The Best Years of Our Lives', 'Bully', 'Elling', 'Mi America', '[REC]', 'Lies in Plain Sight', 'Sharkskin', 'Containment', 'The Timber', 'From Russia with Love', 'The Toxic Avenger Part II', 'Sleeper', 'It Follows', 'Everything You Always Wanted to Know About Sex *But Were Afraid to Ask', 'To Kill a Mockingbird', 'Mad Max 2: The Road Warrior', 'The Legend of Drunken Master', "Boys Don't Cry", 'Silent House', 'The Lives of Others', 'Courageous', 'The Hustler', 'Boom Town', 'The Triplets of Belleville', 'Smoke Signals', 'American Splendor', 'Before Sunset', 'Amores perros', 'Thirteen', "Gentleman's Agreement", "Winter's Bone", 'Touching the Void', "Alexander's Ragtime Band", 'Me and You and Everyone We Know', 'Inside Job', 'We Are Your Friends', 'Ghost Dog: The Way of the Samurai', 'Harsh Times', 'Captive', 'Full Frontal', 'Witchboard', 'Shortbus', 'Waltz with Bashir', 'The Book of Mormon Movie, Volume 1: The Journey', 'No End in Sight', 'The Diary of a Teenage Girl', 'In the Shadow of the Moon', "Meek's Cutoff", 'Inside Deep Throat', 'Dinner Rush', 'Clockwatchers', 'The Virginity Hit', 'Subway', 'House of D', 'Teeth', 'Six-String Samurai', 'Hum To Mohabbat Karega', "It's All Gone Pete Tong", 'Saint John of Las Vegas', '24 7: Twenty Four Seven', 'Stonewall', 'Roadside Romeo', 'This Thing of Ours', 'The Lost Medallion: The Adventures of Billy Stone', 'The Last Five Years', 'The Missing Person', 'Return of the Living Dead 3', 'London', 'Sherrybaby', 'Circle', 'Eden Lake', 'Plush', 'Lesbian Vampire Killers', "Gangster's Paradise: Jerusalema", 'Freeze Frame', 'Grave Encounters', 'Stitches', 'Nine Dead', 'To Be Frank, Sinatra at 100', 'Bananas', 'Supercapitalist', 'Rockaway', 'The Lady from Shanghai', "No Man's Land: The Rise of Reeker", 'Highway', 'Small Apartments', 'Coffee Town', 'The Ghastl

y Love of Johnny X', 'All Is Bright', 'The Torture Chamber of Dr. Sadism', "Straight A's", 'A Funny Thing Happened on the Way to the Forum', 'Slacker Uprising', "The Legend of Hell's Gate: An American Conspiracy", 'The Walking Deceased', 'The Curse of Downers Grove', 'Shark Lake', "River's Edge", 'Northfork', 'The Marine 4: Moving Target', 'Buried', 'Submarine', 'The Square', 'One to Another', 'ABCD (Any Body Can Dance)', 'Man on Wire', 'Abandoned', 'Brotherly Love', 'The Last Exorcism', 'Nowhere Boy', 'A Streetcar Named Desire', 'Dr. Strangelove or: How I Learned to Stop Worrying and Love the Bomb', 'The Crime of Padre Amaro', 'Beasts of the Southern Wild', 'Battle for the Planet of the Apes', 'Songcatcher', 'Higher Ground', 'Vaalu', 'The Greatest Movie Ever Sold', 'Ed and His Dead Mother', 'Travellers and Magicians', "Hang 'em High", 'Deadline - U.S.A.', 'Sublime', "A Beginner's Guide to Snuff", 'Independence Daysaster', 'Dysfunctional Friends', 'Run Lola Run', 'May', 'Against the Wild', 'Under the Same Moon', 'Conquest of the Planet of the Apes', 'In the Bedroom', 'I Spit on Your Grave', 'Happy, Texas', 'My Summer of Love', 'The Lunchbox', 'Yes', "You Can't Take It With You", 'From Here to Eternity', 'She Wore a Yellow Ribbon', 'Grace Unplugged', 'Foolish', 'N-Secure', 'Caramel', 'Out of the Dark', 'The Bubble', 'The Conversation', 'Dil Jo Bhi Kahey...', 'Mississippi Mermaid', 'I Love Your Work', 'Cabin Fever', 'Waitress', 'Bloodsport', 'Mr. Smith Goes to Washington', 'Kids', 'The Squid and the Whale', 'Kissing Jessica Stein', 'Kickboxer: Vengeance', 'Spellbound', 'Exotica', "Buffalo '66", 'Insidious', 'Repo Man', 'Nine Queens', 'The Gatekeepers', 'The Ballad of Jack and Rose', 'The To Do List', 'Killing Zoe', 'The Believer', 'Snow Angels', 'Unsullied', 'Session 9', 'I Want Someone to Eat Cheese With', 'Mooz-lum', 'Hatchet', 'Modern Times', 'Stolen Summer', 'My Name Is Bruce', 'The Salon', 'Road Hard', 'Forty Shades of Blue', 'Amigo', 'Pontypool', 'Trucker', 'Me You and Five Bucks', 'The Lords of Salem', 'Housebound', 'Wal-Mart: The High Cost of Low Price', 'Fetching Cody', 'Last I Heard', 'Closer to the Moon', 'Mutant World', 'Growing Up Smith', 'Checkmate', '#Horror', 'Wind Walkers', 'Snow White and the Seven Dwarfs', 'The Holy Girl', 'Shalako', 'Incident at Loch Ness', 'The Dog Lover', 'GirlHouse', 'The Blue Room', 'House at the End of the Drive', 'Batman', 'Lock, Stock and Two Smoking Barrels', 'The Ballad of Gregorio Cortez', 'The Celebration', 'Trees Lounge', 'Journey from the Fall', 'The Basket', 'Eddie: The Sleepwalking Cannibal', 'Queen of the Mountains', 'Def-Con 4', 'The Hebrew Hammer', "Neal 'n' Nikki", 'The 41–Year–Old Virgin Who Knocked Up Sarah Marshall and Felt Superbad About It', 'Forget Me Not', 'Rebecca', 'Friday the 13th Part 2', 'The Lost Weekend', 'C.H.U.D.', 'Filly Brown', 'The Lion of Judah', 'Niagara', 'How Green Was My Valley', 'Da Sweet Blood of Jesus', 'Sex, Lies, and Videotape', 'Saw', 'Super Troopers', 'The Algerian', 'The Amazing Catfish', 'Monsoon Wedding', 'You Can Count on Me', 'The Trouble with Harry', "But I'm a Cheerleader", 'Home Run', 'Reservoir Dogs', 'The Blue Bird', 'The Good, the Bad and the Ugly', 'The Second Mother', 'Blue Like Jazz', 'Down & Out With The Dolls', 'Pink Ribbons, Inc.', 'Certifiably Jonathan', 'Desire', 'The Blade of Don Juan', 'Grand Theft Parsons', 'Extreme Movie', 'The Charge of the Light Brigade', 'Below Zero', 'Crowsnest', 'Airborne', 'Cotton Comes to Harlem', 'The Wicked Within', 'Bleeding Hearts', 'Waiting...', "Dead Man's Shoes", 'From a Whisper to a Scream', 'Sex With Strangers', "Dracula: Pages from a Virgin's Diary", 'Faith Like Potatoes', 'Beyond the Black Rainbow', 'The Raid', 'The Dead Undead', 'The Vatican Exorcisms', 'Casablanca', 'Lake Mungo', 'Rocket Singh: Salesman of the Year', 'Silent Running', 'Rocky', 'The Sleepwalker', 'Tom Jones', 'Unfriended', 'Taxi Driver', 'The Howling', 'Dr. No', 'Chernobyl Diaries', 'Hellraiser', "God's Not Dead 2", 'Cry_Wolf', 'Godzilla 2000', 'Blue Valentine', 'Transamerica', 'The Devil Inside', 'Beyond the Valley of the Dolls', 'Love Me Tender', 'An Inconvenient Truth', 'Sands of Iwo Jima', 'Shine a Light', 'The Green Inferno', 'Departure', 'The Sessions', 'Food, Inc.', 'October Baby', 'Next Stop Wonderland', 'The Skeleton Twins', 'Martha Marcy May Marlene', 'Obvious Child', 'Frozen River', '20 Feet from Stardom', 'Two Girls and a Guy', 'Walking and Talking', 'Who Killed the Electric Car?', 'The Broken Hearts Club: A Romantic Comedy', 'Bubba Ho-tep', 'Slam', 'Brigham City', 'Fiza', 'Orgazmo', 'All the Real Girls', 'Dream with the Fishes', 'Blue Car', 'Palo Alto', 'Ajami', 'Wristcutters: A Love Story', 'I

Origins', 'The Battle of Shaker Heights', 'The Act of Killing', 'Taxi to the Dark Si
de', 'Once in a Lifetime: The Extraordinary Story of the New York Cosmos', 'Guiana 1
838', 'Lisa Picard Is Famous', 'Antarctica: A Year on Ice', 'A LEGO Brickumentary',
'Hardflip', 'Chocolate: Deep Dark Secrets', 'The House of the Devil', 'The Perfect H
ost', 'Safe Men', 'Speedway Junky', 'The Last Big Thing', 'The Specials', '16 to Lif
e', 'Alone With Her', 'Creative Control', 'Special', 'Sparkler', 'The Helix... Loade
d', 'In Her Line of Fire', 'The Jimmy Show', 'Heli', 'Karachi se Lahore', 'Loving An
nabelle', 'Hits', 'Jimmy and Judy', 'Frat Party', "The Party's Over", 'Proud', 'The
Poker House', 'Childless', 'ZMD: Zombies of Mass Destruction', 'Snow White: A Deadly
Summer', 'Hidden Away', 'My Last Day Without You', 'Steppin: The Movie', "Doc Hollid
ay's Revenge", 'Black Rock', 'Truth or Dare', 'The Pet', 'Bang Bang Baby', 'Fear Cli
nic', 'Zombie Hunter', 'A Fine Step', 'Charly', 'Banshee Chapter', 'Ask Me Anythin
g', 'And Then Came Love', 'Food Chains', 'On the Waterfront', 'L!fe Happens', '4 Mon
ths, 3 Weeks and 2 Days', 'The Horror Network Vol. 1', 'Hard Candy', 'The Quiet', 'C
ircumstance', 'Fruitvale Station', 'The Brass Teapot', 'Bambi', 'The Hammer', 'Latte
r Days', 'Elza', '1982', 'For a Good Time, Call...', 'Celeste & Jesse Forever', 'Tim
e Changer', 'London to Brighton', 'American Hero', 'Windsor Drive', 'A Separation',
'Crying with Laughter', 'Welcome to the Dollhouse', 'Ruby in Paradise', 'Raising Vic
tor Vargas', "Pandora's Box", 'Harrison Montgomery', 'Live-In Maid', 'Deterrence',
'The Mudge Boy', 'The Young Unknowns', 'Not Cool', 'Dead Snow', 'Saints and Soldier
s', 'Vessel', 'American Graffiti', 'Iraq for Sale: The War Profiteers', 'Aqua Teen H
unger Force Colon Movie Film for Theaters', 'Safety Not Guaranteed', 'Kevin Hart: La
ugh at My Pain', 'Kill List', 'The Innkeepers', 'The Conformist', 'Interview with th
e Assassin', 'Donkey Punch', 'All the Boys Love Mandy Lane', 'Bled', 'High Noon', 'H
oop Dreams', 'Rize', 'L.I.E.', 'The Sisterhood of Night', 'B-Girl', 'Half Nelson',
'Naturally Native', 'Hav Plenty', 'Adulterers', 'Escape from Tomorrow', 'Starsucker
s', 'The Hadza: Last of the First', 'After', 'Treachery', 'Walter', 'Top Hat', 'The
Blair Witch Project', 'Woodstock', 'The Kentucky Fried Movie', 'Mercy Streets', 'Car
ousel of Revenge', 'Broken Vessels', 'Water & Power', 'They Will Have to Kill Us Fir
st', 'Light from the Darkroom', 'The Country Doctor', "The Maid's Room", "A Hard Da
y's Night", 'The Harvest (La Cosecha)', 'Love Letters', 'Juliet and Alfa Romeo', 'Fi
reproof', 'Faith Connections', 'Benji', 'Open Water', 'High Road', 'Kingdom of the S
piders', 'Mad Hot Ballroom', 'The Station Agent', 'To Save A Life', 'Wordplay', 'Bey
ond the Mat', 'The Singles Ward', 'Osama', 'Sholem Aleichem: Laughing In The Darknes
s', 'Groove', 'The R.M.', 'Twin Falls Idaho', 'Mean Creek', 'Hurricane Streets', 'Ne
ver Again', 'Civil Brand', 'Lonesome Jim', 'Drinking Buddies', 'Deceptive Practice:
The Mysteries and Mentors of Ricky Jay', 'Seven Samurai', 'The Other Dream Team', 'J
ohnny Suede', 'Finishing The Game', 'Rubber', 'Kiss the Bride', 'The Slaughter Rul
e', 'Monsters', 'The Californians', 'The Living Wake', 'Detention of the Dead', 'Cra
zy Stone', 'Scott Walker: 30 Century Man', 'Everything Put Together', 'Good Kill',
'The Outrageous Sophie Tucker', 'Now Is Good', 'Girls Gone Dead', 'America Is Still
the Place', 'Subconscious', 'Enter Nowhere', 'El Rey de Najayo', 'Fight to the Finis
h', "Alleluia! The Devil's Carnival", 'The Sound and the Shadow', 'Rodeo Girl', 'Bor
n to Fly: Elizabeth Streb vs. Gravity', 'The Little Ponderosa Zoo', 'The Toxic Aveng
er', 'Straight Out of Brooklyn', 'Bloody Sunday', 'Diamond Ruff', 'Conversations wit
h Other Women', 'Poultrygeist: Night of the Chicken Dead', 'Mutual Friends', '42nd S
treet', 'Rise of the Entrepreneur: The Search for a Better Way', 'Metropolitan', 'As
It Is in Heaven', 'Roadside', 'Napoleon Dynamite', 'Blue Ruin', 'Paranormal Activit
y', 'Dogtown and Z-Boys', 'Monty Python and the Holy Grail', 'Quinceañera', 'Gory Go
ry Hallelujah', 'Tarnation', 'I Want Your Money', 'Love in the Time of Monsters', 'T
he Beyond', 'What Happens in Vegas', 'The Dark Hours', 'My Beautiful Laundrette', 'F
abled', 'Show Me', 'Cries and Whispers', 'Intolerance', 'Trekkies', 'The Broadway Me
lody', 'The Evil Dead', 'Maniac', 'Censored Voices', 'Murderball', 'American Ninja
2: The Confrontation', '51 Birch Street', 'Rotor DR1', '12 Angry Men', 'My Dog Tuli
p', 'It Happened One Night', 'Dogtooth', 'Tupac: Resurrection', 'Tumbleweeds', 'The
Prophecy', "When the Cat's Away", 'Pieces of April', 'The Big Swap', 'Old Joy', 'Wen

```
dy and Lucy', '3 Backyards', 'Pierrot le Fou', 'Sisters in Law', 'Ayurveda: Art of B
eing', 'Nothing But a Man', 'First Love, Last Rites', 'Fighting Tommy Riley', 'Royal
Kill', 'The Looking Glass', 'Death Race 2000', 'Locker 13', 'Midnight Cabaret', "And
erson's Cross", 'Bizarre', 'Graduation Day', 'Some Guy Who Kills People', 'Complianc
e', 'Chasing Amy', 'Lovely & Amazing', 'Death Calls', 'Better Luck Tomorrow', 'The I
ncredibly True Adventure of Two Girls In Love', 'Chuck & Buck', 'American Desi', "Am
idst the Devil's Wings", 'Cube', 'Love and Other Catastrophes', 'I Married a Strange
Person!', 'November', 'Like Crazy', 'Teeth and Blood', 'Sugar Town', 'The Motel', 'T
he Canyons', 'On the Outs', 'Shotgun Stories', 'Exam', 'The Sticky Fingers of Time',
'Sunday School Musical', 'Rust', 'Ink', 'The Christmas Bunny', 'Butterfly', 'UnDivid
ed', 'The Frozen', 'Horse Camp', 'Give Me Shelter', 'The Big Parade', 'Little Big To
p', 'Along the Roadside', 'Bronson', 'Western Religion', 'Burn', 'Urbania', 'The Ste
wardesses', 'The Beast from 20,000 Fathoms', 'Mad Max', 'Swingers', 'A Fistful of Do
llars', 'She Done Him Wrong', 'Short Cut to Nirvana: Kumbh Mela', 'The Grace Card',
'Middle of Nowhere', 'Three', 'The Business of Fancydancing', 'Call + Response', 'Ma
levolence', 'Reality Show', 'Super Hybrid', 'Baghead', 'American Beast', 'The Case o
f the Grinning Cat', 'Ordet', 'Good Dick', 'The Man from Earth', 'The Trials Of Darr
yl Hunt', 'Samantha: An American Girl Holiday', 'Yesterday Was a Lie', 'Theresa Is a
Mother', 'H.', 'Archaeology of a Woman', 'Children of Heaven', 'Weekend', "She's Got
ta Have It", 'Butterfly Girl', 'The World Is Mine', 'Another Earth', "Sweet Sweetbac
k's Baadasssss Song", 'Perfect Cowboy', 'Tadpole', 'Once', 'The Woman Chaser', 'The
Horse Boy', 'When the Lights Went Out', 'Heroes of Dirt', 'A Charlie Brown Christma
s', 'Antarctic Edge: 70° South', 'Aroused', 'Top Spin', 'Roger & Me', 'An American i
n Hollywood', 'Sound of My Voice', 'The Blood of My Brother: A Story of Death in Ira
q', "Your Sister's Sister", "A Dog's Breakfast", 'The Married Woman', 'The Birth of
a Nation', 'The Work and The Story', 'Facing the Giants', 'The Gallows', 'Eraserhea
d', 'Hollywood Shuffle', 'The Mighty', 'Penitentiary', 'The Lost Skeleton of Cadavr
a', "Dude Where's My Dog?", 'Cheap Thrills', 'Indie Game: The Movie', 'Straighthead
s', 'Open Secret', 'Echo Dr.', 'The Night Visitor', 'The Past Is a Grotesque Anima
l', 'Peace, Propaganda & the Promised Land', 'Pi', "I Love You, Don't Touch Me!", '2
0 Dates', 'Queen Crab', 'Super Size Me', 'The FP', 'Happy Christmas', "The Brain Tha
t Wouldn't Die", 'Tiger Orange', 'Supporting Characters', 'Absentia', 'The Brothers
McMullen', 'The Dirties', 'Gabriela', 'Tiny Furniture', 'Hayride', 'The Naked Ape',
'Counting', 'The Call of Cthulhu', 'Bending Steel', 'The Signal', 'The Image Revolut
ion', 'This Is Martin Bonner', 'A True Story', 'George Washington', 'Smiling Fish &
Goat On Fire', 'Dawn of the Crescent Moon', 'Raymond Did It', 'The Last Waltz', 'Ru
n, Hide, Die', 'The Exploding Girl', "The Legend of God's Gun", 'Mutual Appreciatio
n', 'Her Cry: La Llorona Investigation', 'Down Terrace', 'Clerks', 'Pink Narcissus',
'Funny Ha Ha', 'In the Company of Men', 'Manito', 'Rampage', 'Slacker', 'Dutch Kill
s', 'Dry Spell', 'Flywheel', 'Backmask', 'The Puffy Chair', 'Stories of Our Lives',
'Breaking Upwards', 'All Superheroes Must Die', 'Pink Flamingos', 'Clean', 'The Circ
le', 'Tin Can Man', 'Cure', 'On The Downlow', 'Sanctuary: Quite a Conundrum', 'Ban
g', 'Primer', 'Cavite', 'El Mariachi', 'Newlyweds', 'Signed, Sealed, Delivered', 'Sh
anghai Calling', 'My Date with Drew']
```

In [20]:

```
# finding the close match for the movie name given by the user

find_close_match = difflib.get_close_matches(movie_name, list_of_all_titles)
print(find_close_match)
```

```
['Batman', 'Batman', 'Catwoman']
```

In [21]:

```
close_match = find_close_match[0]
print(close_match)
```

```
Batman
```

In [22]:
```
# finding the index of the movie with title

index_of_the_movie = movies_data[movies_data.title == close_match]['index'].values[
print(index_of_the_movie)
```

```
1359
```

In [23]:
```
# getting a list of similar movies

similarity_score = list(enumerate(similarity[index_of_the_movie]))
print(similarity_score)
```

```
[(0, 0.02531512269737111), (1, 0.04983293064399152), (2, 0.013599520029326722), (3,
0.20438773732168222), (4, 0.024929726723526918), (5, 0.11533013884014888), (6, 0.0),
(7, 0.005521938648290493), (8, 0.03093493520052642), (9, 0.0897616996163815), (10,
0.11240850198526249), (11, 0.011808001891564552), (12, 0.03825857693295372), (13, 0.
01326374702786566), (14, 0.12815026535769386), (15, 0.010432758242497996), (16, 0.00
5235971636780309), (17, 0.01601669810036477), (18, 0.04153881551585253), (19, 0.0296
8626711977731), (20, 0.025210054010122055), (21, 0.004591239318101996), (22, 0.01792
276874908081), (23, 0.024870438166594427), (24, 0.04186550745628901), (25, 0.0), (2
6, 0.005699120907668437), (27, 0.013106673998149269), (28, 0.014389581335300713), (2
9, 0.005723805699010054), (30, 0.11336219425554919), (31, 0.02290308453698018), (32,
0.0899394677673743), (33, 0.005487629188043741), (34, 0.0), (35, 0.02579241665117427
3), (36, 0.04323488875573723), (37, 0.022073100012780275), (38, 0.01673525129380482
7), (39, 0.03733968312911064), (40, 0.029604400185067402), (41, 0.1065015460468871
8), (42, 0.07244022042935871), (43, 0.023689369843914104), (44, 0.00589686570508782
9), (45, 0.004962729367343318), (46, 0.05119279389322904), (47, 0.00605696785900382
8), (48, 0.01680183887529584), (49, 0.007462062943836138), (50, 0.02447364757740373
7), (51, 0.004490688090626269), (52, 0.03018245470107544), (53, 0.02485276046737457
5), (54, 0.038282186098375935), (55, 0.005755274570173456), (56, 0.00613609490426745
4), (57, 0.009917061156672187), (58, 0.025146268236415702), (59, 0.01358861630969206
1), (60, 0.0), (61, 0.01635015002611376), (62, 0.005416634534426565 5), (63, 0.018338
89030024455), (64, 0.028404894035231252), (65, 0.1775581506611392), (66, 0.017948741
60703813), (67, 0.0), (68, 0.005130123069684274), (69, 0.01618329334393077), (70, 0.
005371331822946504), (71, 0.021555882140030593), (72, 0.12266571244563478), (73, 0.0
4415940718548016), (74, 0.005495141965586583), (75, 0.022057119691207824), (76, 0.00
5516189273301243), (77, 0.008716823552559781), (78, 0.012046810653425794), (79, 0.02
9113928626152034), (80, 0.02148467123933299), (81, 0.02794528136793173), (82, 0.0057
6827305044206), (83, 0.019994216200751452), (84, 0.013308886706222196), (85, 0.01714
8934252787445), (86, 0.009327381057907095), (87, 0.0), (88, 0.014079698000279712),
(89, 0.025715410181997155), (90, 0.02910820842351401), (91, 0.005291140630302694),
(92, 0.011341251045199981), (93, 0.02241215665416583), (94, 0.005318310294658278),
(95, 0.02564889421647799), (96, 0.03097291388389124), (97, 0.004252359556894674), (9
8, 0.032193726743405454), (99, 0.026147609016141292), (100, 0.04263848763117278), (1
01, 0.0359840483687721), (102, 0.015251059855876558), (103, 0.054354666030584614),
(104, 0.005589170793477882), (105, 0.02119994427869696), (106, 0.01148195449626371
1), (107, 0.01737021995321565), (108, 0.023241981578083354), (109, 0.021776146296002
157), (110, 0.01663593844713563), (111, 0.025868460781820198), (112, 0.0142261977951
6028), (113, 0.039521340648464964), (114, 0.02742976146472843), (115, 0.029205437008
730902), (116, 0.03348693881888547), (117, 0.060492562884146284), (118, 0.0101850516
15491632), (119, 0.19262528757150407), (120, 0.008755244893174786), (121, 0.03217909
8673227186), (122, 0.015566153510035626), (123, 0.013352335707811847), (124, 0.00792
1651394546356), (125, 0.013144997885444097), (126, 0.023353967193491126), (127, 0.00
57600906412345766), (128, 0.0), (129, 0.01657534659515722), (130, 0.0), (131, 0.0272
746554799872), (132, 0.008212602697406698), (133, 0.0652059210043185), (134, 0.00516
25316423586885), (135, 0.02868261916813157), (136, 0.0), (137, 0.0307773744877348),
(138, 0.015637725129214016), (139, 0.03565072498219697), (140, 0.01481722217710936
2), (141, 0.0), (142, 0.0), (143, 0.01125008459005959), (144, 0.0), (145, 0.0), (14
6, 0.0), (147, 0.004926143004899927), (148, 0.02654830260044715), (149, 0.0408593843
2174789), (150, 0.03821300742584273), (151, 0.005321016206774113), (152, 0.033314462
70419187), (153, 0.005104759932632073), (154, 0.020861732883614623), (155, 0.0), (15
6, 0.043192368499795894), (157, 0.005748857276590655), (158, 0.015178542794596593),
(159, 0.03514905655970912), (160, 0.025456109332957284), (161, 0.01824859863993154
3), (162, 0.017508564372075104), (163, 0.09490815555089474), (164, 0.057663528142752
97), (165, 0.015898883695491906), (166, 0.006403384496392318), (167, 0.0055361916292
12126), (168, 0.015630626662545388), (169, 0.005393553666615686), (170, 0.0432624527
04246564), (171, 0.034003432577698174), (172, 0.019057091235472567), (173, 0.0), (17
4, 0.05055026688169773), (175, 0.02848895697576779), (176, 0.008199725823514694), (1
```

77, 0.020029123423841614), (178, 0.009357441930127653), (179, 0.009209159657810564),
(180, 0.005863964523086625), (181, 0.021652893869808187), (182, 0.0235365697881937
7), (183, 0.027644775352911852), (184, 0.03493917823417467), (185, 0.020383042371195
83), (186, 0.04459616327625059), (187, 0.01553413453427916), (188, 0.005544672081248
338), (189, 0.02948375476268129), (190, 0.00940060870625009), (191, 0.01907580125132
0162), (192, 0.0), (193, 0.005764152853658987), (194, 0.03175671421104142), (195, 0.
01867767015552487), (196, 0.050872350184735876), (197, 0.02924749073264552), (198,
0.033759003475800534), (199, 0.04127383784609175), (200, 0.019672291888541544), (20
1, 0.009612871394229363), (202, 0.0), (203, 0.013006617597419886), (204, 0.022130509
45729524), (205, 0.02298215684826604), (206, 0.015104619651285126), (207, 0.07517731
687148192), (208, 0.02419897314958249), (209, 0.0056380653548724095), (210, 0.257379
99820859625), (211, 0.029823908263710685), (212, 0.02324123457380759), (213, 0.04328
647825815964), (214, 0.009119664204350115), (215, 0.0773178135949982), (216, 0.01330
8904195041413), (217, 0.06365812644026905), (218, 0.01820775128152958), (219, 0.0438
72578197580316), (220, 0.024925248865578585), (221, 0.020074947827866736), (222, 0.0
056619673562304604), (223, 0.04298330016886992), (224, 0.06381093636174208), (225,
0.006093588113836095), (226, 0.08009811585333676), (227, 0.005390690104302215), (22
8, 0.005143517638008635), (229, 0.014714592345297876), (230, 0.00429551502513737),
(231, 0.0), (232, 0.014866095621163873), (233, 0.00448643551001502), (234, 0.0203329
27975252314), (235, 0.010103203247825286), (236, 0.005512252714746335), (237, 0.0134
07097722261084), (238, 0.02426903109788739), (239, 0.009026133708670711), (240, 0.01
459086841112044), (241, 0.015876309856328475), (242, 0.02596280025216613), (243, 0.0
2970175929625292), (244, 0.004606293916228297), (245, 0.021091411367858307), (246,
0.0050868765893401805), (247, 0.015604601725429042), (248, 0.03215955008792736), (24
9, 0.010041233622224933), (250, 0.024614411005019904), (251, 0.02227356932657924),
(252, 0.022527942902764643), (253, 0.004581957148163642), (254, 0.0292393766257729
6), (255, 0.0), (256, 0.008859149559609284), (257, 0.015269302690415787), (258, 0.01
0689688422080963), (259, 0.020694082901925582), (260, 0.005392803407541181), (261,
0.016436043710683648), (262, 0.016008532494180345), (263, 0.005318416540504533), (26
4, 0.01797732495165702), (265, 0.04045462921658721), (266, 0.005529500831600524), (2
67, 0.03471266703488136), (268, 0.039340417700922385), (269, 0.03592596428678005),
(270, 0.02047072608286063), (271, 0.04205829367344873), (272, 0.03156561287301673),
(273, 0.012493730297180941), (274, 0.005856357216856913), (275, 0.0230140488403933
4), (276, 0.030035534861912905), (277, 0.005460965070966977), (278, 0.12996260715124
025), (279, 0.026139750727758506), (280, 0.026059654002839903), (281, 0.015966624687
11029), (282, 0.004885034999473712), (283, 0.006545138429275421), (284, 0.0100474417
39921288), (285, 0.03771237430169859), (286, 0.005489881771352674), (287, 0.02273805
394855659), (288, 0.0), (289, 0.01043240827881478), (290, 0.01599797560237545), (29
1, 0.03785527048219344), (292, 0.029637035409404967), (293, 0.03850735689533657), (2
94, 0.023311791752251575), (295, 0.029150304221924946), (296, 0.026459567913143592),
(297, 0.015718856581113386), (298, 0.00819747201126473), (299, 0.10154632458465614),
(300, 0.011906750449291664), (301, 0.0), (302, 0.010708044933990125), (303, 0.098839
37789831629), (304, 0.005147368420567504), (305, 0.04876645858500153), (306, 0.01399
3128616778198), (307, 0.005291568896276649), (308, 0.01892387912891352), (309, 0.012
447534711356482), (310, 0.014099823434543982), (311, 0.011388881674615162), (312, 0.
004786517934094848), (313, 0.007900242606542415), (314, 0.03779388023290209), (315,
0.030148920668698065), (316, 0.027439128948452064), (317, 0.0), (318, 0.024843700685
77551), (319, 0.03973736386791702), (320, 0.00547701853389258), (321, 0.008500313679
377706), (322, 0.016956729892305192), (323, 0.0607910339666117), (324, 0.03378259844
361217), (325, 0.0), (326, 0.01988992886308015), (327, 0.035499571393971446), (328,
0.015678536654377837), (329, 0.03350335414928485), (330, 0.016471203129028878), (33
1, 0.03519813630133888), (332, 0.020559468783685246), (333, 0.0), (334, 0.0228339763
32210133), (335, 0.005426700203861826), (336, 0.0192215086758088), (337, 0.003576033
7459753606), (338, 0.01007104124077427), (339, 0.03274058500809053), (340, 0.0179040
03766269105), (341, 0.010410592276632605), (342, 0.05232110588872813), (343, 0.05575
4153391077026), (344, 0.004527095216613905), (345, 0.014815714442063393), (346, 0.00

7930631777048608), (347, 0.0), (348, 0.009826441016833736), (349, 0.0102640529458567
3), (350, 0.022061634090390726), (351, 0.08174380966331447), (352, 0.016325318819716
01), (353, 0.028830612918124995), (354, 0.028092556012041955), (355, 0.0057178695253
29317), (356, 0.013682520917539964), (357, 0.021209560048322404), (358, 0.0495301716
63648544), (359, 0.0), (360, 0.0), (361, 0.01399571574610273), (362, 0.0058741358905
34837), (363, 0.0), (364, 0.04627287142241801), (365, 0.0), (366, 0.0545155291365160
35), (367, 0.016323992554575013), (368, 0.009286540367566893), (369, 0.0275549918668
46026), (370, 0.02378016286738358), (371, 0.01132543208956635), (372, 0.048009596046
192184), (373, 0.03668963121270457), (374, 0.020433371520806647), (375, 0.0), (376,
0.013490994569417328), (377, 0.012446451794775805), (378, 0.0), (379, 0.013866184674
91552), (380, 0.020445412922726842), (381, 0.019973245359529393), (382, 0.0087272178
72332388), (383, 0.013425762087526855), (384, 0.01743346319543433), (385, 0.0), (38
6, 0.02965908189547961), (387, 0.023127653059049586), (388, 0.02934921712825725), (3
89, 0.015121335493388541), (390, 0.01091824586575766), (391, 0.009929634552399723),
(392, 0.0051707235325743955), (393, 0.03926398816349792), (394, 0.00549157207785756
7), (395, 0.01769894802791771), (396, 0.04306307563001721), (397, 0.0), (398, 0.0165
7709368003764), (399, 0.0), (400, 0.022379891670723233), (401, 0.0), (402, 0.0140812
91604134655), (403, 0.032827053531317485), (404, 0.0), (405, 0.017228008857348275),
(406, 0.034700585101405584), (407, 0.022082253077502827), (408, 0.02595533830377956
5), (409, 0.0057873791531853295), (410, 0.01842626297999868), (411, 0.02566569980689
7826), (412, 0.013282417846038267), (413, 0.020320877783699225), (414, 0.00755829445
9328721), (415, 0.05019472209444976), (416, 0.01950646882338211), (417, 0.0715487180
0648291), (418, 0.0), (419, 0.03227370424411756), (420, 0.01917965793095018), (421,
0.022237855115106564), (422, 0.06660418021192085), (423, 0.022164506254997107), (42
4, 0.005363999198083402), (425, 0.015709173817807263), (426, 0.05387121165126171),
(427, 0.008950276812039562), (428, 0.4311643836232694), (429, 0.007639094433465508),
(430, 0.0), (431, 0.01955906316172861), (432, 0.005354651274163252), (433, 0.0242845
94182904715), (434, 0.030463968157348382), (435, 0.011194417840764729), (436, 0.0),
(437, 0.015907779911695567), (438, 0.10277392559139559), (439, 0.0), (440, 0.0094766
61648551307), (441, 0.01933815623254388), (442, 0.019972564229175732), (443, 0.00972
5944990768265), (444, 0.0404675132234011), (445, 0.0), (446, 0.043751045863748145),
(447, 0.05428674675430024), (448, 0.031068559001450127), (449, 0.02617936780109152
5), (450, 0.009793441405616657), (451, 0.010633205817964796), (452, 0.02545015187630
1928), (453, 0.013863948458613266), (454, 0.01514420210517302), (455, 0.005320094859
069555), (456, 0.02234564140242137), (457, 0.01486533530711603), (458, 0.01386482526
9116102), (459, 0.0), (460, 0.0), (461, 0.0), (462, 0.0), (463, 0.01948577981441077
7), (464, 0.0), (465, 0.0), (466, 0.015672308326365784), (467, 0.01838115290265536
5), (468, 0.02251625813538535), (469, 0.012973204775475542), (470, 0.040142637072399
304), (471, 0.009238062231059729), (472, 0.015751681648498644), (473, 0.132170757147
94208), (474, 0.0), (475, 0.01208595098449301), (476, 0.035098251043391517), (477,
0.009475612365586384), (478, 0.0053886847672649), (479, 0.020305875414362677), (480,
0.07657647328497882), (481, 0.013549442607797264), (482, 0.0), (483, 0.0206789817215
70948), (484, 0.04279273956916554), (485, 0.01752986107581509), (486, 0.015608380179
554501), (487, 0.004688304126411212), (488, 0.0184307721862186), (489, 0.01276794594
5677046), (490, 0.005263014548414424), (491, 0.0047044149113152686), (492, 0.0), (49
3, 0.014182387449480029), (494, 0.04150345826328174), (495, 0.023187014523033456),
(496, 0.0), (497, 0.019131469790203838), (498, 0.0), (499, 0.030802219590188137), (5
00, 0.021931087102158506), (501, 0.020194433221440527), (502, 0.0), (503, 0.02855833
5212354184), (504, 0.0), (505, 0.03293303935987978), (506, 0.01040546758365313), (50
7, 0.005304626448524463), (508, 0.005435696902570209), (509, 0.0), (510, 0.041351079
19955205), (511, 0.005574324286548584), (512, 0.013940997040424058), (513, 0.0505558
30048219536), (514, 0.006629023158113295), (515, 0.007771177787962552), (516, 0.0269
48099229211347), (517, 0.008449304195879297), (518, 0.04743670355350071), (519, 0.00
697827718224983), (520, 0.0), (521, 0.014661275300078583), (522, 0.0), (523, 0.01444
8825923945762), (524, 0.009841838523789637), (525, 0.031164697303354742), (526, 0.
0), (527, 0.02149067133338399), (528, 0.03132786516195961), (529, 0.0048484211877960

6), (530, 0.005930164405386325), (531, 0.013124236295722512), (532, 0.0), (533, 0.02
0607458605063626), (534, 0.023985316166857368), (535, 0.013063006814410263), (536,
0.024709079565036842), (537, 0.02427768193470615), (538, 0.007693854925330783), (53
9, 0.015620166799028124), (540, 0.006985241536664359), (541, 0.0043225778169406195),
(542, 0.03116779967343339), (543, 0.013606161618896832), (544, 0.01407447274261928
8), (545, 0.0), (546, 0.04812265039724559), (547, 0.03811071415071551), (548, 0.0073
68097391411546), (549, 0.03234776276810046), (550, 0.0), (551, 0.0), (552, 0.0), (55
3, 0.015596772165942518), (554, 0.027336465415042144), (555, 0.028835549065099268),
(556, 0.031050097534936583), (557, 0.00887802079415849 3), (558, 0.00785446642436289
3), (559, 0.012146856076258003), (560, 0.026948105474084955), (561, 0.0), (562, 0.
0), (563, 0.020101194795755693), (564, 0.0), (565, 0.02228676521747925), (566, 0.007
898156343199089), (567, 0.010326604547687785), (568, 0.00521079049016164), (569, 0.
0), (570, 0.005399126993304583), (571, 0.03166578323334291), (572, 0.021304857336037
212), (573, 0.005702788084032725), (574, 0.03175091979690506), (575, 0.0099541812719
95429), (576, 0.009371555126477136), (577, 0.01313342138818549), (578, 0.05118048264
656639), (579, 0.004063687013663969), (580, 0.005132187938716139), (581, 0.060287457
625623146), (582, 0.013506406082665956), (583, 0.07617125404896875), (584, 0.0948455
9891817565), (585, 0.011504377423919045), (586, 0.02114556118401161), (587, 0.059954
145662823075), (588, 0.0337808230436831), (589, 0.016147977558358416), (590, 0.01268
9566244122208), (591, 0.020933909686590944), (592, 0.015503618061031099), (593, 0.
0), (594, 0.005463707592966443), (595, 0.00488185450288807), (596, 0.005581198270481
37), (597, 0.03485199393305005), (598, 0.0), (599, 0.0), (600, 0.01678281217245045),
(601, 0.005876017634106431), (602, 0.005978906295761177), (603, 0.02360855965242124
3), (604, 0.005287943847300146), (605, 0.011649660850616457), (606, 0.03537493558252
818), (607, 0.005396861321667464), (608, 0.04271399235417937), (609, 0.0056220550829
06893), (610, 0.004863843383905942), (611, 0.0048829783439722175), (612, 0.034316930
516211534), (613, 0.029439799478706983), (614, 0.014294918009758842), (615, 0.020097
533839594256), (616, 0.0), (617, 0.024580834806846426), (618, 0.021578154359445205),
(619, 0.0), (620, 0.035661449502610534), (621, 0.004894557097422808), (622, 0.005509
171161384699), (623, 0.0), (624, 0.023256724026578128), (625, 0.008928547813392097),
(626, 0.009848353733926413), (627, 0.025374921909953645), (628, 0.00809386349927553
4), (629, 0.010848660624759119), (630, 0.007572650679447448), (631, 0.01915349837104
2826), (632, 0.0), (633, 0.030822554147819863), (634, 0.02228742829296616), (635, 0.
02511799848844787), (636, 0.04807695168680448), (637, 0.0), (638, 0.0396640012815206
2), (639, 0.0), (640, 0.005188852949417256), (641, 0.0), (642, 0.01957765478502748
4), (643, 0.014055241400417404), (644, 0.03288495795270681), (645, 0.005550054924716
862), (646, 0.03263267753825964), (647, 0.024758356045488268), (648, 0.0081962150560
85683), (649, 0.0), (650, 0.012065445896039924), (651, 0.014891096654578048), (652,
0.015607602337934932), (653, 0.0053614195427355614), (654, 0.014802947264353612), (6
55, 0.00578722187675423), (656, 0.01715026457715706), (657, 0.014698218217661397),
(658, 0.018314827903677736), (659, 0.01324817479973928), (660, 0.00689088652767348
5), (661, 0.03284314843410831), (662, 0.06458915958920407), (663, 0.0097174567214167
1), (664, 0.024311299603893725), (665, 0.044432466036149254), (666, 0.03570198060032
459), (667, 0.007758480670631074), (668, 0.015867133047224584), (669, 0.008476776832
315623), (670, 0.03249259788439775), (671, 0.008139658583143746), (672, 0.0280805163
96443968), (673, 0.026391236015151927), (674, 0.03582540316592312), (675, 0.01787172
110914862), (676, 0.022896476422471015), (677, 0.022081994233331022), (678, 0.026284
36556703755), (679, 0.015105481998467923), (680, 0.013446993996021294), (681, 0.0854
3127346751898), (682, 0.0), (683, 0.01878342966515282), (684, 0.006869323823112695),
(685, 0.0054155705046102535), (686, 0.01742628185812934), (687, 0.00513958484928233
2), (688, 0.0), (689, 0.010230918975361563), (690, 0.03851770236770982), (691, 0.031
49943434722697), (692, 0.007853698558792655), (693, 0.018825644668711033), (694, 0.0
4587601234144584), (695, 0.026427326406438738), (696, 0.017964598586746865), (697,
0.008979715217794244), (698, 0.009562708219702466), (699, 0.0), (700, 0.013074717794
68753), (701, 0.0), (702, 0.023585376006270774), (703, 0.0), (704, 0.0), (705, 0.0),
(706, 0.0199623864244022), (707, 0.0), (708, 0.02094691260841528), (709, 0.0), (710,

0.027599939819980525), (711, 0.0112090396051443), (712, 0.0), (713, 0.0), (714, 0.02
702036550624075), (715, 0.04120980984410683), (716, 0.01502579758103098), (717, 0.01
700032147721337), (718, 0.01914861485789185), (719, 0.025098893938504195), (720, 0.0
08818675724175216), (721, 0.032654895212822485), (722, 0.00955041484262565), (723,
0.007322232782478983), (724, 0.012904924977953914), (725, 0.02015703632859361), (72
6, 0.01720298216635047), (727, 0.0), (728, 0.022722227379584077), (729, 0.0), (730,
0.007505541392708739), (731, 0.0714748582813296), (732, 0.015874927643149427), (733,
0.021511382902596524), (734, 0.0), (735, 0.011203923491842288), (736, 0.027711499197
496867), (737, 0.005390453411546613), (738, 0.0), (739, 0.06293449312413521), (740,
0.025316251525823445), (741, 0.030197133885458017), (742, 0.009515795367247296), (74
3, 0.026294183527874625), (744, 0.024328443396996222), (745, 0.014296427336469887),
(746, 0.03230474115794068), (747, 0.013342619881024961), (748, 0.04215680173293189),
(749, 0.0), (750, 0.01820941311825481), (751, 0.0077557918109426955), (752, 0.015818
182406409552), (753, 0.13216136203404205), (754, 0.011638460113210685), (755, 0.0587
05703664613695), (756, 0.0075140585682151465), (757, 0.02504044763241721), (758, 0.0
17760025247642412), (759, 0.01799579972299672), (760, 0.019689159848395556), (761,
0.028394240042822164), (762, 0.04094856007215815), (763, 0.03362588358461441), (764,
0.011441669954161434), (765, 0.0), (766, 0.03980676672275263), (767, 0.0221762306038
47574), (768, 0.016231639482474778), (769, 0.026654856838325223), (770, 0.0), (771,
0.009522138250825733), (772, 0.01727299383550612), (773, 0.0060912321987746185), (77
4, 0.07520276566249046), (775, 0.01700841750112397), (776, 0.011474268733619225), (7
77, 0.036960669106148224), (778, 0.0), (779, 0.05259835223293688), (780, 0.0), (781,
0.035262539448534726), (782, 0.04684563840327667), (783, 0.0), (784, 0.0173923402411
55695), (785, 0.01401254306820612), (786, 0.017540064752089086), (787, 0.01195528048
4410663), (788, 0.03219195497258179), (789, 0.008346563812418268), (790, 0.023255012
007745723), (791, 0.045377002143510065), (792, 0.009988465935533025), (793, 0.0), (7
94, 0.018516917064486584), (795, 0.008189520966561724), (796, 0.0), (797, 0.01080633
614892827), (798, 0.0), (799, 0.028485737447099754), (800, 0.06953224882098635), (80
1, 0.0), (802, 0.007997915371082898), (803, 0.02322558415268294), (804, 0.0132952025
87327762), (805, 0.01487731647798404), (806, 0.03439004428669593), (807, 0.014812082
647965255), (808, 0.005373539761798156), (809, 0.031082492523462457), (810, 0.031267
958136639976), (811, 0.008666754761311112), (812, 0.013532913444182676), (813, 0.144
14128303962467), (814, 0.009917013842729685), (815, 0.019008907792410646), (816, 0.
0), (817, 0.0), (818, 0.005623551909990202), (819, 0.0), (820, 0.01100659961525617
8), (821, 0.03192814897030172), (822, 0.02751652787278477), (823, 0.0053786801080690
31), (824, 0.0), (825, 0.035696287623329465), (826, 0.05858774897651223), (827, 0.05
8651554758132156), (828, 0.022784980087159606), (829, 0.009694072031898334), (830,
0.04368824829368255), (831, 0.04140806316101746), (832, 0.04420586787560895), (833,
0.021656424197982804), (834, 0.005518430748913458 5), (835, 0.0), (836, 0.05743473075
9063225), (837, 0.0), (838, 0.015056916048470145), (839, 0.015031364666698273), (84
0, 0.01234700492015121), (841, 0.029695076657601), (842, 0.03141126738894341), (843,
0.0), (844, 0.013035984977621692), (845, 0.0), (846, 0.031431396655688656), (847, 0.
01609259766631261), (848, 0.014848207460403377), (849, 0.0077578229151192305), (850,
0.005332021757716764), (851, 0.01883109842733956), (852, 0.0260437875929925), (853,
0.0), (854, 0.020651091114809082), (855, 0.007905395000716884), (856, 0.030627841018
69689), (857, 0.00898777354115968), (858, 0.026600054273991522), (859, 0.01406718899
5630914), (860, 0.0), (861, 0.0), (862, 0.025889571270560203), (863, 0.0056050268503
635555), (864, 0.015095832941339711), (865, 0.016343451888915356), (866, 0.014754508
2691226), (867, 0.04611844113646047), (868, 0.0), (869, 0.037335523784109315), (870,
0.10722005407914785), (871, 0.022845843248684217), (872, 0.0), (873, 0.0222942942725
84284), (874, 0.016149801713601915), (875, 0.009174571216758421), (876, 0.0076837729
686955744), (877, 0.017236054816731523), (878, 0.0), (879, 0.016483127455268996), (8
80, 0.06391064887633026), (881, 0.026720355420258073), (882, 0.01426290745322866),
(883, 0.016055388176434853), (884, 0.02069524992863983), (885, 0.0), (886, 0.0175971
53024730687), (887, 0.0), (888, 0.005247979592649513), (889, 0.0), (890, 0.0), (891,
0.0), (892, 0.016697122039663805), (893, 0.012315931792271875), (894, 0.014344007397

738027), (895, 0.0), (896, 0.034300388296503094), (897, 0.009201581335226128), (898, 0.020345862019732805), (899, 0.020258539852295893), (900, 0.0), (901, 0.02256589929032795), (902, 0.0), (903, 0.02228764760119481), (904, 0.0766581543058702), (905, 0.01961555455800855), (906, 0.0), (907, 0.0), (908, 0.0), (909, 0.0), (910, 0.008562719824375384), (911, 0.013533002154688875), (912, 0.0), (913, 0.0), (914, 0.01266440931111332), (915, 0.014172443789618369), (916, 0.0), (917, 0.022329315902897218), (918, 0.02937013579894319), (919, 0.013982166900445563), (920, 0.04162113635621227), (921, 0.0), (922, 0.014022847549478943), (923, 0.026737792232875084), (924, 0.009004915710337635), (925, 0.0), (926, 0.0), (927, 0.02274077420912185), (928, 0.010956482890816462), (929, 0.028892795434023986), (930, 0.02092726424669351), (931, 0.02032983426503213), (932, 0.011805561898974404), (933, 0.023829904876997883), (934, 0.024963375177119905), (935, 0.0636982726513143), (936, 0.01973580193674106), (937, 0.01482783282084929), (938, 0.018134612364586883), (939, 0.0440147245600844), (940, 0.0), (941, 0.028759980051565475), (942, 0.009198503282832091), (943, 0.020796079676191383), (944, 0.008039673761352435), (945, 0.005304072052753753), (946, 0.04690419049379716), (947, 0.03256091563095445), (948, 0.01650860552222675), (949, 0.008816437791937597), (950, 0.011552700370816793), (951, 0.005270744600501487), (952, 0.014337502799936806), (953, 0.011101262555396126), (954, 0.0), (955, 0.02198916642979698), (956, 0.01548572385037909), (957, 0.0), (958, 0.062408672993923664), (959, 0.028598183747496796), (960, 0.0), (961, 0.0), (962, 0.01321831626070123), (963, 0.08093607979181264), (964, 0.016913919497684303), (965, 0.007287993438683428), (966, 0.0), (967, 0.02486759973259156), (968, 0.04550911170536931), (969, 0.05965536040402045), (970, 0.030630405719016898), (971, 0.014467317678648858), (972, 0.015736596850609894), (973, 0.029574286842999085), (974, 0.014645469648839992), (975, 0.008341873263310436), (976, 0.00578993808670283), (977, 0.009952578882221985), (978, 0.04051553636656369), (979, 0.014432562751749999), (980, 0.034991247559081506), (981, 0.035531017419820225), (982, 0.022129547246681848), (983, 0.006931879856714528 5), (984, 0.014374169382007224), (985, 0.006133163108993261), (986, 0.0), (987, 0.008760570772492102), (988, 0.013691169261071875), (989, 0.004924861442996708), (990, 0.013755877372139326), (991, 0.0), (992, 0.03175309929532447), (993, 0.025012133285939552), (994, 0.05183090711955347 4), (995, 0.009864622919803969), (996, 0.00757649862289347), (997, 0.0), (998, 0.031294961134766645), (999, 0.028046435773860068), (1000, 0.02326487032397238), (1001, 0.05239940922199486), (1002, 0.1005262424012352), (1003, 0.011136545861787504), (1004, 0.0), (1005, 0.023749380413372243), (1006, 0.04685486296251217), (1007, 0.0), (1008, 0.00948031654424961), (1009, 0.04799624299749351), (1010, 0.008712524730554715), (1011, 0.0), (1012, 0.005761450271103945), (1013, 0.016580937261998246), (1014, 0.0), (1015, 0.028522202797192253), (1016, 0.03947107235591406), (1017, 0.13713281929528845), (1018, 0.03810064751949809), (1019, 0.0), (1020, 0.0156319885403262), (1021, 0.007900666830026687), (1022, 0.020314123936655725), (1023, 0.0), (1024, 0.0427319011984207 96), (1025, 0.024326126301632486), (1026, 0.03202773810680668), (1027, 0.03836958295913614), (1028, 0.0), (1029, 0.03382335917897186), (1030, 0.02424688609693220 8), (1031, 0.03247038570649348), (1032, 0.0), (1033, 0.015346049259993975), (1034, 0.05465360864260904), (1035, 0.11197582745207195), (1036, 0.0169158005768121), (1037, 0.0), (1038, 0.03304611511529365), (1039, 0.0), (1040, 0.03259510124285758), (1041, 0.011585739784242965), (1042, 0.010938409138879545), (1043, 0.06148903574491431), (1044, 0.00483235694450333), (1045, 0.0091868649473202), (1046, 0.02619576886066415 2), (1047, 0.005741385211568459), (1048, 0.08353639093976094), (1049, 0.031648750799709795), (1050, 0.03858617259693438), (1051, 0.00793784732812935), (1052, 0.03349553822594242), (1053, 0.02096379469007401), (1054, 0.0), (1055, 0.018456087030565557), (1056, 0.004797228573047399), (1057, 0.008637735007822455), (1058, 0.03442714619802041), (1059, 0.018999183607856466), (1060, 0.0), (1061, 0.008960694446576207), (1062, 0.0), (1063, 0.04028775499564254), (1064, 0.0184573138577872), (1065, 0.0), (1066, 0.0), (1067, 0.0), (1068, 0.0), (1069, 0.0), (1070, 0.014711569917009434), (1071, 0.0), (1072, 0.0), (1073, 0.022495004724559745), (1074, 0.024831680205808962), (1075, 0.02386223268207443), (1076, 0.0940401999293739), (1077, 0.004292800055937971), (1078, 0.028131182878539572), (1079, 0.0048429569993686675), (1080, 0.00991772298510242

2), (1081, 0.029366575006229675), (1082, 0.037454814107502454), (1083, 0.00561123378
6916412), (1084, 0.012794674636422042), (1085, 0.10023066759543668), (1086, 0.035124
66679352937), (1087, 0.07653280604972879), (1088, 0.0), (1089, 0.03057820211648519),
(1090, 0.0), (1091, 0.025051791391380876), (1092, 0.03674322166332937), (1093, 0.005
11614242192896), (1094, 0.02046026827116788), (1095, 0.02157535030315407), (1096, 0.
012217180267611535), (1097, 0.051812132668283176), (1098, 0.009026025569565293), (10
99, 0.00798657052156911), (1100, 0.012318282989797822), (1101, 0.00498673369798524
4), (1102, 0.016543564002150606), (1103, 0.020550784118254034), (1104, 0.00490599730
9755556), (1105, 0.007507520950666183), (1106, 0.0049229631964249594), (1107, 0.0),
(1108, 0.02467378377900406), (1109, 0.027724036021638455), (1110, 0.0471083924967759
1), (1111, 0.014657067953291992), (1112, 0.02917610781736512), (1113, 0.044857002035
258026), (1114, 0.014958503157539604), (1115, 0.010045155534122305), (1116, 0.0), (1
117, 0.013209757980126156), (1118, 0.0228065418199093), (1119, 0.01794367291124868),
(1120, 0.05950894940289865), (1121, 0.008294242471426677), (1122, 0.0220283158167087
6), (1123, 0.0), (1124, 0.038948480563430844), (1125, 0.048928847177323063), (1126,
0.0), (1127, 0.027921419084988117), (1128, 0.0), (1129, 0.010999863723701143), (113
0, 0.0), (1131, 0.02909138439562463), (1132, 0.023580888972459302), (1133, 0.0140909
54994041284), (1134, 0.01848419862286736), (1135, 0.0077783141153367945), (1136, 0.0
046405633039279534), (1137, 0.012393198676248784), (1138, 0.012012382378604771), (11
39, 0.01604538145332697), (1140, 0.011144195561107666), (1141, 0.0897534119962937),
(1142, 0.0), (1143, 0.015935447827777442), (1144, 0.05196123866300828), (1145, 0.0),
(1146, 0.02818822002210506), (1147, 0.023191479743118984), (1148, 0.0601076516918592
9), (1149, 0.008531038938045889), (1150, 0.00906537582338922), (1151, 0.014558045430
265763), (1152, 0.015836772148492755), (1153, 0.025722455536786062), (1154, 0.009414
801647695524), (1155, 0.004811608235094448), (1156, 0.017898578279374487), (1157, 0.
0), (1158, 0.005532939602783096), (1159, 0.04666436412856402), (1160, 0.016500915655
068834), (1161, 0.008632937910957851), (1162, 0.0300113239739253), (1163, 0.03787797
511689461), (1164, 0.0), (1165, 0.043492600825020586), (1166, 0.011416736755473642),
(1167, 0.0), (1168, 0.0), (1169, 0.009424153020576552), (1170, 0.01782500402095107),
(1171, 0.009012024986638572), (1172, 0.032801566418994364), (1173, 0.0), (1174, 0.05
648892011570573), (1175, 0.030578352149970636), (1176, 0.022893457942976626), (1177,
0.0335356211125964), (1178, 0.013818194169235676), (1179, 0.009675526837040626), (11
80, 0.02698732438479501), (1181, 0.02957607607291457), (1182, 0.015935098832651835),
(1183, 0.05247200231886881), (1184, 0.03134798916087379), (1185, 0.00824853398426471
4), (1186, 0.008360515643054322), (1187, 0.026063767446498107), (1188, 0.03685944318
062393), (1189, 0.0), (1190, 0.00983133783005054), (1191, 0.03800587212380693), (119
2, 0.04076275116572928), (1193, 0.00469657559411346), (1194, 0.02217266864060887),
(1195, 0.018532073439352398), (1196, 0.03230224759674958), (1197, 0.0085694899495392
94), (1198, 0.0), (1199, 0.004644005471996142), (1200, 0.005104806007917166), (1201,
0.017434336362412362), (1202, 0.027910207620286995), (1203, 0.018025072768320487),
(1204, 0.018014847869633022), (1205, 0.0), (1206, 0.011233965487079353), (1207, 0.02
8933812704153447), (1208, 0.0), (1209, 0.013767757831726598), (1210, 0.0), (1211, 0.
0), (1212, 0.00713363590404558), (1213, 0.0380022147669937), (1214, 0.03473740532650
401), (1215, 0.0351963359206349), (1216, 0.03786025128668011), (1217, 0.034551565291
79429), (1218, 0.0), (1219, 0.0), (1220, 0.03272010843532369), (1221, 0.008123758336
034967), (1222, 0.009897545148202538), (1223, 0.059922058269882), (1224, 0.0), (122
5, 0.01755756885679377), (1226, 0.007504796955659943), (1227, 0.030595747837277358),
(1228, 0.020779494190111946), (1229, 0.0), (1230, 0.02624860488649873), (1231, 0.0),
(1232, 0.03003882024706265), (1233, 0.04066036452760822), (1234, 0.0313019214148565
6), (1235, 0.0), (1236, 0.06921697670526414), (1237, 0.023748477729621457), (1238,
0.019449288579735782), (1239, 0.03526731054775968), (1240, 0.008663063811007912), (1
241, 0.08609227902838198), (1242, 0.0), (1243, 0.050700350160624824), (1244, 0.02329
1964886609697), (1245, 0.0336810882884475), (1246, 0.010619550176089785), (1247, 0.0
53581002184245344), (1248, 0.0), (1249, 0.005656863725235057), (1250, 0.027270723941
12056), (1251, 0.0), (1252, 0.0), (1253, 0.017244356378755998), (1254, 0.02263332376
324078), (1255, 0.007689324296503561), (1256, 0.02857893319033331), (1257, 0.0237411

33133852284), (1258, 0.0), (1259, 0.046382514953439494), (1260, 0.01293307183144992
4), (1261, 0.0), (1262, 0.0), (1263, 0.00824179209911462), (1264, 0.0154294018476275
16), (1265, 0.04320191547401773), (1266, 0.015580760946057096), (1267, 0.00742016381
7329874), (1268, 0.0), (1269, 0.015361647906030014), (1270, 0.021272374341004117),
(1271, 0.0214807964615722), (1272, 0.024788436669427188), (1273, 0.00580799938128251
1), (1274, 0.008809322091417276), (1275, 0.017097624635175872), (1276, 0.00866954282
4084029), (1277, 0.013849161288857003), (1278, 0.0488360430519503), (1279, 0.0226205
57935525098), (1280, 0.03666727665729252), (1281, 0.01160519512710677), (1282, 0.021
236119450291566), (1283, 0.01941897961500091), (1284, 0.00558426838838599), (1285,
0.005629935367298657), (1286, 0.0335424833160683), (1287, 0.010479910941085361), (12
88, 0.007155601262555008), (1289, 0.037943844240752506), (1290, 0.00483446925672430
5), (1291, 0.012822320920255028), (1292, 0.054451481344167885), (1293, 0.06301875726
076642), (1294, 0.01390253701229965), (1295, 0.0), (1296, 0.10584478077207024), (129
7, 0.0), (1298, 0.011650842289952557), (1299, 0.017645987253240195), (1300, 0.017510
638080635035), (1301, 0.022244056799122433), (1302, 0.005545870742472661), (1303, 0.
05585861136454683), (1304, 0.016727586039875213), (1305, 0.022180881228978204), (130
6, 0.012668417213024122), (1307, 0.0), (1308, 0.018447048671171393), (1309, 0.008208
236632796575), (1310, 0.004781821804758146), (1311, 0.03866179996389761), (1312, 0.0
27733575234789337), (1313, 0.013019787580348535), (1314, 0.01931834178864059), (131
5, 0.008197753361605725), (1316, 0.04929772852512586), (1317, 0.005648048512342407),
(1318, 0.008526103682848115), (1319, 0.005340388672171497), (1320, 0.023941458040203
61), (1321, 0.005640488690272382), (1322, 0.01986408056030938), (1323, 0.0), (1324,
0.02534185104536406), (1325, 0.0), (1326, 0.0), (1327, 0.016706318737225168), (1328,
0.01559789092562981), (1329, 0.0), (1330, 0.0), (1331, 0.029667946796972697), (1332,
0.0047864257428798076), (1333, 0.0), (1334, 0.032720866061708175), (1335, 0.00971794
20605383), (1336, 0.008713114481184604), (1337, 0.030063073922606327), (1338, 0.0159
6364374262817), (1339, 0.027037333574238043), (1340, 0.007924608955632453), (1341,
0.010085586585247039), (1342, 0.038950542962535556), (1343, 0.0679178458251457), (13
44, 0.0), (1345, 0.0), (1346, 0.017622689430631954), (1347, 0.005120639902567997),
(1348, 0.03954765666933953), (1349, 0.00841102396954277), (1350, 0.00913659883569761
3), (1351, 0.018211622978689647), (1352, 0.008601196207514825), (1353, 0.0), (1354,
0.022712737635219946), (1355, 0.0), (1356, 0.0), (1357, 0.006434786309166769), (135
8, 0.04131304321764742), (1359, 1.0), (1360, 0.0), (1361, 0.019833133965028015), (13
62, 0.030029774381100886), (1363, 0.057179260936214775), (1364, 0.0334963491112169
2), (1365, 0.0), (1366, 0.019993313149369932), (1367, 0.00850499609895086), (1368,
0.04197806701094106), (1369, 0.0433392780847158), (1370, 0.02085133594949017), (137
1, 0.0), (1372, 0.017666804315864287), (1373, 0.03573361307149581), (1374, 0.0727651
2150372845), (1375, 0.02766969073233455), (1376, 0.0), (1377, 0.025618729675440363),
(1378, 0.0), (1379, 0.008290334565682565), (1380, 0.02296003928356085), (1381, 0.009
900008053882567), (1382, 0.047751731789872645), (1383, 0.0), (1384, 0.02095407735033
2955), (1385, 0.0), (1386, 0.0), (1387, 0.0), (1388, 0.008586463674760108), (1389,
0.01500270486904054), (1390, 0.036918948858573365), (1391, 0.0), (1392, 0.0366019472
2613763), (1393, 0.0), (1394, 0.02980985823371892), (1395, 0.005481974450441863), (1
396, 0.0), (1397, 0.03552015136100155), (1398, 0.005907813640041784), (1399, 0.00566
6944437724892), (1400, 0.037800196373694084), (1401, 0.0), (1402, 0.0109828236019014
55), (1403, 0.03761948767754858), (1404, 0.017851738612026158), (1405, 0.0), (1406,
0.007703678119017641), (1407, 0.017147499642535726), (1408, 0.025636739337390285),
(1409, 0.03777411306063175), (1410, 0.008554984992405472), (1411, 0.0293471417476942
26), (1412, 0.028779662134377625), (1413, 0.018031095825940046), (1414, 0.0), (1415,
0.031008445272316047), (1416, 0.0), (1417, 0.04540145189851705), (1418, 0.0336834986
7931224), (1419, 0.022886057045970763), (1420, 0.03123619410259606), (1421, 0.071453
87858299798), (1422, 0.019508606032306208), (1423, 0.011379127789045778), (1424, 0.
0), (1425, 0.020498630997590347), (1426, 0.02020687227786065), (1427, 0.019692892642
776887), (1428, 0.020219277065378273), (1429, 0.008297681873716949), (1430, 0.018302
01281564051), (1431, 0.03765706687564924), (1432, 0.041097611614166765), (1433, 0.01
640852485747147), (1434, 0.06650848491705977), (1435, 0.0244195061602232), (1436, 0.

0), (1437, 0.004772410004958566), (1438, 0.01450372255063764), (1439, 0.042935086671
72626), (1440, 0.014012766584761716), (1441, 0.025309749409157523), (1442, 0.0187466
23576687924), (1443, 0.030218441046223374), (1444, 0.0), (1445, 0.01323382482272488
3), (1446, 0.01073375076241159), (1447, 0.005065863956775308), (1448, 0.013931988276
047544), (1449, 0.01051049246276645), (1450, 0.01200294191627159), (1451, 0.03767982
1179759584), (1452, 0.0), (1453, 0.030922810477248576), (1454, 0.0), (1455, 0.0), (1
456, 0.009982392031103189), (1457, 0.012659535700538575), (1458, 0.0474449198294579
1), (1459, 0.013270934195827033), (1460, 0.0), (1461, 0.036580189662531445), (1462,
0.006405688609516683), (1463, 0.011555368030724987), (1464, 0.010284505669030521),
(1465, 0.005185230062999871), (1466, 0.028320021861633106), (1467, 0.026217886953468
07), (1468, 0.02263157217854836), (1469, 0.043763468277525225), (1470, 0.02169221680
887711), (1471, 0.0), (1472, 0.018257957400084183), (1473, 0.02752364297483371), (14
74, 0.10078843818576026), (1475, 0.0), (1476, 0.0), (1477, 0.10199709558322863), (14
78, 0.07074037978729494), (1479, 0.01479317528968298), (1480, 0.004833616221008409),
(1481, 0.009800395041701682), (1482, 0.013348013447405924), (1483, 0.0), (1484, 0.01
9377355227446973), (1485, 0.0), (1486, 0.012986286768971267), (1487, 0.0), (1488, 0.
034759677654723334), (1489, 0.0), (1490, 0.02869088694045427), (1491, 0.005139455403
447934), (1492, 0.0257839261152216), (1493, 0.0), (1494, 0.010241691794116957), (149
5, 0.032556584487681256), (1496, 0.0), (1497, 0.034141018029078124), (1498, 0.016217
518358364266), (1499, 0.008864783686100112), (1500, 0.022444863543466864), (1501, 0.
008834379100895198), (1502, 0.00889347569647855), (1503, 0.03922376831304132), (150
4, 0.03248624217891103), (1505, 0.014117239215771592), (1506, 0.059475397637864434),
(1507, 0.021367076059543886), (1508, 0.011824989606273715), (1509, 0.0), (1510, 0.00
539988610208904), (1511, 0.0), (1512, 0.14705162654306442), (1513, 0.052881899233813
89), (1514, 0.015527553951531551), (1515, 0.04371327390526024), (1516, 0.01681427633
276236), (1517, 0.018543716643440747), (1518, 0.013480981994518023), (1519, 0.022800
943234727596), (1520, 0.0), (1521, 0.0), (1522, 0.026846566547207188), (1523, 0.0201
3780616342415), (1524, 0.022652613152636603), (1525, 0.009157339065228243), (1526,
0.0), (1527, 0.022850734653789555), (1528, 0.006094931444203884), (1529, 0.019221268
36497642), (1530, 0.0), (1531, 0.01973666357808295), (1532, 0.0), (1533, 0.021349851
527078183), (1534, 0.0), (1535, 0.018070538742894514), (1536, 0.0), (1537, 0.0167670
59622503695), (1538, 0.02020610747068459), (1539, 0.0), (1540, 0.0), (1541, 0.025113
263910305602), (1542, 0.012509475622448047), (1543, 0.02151162704012102), (1544, 0.
0), (1545, 0.008106087729578507), (1546, 0.028197686626645405), (1547, 0.01840224832
923179), (1548, 0.0223742017218123), (1549, 0.009703900509859342), (1550, 0.00720572
3586617939), (1551, 0.007491678996788645), (1552, 0.034257172964755264), (1553, 0.00
7158359547476221), (1554, 0.009013976222033116), (1555, 0.0), (1556, 0.0234737030312
52284), (1557, 0.0), (1558, 0.038146343919781674), (1559, 0.013785794831521148), (15
60, 0.008825888751138114), (1561, 0.05300836086982752), (1562, 0.00837606522606324
1), (1563, 0.012828588064996519), (1564, 0.02461725209610294), (1565, 0.017407385132
509796), (1566, 0.06940034499958737), (1567, 0.0), (1568, 0.005030433989265982), (15
69, 0.01465181323774375), (1570, 0.009061741366227418), (1571, 0.0174267609054211),
(1572, 0.021142460933170244), (1573, 0.012317650873172716), (1574, 0.037530047757129
27), (1575, 0.02814387701249745), (1576, 0.008909604562719449), (1577, 0.03719006928
4506376), (1578, 0.034549399484208515), (1579, 0.03581020933631408), (1580, 0.0), (1
581, 0.007108888278063722), (1582, 0.0), (1583, 0.01163216536219096), (1584, 0.00892
2551492771418), (1585, 0.015606251271792982), (1586, 0.005513418879816727), (1587,
0.0), (1588, 0.012520420418828145), (1589, 0.03100537870922127), (1590, 0.0), (1591,
0.008993917973956286), (1592, 0.03015243135727702), (1593, 0.0), (1594, 0.0603608083
986681), (1595, 0.02541498779945982), (1596, 0.021529974960177826), (1597, 0.0189101
0136816346), (1598, 0.008211232625811155), (1599, 0.011080660620207674), (1600, 0.04
9336396895483095), (1601, 0.0), (1602, 0.0), (1603, 0.011132776533034102), (1604, 0.
0), (1605, 0.04300799644485527), (1606, 0.025204389615138215), (1607, 0.014061167248
561286), (1608, 0.020125990512872644), (1609, 0.00784647007888458), (1610, 0.0053693
26219846216), (1611, 0.04112474398198277), (1612, 0.022943313874495386), (1613, 0.
0), (1614, 0.0), (1615, 0.01297770565525629), (1616, 0.004847094862544482), (1617,

0.0053964852496758515), (1618, 0.02122657151451314), (1619, 0.023674792256187502),
(1620, 0.0), (1621, 0.009394793734920446), (1622, 0.0072445371562285575), (1623, 0.0
09712381823606215), (1624, 0.03752419572986529), (1625, 0.012266285368827462), (162
6, 0.020397431679884087), (1627, 0.008299623985191733), (1628, 0.01477286695850043
6), (1629, 0.06407184052001345), (1630, 0.024152976998750332), (1631, 0.018306258356
871125), (1632, 0.02415532971118427), (1633, 0.014610411586649176), (1634, 0.0), (16
35, 0.02974846766584079), (1636, 0.005776708803441586), (1637, 0.02397936733691121
2), (1638, 0.0), (1639, 0.00688898481806359), (1640, 0.0), (1641, 0.0082084368593815
08), (1642, 0.0), (1643, 0.035529457558863255), (1644, 0.0284051403211077), (1645,
0.005031870998304777), (1646, 0.0), (1647, 0.014538793312907415), (1648, 0.009446061
997133983), (1649, 0.0), (1650, 0.009552085915181458), (1651, 0.032880834361713684),
(1652, 0.03993685756670881), (1653, 0.04545837835897055), (1654, 0.0), (1655, 0.0459
4587261868763), (1656, 0.008260393599957882), (1657, 0.01288095757721532), (1658, 0.
0128506534095462), (1659, 0.01768280470469291), (1660, 0.015244375297957365), (1661,
0.03172097115172967), (1662, 0.0), (1663, 0.02353232186537981), (1664, 0.01938077852
5990256), (1665, 0.021434439428366), (1666, 0.006121681872805396), (1667, 0.00691303
8735809923), (1668, 0.02797195137612423), (1669, 0.023147894763598367), (1670, 0.028
79239276386738), (1671, 0.013203531345148078), (1672, 0.012543682946555779), (1673,
0.0), (1674, 0.04859943492495315), (1675, 0.025041519709610438), (1676, 0.0072165552
84997412), (1677, 0.0), (1678, 0.0210044775606081), (1679, 0.0242445472580111), (168
0, 0.033016471643636004), (1681, 0.0076795369276497815), (1682, 0.0303294315535652
1), (1683, 0.0), (1684, 0.0), (1685, 0.05279678377370657), (1686, 0.0142273329542451
64), (1687, 0.025927763570016374), (1688, 0.007844204355925711), (1689, 0.0257764026
9568751), (1690, 0.0), (1691, 0.0), (1692, 0.03823028320857572), (1693, 0.0092267518
00888274), (1694, 0.011067411156849118), (1695, 0.011665180329931612), (1696, 0.0085
29304605176043), (1697, 0.004806718483768213), (1698, 0.0), (1699, 0.041774080788419
76), (1700, 0.0), (1701, 0.018458944281127387), (1702, 0.0051904936226976435), (170
3, 0.0), (1704, 0.0), (1705, 0.009382041231411158), (1706, 0.017769568833652126), (1
707, 0.022391752195013968), (1708, 0.012537489703183169), (1709, 0.0), (1710, 0.0),
(1711, 0.009076622623200492), (1712, 0.02004263383622975), (1713, 0.0052788304698636
47), (1714, 0.023259362870364012), (1715, 0.010511793570983457), (1716, 0.0), (1717,
0.01828553955129434), (1718, 0.009201847044412254), (1719, 0.013773664037395458), (1
720, 0.07785389662462186), (1721, 0.021242657560316115), (1722, 0.0171434874693426
8), (1723, 0.0), (1724, 0.0), (1725, 0.0), (1726, 0.03080907028115161), (1727, 0.014
639101750610525), (1728, 0.015600503534506037), (1729, 0.0), (1730, 0.0), (1731, 0.0
256420372576307), (1732, 0.00958640864870957), (1733, 0.0058969097159748), (1734, 0.
026818297763628863), (1735, 0.011636958444510006), (1736, 0.007935721003861168), (17
37, 0.02175420736919535), (1738, 0.0), (1739, 0.008650953199043365), (1740, 0.089093
61212904546), (1741, 0.005446897459944766), (1742, 0.012841432896343572), (1743, 0.0
048745050123190848), (1744, 0.008940137353045371), (1745, 0.020957586466537226), (174
6, 0.015557358009521903), (1747, 0.02712577089309213 2), (1748, 0.01637881312634858),
(1749, 0.009012775603607633), (1750, 0.012683369244718658), (1751, 0.008905604206778
191), (1752, 0.006062587214095426), (1753, 0.012468725762600427), (1754, 0.0), (175
5, 0.0), (1756, 0.0), (1757, 0.02541814734408382), (1758, 0.04003917737948752), (175
9, 0.00967239657720542), (1760, 0.0), (1761, 0.016892274395189356), (1762, 0.0095691
50570978192), (1763, 0.008630437830550624), (1764, 0.019117169709239306), (1765, 0.0
26470240858585455), (1766, 0.0), (1767, 0.00925444509582254), (1768, 0.0239909056484
39757), (1769, 0.006365628221953609), (1770, 0.022181256487692737), (1771, 0.0793942
4641304792), (1772, 0.0), (1773, 0.006058671554193209), (1774, 0.0), (1775, 0.033504
89539535297), (1776, 0.0), (1777, 0.005865537480103254), (1778, 0.00779018876808297
3), (1779, 0.029459817144237448), (1780, 0.021495432172595943), (1781, 0.01646201780
6447998), (1782, 0.056308948618240914), (1783, 0.015125416576253315), (1784, 0.0),
(1785, 0.03713984750749881), (1786, 0.013756186572400164), (1787, 0.0323203106416629
1), (1788, 0.022565812244005345), (1789, 0.019187973527433757), (1790, 0.04855656948
2376466), (1791, 0.017205808939932586), (1792, 0.03958339043802276), (1793, 0.034882
82343738186), (1794, 0.03735548424081349), (1795, 0.018098314488782578), (1796, 0.02

5755866711754498), (1797, 0.0), (1798, 0.008421812224210398), (1799, 0.0402862809130
8526), (1800, 0.022727310755579308), (1801, 0.0), (1802, 0.039568680994339814), (180
3, 0.09486132363970294), (1804, 0.03564599197166588), (1805, 0.008628890947105694),
(1806, 0.019898305087949535), (1807, 0.01704209471712633), (1808, 0.0), (1809, 0.0),
(1810, 0.0), (1811, 0.007382294067259588), (1812, 0.019278369520200022), (1813, 0.07
02107976791396), (1814, 0.013890288225782392), (1815, 0.005178792908771225), (1816,
0.032128349965699056), (1817, 0.011456444443825528), (1818, 0.019362801191886492),
(1819, 0.020337825225187404), (1820, 0.01675024411032386), (1821, 0.0098347915709271
45), (1822, 0.03617576829397694), (1823, 0.0), (1824, 0.0378981012798182), (1825, 0.
005092744533513066), (1826, 0.025205395499702174), (1827, 0.0644590393699526), (182
8, 0.016407214488317088), (1829, 0.007396501854834192), (1830, 0.04049210660090152),
(1831, 0.0), (1832, 0.00843057883447386), (1833, 0.0), (1834, 0.01976863725904065),
(1835, 0.0), (1836, 0.0052912791828490634), (1837, 0.04217668058421618), (1838, 0.03
211193330356925), (1839, 0.018061614132809926), (1840, 0.0), (1841, 0.0), (1842, 0.0
2161818377017973), (1843, 0.01737128670155446), (1844, 0.005697925012441475), (1845,
0.00965050855111265), (1846, 0.05858161919230411), (1847, 0.042712112247782), (1848,
0.020732962869425876), (1849, 0.019902901381266977), (1850, 0.023527216513489886),
(1851, 0.05137565980020797), (1852, 0.011855724053823681), (1853, 0.0223108647307386
32), (1854, 0.005054757724731262), (1855, 0.0), (1856, 0.02499576497226054), (1857,
0.02994956194695472), (1858, 0.06975691386608407), (1859, 0.008217356184935931), (18
60, 0.014384186580270561), (1861, 0.01957254411962333), (1862, 0.0), (1863, 0.042031
90343734929), (1864, 0.0), (1865, 0.009947359302382813), (1866, 0.0), (1867, 0.0),
(1868, 0.013386345355129746), (1869, 0.022547764366525922), (1870, 0.024679406821881
085), (1871, 0.0), (1872, 0.005561831391579429), (1873, 0.04134639130536735), (1874,
0.007904344677483348), (1875, 0.0), (1876, 0.0), (1877, 0.011109943904804038), (187
8, 0.025147809940367815), (1879, 0.02935651787512665), (1880, 0.005514445043253252),
(1881, 0.04382449735381551), (1882, 0.008850653355231598), (1883, 0.0172180144969415
4), (1884, 0.04898143796896101), (1885, 0.0), (1886, 0.05135926016803144), (1887, 0.
0), (1888, 0.0), (1889, 0.02524835968971871), (1890, 0.041713935503622346), (1891,
0.0), (1892, 0.011371550377381502), (1893, 0.031729419913026816), (1894, 0.023451828
48817197), (1895, 0.0), (1896, 0.017648617559057802), (1897, 0.035985601534210425),
(1898, 0.0), (1899, 0.022681210769162134), (1900, 0.01739583785738129), (1901, 0.0),
(1902, 0.06175659279873491), (1903, 0.0), (1904, 0.008248325920015233), (1905, 0.008
543211535626222), (1906, 0.0), (1907, 0.021249325849423597), (1908, 0.00495746534250
9351), (1909, 0.007995232460302829), (1910, 0.024075304155847868), (1911, 0.0), (191
2, 0.019251555136440564), (1913, 0.01968621374115141), (1914, 0.03613944838100905),
(1915, 0.016097319841717375), (1916, 0.015762545444655517), (1917, 0.0), (1918, 0.04
544963560382887), (1919, 0.0), (1920, 0.005884147455237689), (1921, 0.02528183097818
1936), (1922, 0.0), (1923, 0.007531840305792245), (1924, 0.010416920962543878), (192
5, 0.031469921240613145), (1926, 0.008734337441492154), (1927, 0.03150432010600775
4), (1928, 0.0), (1929, 0.05267613017000264), (1930, 0.011866643277531164), (1931,
0.019528054525154097), (1932, 0.017504834537351913), (1933, 0.006029926885612366),
(1934, 0.0), (1935, 0.03768693515944097), (1936, 0.02195382896066036), (1937, 0.0166
43692779351245), (1938, 0.0), (1939, 0.03384359049313771), (1940, 0.0), (1941, 0.0),
(1942, 0.009289478917794802), (1943, 0.029039093987303304), (1944, 0.030380938402390
29), (1945, 0.013419865151133454), (1946, 0.022231660339560783), (1947, 0.0193385063
93074367), (1948, 0.007135786975592431), (1949, 0.0), (1950, 0.0), (1951, 0.0), (195
2, 0.025603899893977516), (1953, 0.005717713486632421), (1954, 0.02777665774401352
8), (1955, 0.011214231219779627), (1956, 0.0055754579569536455), (1957, 0.0172066346
61685045), (1958, 0.009543457590741138), (1959, 0.03694055596901151), (1960, 0.01212
2863594735973), (1961, 0.0), (1962, 0.007365899595637163), (1963, 0.0580852636274594
16), (1964, 0.06641267452470394), (1965, 0.0), (1966, 0.015416505847956232), (1967,
0.02689508947853423), (1968, 0.0), (1969, 0.02481791143280848), (1970, 0.02199812417
6773493), (1971, 0.006677196039465772), (1972, 0.0), (1973, 0.013340100815097008),
(1974, 0.016841460497583626), (1975, 0.009956953571886654), (1976, 0.050266582973901
706), (1977, 0.04919724187128134), (1978, 0.016225781808052273), (1979, 0.0), (1980,

0.008697317750059086), (1981, 0.026229566109401775), (1982, 0.025472347325829457),
(1983, 0.011279617573603204), (1984, 0.012732449525331598), (1985, 0.0), (1986, 0.01
4934726112894302), (1987, 0.01695800236961683), (1988, 0.0), (1989, 0.04159987031212
476), (1990, 0.014558528630020558), (1991, 0.032358846958819215), (1992, 0.034138506
44041443), (1993, 0.008051298461780137), (1994, 0.0), (1995, 0.03942957756973021),
(1996, 0.02294561494723391), (1997, 0.0), (1998, 0.0), (1999, 0.040834017589824856),
(2000, 0.0), (2001, 0.034544979886007694), (2002, 0.025039274493230384), (2003, 0.01
8252711034050545), (2004, 0.04008942140881579), (2005, 0.0), (2006, 0.0), (2007, 0.0
2623971423777043), (2008, 0.015982728574063187), (2009, 0.0), (2010, 0.0), (2011, 0.
0), (2012, 0.0), (2013, 0.004356932703326003), (2014, 0.025349207709217652), (2015,
0.03696916006610588), (2016, 0.011212873969711067), (2017, 0.009922309775577952), (2
018, 0.020982624753504646), (2019, 0.02620975823102622), (2020, 0.02865455052450085
7), (2021, 0.01644214799152987), (2022, 0.02133537645385125), (2023, 0.0046999340959
0895), (2024, 0.008504331010860676), (2025, 0.03892760176597966), (2026, 0.049807132
061791326), (2027, 0.00953146635438233), (2028, 0.0), (2029, 0.054354817322176), (20
30, 0.0), (2031, 0.008349646169041549), (2032, 0.0), (2033, 0.014408137063061224),
(2034, 0.044983445730573966), (2035, 0.04454105154294242), (2036, 0.0415110702206215
95), (2037, 0.00982907857108947), (2038, 0.007761358710588314), (2039, 0.0), (2040,
0.008526911385754945), (2041, 0.0), (2042, 0.03087894974545649), (2043, 0.0136618607
98531617), (2044, 0.0), (2045, 0.0), (2046, 0.022094312947349776), (2047, 0.08005984
967911549), (2048, 0.02733570859398382), (2049, 0.01749051128727583), (2050, 0.01230
7645119260018), (2051, 0.0), (2052, 0.03302849529816566), (2053, 0.0), (2054, 0.0303
08363102700166), (2055, 0.0543643162789638), (2056, 0.0099972787446526), (2057, 0.02
370800340092297), (2058, 0.0), (2059, 0.0), (2060, 0.018013744482188336), (2061, 0.0
3474280688013156), (2062, 0.03749317710981428), (2063, 0.02200818844793445), (2064,
0.0), (2065, 0.0), (2066, 0.05609985436718217), (2067, 0.0), (2068, 0.02167841015806
747), (2069, 0.03242684366658055), (2070, 0.06136751707477053), (2071, 0.00987938611
7380753), (2072, 0.0076435028807289845), (2073, 0.0), (2074, 0.012125007913738632),
(2075, 0.005219967962226397), (2076, 0.0), (2077, 0.0), (2078, 0.02588070604051220
8), (2079, 0.0), (2080, 0.008865215351355408), (2081, 0.00989865939708576), (2082,
0.014648987924051421), (2083, 0.016832288661381645), (2084, 0.005235231116409664),
(2085, 0.028594412014338495), (2086, 0.024980719858491693), (2087, 0.0), (2088, 0.01
0247263576363835), (2089, 0.014334077242662028), (2090, 0.02273945608357307), (2091,
0.022517506395989453), (2092, 0.023759327990221575), (2093, 0.005591394414196888),
(2094, 0.02583515634206981), (2095, 0.0), (2096, 0.015349580869854933), (2097, 0.0),
(2098, 0.018611172285554166), (2099, 0.0406793543088989), (2100, 0.0346021409836678
7), (2101, 0.0), (2102, 0.018206376272252897), (2103, 0.01803548528171557), (2104,
0.03696327911930247), (2105, 0.025598981817030915), (2106, 0.0), (2107, 0.0), (2108,
0.08391604344392495), (2109, 0.0), (2110, 0.00790637975131982), (2111, 0.02123574489
107766), (2112, 0.0), (2113, 0.010513386242058848), (2114, 0.05937475363169126), (21
15, 0.008276007050561949), (2116, 0.00864082945881617), (2117, 0.00908103786664499
2), (2118, 0.0), (2119, 0.0), (2120, 0.00917944002843519), (2121, 0.0), (2122, 0.005
962649978188197), (2123, 0.07049506017980059), (2124, 0.0073684666604114536), (2125,
0.0), (2126, 0.037902189006431476), (2127, 0.010137987118395897), (2128, 0.0), (212
9, 0.00518710618679695), (2130, 0.007401088606339198), (2131, 0.0071396846732225),
(2132, 0.0), (2133, 0.008253340076687482), (2134, 0.0), (2135, 0.0), (2136, 0.021530
087422208202), (2137, 0.0), (2138, 0.013870745401339092), (2139, 0.0), (2140, 0.0131
50997355821741), (2141, 0.010677538797630138), (2142, 0.02177599767565398), (2143,
0.0), (2144, 0.0), (2145, 0.03923975791495391), (2146, 0.0), (2147, 0.02048563411582
7676), (2148, 0.024618857390416513), (2149, 0.008525481441929175), (2150, 0.00964169
8502016457), (2151, 0.00849022313407311), (2152, 0.008892484123575494), (2153, 0.009
627329387371474), (2154, 0.014261090045043556), (2155, 0.013334535307348962), (2156,
0.013687613871801152), (2157, 0.04883326969652898), (2158, 0.009103888293947683), (2
159, 0.041007751621469475), (2160, 0.032491344751043524), (2161, 0.01347070797175579
6), (2162, 0.0), (2163, 0.024502747033782368), (2164, 0.0070879422218198115), (2165,
0.009736520709247952), (2166, 0.01071754139605724), (2167, 0.017558988362293984), (2

168, 0.01842226613972403), (2169, 0.02457124722412299), (2170, 0.02263953778195152
2), (2171, 0.01045506002247013), (2172, 0.0), (2173, 0.009001634129550014), (2174,
0.0), (2175, 0.01842159174913272), (2176, 0.020377266779787042), (2177, 0.0190305950
19174895), (2178, 0.023917814759109417), (2179, 0.03212235640900879), (2180, 0.01605
0011794654435), (2181, 0.011497940963532812), (2182, 0.019590771070673543), (2183,
0.0), (2184, 0.018287953127304444), (2185, 0.007963298199192347), (2186, 0.008081774
148035419), (2187, 0.031307931554417225), (2188, 0.0064030161335636914), (2189, 0.00
9997577637209175), (2190, 0.0), (2191, 0.04727771543208084), (2192, 0.02096614214082
8384), (2193, 0.03806330499832303), (2194, 0.004846742555154652), (2195, 0.019852604
75034096), (2196, 0.012586639576815585), (2197, 0.02274164651064088), (2198, 0.00561
9890180753037), (2199, 0.019457325176647743), (2200, 0.011845484541211246), (2201,
0.006296412789100745), (2202, 0.028471410994612575), (2203, 0.02152236314601559), (2
204, 0.03014929092685537), (2205, 0.0472730752732396), (2206, 0.009825180774650782),
(2207, 0.00597956077369272), (2208, 0.0), (2209, 0.012005213997481702), (2210, 0.028
023565952007147), (2211, 0.0), (2212, 0.024949799481613175), (2213, 0.0), (2214, 0.0
19156430265523303), (2215, 0.02176934598333656), (2216, 0.026278164821076228), (221
7, 0.0), (2218, 0.01411445087267167), (2219, 0.0), (2220, 0.00857074760528601), (222
1, 0.053576821197112795), (2222, 0.026385699481707194), (2223, 0.05960804990085404),
(2224, 0.0), (2225, 0.023881009027613864), (2226, 0.0), (2227, 0.00995776038635303
6), (2228, 0.019745654355638576), (2229, 0.011804487037548588), (2230, 0.01642259007
961301), (2231, 0.00973878582136455), (2232, 0.0), (2233, 0.028057445099759312), (22
34, 0.036147146365960586), (2235, 0.0), (2236, 0.0), (2237, 0.02304919835869488), (2
238, 0.008078951363265828), (2239, 0.010263889731607297), (2240, 0.0417481532184128
5), (2241, 0.0), (2242, 0.03038984108920955), (2243, 0.0), (2244, 0.0146558513117651
8), (2245, 0.009065728286658787), (2246, 0.0), (2247, 0.029882527026796557), (2248,
0.0), (2249, 0.017668261050699227), (2250, 0.01363497500537029), (2251, 0.0195660430
91467852), (2252, 0.0), (2253, 0.013431428442066351), (2254, 0.0), (2255, 0.04074315
087193084), (2256, 0.0), (2257, 0.0), (2258, 0.017284678826654527), (2259, 0.0), (22
60, 0.02819822542529289), (2261, 0.030713594386319557), (2262, 0.00953954924016600
3), (2263, 0.017908675281275585), (2264, 0.005321789201754367), (2265, 0.07480103949
044059), (2266, 0.012089392779855757), (2267, 0.0), (2268, 0.01757890626814134), (22
69, 0.008177598703837795), (2270, 0.0), (2271, 0.0), (2272, 0.0), (2273, 0.0), (227
4, 0.0), (2275, 0.026113741510890576), (2276, 0.015494388556985081), (2277, 0.012769
727612697677), (2278, 0.0), (2279, 0.0), (2280, 0.028090418960616013), (2281, 0.0183
57336384289446), (2282, 0.0070413578597557916), (2283, 0.007223035842208167), (2284,
0.060096055197682186), (2285, 0.05072283634197921), (2286, 0.0), (2287, 0.0), (2288,
0.014575587616796275), (2289, 0.037911143150644935), (2290, 0.009788448576869925),
(2291, 0.03559235476057287), (2292, 0.0), (2293, 0.005314507202980626), (2294, 0.015
996749378376703), (2295, 0.013286483189991219), (2296, 0.022423453660892587), (2297,
0.0), (2298, 0.0), (2299, 0.0), (2300, 0.015118484445708492), (2301, 0.0480852275782
0377), (2302, 0.02026149813892408), (2303, 0.0317129075161984), (2304, 0.02190540227
2654223), (2305, 0.02148815680098999), (2306, 0.00798002295305227), (2307, 0.0265490
4429489527), (2308, 0.017505560722870284), (2309, 0.00593627613231451), (2310, 0.037
03116569894049), (2311, 0.028213272868602368), (2312, 0.0), (2313, 0.118343329887697
2), (2314, 0.017902590655836777), (2315, 0.0), (2316, 0.0), (2317, 0.020508541342956
55), (2318, 0.008028620507667924), (2319, 0.0), (2320, 0.04373758512234605), (2321,
0.05148717305895325), (2322, 0.03169646736621697), (2323, 0.01647336922826021), (232
4, 0.024745694280172124), (2325, 0.0), (2326, 0.010122271011278966), (2327, 0.004888
511404914587), (2328, 0.00938957453283168), (2329, 0.007966809564085051), (2330, 0.0
20915467859018835), (2331, 0.049704567350787016), (2332, 0.0), (2333, 0.030210844530
69152), (2334, 0.05526612036216298), (2335, 0.007889334231240852), (2336, 0.02219597
296322629), (2337, 0.0), (2338, 0.033999037925691006), (2339, 0.01658709142666543),
(2340, 0.06378589921415115), (2341, 0.02323731807251312), (2342, 0.03579302583003756
6), (2343, 0.03404904437551764), (2344, 0.0), (2345, 0.009897449029082779), (2346,
0.019349639936663277), (2347, 0.0), (2348, 0.007959372393462219), (2349, 0.036106454
06033575), (2350, 0.013811241044855878), (2351, 0.031677310143627804), (2352, 0.0059

14269667336131), (2353, 0.0), (2354, 0.018327080030753545), (2355, 0.019737413682336
506), (2356, 0.02158623381877749), (2357, 0.04218005116739609), (2358, 0.00863700037
0617256), (2359, 0.0), (2360, 0.02788330183434064), (2361, 0.008789661272089458), (2
362, 0.051761187642313536), (2363, 0.01674426403610908), (2364, 0.02820601988246750
8), (2365, 0.0853438442958511), (2366, 0.04337057394012591), (2367, 0.02302019666880
139), (2368, 0.06064463603285601), (2369, 0.0), (2370, 0.0), (2371, 0.02085862264652
7342), (2372, 0.022397070739033156), (2373, 0.0), (2374, 0.0), (2375, 0.020430749249
052808), (2376, 0.01699964381880189), (2377, 0.01767047550685913), (2378, 0.02455015
3498530215), (2379, 0.0), (2380, 0.013292623127802008), (2381, 0.11752380293327759),
(2382, 0.0), (2383, 0.0766994239491908), (2384, 0.020204105912477472), (2385, 0.0),
(2386, 0.005133981393820354), (2387, 0.03902280806428206), (2388, 0.0123927787463485
84), (2389, 0.02831126068370613), (2390, 0.01602751841314606), (2391, 0.010321920352
559162), (2392, 0.020111496579386233), (2393, 0.00824847281976314), (2394, 0.0153439
54673640987), (2395, 0.017356394814809942), (2396, 0.0), (2397, 0.007523103640565219
5), (2398, 0.036602706025905735), (2399, 0.022973571750498186), (2400, 0.01420103279
0993107), (2401, 0.0), (2402, 0.0), (2403, 0.025548645238506567), (2404, 0.007876278
023481365), (2405, 0.017037531044987925), (2406, 0.0), (2407, 0.022324214699764172),
(2408, 0.008194961163281255), (2409, 0.009217842955270813), (2410, 0.0), (2411, 0.
0), (2412, 0.013316977347882984), (2413, 0.014981158100012361), (2414, 0.01960135724
5267757), (2415, 0.033300074563604315), (2416, 0.01128592183734534), (2417, 0.025198
045508376697), (2418, 0.0), (2419, 0.0), (2420, 0.009054142209635392), (2421, 0.0),
(2422, 0.0), (2423, 0.011200377995020383), (2424, 0.007414025241205139), (2425, 0.02
7641447857352862), (2426, 0.009343130391775228), (2427, 0.028784922682624833), (242
8, 0.007779745653856696), (2429, 0.011472623353608848), (2430, 0.04470385235806187),
(2431, 0.03540154952130582), (2432, 0.0), (2433, 0.08365489821337185), (2434, 0.0),
(2435, 0.027910142510684136), (2436, 0.0), (2437, 0.010819572876558938), (2438, 0.01
115940783588043), (2439, 0.02052714041748429), (2440, 0.006773356509660382), (2441,
0.008221889632725188), (2442, 0.029462613894081127), (2443, 0.0), (2444, 0.047777216
36791725), (2445, 0.019579106676952403), (2446, 0.028949760447109608), (2447, 0.0),
(2448, 0.028435360339650977), (2449, 0.013765745545092837), (2450, 0.013888493092364
501), (2451, 0.0), (2452, 0.025848180826074527), (2453, 0.0), (2454, 0.0399943628781
3538), (2455, 0.017103881419806684), (2456, 0.02567848145200457), (2457, 0.008522131
723674714), (2458, 0.0), (2459, 0.0), (2460, 0.0), (2461, 0.01393657476210283), (246
2, 0.008716013163586033), (2463, 0.008000617601678538), (2464, 0.0), (2465, 0.009217
588136556633), (2466, 0.03592380627189738), (2467, 0.007662022201831444), (2468, 0.0
0875983390739302), (2469, 0.0), (2470, 0.05224177322812685), (2471, 0.01310108633539
3194), (2472, 0.0), (2473, 0.022144469585115324), (2474, 0.05270796628302849), (247
5, 0.0), (2476, 0.0), (2477, 0.0), (2478, 0.014947312664113824), (2479, 0.0), (2480,
0.019214151689850353), (2481, 0.0), (2482, 0.07053390554831772), (2483, 0.0079637672
98913118), (2484, 0.007888845267434868), (2485, 0.02361058097952176), (2486, 0.00478
2269736584759), (2487, 0.010091099831675353), (2488, 0.018034608553201634), (2489,
0.008467067925442032), (2490, 0.016171779351878784), (2491, 0.019680944200211176),
(2492, 0.07720993930846724), (2493, 0.0), (2494, 0.0), (2495, 0.008136252616856646),
(2496, 0.03527888894762726), (2497, 0.0), (2498, 0.0), (2499, 0.010337581497730581),
(2500, 0.0), (2501, 0.0), (2502, 0.008021036122187287), (2503, 0.0), (2504, 0.009877
157553128763), (2505, 0.0), (2506, 0.030987042258130218), (2507, 0.0183164785614545
4), (2508, 0.009385714421556335), (2509, 0.0), (2510, 0.02440239443065885), (2511,
0.009015120299662442), (2512, 0.0), (2513, 0.036609829215355764), (2514, 0.073373373
85697332), (2515, 0.004190295521719535), (2516, 0.0), (2517, 0.0), (2518, 0.0), (251
9, 0.0), (2520, 0.012424463940706334), (2521, 0.004468120464552313), (2522, 0.023679
65149206115), (2523, 0.0), (2524, 0.01963042736661027), (2525, 0.00837364100997175
4), (2526, 0.0), (2527, 0.015074790578738399), (2528, 0.0), (2529, 0.017841390817484
884), (2530, 0.13737322473729185), (2531, 0.01908054548694253), (2532, 0.02990985019
8565756), (2533, 0.0), (2534, 0.01792085913649957), (2535, 0.008564713539833924), (2
536, 0.025909910840919887), (2537, 0.0), (2538, 0.019608409148003356), (2539, 0.0290
8320534287104), (2540, 0.03801305588394782), (2541, 0.0), (2542, 0.01049745827226431

3), (2543, 0.04475496708026995), (2544, 0.016013790702558105), (2545, 0.035133850822
99476), (2546, 0.01671720487864706), (2547, 0.0), (2548, 0.009737526545616474), (254
9, 0.007740637190366689), (2550, 0.022820806370494315), (2551, 0.0), (2552, 0.0), (2
553, 0.039753973824415326), (2554, 0.019896754083615294), (2555, 0.01777469527973126
3), (2556, 0.031175182833051644), (2557, 0.016883930202002184), (2558, 0.01232377502
8799834), (2559, 0.0), (2560, 0.05724815289422347), (2561, 0.0078065887356805), (256
2, 0.03431029867512571), (2563, 0.017304789987042445), (2564, 0.013786292530148845),
(2565, 0.005342303208770424), (2566, 0.0), (2567, 0.026906876494602393), (2568, 0.01
916401166234881), (2569, 0.00791314249965777), (2570, 0.0), (2571, 0.021070133100436
563), (2572, 0.04547730814514499), (2573, 0.02821981714658567), (2574, 0.05066807964
7379946), (2575, 0.019330292136337187), (2576, 0.0), (2577, 0.03457971413009998), (2
578, 0.013836301560673045), (2579, 0.02208403983258324), (2580, 0.03453077615504799
4), (2581, 0.028689781019858), (2582, 0.023885926014894625), (2583, 0.02302420727837
438), (2584, 0.026189207628373564), (2585, 0.030932869601498297), (2586, 0.054729191
08104719), (2587, 0.02925657372431103), (2588, 0.0), (2589, 0.0), (2590, 0.0), (259
1, 0.011140185255314298), (2592, 0.03585250384637262), (2593, 0.007326690338957168),
(2594, 0.0), (2595, 0.0), (2596, 0.014035198521420893), (2597, 0.02547058832478942),
(2598, 0.02104956192007803), (2599, 0.0), (2600, 0.0), (2601, 0.0), (2602, 0.0129182
37121065816), (2603, 0.009766709666007716), (2604, 0.02179497168763226), (2605, 0.
0), (2606, 0.013618326126692283), (2607, 0.007425068429264804), (2608, 0.0), (2609,
0.03743491625964058), (2610, 0.01894034816392277), (2611, 0.009249984214177734), (26
12, 0.01793171710312719), (2613, 0.021428768775321853), (2614, 0.02359533164246350
3), (2615, 0.02721812566439879), (2616, 0.006262293613899253), (2617, 0.0), (2618,
0.0), (2619, 0.013613724440774687), (2620, 0.0), (2621, 0.02015357031000619), (2622,
0.022165024393882717), (2623, 0.006406800582658762), (2624, 0.011290104322949144),
(2625, 0.021818440733496373), (2626, 0.0072432969922821035), (2627, 0.01987078865281
5257), (2628, 0.016307518156685394), (2629, 0.0), (2630, 0.02148728643075887), (263
1, 0.06504777397079482), (2632, 0.018086621195293473), (2633, 0.031882780925649314),
(2634, 0.0), (2635, 0.07612157059438401), (2636, 0.0), (2637, 0.0), (2638, 0.0184805
8822578971), (2639, 0.011904450281487955), (2640, 0.02193353948740738), (2641, 0.004
567764771982903), (2642, 0.007850132168961967), (2643, 0.0), (2644, 0.00527961604246
8476), (2645, 0.033435541045359174), (2646, 0.015428457714895445), (2647, 0.04728576
1345967625), (2648, 0.0), (2649, 0.008965839130655243), (2650, 0.00575734175785355),
(2651, 0.009984038131396575), (2652, 0.011940911265721117), (2653, 0.0), (2654, 0.01
7122195155005328), (2655, 0.11270953916613297), (2656, 0.0), (2657, 0.02146595747035
468), (2658, 0.0), (2659, 0.027795956225076), (2660, 0.007466209978199735), (2661,
0.007767515016351138), (2662, 0.0), (2663, 0.0), (2664, 0.0), (2665, 0.0), (2666, 0.
0), (2667, 0.03406328359556072), (2668, 0.0), (2669, 0.007800982860591928), (2670,
0.008922632722430955), (2671, 0.0), (2672, 0.0), (2673, 0.0), (2674, 0.0111765735045
69326), (2675, 0.0215419178433829), (2676, 0.03491978602597076), (2677, 0.0065803984
17854229), (2678, 0.005049462851715121), (2679, 0.0), (2680, 0.0), (2681, 0.02178027
511331353), (2682, 0.0), (2683, 0.005192921868578381), (2684, 0.0), (2685, 0.1098042
1440315517), (2686, 0.0), (2687, 0.02673082814336546), (2688, 0.04158228311037358),
(2689, 0.021870611615505447), (2690, 0.0), (2691, 0.0), (2692, 0.04079133949350325
5), (2693, 0.010720236185077343), (2694, 0.0), (2695, 0.023017457919675435), (2696,
0.0), (2697, 0.0), (2698, 0.0), (2699, 0.0), (2700, 0.0169649361873103), (2701, 0.
0), (2702, 0.018256206263264237), (2703, 0.015063769446971059), (2704, 0.02775995351
3415186), (2705, 0.0), (2706, 0.0), (2707, 0.0), (2708, 0.035550217859991934), (270
9, 0.0), (2710, 0.0), (2711, 0.0), (2712, 0.005215963147795358), (2713, 0.0435002351
0784011), (2714, 0.0), (2715, 0.03881810617499204), (2716, 0.00975131439542477), (27
17, 0.0), (2718, 0.0), (2719, 0.008813526034982797), (2720, 0.008815825767421964),
(2721, 0.016841752055711694), (2722, 0.007956938647448092), (2723, 0.0), (2724, 0.
0), (2725, 0.0), (2726, 0.02213971063872107), (2727, 0.01480086453141679), (2728, 0.
006793651759084413), (2729, 0.00915801220209005), (2730, 0.03924194624615451), (273
1, 0.05719320121356585), (2732, 0.017563111403640516), (2733, 0.0), (2734, 0.0086254
60195998715), (2735, 0.0), (2736, 0.026887005543794428), (2737, 0.0), (2738, 0.01879

3617078935836), (2739, 0.007688456357432549), (2740, 0.08940625662471981), (2741, 0.
03033300660698405), (2742, 0.029318959907519586), (2743, 0.0), (2744, 0.0), (2745,
0.009789391153187966), (2746, 0.021128667759444206), (2747, 0.021913261649682676),
(2748, 0.0), (2749, 0.033377101098615664), (2750, 0.006514174750681592), (2751, 0.
0), (2752, 0.008286110254383104), (2753, 0.09951168410001496), (2754, 0.0), (2755,
0.026544621098285273), (2756, 0.02824916218904628), (2757, 0.023338152973196875), (2
758, 0.017146272976666894), (2759, 0.02579331492350329), (2760, 0.0), (2761, 0.02256
3717532086677), (2762, 0.007023964134395655), (2763, 0.029478615118976897), (2764,
0.0), (2765, 0.018715792154415774), (2766, 0.013065575421491586), (2767, 0.007919744
59741216), (2768, 0.0), (2769, 0.02841372641299809), (2770, 0.0), (2771, 0.031944743
116527737), (2772, 0.0), (2773, 0.03935607751611092), (2774, 0.006823794615014855),
(2775, 0.013310526377249988), (2776, 0.0), (2777, 0.03953499009970756), (2778, 0.019
10496071162926), (2779, 0.022805851491782297), (2780, 0.04217313387955453), (2781,
0.005033207383864078), (2782, 0.0), (2783, 0.023570970576071857), (2784, 0.009909178
075821692), (2785, 0.005337151745043536), (2786, 0.008581720572390315), (2787, 0.0),
(2788, 0.02089255347403614), (2789, 0.05732486796919024), (2790, 0.02952446459302594
5), (2791, 0.008538822884785868), (2792, 0.03847009120506302), (2793, 0.081134611861
71188), (2794, 0.008458868150231419), (2795, 0.007649638620855087), (2796, 0.0391747
52591650465), (2797, 0.016890080722243156), (2798, 0.017356878376739125), (2799, 0.0
17212753272364312), (2800, 0.008468968785837246), (2801, 0.007192450337442204), (280
2, 0.016970505659059738), (2803, 0.004599393040846562), (2804, 0.00784840084231840
1), (2805, 0.09653055839236144), (2806, 0.004157112526164853), (2807, 0.008409084626
136808), (2808, 0.01345961312079089), (2809, 0.01939930786334127), (2810, 0.0), (281
1, 0.0), (2812, 0.032407014391467245), (2813, 0.051583845389576416), (2814, 0.030917
21252913968), (2815, 0.02811290499833126), (2816, 0.016310509425586678), (2817, 0.
0), (2818, 0.03580841573281995), (2819, 0.011851542733676244), (2820, 0.027579551411
52403), (2821, 0.0), (2822, 0.06523623085813704), (2823, 0.03205158672321503), (282
4, 0.05114689340419648), (2825, 0.0), (2826, 0.07908804353048146), (2827, 0.00530975
9852241237), (2828, 0.007477424663806525), (2829, 0.028501651318413403), (2830, 0.05
290889080627135), (2831, 0.008697112855002414), (2832, 0.008177060740486486), (2833,
0.021656480065505173), (2834, 0.0), (2835, 0.030814649305162693), (2836, 0.013265241
723364888), (2837, 0.01045410620160236), (2838, 0.0), (2839, 0.015064075176666195),
(2840, 0.0), (2841, 0.0), (2842, 0.009301902472823349), (2843, 0.0), (2844, 0.0), (2
845, 0.0), (2846, 0.010600044798819248), (2847, 0.0), (2848, 0.020125541058772435),
(2849, 0.03719209263660313), (2850, 0.0), (2851, 0.03658903233167636), (2852, 0.0),
(2853, 0.010811542554524215), (2854, 0.0), (2855, 0.0), (2856, 0.01059756590299132
7), (2857, 0.0), (2858, 0.11413652180839844), (2859, 0.01925992784410293), (2860, 0.
019375424421008827), (2861, 0.01834613911499412), (2862, 0.04603206010258901), (286
3, 0.00513609193782345), (2864, 0.029869445351926205), (2865, 0.007830828833211092),
(2866, 0.0), (2867, 0.007167015455204336), (2868, 0.011001105993800987), (2869, 0.02
4899792820574917), (2870, 0.0), (2871, 0.04860319935980385), (2872, 0.00970400985718
9484), (2873, 0.0), (2874, 0.010493159812379384), (2875, 0.016307131246229324), (287
6, 0.021078169585329223), (2877, 0.0), (2878, 0.05515185050930864), (2879, 0.0206253
05508139076), (2880, 0.0), (2881, 0.0), (2882, 0.0), (2883, 0.0), (2884, 0.031566493
567821485), (2885, 0.008858374271896388), (2886, 0.0), (2887, 0.02366854162109285),
(2888, 0.014170908630617039), (2889, 0.018268303863125872), (2890, 0.0), (2891, 0.02
071016503660198), (2892, 0.047332554693430715), (2893, 0.0), (2894, 0.02174863664458
3534), (2895, 0.0), (2896, 0.0), (2897, 0.0687325954457757), (2898, 0.0), (2899, 0.0
2924126018824487), (2900, 0.0), (2901, 0.00860696417898728), (2902, 0.02046638767894
536), (2903, 0.0), (2904, 0.008983219882562277), (2905, 0.006674882002410663), (290
6, 0.008689484878656518), (2907, 0.01825563560141021), (2908, 0.0), (2909, 0.0210989
9570669337), (2910, 0.0), (2911, 0.007887976175178972), (2912, 0.01335291975436016),
(2913, 0.0), (2914, 0.02064456738632578), (2915, 0.027264879644821115), (2916, 0.0),
(2917, 0.0), (2918, 0.005265128699522157), (2919, 0.00895625406654035), (2920, 0.0),
(2921, 0.01635456215161986), (2922, 0.020041619935676415), (2923, 0.0), (2924, 0.0),
(2925, 0.0), (2926, 0.021922542409662633), (2927, 0.026660957131927718), (2928, 0.00

9816670186017134), (2929, 0.035141199092156336), (2930, 0.0), (2931, 0.0266790675348
60434), (2932, 0.007644523066420075), (2933, 0.00732790404069939), (2934, 0.0), (293
5, 0.0), (2936, 0.0), (2937, 0.0), (2938, 0.017750792658544044), (2939, 0.0203997813
8276963), (2940, 0.016011846071586327), (2941, 0.008022430308148016), (2942, 0.0),
(2943, 0.0), (2944, 0.0195305334945897), (2945, 0.0), (2946, 0.04230893080456544),
(2947, 0.008287722643896864), (2948, 0.008125166438300345), (2949, 0.016623544455159
98), (2950, 0.008104747236162288), (2951, 0.0), (2952, 0.021417697113726212), (2953,
0.014302492069185551), (2954, 0.0), (2955, 0.0), (2956, 0.0), (2957, 0.0083845866310
18164), (2958, 0.016643740533079855), (2959, 0.021189971123077016), (2960, 0.0088929
9840815021), (2961, 0.028844440036226753), (2962, 0.0), (2963, 0.0), (2964, 0.023996
80854553707), (2965, 0.008327123080817546), (2966, 0.0), (2967, 0.0091732215244996
8), (2968, 0.0), (2969, 0.0), (2970, 0.0), (2971, 0.015574242828561253), (2972, 0.00
6384021715683351), (2973, 0.010002826633673762), (2974, 0.0), (2975, 0.0183527834873
21334), (2976, 0.0), (2977, 0.014846762409476362), (2978, 0.02124140768447708), (297
9, 0.008532131421870018), (2980, 0.005897718848800492), (2981, 0.0), (2982, 0.043734
275425904495), (2983, 0.06841359888033002), (2984, 0.0), (2985, 0.0151528359226867
3), (2986, 0.01842497656229629), (2987, 0.0), (2988, 0.0), (2989, 0.0), (2990, 0.010
560021439008917), (2991, 0.016033890831893882), (2992, 0.0), (2993, 0.01667372101197
228), (2994, 0.007881245256192233), (2995, 0.005250568386014171), (2996, 0.005124339
139704501), (2997, 0.0), (2998, 0.009033248196177252), (2999, 0.029063516731651903),
(3000, 0.009698613823861262), (3001, 0.026988659324938612), (3002, 0.0), (3003, 0.01
7448923729866266), (3004, 0.010942476625300828), (3005, 0.0), (3006, 0.0188669395798
86197), (3007, 0.004213605300378606), (3008, 0.01665085230894172), (3009, 0.03228319
3625339386), (3010, 0.045936908711060494), (3011, 0.016646333073262944), (3012, 0.01
2388963363858384), (3013, 0.00986218617760741), (3014, 0.03934397767922128), (3015,
0.0), (3016, 0.00888114917044104), (3017, 0.00830934872982811), (3018, 0.00712615041
8614115), (3019, 0.008189699647710073), (3020, 0.02373838630851025), (3021, 0.005979
733456284166), (3022, 0.0), (3023, 0.0), (3024, 0.0), (3025, 0.0), (3026, 0.0), (302
7, 0.0), (3028, 0.0), (3029, 0.004906763874052582), (3030, 0.030503590832391892), (3
031, 0.05608432418691643), (3032, 0.006472934338084550 4), (3033, 0.0688248369448167
5), (3034, 0.0), (3035, 0.020342589588662453), (3036, 0.0), (3037, 0.0), (3038, 0.
0), (3039, 0.0), (3040, 0.01304201955875002), (3041, 0.0), (3042, 0.0181067492905780
88), (3043, 0.026138123902603193), (3044, 0.009549390973542593), (3045, 0.0073146364
74626293), (3046, 0.04541372738982333), (3047, 0.05227719423322646), (3048, 0.007788
254002700383), (3049, 0.016267715419819664), (3050, 0.027322690830130215), (3051, 0.
016117199385612204), (3052, 0.007353578328557356), (3053, 0.0614937693536529), (305
4, 0.0), (3055, 0.01904268391790459), (3056, 0.017070664226139093), (3057, 0.0), (30
58, 0.0), (3059, 0.0), (3060, 0.0), (3061, 0.0), (3062, 0.012751866613785096), (306
3, 0.01571609532485843), (3064, 0.0), (3065, 0.0), (3066, 0.009629825534849208), (30
67, 0.0), (3068, 0.03807974405788157), (3069, 0.014711134738463055), (3070, 0.007273
009934693388), (3071, 0.0), (3072, 0.027773070148091546), (3073, 0.01965308421332278
4), (3074, 0.011963396431367183), (3075, 0.07656011925039205), (3076, 0.008140345973
126186), (3077, 0.005389591352749593), (3078, 0.024143378199026195), (3079, 0.0), (3
080, 0.02593252628831628), (3081, 0.0), (3082, 0.0), (3083, 0.024349835998103662),
(3084, 0.0), (3085, 0.008146770991792316), (3086, 0.029973002426384506), (3087, 0.00
7901869200902557), (3088, 0.02636032002322339), (3089, 0.005634743987543985), (3090,
0.007478303174522463), (3091, 0.00615466903461918), (3092, 0.008731537672391303), (3
093, 0.013820648765719666), (3094, 0.018563964022973518), (3095, 0.01783907232877769
4), (3096, 0.0), (3097, 0.03304721251096088), (3098, 0.00558599820155528), (3099, 0.
004339620837882695), (3100, 0.0), (3101, 0.00830792328482796), (3102, 0.0), (3103,
0.0), (3104, 0.0), (3105, 0.0), (3106, 0.02983696085018368), (3107, 0.0), (3108, 0.0
3311047873174014), (3109, 0.0), (3110, 0.0), (3111, 0.028713145888641392), (3112, 0.
03254416713480706), (3113, 0.0), (3114, 0.02182121672994405), (3115, 0.0), (3116, 0.
022588030355625822), (3117, 0.0), (3118, 0.0), (3119, 0.0), (3120, 0.031549723069320
67), (3121, 0.0), (3122, 0.0), (3123, 0.006976892297563276), (3124, 0.00766508096606
8965), (3125, 0.030366984987502108), (3126, 0.01127628619865739), (3127, 0.018821130

422907575), (3128, 0.0059595990162209875), (3129, 0.0), (3130, 0.0), (3131, 0.0), (3
132, 0.00780453779510678), (3133, 0.004312512451717949), (3134, 0.0), (3135, 0.0),
(3136, 0.02776827173916815), (3137, 0.0), (3138, 0.021266897016741428), (3139, 0.0),
(3140, 0.0), (3141, 0.017986041094174607), (3142, 0.029832128183312788), (3143, 0.01
8689940200818402), (3144, 0.014258042751768613), (3145, 0.019566763735771934), (314
6, 0.046692892879965475), (3147, 0.008337693254790833), (3148, 0.07051443394414272),
(3149, 0.0), (3150, 0.0), (3151, 0.022390879571115428), (3152, 0.0), (3153, 0.005538
209502594359), (3154, 0.0060409568104516816), (3155, 0.0), (3156, 0.0), (3157, 0.010
03521087711719), (3158, 0.022577085131124948), (3159, 0.0), (3160, 0.017373493413739
25), (3161, 0.013945103609747201), (3162, 0.03105149143564838), (3163, 0.0), (3164,
0.0), (3165, 0.017978357336314768), (3166, 0.0), (3167, 0.0), (3168, 0.0361343664429
4308), (3169, 0.006760329558070126), (3170, 0.012091943321009983), (3171, 0.0), (317
2, 0.010644153982213873), (3173, 0.016577061976447954), (3174, 0.0), (3175, 0.006900
172394531548), (3176, 0.008887594052887466), (3177, 0.019419667468474983), (3178, 0.
0), (3179, 0.007269336091764261), (3180, 0.014386788885050529), (3181, 0.01035974906
3065688), (3182, 0.0), (3183, 0.020047139404423597), (3184, 0.05375836747802638), (3
185, 0.0), (3186, 0.01992840065415868), (3187, 0.0), (3188, 0.0), (3189, 0.021610385
23360965), (3190, 0.0), (3191, 0.0), (3192, 0.019491481444918187), (3193, 0.0), (319
4, 0.0), (3195, 0.008283658812848118), (3196, 0.0), (3197, 0.007140969295830423), (3
198, 0.0), (3199, 0.008937155236889387), (3200, 0.0), (3201, 0.01394499404557601),
(3202, 0.02953841408185413), (3203, 0.02916587845601703), (3204, 0.0), (3205, 0.0065
97564676605912), (3206, 0.0), (3207, 0.017678045944648724), (3208, 0.053493572015587
01), (3209, 0.020878623482585897), (3210, 0.0), (3211, 0.050277673460301965), (3212,
0.0), (3213, 0.0), (3214, 0.009591132244567663), (3215, 0.0), (3216, 0.0), (3217, 0.
0), (3218, 0.041931442770066975), (3219, 0.008286258584938411), (3220, 0.01995376318
2146554), (3221, 0.0048413563254619384), (3222, 0.0), (3223, 0.0), (3224, 0.02386332
9128368506), (3225, 0.010688226271161332), (3226, 0.0), (3227, 0.01418825938825486
4), (3228, 0.0), (3229, 0.028189712953565225), (3230, 0.0), (3231, 0.004334164872270
242), (3232, 0.04522567127593379), (3233, 0.0), (3234, 0.015695954365573153), (3235,
0.0), (3236, 0.02722649248697842), (3237, 0.019203830253771155), (3238, 0.0218686319
7915005), (3239, 0.039305181294137834), (3240, 0.0), (3241, 0.017402116829586093),
(3242, 0.07115649368602578), (3243, 0.04250479555235183), (3244, 0.029144017553061),
(3245, 0.008410226317822496), (3246, 0.015391822501281115), (3247, 0.027219697341201
53), (3248, 0.0), (3249, 0.0), (3250, 0.047684927328473056), (3251, 0.02409472858101
4403), (3252, 0.019235461351458108), (3253, 0.01641136858084838), (3254, 0.008278851
327421167), (3255, 0.007383222319211602), (3256, 0.006806007187921874), (3257, 0.006
974398640630545), (3258, 0.0), (3259, 0.011582072402653344), (3260, 0.02173814978824
745), (3261, 0.0), (3262, 0.0), (3263, 0.007259342935721644), (3264, 0.0), (3265, 0.
0), (3266, 0.00839810809188871), (3267, 0.06484032054628708), (3268, 0.0), (3269, 0.
0), (3270, 0.008756740049841881), (3271, 0.0), (3272, 0.0), (3273, 0.014848491780132
991), (3274, 0.004449249065720359), (3275, 0.006043744372866129), (3276, 0.017657375
993137867), (3277, 0.015697726113890545), (3278, 0.0), (3279, 0.00893119971888063),
(3280, 0.018253732010575077), (3281, 0.017858772403159282), (3282, 0.010618334986523
918), (3283, 0.0), (3284, 0.02874741487048716), (3285, 0.021501094123177543), (3286,
0.01680281277180236), (3287, 0.04087820747828064), (3288, 0.0), (3289, 0.0), (3290,
0.013821778560342097), (3291, 0.02367872810724211), (3292, 0.0), (3293, 0.0293621701
19061767), (3294, 0.027082458320694926), (3295, 0.0), (3296, 0.007180469313973745),
(3297, 0.06256549822883029), (3298, 0.0), (3299, 0.006442310376974764), (3300, 0.0),
(3301, 0.0086379864659801), (3302, 0.00935185632297031), (3303, 0.01544992367191985
4), (3304, 0.007032911169183759), (3305, 0.012598033238040054), (3306, 0.04688945080
31419), (3307, 0.007240862847337393), (3308, 0.020315404280067877), (3309, 0.0114199
85499892791), (3310, 0.0), (3311, 0.0), (3312, 0.021677066955678212), (3313, 0.00812
7061468231849), (3314, 0.0), (3315, 0.03145725933153994), (3316, 0.01352713156212352
6), (3317, 0.0), (3318, 0.021779899180964468), (3319, 0.0), (3320, 0.009097950977250
642), (3321, 0.0355284881177735), (3322, 0.007453144952559375), (3323, 0.07093837703
816055), (3324, 0.0), (3325, 0.04388566680159247), (3326, 0.08864740283572532), (332

7, 0.0), (3328, 0.0), (3329, 0.022281683124023476), (3330, 0.009331447169828247), (3
331, 0.0), (3332, 0.03483725220844276), (3333, 0.03706615486266431), (3334, 0.033802
904228006136), (3335, 0.014930148564769144), (3336, 0.013823729288610524), (3337, 0.
024604373831795614), (3338, 0.033694102904873416), (3339, 0.009519848470270773), (33
40, 0.0), (3341, 0.0), (3342, 0.0), (3343, 0.005396837478246529), (3344, 0.0), (334
5, 0.013126294183100427), (3346, 0.007812482455297258), (3347, 0.03577129882601757),
(3348, 0.041615652836445444), (3349, 0.020172739288629046), (3350, 0.086550546231693
36), (3351, 0.039395968822721505), (3352, 0.02160092649701403), (3353, 0.0), (3354,
0.02266755339048144), (3355, 0.06163725811314999), (3356, 0.0), (3357, 0.00693568445
3099439), (3358, 0.006320899411055797), (3359, 0.04422151373766265), (3360, 0.009416
067423415553), (3361, 0.026916611287387024), (3362, 0.018964735871980654), (3363, 0.
0), (3364, 0.011161681231402118), (3365, 0.0), (3366, 0.0), (3367, 0.0), (3368, 0.03
993230481575126), (3369, 0.0), (3370, 0.01174969014663075), (3371, 0.023224810684275
33), (3372, 0.06340151915093728), (3373, 0.006686714282916756), (3374, 0.01246799472
917649), (3375, 0.0477623719685448), (3376, 0.009140309631343506), (3377, 0.02796278
607354603), (3378, 0.007728521599656517 5), (3379, 0.013879795869627953), (3380, 0.
0), (3381, 0.008576967414987828), (3382, 0.02466671237286574), (3383, 0.016919907662
983753), (3384, 0.0), (3385, 0.0276423993198935), (3386, 0.00857055808391046), (338
7, 0.010076597968378795), (3388, 0.0), (3389, 0.021384333278687044), (3390, 0.032681
44436260144), (3391, 0.01424465592632181), (3392, 0.0), (3393, 0.0), (3394, 0.020314
95832959946), (3395, 0.0), (3396, 0.0), (3397, 0.00936677968908261), (3398, 0.0), (3
399, 0.010649995796497485), (3400, 0.02893376136077479), (3401, 0.01611945425942493
5), (3402, 0.018533365215852), (3403, 0.0), (3404, 0.03822017262721622), (3405, 0.04
9614662138495455), (3406, 0.021846087600919415), (3407, 0.0), (3408, 0.0274753764276
06172), (3409, 0.04442588575633951), (3410, 0.0), (3411, 0.0), (3412, 0.014224190587
59691), (3413, 0.007470593693982962), (3414, 0.014641821376556615), (3415, 0.0), (34
16, 0.011596915149669994), (3417, 0.022871456299014305), (3418, 0.00649763393824693
1), (3419, 0.0), (3420, 0.01557177233324683), (3421, 0.007016599330783714), (3422,
0.0), (3423, 0.03852331888100041), (3424, 0.03604919313010584), (3425, 0.00635937151
5884716), (3426, 0.0), (3427, 0.03472659932407464), (3428, 0.008342685956353733), (3
429, 0.0), (3430, 0.01965708350871151), (3431, 0.008586161946688004), (3432, 0.0),
(3433, 0.008463161571889338), (3434, 0.02901340180892563), (3435, 0.0), (3436, 0.008
780033224539123), (3437, 0.007111197451608696), (3438, 0.0), (3439, 0.04566988215905
9386), (3440, 0.014825296522977274), (3441, 0.01611281974764695), (3442, 0.008058412
563203481), (3443, 0.0), (3444, 0.008572991281966036), (3445, 0.014053682771515474),
(3446, 0.033915528108467576), (3447, 0.007408435854578011), (3448, 0.010435873127939
846), (3449, 0.01718603232909577 2), (3450, 0.03602193097919057), (3451, 0.0330266093
27336774), (3452, 0.0), (3453, 0.0), (3454, 0.007094386629024889), (3455, 0.01910717
1350900877), (3456, 0.019849518552316366), (3457, 0.0), (3458, 0.00822599188188811
3), (3459, 0.016993000139441404), (3460, 0.012100606254904103), (3461, 0.0), (3462,
0.01701323485492115), (3463, 0.013187342939057281), (3464, 0.027750303051607812), (3
465, 0.030494070493586975), (3466, 0.08668912224324149), (3467, 0.00668577428860996
4), (3468, 0.0), (3469, 0.0178274960403451), (3470, 0.026042314663075636), (3471, 0.
0), (3472, 0.013525842339425298), (3473, 0.007809192211093686), (3474, 0.01230464518
8135344), (3475, 0.03975043342779548), (3476, 0.0), (3477, 0.03440929074610513), (34
78, 0.0), (3479, 0.031382489691811226), (3480, 0.018213793264644982), (3481, 0.02204
2587180242275), (3482, 0.01246814374056671), (3483, 0.00790431840179359), (3484, 0.
0), (3485, 0.015624499474612745), (3486, 0.01092473494325089), (3487, 0.0), (3488,
0.009045747024561446), (3489, 0.02267651371268215), (3490, 0.022631742989093227), (3
491, 0.0), (3492, 0.038595106128332404), (3493, 0.0), (3494, 0.01610171812656732),
(3495, 0.01810066761766072), (3496, 0.0), (3497, 0.0), (3498, 0.0), (3499, 0.0284263
4049699244), (3500, 0.008192139047744663), (3501, 0.02183859071095311), (3502, 0.0),
(3503, 0.020021537199765658), (3504, 0.01857314345513272), (3505, 0.0208775388854622
88), (3506, 0.00845778611946717), (3507, 0.02445344878453156), (3508, 0.008169847411
70989), (3509, 0.0064644170583374825), (3510, 0.024536475046640752), (3511, 0.017606
26410453416), (3512, 0.006863633391340462), (3513, 0.0), (3514, 0.0083912984972322

7), (3515, 0.025942054980027687), (3516, 0.004730667189624703), (3517, 0.0), (3518,
0.0423025122877909), (3519, 0.0), (3520, 0.015309141745697401), (3521, 0.0), (3522,
0.005883926218205288), (3523, 0.0056288147349748945), (3524, 0.0), (3525, 0.0), (352
6, 0.007961696799759355), (3527, 0.012499929262499137), (3528, 0.01725994632174368
5), (3529, 0.016442639263234096), (3530, 0.0), (3531, 0.021274974632137365), (3532,
0.008048703932781206), (3533, 0.015232340942432473), (3534, 0.0), (3535, 0.0), (353
6, 0.0), (3537, 0.008217329852494376), (3538, 0.03765221577616237), (3539, 0.0), (35
40, 0.0), (3541, 0.007881526207729913), (3542, 0.0), (3543, 0.0), (3544, 0.033133502
30211917), (3545, 0.0054943361402165596), (3546, 0.034330301128659875), (3547, 0.0),
(3548, 0.013417205086468337), (3549, 0.008233676057789657), (3550, 0.063930638084499
93), (3551, 0.0), (3552, 0.018072052315689015), (3553, 0.009632270526504868), (3554,
0.010488878112705529), (3555, 0.0), (3556, 0.009791365805198233), (3557, 0.0), (355
8, 0.009034158580559544), (3559, 0.017965690757809898), (3560, 0.0), (3561, 0.017858
28459217152), (3562, 0.020688125901358635), (3563, 0.009297030069886943), (3564, 0.0
6748597328266652), (3565, 0.0), (3566, 0.008057149668978525), (3567, 0.0162358763962
5799), (3568, 0.008552306480355749), (3569, 0.010703892723267387), (3570, 0.0), (357
1, 0.008653717143061106), (3572, 0.0), (3573, 0.0), (3574, 0.01807640205053272), (35
75, 0.0238854996279511), (3576, 0.007092370959654787), (3577, 0.0), (3578, 0.0), (35
79, 0.040663397551497095), (3580, 0.010214933653245375), (3581, 0.00813054623944557
9), (3582, 0.00845196790553298), (3583, 0.0), (3584, 0.01270363781105749), (3585, 0.
0), (3586, 0.017337549133132416), (3587, 0.0), (3588, 0.007320581378888757), (3589,
0.0), (3590, 0.01692262346324305), (3591, 0.02129475754254638), (3592, 0.01751422737
152352), (3593, 0.017922425249429833), (3594, 0.006825992736765534), (3595, 0.039893
01472654164), (3596, 0.03280142148120986), (3597, 0.020193550672841114), (3598, 0.
0), (3599, 0.03103017019524875), (3600, 0.018487660289643285), (3601, 0.0), (3602,
0.028534327636764778), (3603, 0.03347660190901597), (3604, 0.02233947474424601), (36
05, 0.013502509610412616), (3606, 0.005230724421820102), (3607, 0.05735572092620986
5), (3608, 0.00913650731444795), (3609, 0.01857762198825906), (3610, 0.0), (3611, 0.
01933894479238951), (3612, 0.0), (3613, 0.01070732887552611), (3614, 0.0), (3615, 0.
005710027398240209), (3616, 0.014854673840520144), (3617, 0.012981790080614633), (36
18, 0.0062506823878716155), (3619, 0.029870418592994934), (3620, 0.00834588501107702
6), (3621, 0.0), (3622, 0.017223651182254985), (3623, 0.014930920496193798), (3624,
0.04322362197604023), (3625, 0.006812849225333099), (3626, 0.015479964272307519), (3
627, 0.008367735030674645), (3628, 0.03414585325726132), (3629, 0.0), (3630, 0.09220
755061184922), (3631, 0.017053704667855607), (3632, 0.02284431799979458), (3633, 0.
0), (3634, 0.047534105702211), (3635, 0.01671879606662281), (3636, 0.015922809409126
06), (3637, 0.06850035824377265), (3638, 0.009162758847320866), (3639, 0.00751147660
8410464), (3640, 0.01713700037066372), (3641, 0.0), (3642, 0.0), (3643, 0.0264398854
4473405), (3644, 0.02246803151051133), (3645, 0.006639063050271446), (3646, 0.020441
997901347927), (3647, 0.007885521427532032), (3648, 0.03385031833272621), (3649, 0.
0), (3650, 0.005911453517416629), (3651, 0.019769349550181767), (3652, 0.03354018114
662275), (3653, 0.008599111860640826), (3654, 0.0), (3655, 0.0), (3656, 0.0), (3657,
0.0081332533465417), (3658, 0.007559612572913113), (3659, 0.027135351125975853), (36
60, 0.018415786600251196), (3661, 0.011402765944382625), (3662, 0.02868351699418072
3), (3663, 0.0), (3664, 0.009611710582400939), (3665, 0.009530866153955375), (3666,
0.0), (3667, 0.024524349708350518), (3668, 0.022697534143570702), (3669, 0.021598348
634490647), (3670, 0.0), (3671, 0.010888624412790326), (3672, 0.024403838937443217),
(3673, 0.0), (3674, 0.0), (3675, 0.0), (3676, 0.0), (3677, 0.0), (3678, 0.0275000123
70676348), (3679, 0.030664661577073798), (3680, 0.03665975638251558), (3681, 0.0),
(3682, 0.008574932443123913), (3683, 0.008502094100846165), (3684, 0.008124052215293
338), (3685, 0.06012900367722529), (3686, 0.0), (3687, 0.0), (3688, 0.02068346654806
1143), (3689, 0.005554347904035231), (3690, 0.0), (3691, 0.019393907601754353), (369
2, 0.0), (3693, 0.008410083577209045), (3694, 0.03688229980459641), (3695, 0.0226261
25255064973), (3696, 0.0), (3697, 0.0), (3698, 0.015902367834127244), (3699, 0.02243
3763024109387), (3700, 0.03509912768150969), (3701, 0.008498831447368757), (3702, 0.
0), (3703, 0.02247095838252873), (3704, 0.021777945620316614), (3705, 0.0), (3706,

0.019885823080128526), (3707, 0.005315193390659952 5), (3708, 0.0), (3709, 0.013570805246580792), (3710, 0.02306355003939454), (3711, 0.03355446024770533), (3712, 0.0), (3713, 0.0), (3714, 0.01731356656461907), (3715, 0.004996217147975443), (3716, 0.00726935043960703 1), (3717, 0.02914875131896051), (3718, 0.0), (3719, 0.06293374984621716), (3720, 0.0), (3721, 0.0), (3722, 0.004389894690296252), (3723, 0.0), (3724, 0.0256217270399132), (3725, 0.013481878781048548), (3726, 0.028082931495633824), (3727, 0.008431439219758243), (3728, 0.0), (3729, 0.0), (3730, 0.016857903197132696), (3731, 0.0), (3732, 0.0), (3733, 0.020802578265744335), (3734, 0.021286263111711556), (3735, 0.03181537157490147), (3736, 0.0), (3737, 0.018229791574562133), (3738, 0.0), (3739, 0.034748000857707885), (3740, 0.0052455037587596474), (3741, 0.008834681349579099), (3742, 0.0), (3743, 0.030130962666952157), (3744, 0.019342345853627447), (3745, 0.023895805485874087), (3746, 0.0), (3747, 0.024192468481663364), (3748, 0.014758536281916954), (3749, 0.005165716221369657), (3750, 0.008708949270069152), (3751, 0.0), (3752, 0.009556287096574403), (3753, 0.05128002013987719), (3754, 0.023185673568415097), (3755, 0.0204259675670689), (3756, 0.0339415249175436), (3757, 0.02892468693879513), (3758, 0.061448143793647665), (3759, 0.008906941759391942), (3760, 0.00881366114865915), (3761, 0.0), (3762, 0.0399034538062805), (3763, 0.0), (3764, 0.03263074142271305), (3765, 0.01627254257681505), (3766, 0.05540022211096728), (3767, 0.008924923476056553), (3768, 0.0), (3769, 0.018421063882608835), (3770, 0.032831081309707265), (3771, 0.02262085936625541), (3772, 0.018837726693303164), (3773, 0.0), (3774, 0.04860532649135153), (3775, 0.036473830860536974), (3776, 0.007162240987433001), (3777, 0.0117574180190466), (3778, 0.028259274724955118), (3779, 0.01069818693855462), (3780, 0.017224564830784668), (3781, 0.0), (3782, 0.03158749143043442), (3783, 0.0), (3784, 0.015380935132776236), (3785, 0.00787323193390393), (3786, 0.01052345684812332), (3787, 0.0), (3788, 0.0), (3789, 0.016691119781322223), (3790, 0.007567434618230866), (3791, 0.01873420829635461), (3792, 0.0), (3793, 0.0), (3794, 0.0), (3795, 0.015039873261524506), (3796, 0.025465562245690885), (3797, 0.00855946207001766 6), (3798, 0.010669295733820213), (3799, 0.013273167504061956), (3800, 0.011661978490169197), (3801, 0.0), (3802, 0.01729044944538959), (3803, 0.010651264325911588), (3804, 0.00410361069412517), (3805, 0.0), (3806, 0.01005868194329982), (3807, 0.0), (3808, 0.011719095056729537), (3809, 0.0), (3810, 0.0), (3811, 0.0), (3812, 0.00804176299433157), (3813, 0.022149358102777045), (3814, 0.0), (3815, 0.02045012568681413), (3816, 0.005668891685470773), (3817, 0.007571705029118976), (3818, 0.0), (3819, 0.0585620788359174), (3820, 0.009058295697759775), (3821, 0.007884907756170689), (3822, 0.0), (3823, 0.010357696787555047), (3824, 0.008580527846624716), (3825, 0.0), (3826, 0.02524187742166132), (3827, 0.0), (3828, 0.02100589059131794), (3829, 0.03494233684952769), (3830, 0.005759821102857798), (3831, 0.02125017551073208), (3832, 0.015801123609065645), (3833, 0.04081283854681531), (3834, 0.026161168557652853), (3835, 0.0), (3836, 0.026785282913353692), (3837, 0.036541655221875094), (3838, 0.0), (3839, 0.00717368263340481), (3840, 0.01882685732596795), (3841, 0.034062757747623), (3842, 0.005511321634844242), (3843, 0.0), (3844, 0.007135615257177114), (3845, 0.025976878807820743), (3846, 0.0), (3847, 0.028991423943799232), (3848, 0.022635179749546985), (3849, 0.013733865912462931), (3850, 0.007528921749279049), (3851, 0.05023263968999226), (3852, 0.0), (3853, 0.0), (3854, 0.08270463326994387), (3855, 0.018501239589191025), (3856, 0.01643994732934199), (3857, 0.018117900178294826), (3858, 0.030037578257637323), (3859, 0.025167591128221548), (3860, 0.015231041516259713), (3861, 0.016931957473167437), (3862, 0.0), (3863, 0.0), (3864, 0.008110710132581238), (3865, 0.02531517904992564), (3866, 0.034444390226544004), (3867, 0.008184928190010737), (3868, 0.027909785585801657), (3869, 0.022456094618819183), (3870, 0.007713664152154154), (3871, 0.0), (3872, 0.012531591919830479), (3873, 0.05986413704367761), (3874, 0.02692710348454094), (3875, 0.009257668796753481), (3876, 0.006835118095338313), (3877, 0.0), (3878, 0.021300307628795336), (3879, 0.0073768289657509525), (3880, 0.006417651516985444), (3881, 0.0), (3882, 0.005927301503942796), (3883, 0.0428439235519079), (3884, 0.0042220292169891285), (3885, 0.008897098892136657), (3886, 0.010196796300999175), (3887, 0.024040298510155534), (3888, 0.012010429915946854), (3889, 0.024820831202536156), (3890, 0.03324173900573309), (3891, 0.04337769563404051), (3892,

0.0), (3893, 0.0476952660288304), (3894, 0.0), (3895, 0.006938996010822264), (3896,
0.018216900910687044), (3897, 0.02075711274826168), (3898, 0.027514066275856085), (3
899, 0.030369122796613413), (3900, 0.01588959700678226), (3901, 0.04439155572373064
4), (3902, 0.0), (3903, 0.019991996083841806), (3904, 0.0), (3905, 0.050041144318888
474), (3906, 0.028300947435725355), (3907, 0.010037362991773602), (3908, 0.029400340
60581049), (3909, 0.0), (3910, 0.016746274440802775), (3911, 0.022525889537273178),
(3912, 0.016123638025176065), (3913, 0.0), (3914, 0.015977552271821523), (3915, 0.01
0635703769647691), (3916, 0.013838320156849), (3917, 0.0), (3918, 0.0182481718571794
5), (3919, 0.03721123755911123), (3920, 0.013332434573706437), (3921, 0.046048612163
11124), (3922, 0.0), (3923, 0.0), (3924, 0.0), (3925, 0.00888413575941073), (3926,
0.0), (3927, 0.0), (3928, 0.030302148725765823), (3929, 0.012655154743902157), (393
0, 0.008403319137724874), (3931, 0.0), (3932, 0.010732711442709235), (3933, 0.0), (3
934, 0.02032392460108762), (3935, 0.038656911129619415), (3936, 0.01652587928619184
8), (3937, 0.0), (3938, 0.0), (3939, 0.0285325923124759), (3940, 0.02312027452704457
4), (3941, 0.0056565604433577625), (3942, 0.00518697935251437), (3943, 0.00638639864
4832711), (3944, 0.014094405511030984), (3945, 0.027900819315858597), (3946, 0.02779
9019657750428), (3947, 0.016687819615777998), (3948, 0.06344759582511454), (3949, 0.
0), (3950, 0.0), (3951, 0.03825204105087867), (3952, 0.023128197320152487), (3953,
0.022829932411491248), (3954, 0.01979968882488026), (3955, 0.0), (3956, 0.0488346922
23244075), (3957, 0.014291579039763224), (3958, 0.0), (3959, 0.01889228310844543),
(3960, 0.0), (3961, 0.0), (3962, 0.0054266107758282065), (3963, 0.0), (3964, 0.02248
7871705361817), (3965, 0.023966007445392082), (3966, 0.03183722705641594), (3967, 0.
0), (3968, 0.0), (3969, 0.0), (3970, 0.007839233396686473), (3971, 0.056621348502977
2), (3972, 0.02483274145449593), (3973, 0.00926919826833974), (3974, 0.0060957348614
268676), (3975, 0.032607713077536946), (3976, 0.01380281189024388), (3977, 0.0), (39
78, 0.021804884294299585), (3979, 0.01677076117193053), (3980, 0.0), (3981, 0.019414
843011404684), (3982, 0.005220591162090804), (3983, 0.014630136717243532), (3984, 0.
07151644177116366), (3985, 0.0), (3986, 0.06345431292205245), (3987, 0.0088213271478
68901), (3988, 0.026833284021455805), (3989, 0.0), (3990, 0.0), (3991, 0.03288481417
787307), (3992, 0.0), (3993, 0.0058733065436512135), (3994, 0.0), (3995, 0.0), (399
6, 0.0), (3997, 0.0), (3998, 0.016964648384755317), (3999, 0.0), (4000, 0.0), (4001,
0.0), (4002, 0.0), (4003, 0.014330406393226287), (4004, 0.005830939307813409), (400
5, 0.0), (4006, 0.0), (4007, 0.0), (4008, 0.030063298479964348), (4009, 0.0141549272
79777796), (4010, 0.03538153532993036), (4011, 0.044514698024180566), (4012, 0.02988
7103619596282), (4013, 0.03982906877176247), (4014, 0.008283002314604746), (4015, 0.
019568845753775002), (4016, 0.0), (4017, 0.020932291299746476), (4018, 0.0), (4019,
0.022977782980952967), (4020, 0.009472162943483471), (4021, 0.0), (4022, 0.029451517
91400584), (4023, 0.0), (4024, 0.0075136753791302845), (4025, 0.0), (4026, 0.0), (40
27, 0.052924009478107825), (4028, 0.03566254200133823), (4029, 0.0), (4030, 0.019340
416223934877), (4031, 0.0), (4032, 0.015237863014281771), (4033, 0.01439762839218748
3), (4034, 0.0), (4035, 0.0), (4036, 0.0), (4037, 0.024222956839934705), (4038, 0.02
61385622310165), (4039, 0.01819749655557526), (4040, 0.008423010439333191), (4041,
0.04498378705481512), (4042, 0.005241060083805741), (4043, 0.0), (4044, 0.0), (4045,
0.04255938296352993), (4046, 0.0), (4047, 0.02382274621901415), (4048, 0.01022994417
8983711), (4049, 0.009017544483032625), (4050, 0.04135800652181796), (4051, 0.026832
076226780934), (4052, 0.009137680103957734), (4053, 0.018685919921167877), (4054, 0.
020476113809639852), (4055, 0.014009950626018734), (4056, 0.0), (4057, 0.0), (4058,
0.0), (4059, 0.0), (4060, 0.0), (4061, 0.005804761630695971), (4062, 0.0227407754052
13007), (4063, 0.03770244303501354), (4064, 0.0), (4065, 0.05595732479948473), (406
6, 0.0), (4067, 0.019845494805572288), (4068, 0.0), (4069, 0.0), (4070, 0.0), (4071,
0.019026323862686198), (4072, 0.005713835444529831), (4073, 0.037109376366817104),
(4074, 0.0), (4075, 0.0), (4076, 0.00828719733309934), (4077, 0.021238396891828812),
(4078, 0.020039476478961014), (4079, 0.04008228736984204), (4080, 0.0079777014559669
8), (4081, 0.012072264272219402), (4082, 0.0), (4083, 0.0), (4084, 0.021512587003887
043), (4085, 0.0), (4086, 0.0), (4087, 0.012210164394927153), (4088, 0.0340664940679
9629), (4089, 0.0), (4090, 0.00718094406158436), (4091, 0.0), (4092, 0.0), (4093, 0.

005701108578900813), (4094, 0.021480582162419812), (4095, 0.0), (4096, 0.00646546465
9595971), (4097, 0.010097053147082874), (4098, 0.021286105142155894), (4099, 0.02074
048248269452), (4100, 0.02811374067687117), (4101, 0.0), (4102, 0.0), (4103, 0.0),
(4104, 0.0), (4105, 0.029236208145831964), (4106, 0.023752152836940085), (4107, 0.
0), (4108, 0.019068693272798268), (4109, 0.008605110916871304), (4110, 0.03472495967
479615), (4111, 0.006359807876199569), (4112, 0.0), (4113, 0.0), (4114, 0.0049336910
40317982), (4115, 0.0), (4116, 0.0), (4117, 0.0334481646961512), (4118, 0.0), (4119,
0.01789455089260772), (4120, 0.025206219807988355), (4121, 0.0), (4122, 0.0), (4123,
0.0), (4124, 0.005948704223365627), (4125, 0.010501408641802424), (4126, 0.0), (412
7, 0.03615009113144703), (4128, 0.0), (4129, 0.0059128410836825004), (4130, 0.0), (4
131, 0.01938850094274776), (4132, 0.06519615508825678), (4133, 0.0), (4134, 0.007507
165002268491), (4135, 0.026715223763690975), (4136, 0.00963081758787016), (4137, 0.
0), (4138, 0.009451457533864858), (4139, 0.032159864806466675), (4140, 0.0), (4141,
0.0), (4142, 0.04008505369536294), (4143, 0.006172798970472118), (4144, 0.0218520980
5770838), (4145, 0.019476251294938643), (4146, 0.014471031507595471), (4147, 0.01033
4556435606615), (4148, 0.0), (4149, 0.00935405927823007), (4150, 0.01511544009426098
4), (4151, 0.025495013894241345), (4152, 0.011796506006603412), (4153, 0.07127507259
646959), (4154, 0.043133043174220106), (4155, 0.013732607977739997), (4156, 0.020822
643470693958), (4157, 0.0), (4158, 0.05704977182229488), (4159, 0.0498265949039700
5), (4160, 0.04108274776237279), (4161, 0.007121979634927727), (4162, 0.0), (4163,
0.017833374959496953), (4164, 0.029219991082797412), (4165, 0.0), (4166, 0.0), (416
7, 0.02657907615836804), (4168, 0.0), (4169, 0.0), (4170, 0.009522611433408934), (41
71, 0.0), (4172, 0.030843848670895114), (4173, 0.007856250731607695), (4174, 0.00575
3848393351417), (4175, 0.018330668281005526), (4176, 0.06219157466177185), (4177, 0.
0418810790043094), (4178, 0.0), (4179, 0.014882003497418519), (4180, 0.0094302398782
1648), (4181, 0.020100199966455613), (4182, 0.020933494176032633), (4183, 0.11353876
531321921), (4184, 0.04532745476604422), (4185, 0.0), (4186, 0.0), (4187, 0.00609412
87842083405), (4188, 0.0), (4189, 0.0404943134516505), (4190, 0.0), (4191, 0.0209812
2572166964), (4192, 0.008041477453170473), (4193, 0.04049747079761369), (4194, 0.0),
(4195, 0.007913014458346867), (4196, 0.006702064358825231), (4197, 0.008263699284628
442), (4198, 0.0), (4199, 0.0), (4200, 0.059506549998311), (4201, 0.0180110107922959
28), (4202, 0.007351418661532418), (4203, 0.04993872226218514), (4204, 0.00783318181
4846973), (4205, 0.008963532932010123), (4206, 0.0), (4207, 0.0), (4208, 0.0), (420
9, 0.02946035471536927), (4210, 0.0), (4211, 0.04201447494129147), (4212, 0.0), (421
3, 0.009741310905190287), (4214, 0.012687765261049002), (4215, 0.01180893978586950
2), (4216, 0.0078703182496836), (4217, 0.009058096224510323), (4218, 0.0), (4219, 0.
0), (4220, 0.004670831880857795), (4221, 0.054284510472713446), (4222, 0.01779437489
0383248), (4223, 0.006098225314203334), (4224, 0.00882742049201765), (4225, 0.017003
891296486277), (4226, 0.006240093037830945), (4227, 0.0), (4228, 0.0361819488075222
6), (4229, 0.0), (4230, 0.013776034714659435), (4231, 0.015456706378099405), (4232,
0.0190011688590868), (4233, 0.017641359488570207), (4234, 0.0), (4235, 0.0), (4236,
0.0), (4237, 0.008261552707641099), (4238, 0.016386604363445457), (4239, 0.029309778
727981246), (4240, 0.0), (4241, 0.01039320093067048), (4242, 0.0), (4243, 0.0), (424
4, 0.0), (4245, 0.0), (4246, 0.03351513059708041), (4247, 0.0), (4248, 0.0), (4249,
0.00835116343896571), (4250, 0.0), (4251, 0.023296212848429303), (4252, 0.0290570525
08361145), (4253, 0.0), (4254, 0.029523893148616673), (4255, 0.04483395822380165),
(4256, 0.042238006591922735), (4257, 0.0), (4258, 0.0063453053181198255), (4259, 0.0
23576193859579795), (4260, 0.0), (4261, 0.023373775267356657), (4262, 0.0), (4263,
0.00743221492616897), (4264, 0.008958424529064913), (4265, 0.006771129380892885), (4
266, 0.0), (4267, 0.09468285761112293), (4268, 0.007691007015220353), (4269, 0.0),
(4270, 0.0), (4271, 0.0), (4272, 0.008545807980274156), (4273, 0.02903996761558035
4), (4274, 0.0), (4275, 0.005099362409852739), (4276, 0.03956371495025973), (4277,
0.0075743121274934565), (4278, 0.009464212926195429), (4279, 0.022024514850632875),
(4280, 0.0), (4281, 0.008643007914679396), (4282, 0.009081174337117321), (4283, 0.01
4332810491009864), (4284, 0.024897717747778072), (4285, 0.03687889526988989), (4286,
0.03180965052205108), (4287, 0.006529840059985693), (4288, 0.01816339886505262), (42

89, 0.0), (4290, 0.0), (4291, 0.025602300444489825), (4292, 0.007597582997653554),
(4293, 0.0), (4294, 0.0), (4295, 0.030943302946593594), (4296, 0.0), (4297, 0.019576
47368273257), (4298, 0.0), (4299, 0.0), (4300, 0.05132829819610413), (4301, 0.012955
881810923449), (4302, 0.0071696254758530035), (4303, 0.0), (4304, 0.0), (4305, 0.0),
(4306, 0.0), (4307, 0.0), (4308, 0.0), (4309, 0.0), (4310, 0.007464709026905589), (4
311, 0.042376840774935), (4312, 0.03944516853312812), (4313, 0.037920735335717756),
(4314, 0.0), (4315, 0.0047238512062288), (4316, 0.004742445456975398), (4317, 0.0084
18881831423225), (4318, 0.0), (4319, 0.025380193282361255), (4320, 0.017048585347335
086), (4321, 0.008864708440215053), (4322, 0.0), (4323, 0.0), (4324, 0.0195430543011
00492), (4325, 0.02360830506485523), (4326, 0.02157391943089125), (4327, 0.013728346
759044576), (4328, 0.010611956636390087), (4329, 0.0428452770884991), (4330, 0.01030
8400914699724), (4331, 0.007280559350666071), (4332, 0.0), (4333, 0.0132544675844073
53), (4334, 0.0), (4335, 0.0), (4336, 0.0), (4337, 0.015789086912455935), (4338, 0.
0), (4339, 0.04173266823829371), (4340, 0.006910398014265468), (4341, 0.0), (4342,
0.0), (4343, 0.007823808990535185), (4344, 0.004501719528877708), (4345, 0.010177358
789610852), (4346, 0.01340905478827646), (4347, 0.0), (4348, 0.017625826538003934),
(4349, 0.005501383169600896), (4350, 0.03752802763522038), (4351, 0.0049155466951480
99), (4352, 0.01559907026918832), (4353, 0.01355571485898728), (4354, 0.0), (4355,
0.0177893788924605), (4356, 0.030717641857640622), (4357, 0.017618608497090134), (43
58, 0.0), (4359, 0.0), (4360, 0.018431962768543402), (4361, 0.0), (4362, 0.0), (436
3, 0.0), (4364, 0.0), (4365, 0.00849963016960993), (4366, 0.0), (4367, 0.01042329527
717722), (4368, 0.03609669496045223), (4369, 0.0), (4370, 0.00994239424526903), (437
1, 0.008933780040529984), (4372, 0.02479895124378561), (4373, 0.008549387756846594),
(4374, 0.0), (4375, 0.019193832804282808), (4376, 0.021801046159450426), (4377, 0.00
7950081240470907), (4378, 0.02721743863977713), (4379, 0.018484048859335078), (4380,
0.020482528702467175), (4381, 0.007628986310298959), (4382, 0.013895426737584378),
(4383, 0.014295825600831508), (4384, 0.0), (4385, 0.0), (4386, 0.02881619055847516
7), (4387, 0.0), (4388, 0.045716357235399775), (4389, 0.0), (4390, 0.007548256844989
767), (4391, 0.021806824652795005), (4392, 0.03155180196283023), (4393, 0.0151143946
4674318), (4394, 0.0068369326462402735), (4395, 0.007083834483934841), (4396, 0.0),
(4397, 0.0067470530740054956), (4398, 0.0), (4399, 0.0626817291661942), (4400, 0.0),
(4401, 0.030593707314490928), (4402, 0.004703132375248852), (4403, 0.0), (4404, 0.00
6538351925547434), (4405, 0.0), (4406, 0.0), (4407, 0.0), (4408, 0.02747295622250058
7), (4409, 0.0), (4410, 0.0), (4411, 0.07357327641709638), (4412, 0.0), (4413, 0.0),
(4414, 0.010415766262004633), (4415, 0.023918356803842442), (4416, 0.0), (4417, 0.
0), (4418, 0.03058324536558106), (4419, 0.0), (4420, 0.0), (4421, 0.0264013272952018
06), (4422, 0.0), (4423, 0.0), (4424, 0.0), (4425, 0.006175091089703483), (4426, 0.
0), (4427, 0.01927266798882914), (4428, 0.03198083239514259), (4429, 0.0), (4430, 0.
025620425298163683), (4431, 0.0), (4432, 0.06979240805656593), (4433, 0.0), (4434,
0.0), (4435, 0.0), (4436, 0.0), (4437, 0.0), (4438, 0.022870792953686863), (4439, 0.
05693854482573927), (4440, 0.03718320807338606), (4441, 0.0), (4442, 0.0171691748063
38575), (4443, 0.0), (4444, 0.0), (4445, 0.0), (4446, 0.0), (4447, 0.010183020209924
933), (4448, 0.016752807757106525), (4449, 0.008837224799150863), (4450, 0.006424147
434476346), (4451, 0.0), (4452, 0.0), (4453, 0.0), (4454, 0.01801860702023059), (445
5, 0.0), (4456, 0.0), (4457, 0.0), (4458, 0.0), (4459, 0.0), (4460, 0.00596008809586
97545), (4461, 0.020700562426783264), (4462, 0.0), (4463, 0.0), (4464, 0.0), (4465,
0.005546866060051285), (4466, 0.0), (4467, 0.019924712532179627), (4468, 0.0), (446
9, 0.0), (4470, 0.010500723776802872), (4471, 0.019378120241405016), (4472, 0.027753
517605867153), (4473, 0.039971218006751505), (4474, 0.0), (4475, 0.00861832463609193
8), (4476, 0.008222961450500511), (4477, 0.035509387237202705), (4478, 0.0), (4479,
0.008684982745146274), (4480, 0.0), (4481, 0.009271047852722926), (4482, 0.020168082
889197712), (4483, 0.008411900697427017), (4484, 0.007573511428283609), (4485, 0.0),
(4486, 0.015127570162180718), (4487, 0.00551196594659381), (4488, 0.0088982145914634
28), (4489, 0.01126895407748917), (4490, 0.0), (4491, 0.0), (4492, 0.0), (4493, 0.02
263597819989708), (4494, 0.011164944615909845), (4495, 0.0), (4496, 0.02850937208468
575), (4497, 0.01646975565419904), (4498, 0.030361530753229626), (4499, 0.0075518192

297254864), (4500, 0.0), (4501, 0.010383346262420592), (4502, 0.00939465681159704),
(4503, 0.0), (4504, 0.0), (4505, 0.0), (4506, 0.029887571721278913), (4507, 0.026290
46389058557), (4508, 0.0), (4509, 0.009152777726504628), (4510, 0.0), (4511, 0.0),
(4512, 0.0), (4513, 0.0), (4514, 0.029122337929354876), (4515, 0.04885647202358133),
(4516, 0.0), (4517, 0.0), (4518, 0.048657949715749355), (4519, 0.02764106088415877
3), (4520, 0.0), (4521, 0.0), (4522, 0.027997420183782068), (4523, 0.0), (4524, 0.00
8118910070572088), (4525, 0.010239175091362967), (4526, 0.03573541431954977), (4527,
0.03613588749026028), (4528, 0.016612843255447495), (4529, 0.020808032182087318), (4
530, 0.0), (4531, 0.008753350112336243), (4532, 0.0), (4533, 0.0), (4534, 0.0), (453
5, 0.017535581418998837), (4536, 0.0), (4537, 0.0), (4538, 0.0), (4539, 0.0496307396
99988165), (4540, 0.0), (4541, 0.0), (4542, 0.0), (4543, 0.011220923472811117), (454
4, 0.0), (4545, 0.01651676341407265), (4546, 0.005955046389491045), (4547, 0.0), (45
48, 0.0), (4549, 0.005505685485488944), (4550, 0.0), (4551, 0.0), (4552, 0.023602027
507068947), (4553, 0.0), (4554, 0.02936435854975045), (4555, 0.03592531634931903),
(4556, 0.00823174020504103), (4557, 0.005992759906969406), (4558, 0.0416698097268931
24), (4559, 0.012002694175231901), (4560, 0.0), (4561, 0.0), (4562, 0.0), (4563, 0.0
04318625084217244), (4564, 0.0), (4565, 0.005610193331200937), (4566, 0.0), (4567,
0.0), (4568, 0.0), (4569, 0.020495527614960714), (4570, 0.0), (4571, 0.0), (4572, 0.
00796218261301683), (4573, 0.01727611124121416), (4574, 0.01406379388613152), (4575,
0.0), (4576, 0.00689000185723599), (4577, 0.008968381969534603), (4578, 0.0), (4579,
0.01982926007038003), (4580, 0.03622285430310541), (4581, 0.0), (4582, 0.0), (4583,
0.03277353470453816), (4584, 0.025268834479702604), (4585, 0.03435018649123674), (45
86, 0.0), (4587, 0.0), (4588, 0.0), (4589, 0.0), (4590, 0.0), (4591, 0.0), (4592, 0.
008226455498323862), (4593, 0.0), (4594, 0.02210554483241709), (4595, 0.016563799824
0981), (4596, 0.009245402613121971), (4597, 0.012196765764959186), (4598, 0.0), (459
9, 0.028194480916451325), (4600, 0.008492458601566376), (4601, 0.04494072980366674),
(4602, 0.022307021773752184), (4603, 0.0), (4604, 0.01715868957488465), (4605, 0.014
849068302119297), (4606, 0.011416225215678366), (4607, 0.0), (4608, 0.03296545338243
5755), (4609, 0.0), (4610, 0.02099808219179862), (4611, 0.0), (4612, 0.0), (4613, 0.
010294760716985882), (4614, 0.0), (4615, 0.0), (4616, 0.06403425394321395), (4617,
0.0), (4618, 0.024359280031183518), (4619, 0.0), (4620, 0.006072986416382341), (462
1, 0.07837896649693904), (4622, 0.0), (4623, 0.005508704247561916), (4624, 0.0), (46
25, 0.027035064284016603), (4626, 0.013104790516575593), (4627, 0.0), (4628, 0.0),
(4629, 0.025384493463633688), (4630, 0.05546741612283619), (4631, 0.023144525883136
9), (4632, 0.0), (4633, 0.0), (4634, 0.008548867495514648), (4635, 0.0), (4636, 0.
0), (4637, 0.0), (4638, 0.02605735428282043), (4639, 0.0), (4640, 0.0), (4641, 0.0),
(4642, 0.01910199601430621), (4643, 0.03490846656476736), (4644, 0.0), (4645, 0.0246
919909655625), (4646, 0.03903683259951401), (4647, 0.00933585139697818), (4648, 0.00
7192235318862787), (4649, 0.01740957091634849), (4650, 0.017596256750848464), (4651,
0.0), (4652, 0.0), (4653, 0.0), (4654, 0.01792204070545419), (4655, 0.0), (4656, 0.0
08071877138176855), (4657, 0.0), (4658, 0.0), (4659, 0.008618744250420573), (4660,
0.0), (4661, 0.0), (4662, 0.0), (4663, 0.024163588968421926), (4664, 0.0442273081802
5078), (4665, 0.009671658716015748), (4666, 0.015218842938926289), (4667, 0.0), (466
8, 0.00812127649467453), (4669, 0.0), (4670, 0.04395226544007953), (4671, 0.0), (467
2, 0.03517754294343678), (4673, 0.00794071427576975), (4674, 0.0), (4675, 0.06076579
391953683), (4676, 0.0), (4677, 0.00877423435389705), (4678, 0.009056769495356205),
(4679, 0.013999750215399308), (4680, 0.008905737538919345), (4681, 0.0), (4682, 0.
0), (4683, 0.0), (4684, 0.0), (4685, 0.0), (4686, 0.0), (4687, 0.0), (4688, 0.014258
2410038743), (4689, 0.0), (4690, 0.0), (4691, 0.0), (4692, 0.0), (4693, 0.0), (4694,
0.0), (4695, 0.0), (4696, 0.00899520611993711), (4697, 0.0), (4698, 0.0), (4699, 0.
0), (4700, 0.0), (4701, 0.023113170960263957), (4702, 0.0), (4703, 0.0), (4704, 0.02
755107141247336), (4705, 0.0), (4706, 0.0), (4707, 0.04888669936037911), (4708, 0.00
7520218607396731), (4709, 0.0), (4710, 0.0), (4711, 0.0), (4712, 0.0), (4713, 0.0425
2704567494593), (4714, 0.0), (4715, 0.0), (4716, 0.0), (4717, 0.036263469899302486),
(4718, 0.008141129824638679), (4719, 0.0), (4720, 0.008427405496050828), (4721, 0.
0), (4722, 0.0), (4723, 0.0), (4724, 0.0435288820490115), (4725, 0.0), (4726, 0.0148

```
71171022945306), (4727, 0.019902934749441867), (4728, 0.012756795991799913), (4729,
0.0), (4730, 0.08191403933124478), (4731, 0.0), (4732, 0.02658047073158657), (4733,
0.015414064265363212), (4734, 0.006244465355735435), (4735, 0.02197488302956456), (4
736, 0.0), (4737, 0.0), (4738, 0.0), (4739, 0.03435353572385781), (4740, 0.008218175
296016199), (4741, 0.0), (4742, 0.033473981390933785), (4743, 0.0), (4744, 0.0083426
65834602725), (4745, 0.0), (4746, 0.009250526313751945), (4747, 0.0), (4748, 0.0),
(4749, 0.03303006212149451), (4750, 0.0), (4751, 0.0), (4752, 0.006673211172831979),
(4753, 0.0), (4754, 0.0), (4755, 0.0), (4756, 0.011668526156901298), (4757, 0.030734
02916622718), (4758, 0.0), (4759, 0.03097738725908653), (4760, 0.05854776032580973),
(4761, 0.0), (4762, 0.0), (4763, 0.009510215661582229), (4764, 0.01058093889077306
3), (4765, 0.0), (4766, 0.0), (4767, 0.017105970967803186), (4768, 0.0), (4769, 0.00
7850431581962624), (4770, 0.0), (4771, 0.0), (4772, 0.04319163499952621), (4773, 0.0
19896854448345436), (4774, 0.02459422889255241), (4775, 0.0), (4776, 0.0182535097220
63053), (4777, 0.0), (4778, 0.03133502448379851), (4779, 0.0), (4780, 0.009269068742
749641), (4781, 0.0), (4782, 0.010483456668709399), (4783, 0.0), (4784, 0.0), (4785,
0.0), (4786, 0.0), (4787, 0.009088114682982247), (4788, 0.01614538632223229), (4789,
0.03891584535903817), (4790, 0.0), (4791, 0.0317728220541161), (4792, 0.006978860831
716811), (4793, 0.016136374840089375), (4794, 0.0), (4795, 0.05124223548809905), (47
96, 0.020273113996351778), (4797, 0.0), (4798, 0.009531042329918646), (4799, 0.00743
5200207794489), (4800, 0.0), (4801, 0.010675801655328677), (4802, 0.0)]
```

In [24]:
```
len(similarity_score)
```

Out[24]:
```
4803
```

In [25]:
```
# sorting the movies based on their similarity score

sorted_similar_movies = sorted(similarity_score, key = lambda x:x[1], reverse = Tru
print(sorted_similar_movies)
```

```
[(1359, 1.0), (428, 0.4311643836232694), (210, 0.25737999820859625), (3, 0.204387737
32168222), (119, 0.19262528757150407), (65, 0.1775581506611392), (1512, 0.1470516265
4306442), (813, 0.14414128303962467), (2530, 0.13737322473729185), (1017, 0.13713281
929528845), (473, 0.13217075714794208), (753, 0.13216136203404205), (278, 0.12996260
715124025), (14, 0.12815026535769386), (72, 0.12266571244563478), (2313, 0.118343329
8876972), (2381, 0.11752380293327759), (5, 0.11533013884014888), (2858, 0.1141365218
0839844), (4183, 0.11353876531321921), (30, 0.11336219425554919), (2655, 0.112709539
16613297), (10, 0.11240850198526249), (1035, 0.11197582745207195), (2685, 0.10980421
440315517), (870, 0.10722005407914785), (41, 0.10650154604688718), (1296, 0.10584478
077207024), (438, 0.10277392559139559), (1477, 0.10199709558322863), (299, 0.1015463
2458465614), (1474, 0.10078843818576026), (1002, 0.1005262424012352), (1085, 0.10023
066759543668), (2753, 0.09951168410001496), (303, 0.09883937789831629), (2805, 0.096
53055839236144), (163, 0.09490815555089474), (1803, 0.09486132363970294), (584, 0.09
484559891817565), (4267, 0.09468285761112293), (1076, 0.0940401999293739), (3630, 0.
09220755061184922), (32, 0.0899394677673743), (9, 0.0897616996163815), (1141, 0.0897
534119962937), (2740, 0.08940625662471981), (1740, 0.08909361212904546), (3326, 0.08
864740283572532), (3466, 0.08668912224324149), (3350, 0.08655054623169336), (1241,
0.08609227902838198), (681, 0.08543127346751898), (2365, 0.0853438442958511), (2108,
0.08391604344392495), (2433, 0.08365489821337185), (1048, 0.08353639093976094), (385
4, 0.08270463326994387), (4730, 0.08191403933124478), (351, 0.08174380966331447), (2
793, 0.08113461186171188), (963, 0.08093607979181264), (226, 0.08009811585333676),
(2047, 0.08005984967911549), (1771, 0.07939424641304792), (2826, 0.0790880435304814
6), (4621, 0.07837896649693904), (1720, 0.07785389662462186), (215, 0.07731781359499
82), (2492, 0.07720993930846724), (2383, 0.0766994239491908), (904, 0.07665815430587
02), (480, 0.07657647328497882), (3075, 0.07656011925039205), (1087, 0.0765328060497
2879), (583, 0.07617125404896875), (2635, 0.07612157059438401), (774, 0.075202765662
49046), (207, 0.07517731687148192), (2265, 0.07480103949044059), (4411, 0.0735732764
1709638), (2514, 0.07337337385697332), (1374, 0.07276512150372845), (42, 0.072440220
42935871), (417, 0.07154871800648291), (3984, 0.07151644177116366), (731, 0.07147485
82813296), (1421, 0.07145387858299798), (4153, 0.07127507259646959), (3242, 0.071156
49368602578), (3323, 0.07093837703816055), (1478, 0.07074037978729494), (2482, 0.070
53390554831772), (3148, 0.07051443394414272), (2123, 0.07049506017980059), (1813, 0.
0702107976791396), (4432, 0.06979240805656593), (1858, 0.06975691386608407), (800,
0.06953224882098635), (1566, 0.06940034499958737), (1236, 0.06921697670526414), (303
3, 0.06882483694481675), (2897, 0.0687325954457757), (3637, 0.06850035824377265), (2
983, 0.06841359888033002), (1343, 0.0679178458251457), (3564, 0.06748597328266652),
(422, 0.06660418021192085), (1434, 0.06650848491705977), (1964, 0.0664126745247039
4), (2822, 0.06523623085813704), (133, 0.0652059210043185), (4132, 0.065196155088256
78), (2631, 0.06504777397079482), (3267, 0.06484032054628708), (662, 0.0645891595892
0407), (1827, 0.0644590393699526), (1629, 0.06407184052001345), (4616, 0.06403425394
321395), (3550, 0.06393063808449993), (880, 0.06391064887633026), (224, 0.0638109363
6174208), (2340, 0.06378589921415115), (935, 0.0636982726513143), (217, 0.0636581264
4026905), (3986, 0.06345431292205245), (3948, 0.06344759582511454), (3372, 0.0634015
1915093728), (1293, 0.06301875726076642), (739, 0.06293449312413521), (3719, 0.06293
374984621716), (4399, 0.0626817291661942), (3297, 0.06256549822883029), (958, 0.0624
08672993923664), (4176, 0.06219157466177185), (1902, 0.06175659279873491), (3355, 0.
06163725811314999), (3053, 0.0614937693536529), (1043, 0.06148903574491431), (3758,
0.061448143793647665), (2070, 0.06136751707477053), (323, 0.0607910339666117), (467
5, 0.06076579391953683), (2368, 0.06064463603285601), (117, 0.060492562884146284),
(1594, 0.0603608083986681), (581, 0.060287457625623146), (3685, 0.0601290036772252
9), (1148, 0.06010765169185929), (2284, 0.060096055197682186), (587, 0.0599541456628
23075), (1223, 0.059922058269882), (3873, 0.05986413704367761), (969, 0.059655360404
02045), (2223, 0.05960804990085404), (1120, 0.05950894940289865), (4200, 0.059506549
998311), (1506, 0.059475397637864434), (2114, 0.05937475363169126), (755, 0.05870570
3664613695), (827, 0.058651554758132156), (826, 0.05858774897651223), (1846, 0.05858
161919230411), (3819, 0.0585620788359174), (4760, 0.05854776032580973), (1963, 0.058
```

085263627459416), (164, 0.05766352814275297), (836, 0.057434730759063225), (3607, 0.
057355720926209865), (2789, 0.05732486796919024), (2560, 0.05724815289422347), (273
1, 0.05719320121356585), (1363, 0.057179260936214775), (4158, 0.05704977182229488),
(4439, 0.05693854482573927), (3971, 0.0566213485029772), (1174, 0.0564889201157057
3), (1782, 0.056308948618240914), (2066, 0.05609985436718217), (3031, 0.056084324186
91643), (4065, 0.05595732479948473), (1303, 0.05585861136454683), (343, 0.0557541533
91077026), (4630, 0.05546741612283619), (3766, 0.05540022211096728), (2334, 0.055266
12036216298), (2878, 0.05515185050930864), (2586, 0.05472919108104719), (1034, 0.054
65360864260904), (366, 0.054515529136516035), (1292, 0.054451481344167885), (2055,
0.0543643162789638), (2029, 0.054354817322176), (103, 0.054354666030584614), (447,
0.05428674675430024), (4221, 0.054284510472713446), (426, 0.05387121165126171), (318
4, 0.05375836747802638), (1247, 0.053581002184245344), (2221, 0.053576821197112795),
(3208, 0.05349357201558701), (1561, 0.05300836086982752), (4027, 0.05292400947810782
5), (2830, 0.05290889080627135), (1513, 0.05288189923381389), (1685, 0.0527967837737
0657), (2474, 0.05270796628302849), (1929, 0.05267613017000264), (779, 0.05259835223
293688), (1183, 0.05247200231886881), (1001, 0.05239940922199486), (342, 0.052321105
88872813), (3047, 0.05227719423322646), (2470, 0.05224177322812685), (1144, 0.051961
23866300828), (994, 0.051830907119553474), (1097, 0.051812132668283176), (2362, 0.05
1761187642313536), (2813, 0.051583845389576416), (2321, 0.05148717305895325), (1851,
0.05137565980020797), (1886, 0.05135926016803144), (4300, 0.05132829819610413), (375
3, 0.05128002013987719), (4795, 0.05124223548809905), (46, 0.05119279389322904), (57
8, 0.05118048264656639), (2824, 0.05114689340419648), (196, 0.050872350184735876),
(2285, 0.05072283634197921), (1243, 0.050700350160624824), (2574, 0.0506680796473799
46), (513, 0.050555830048219536), (174, 0.05055026688169773), (3211, 0.0502776734603
01965), (1976, 0.050266582973901706), (3851, 0.05023263968999226), (415, 0.050194722
09444976), (3905, 0.050041144318888474), (4203, 0.04993872226218514), (1, 0.04983293
064399152), (4159, 0.04982659490397005), (2026, 0.049807132061791326), (2331, 0.0497
04567350787016), (4539, 0.049630739699988165), (3405, 0.049614662138495455), (358,
0.049530171663648544), (1600, 0.049336396895483095), (1316, 0.04929772852512586), (1
977, 0.04919724187128134), (1884, 0.04898143796896101), (1125, 0.04892884717732306
3), (4707, 0.04888669936037911), (4515, 0.04885647202358133), (1278, 0.0488360430519
503), (3956, 0.048834692223244075), (2157, 0.04883326969652898), (305, 0.04876645858
500153), (4518, 0.048657949715749355), (3774, 0.04860532649135153), (2871, 0.0486031
9935980385), (1674, 0.04859943492495315), (1790, 0.048556569482376466), (546, 0.0481
2265039724559), (2301, 0.04808522757820377), (636, 0.04807695168680448), (372, 0.048
009596046192184), (1009, 0.04799624299749351), (2444, 0.04777721636791725), (3375,
0.0477623719685448), (1382, 0.047751731789872645), (3893, 0.0476952660288304), (325
0, 0.047684927328473056), (3634, 0.047534105702211), (1458, 0.04744491982945791), (5
18, 0.04743670355350071), (2892, 0.047332554693430715), (2647, 0.04728576134596762
5), (2191, 0.04727771543208084), (2205, 0.0472730752732396), (1110, 0.04710839249677
591), (946, 0.04690419049379716), (3306, 0.0468894508031419), (1006, 0.0468548629625
1217), (782, 0.04684563840327667), (3146, 0.046692892879965475), (1159, 0.0466643641
2856402), (1259, 0.046382514953439494), (364, 0.04627287142241801), (867, 0.04611844
113646047), (3921, 0.04604861216311124), (2862, 0.04603206010258901), (1655, 0.04594
587261868763), (3010, 0.045936908711060494), (694, 0.04587601234144584), (4388, 0.04
5716357235399775), (3439, 0.045669882159059386), (968, 0.04550911170536931), (2572,
0.04547730814514499), (1653, 0.04545837835897055), (1918, 0.04544963560382887), (304
6, 0.04541372738982333), (1417, 0.04540145189851705), (791, 0.045377002143510065),
(4184, 0.04532745476604422), (3232, 0.04522567127593379), (4041, 0.0449837870548151
2), (2034, 0.044983445730573966), (4601, 0.04494072980366674), (1113, 0.044857002035
258026), (4255, 0.04483395822380165), (2543, 0.04475496708026995), (2430, 0.04470385
235806187), (186, 0.04459616327625059), (2035, 0.04454105154294242), (4011, 0.044514
698024180566), (665, 0.044432466036149254), (3409, 0.04442588575633951), (3901, 0.04
4391555723730644), (4664, 0.04422730818025078), (3359, 0.04422151373766265), (832,
0.04420586787560895), (73, 0.04415940718548016), (939, 0.0440147245600844), (4670,
0.04395226544007953), (3325, 0.04388566680159247), (219, 0.043872578197580316), (188

1, 0.04382449735381551), (1469, 0.043763468277525225), (446, 0.043751045863748145),
(2320, 0.04373758512234605), (2982, 0.043734275425904495), (1515, 0.0437132739052602
4), (830, 0.04368824829368255), (4724, 0.0435288820490115), (2713, 0.043500235107840
11), (1165, 0.043492600825020586), (3891, 0.04337769563404051), (2366, 0.04337057394
012591), (1369, 0.0433392780847158), (213, 0.04328647825815964), (170, 0.04326245270
4246564), (36, 0.04323488875573723), (3624, 0.04322362197604023), (1265, 0.043201915
47401773), (156, 0.043192368499795894), (4772, 0.04319163499952621), (4154, 0.043133
043174220106), (396, 0.04306307563001721), (1605, 0.04300799644485527), (223, 0.0429
8330016886992), (1439, 0.04293508667172626), (4329, 0.0428452770884991), (3883, 0.04
28439235519079), (484, 0.04279273956916554), (1024, 0.042731901198420796), (608, 0.0
4271399235417937), (1847, 0.042712112247782), (100, 0.04263848763117278), (4045, 0.0
4255938296352993), (4713, 0.04252704567494593), (3243, 0.04250479555235183), (4311,
0.042376840774935), (2946, 0.04230893080456544), (3518, 0.0423025122877909), (4256,
0.042238006591922735), (2357, 0.04218005116739609), (1837, 0.04217668058421618), (27
80, 0.04217313387955453), (748, 0.04215680173293189), (271, 0.04205829367344873), (1
863, 0.04203190343734929), (4211, 0.04201447494129147), (1368, 0.04197806701094106),
(3218, 0.041931442770066975), (4177, 0.0418810790043094), (24, 0.04186550745628901),
(1699, 0.04177408078841976), (2240, 0.04174815321841285), (4339, 0.0417326682382937
1), (1890, 0.041713935503622346), (4558, 0.041669809726893124), (920, 0.041621136356
21227), (3348, 0.041615652836445444), (1989, 0.04159987031212476), (2688, 0.04158228
311037358), (18, 0.04153881551585253), (2036, 0.041511070220621595), (494, 0.0415034
5826328174), (831, 0.04140806316101746), (4050, 0.04135800652181796), (510, 0.041351
07919955205), (1873, 0.04134639130536735), (1358, 0.04131304321764742), (199, 0.0412
7383784609175), (715, 0.04120980984410683), (1611, 0.04112474398198277), (1432, 0.04
1097611614166765), (4160, 0.04108274776237279), (2159, 0.041007751621469475), (762,
0.04094856007215815), (3287, 0.04087820747828064), (149, 0.04085938432174789), (199
9, 0.040834017589824856), (3833, 0.04081283854681531), (2692, 0.040791339493503255),
(1192, 0.04076275116572928), (2255, 0.04074315087193084), (2099, 0.040679354308898
9), (3579, 0.040663397551497095), (1233, 0.04066036452760822), (978, 0.0405155363665
6369), (4193, 0.04049747079761369), (4189, 0.0404943134516505), (1830, 0.04049210660
090152), (444, 0.0404675132234011), (265, 0.04045462921658721), (1063, 0.04028775499
564254), (1799, 0.04028628091308526), (470, 0.040142637072399304), (2004, 0.04008942
140881579), (4142, 0.04008505369536294), (4079, 0.04008228736984204), (1758, 0.04003
917737948752), (2454, 0.03999436287813538), (4473, 0.039971218006751505), (1652, 0.0
3993685756670881), (3368, 0.03993230481575126), (3762, 0.0399034538062805), (3595,
0.03989301472654164), (4013, 0.03982906877176247), (766, 0.03980676672275263), (255
3, 0.039753973824415326), (3475, 0.03975043342779548), (319, 0.03973736386791702),
(638, 0.03966400128152062), (1792, 0.03958339043802276), (1802, 0.03956868099433981
4), (4276, 0.03956371495025973), (1348, 0.03954765666933953), (2777, 0.0395349900997
0756), (113, 0.039521340648464964), (1016, 0.03947107235591406), (4312, 0.0394451685
3312812), (1995, 0.03942957756973021), (3351, 0.039395968822721505), (2773, 0.039356
07751611092), (3014, 0.03934397767922128), (268, 0.039340417700922385), (3239, 0.039
305181294137834), (393, 0.03926398816349792), (2730, 0.03924194624615451), (2145, 0.
03923975791495391), (1503, 0.03922376831304132), (2796, 0.039174752591650465), (464
6, 0.03903683259951401), (2387, 0.03902280806428206), (1342, 0.038950542962535556),
(1124, 0.038948480563430844), (2025, 0.03892760176597966), (4789, 0.0389158453590381
7), (2715, 0.03881810617499204), (1311, 0.03866179996389761), (3935, 0.0386569111296
19415), (3492, 0.038595106128332404), (1050, 0.03858617259693438), (3423, 0.03852331
888100041), (690, 0.03851770236770982), (293, 0.03850735689533657), (2792, 0.0384700
9120506302), (1027, 0.03836958295913614), (54, 0.038282186098375935), (12, 0.0382585
7693295372), (3951, 0.03825204105087867), (1692, 0.03823028320857572), (3404, 0.0382
2017262721622), (150, 0.03821300742584273), (1558, 0.038146343919781674), (547, 0.03
811071415071551), (1018, 0.03810064751949809), (3068, 0.03807974405788157), (2193,
0.03806330499832303), (2540, 0.03801305588394782), (1191, 0.03800587212380693), (121
3, 0.0380022147669937), (1289, 0.037943844240752506), (4313, 0.037920735335717756),
(2289, 0.037911143150644935), (2126, 0.037902189006431476), (1824, 0.037898101279818

2), (1163, 0.03787797511689461), (1216, 0.03786025128668011), (291, 0.03785527048219
344), (1400, 0.03780019637369408 4), (314, 0.03779388023290209), (1409, 0.03777411306
063175), (285, 0.03771237430169859), (4063, 0.03770244303501354), (1935, 0.037686935
15944097), (1451, 0.03767982117975958 4), (1431, 0.03765706687564924), (3538, 0.03765
221577616237), (1403, 0.03761948767754858), (1574, 0.03753004775712927), (4350, 0.03
752802763522038), (1624, 0.03752419572986529), (2062, 0.03749317710981428), (1082,
0.037454814107502454), (2609, 0.03743491625964058), (1794, 0.03735548424081349), (3
9, 0.03733968312911064), (869, 0.03733552378410931 5), (3919, 0.03721123755911123),
(2849, 0.03719209263660313), (1577, 0.03719006928450637 6), (4440, 0.0371832080733860
6), (1785, 0.03713984750749881), (4073, 0.03710937636681710 4), (3333, 0.037066154862
66431), (2310, 0.03703116569894049), (2015, 0.03696916006610588), (2104, 0.036963279
11930247), (777, 0.03696066910614822 4), (1959, 0.03694055596901151), (1390, 0.036918
948858573365), (3694, 0.03688229980459641), (4285, 0.03687889526988989), (1188, 0.03
685944318062393), (1092, 0.03674322166332937), (373, 0.03668963121270457), (1280, 0.
03666727665729252), (3680, 0.03665975638251558), (2513, 0.03660982921535576 4), (239
8, 0.03660270602590573 5), (1392, 0.03660194722613763), (2851, 0.03658903233167636),
(1461, 0.036580189662531445), (3837, 0.036541655221875094), (3775, 0.036473830860536
974), (4717, 0.036263469899302486), (4580, 0.03622285430310541), (4228, 0.0361819488
0752226), (1822, 0.03617576829397694), (4127, 0.03615009113144703), (2234, 0.0361471
46365960586), (1914, 0.03613944838100905), (4527, 0.03613588749026028), (3168, 0.036
13436644294308), (2349, 0.03610645406033575), (4368, 0.03609669496045223), (3424, 0.
03604919313010584), (3450, 0.03602193097919057), (1897, 0.035985601534210425), (101,
0.0359840483687721), (269, 0.03592596428678005), (4555, 0.03592531634931903), (2466,
0.03592380627189738), (2592, 0.03585250384637262), (674, 0.03582540316592312), (157
9, 0.03581020933631408), (2818, 0.03580841573281995), (2342, 0.035793025830037566),
(3347, 0.03577129882601757), (4526, 0.03573541431954977), (1373, 0.0357336130714958
1), (666, 0.03570198060032459), (825, 0.035696287623329465), (4028, 0.03566254200133
823), (620, 0.035661449502610534), (139, 0.03565072498219697), (1804, 0.035645991971
66588), (2291, 0.03559235476057287), (2708, 0.035550217859991934), (981, 0.035531017
419820225), (1643, 0.035529457558863255), (3321, 0.0355284881177735), (1397, 0.03552
015136100155), (4477, 0.035509387237202705), (327, 0.035499571393971446), (2431, 0.0
3540154952130582), (4010, 0.03538153532993036), (606, 0.03537493558252818), (2496,
0.03527888894762726), (1239, 0.03526731054775968), (781, 0.035262539448534726), (33
1, 0.03519813630133888), (1215, 0.0351963359206349), (4672, 0.03517754294343678), (1
59, 0.03514905655970912), (2929, 0.035141199092156336), (2545, 0.03513385082299476),
(1086, 0.03512466679352937), (3700, 0.03509912768150969), (476, 0.03509825104339151
7), (980, 0.034991247559081506), (3829, 0.03494233684952769), (184, 0.03493917823417
467), (2676, 0.03491978602597076), (4643, 0.03490846656476736), (1793, 0.03488282343
738186), (597, 0.03485199393305005), (3332, 0.03483725220844276), (1488, 0.034759677
654723334), (3739, 0.034748000857707885), (2061, 0.03474280688013156), (1214, 0.0347
3740532650401), (3427, 0.03472659932407464), (4110, 0.03472495967479615), (267, 0.03
471266703488136), (406, 0.034700585101405584), (2100, 0.03460214098366787), (2577,
0.03457971413009998), (1217, 0.03455156529179429), (1578, 0.034549399484208515), (20
01, 0.034544979886007694), (2580, 0.034530776155047994), (3866, 0.03444439022654400
4), (1058, 0.03442714619802041), (3477, 0.03440929074610513), (806, 0.03439004428669
593), (4739, 0.03435353572385781), (4585, 0.03435018649123674), (3546, 0.03433030112
8659875), (612, 0.034316930516211534), (2562, 0.03431029867512571), (896, 0.03430038
8296503094), (1552, 0.034257172964755264), (3628, 0.03414585325726132), (1497, 0.034
141018029078124), (1992, 0.03413850644041443), (4088, 0.03406649406799629), (2667,
0.03406328359556072), (3841, 0.034062757747623), (2343, 0.03404904437551764), (171,
0.034003432577698174), (2338, 0.033999037925691006), (3756, 0.0339415249175436), (34
46, 0.033915528108467576), (3648, 0.03385031833272621), (1939, 0.03384359049313771),
(1029, 0.03382335917897186), (3334, 0.033802904228006136), (324, 0.0337825984436121
7), (588, 0.0337808230436831), (198, 0.033759003475800534), (3338, 0.033694102904873
416), (1418, 0.03368349867931224), (1245, 0.0336810882884475), (763, 0.0336258835846
1441), (3711, 0.03355446024770533), (1286, 0.0335424833160683), (3652, 0.03354018114

662275), (1177, 0.0335356211125964), (4246, 0.03351513059708041), (1775, 0.033504895
39535297), (329, 0.03350335414928485), (1364, 0.03349634911121692), (1052, 0.0334955
3822594242), (116, 0.03348693881888547), (3603, 0.03347660190901597), (4742, 0.03347
3981390933785), (4117, 0.0334481646961512), (2645, 0.033435541045359174), (2749, 0.0
333771010986156 64), (152, 0.03331446270419187), (2415, 0.033300074563604315), (3890,
0.03324173900573309), (3544, 0.03313350230211917), (3108, 0.03311047873174014), (309
7, 0.03304721251096088), (1038, 0.03304611511529365), (4749, 0.03303006212149451),
(2052, 0.03302849529816566), (3451, 0.033026609327336774), (1680, 0.0330164716436360
04), (4608, 0.032965453382435755), (505, 0.03293303935987978), (644, 0.0328849579527
0681), (3991, 0.03288481417787307), (1651, 0.032880834361713684), (661, 0.0328431484
3410831), (3770, 0.032831081309707265), (403, 0.032827053531317485), (1172, 0.032801
566418994364), (3596, 0.03280142148120986), (4583, 0.03277353470453816), (339, 0.032
74058500809053), (1334, 0.032720866061708175), (1220, 0.03272010843532369), (3390,
0.03268144436260144), (721, 0.032654895212822485), (646, 0.03263267753825964), (376
4, 0.03263074142271305), (3975, 0.032607713077536946), (1040, 0.03259510124285758),
(947, 0.03256091563095445), (1495, 0.032556584487681256), (3112, 0.0325441671348070
6), (670, 0.03249259788439775), (2160, 0.032491344751043524), (1504, 0.0324862421789
1103), (1031, 0.03247038570649348), (2069, 0.03242684366658055), (2812, 0.0324070143
91467245), (1991, 0.032358846958819215), (549, 0.03234776276810046), (1787, 0.032320
31064166291), (746, 0.03230474115794068), (1196, 0.03230224759674958), (3009, 0.0322
83193625339386), (419, 0.03227370424411756), (98, 0.032193726743405454), (788, 0.032
19195497258179), (121, 0.032179098673227186), (4139, 0.032159864806466675), (248, 0.
03215955008792736), (1816, 0.032128349965699056), (2179, 0.03212235640900879), (183
8, 0.03211193330356925), (2823, 0.03205158672321503), (1026, 0.03202773810680668),
(4428, 0.03198083239514259), (2771, 0.031944743116527737), (821, 0.0319281489703017
2), (2633, 0.031882780925649314), (3966, 0.03183722705641594), (3735, 0.031815371574
90147), (4286, 0.03180965052205108), (4791, 0.0317728220541161), (194, 0.03175671421
104142), (992, 0.03175309929532447), (574, 0.03175091979690506), (1893, 0.0317294199
13026816), (1661, 0.03172097115172967), (2303, 0.0317129075161984), (2322, 0.0316964
6736621697), (2351, 0.031677310143627804), (571, 0.03166578323334291), (1049, 0.0316
48750799709795), (3782, 0.03158749143043442), (2884, 0.031566493567821485), (272, 0.
03156561287301673), (4392, 0.03155180196283023), (3120, 0.03154972306932067), (1927,
0.031504320106007754), (691, 0.03149943434722697), (1925, 0.031469921240613145), (33
15, 0.03145725933153994), (846, 0.031431396655688656), (842, 0.03141126738894341),
(3479, 0.031382489691811226), (1184, 0.03134798916087379), (4778, 0.0313350244837985
1), (528, 0.03132786516195961), (2187, 0.031307931554417225), (1234, 0.0313019214148
5656), (998, 0.031294961134766645), (810, 0.031267958136639976), (1420, 0.0312361941
0259606), (2556, 0.031175182833051644), (542, 0.03116779967343339), (525, 0.03116469
7303354742), (809, 0.031082492523462457), (448, 0.031068559001450127), (3162, 0.0310
5149143564838), (556, 0.031050097534936583), (3599, 0.03103017019524875), (1415, 0.0
31008445272316047), (1589, 0.03100537870922127), (2506, 0.030987042258130218), (475
9, 0.03097738725908653), (96, 0.03097291388389124), (4295, 0.030943302946593594),
(8, 0.03093493520052642), (2585, 0.030932869601498297), (1453, 0.03092281047724857
6), (2814, 0.03091721252913968), (2042, 0.03087894974545649), (4172, 0.0308438486708
95114), (633, 0.030822554147819863), (2835, 0.030814649305162693), (1726, 0.03080907
028115161), (499, 0.030802219590188137), (137, 0.0307773744877348), (4757, 0.0307340
2916622718), (4356, 0.030717641857640622), (2261, 0.030713594386319557), (3679, 0.03
0664661577073798), (970, 0.030630405719016898), (856, 0.03062784101869689), (1227,
0.030595747837277358), (4401, 0.030593707314490928), (4418, 0.03058324536558106), (1
175, 0.030578352149970636), (1089, 0.03057820211648519), (3030, 0.03050359083239189
2), (3465, 0.030494070493586975), (434, 0.030463968157348382), (2242, 0.030389841089
20955), (1944, 0.03038093840239029), (3899, 0.030369122796613413), (3125, 0.03036698
4987502108), (4498, 0.030361530753229626), (2741, 0.03033300660698405), (1682, 0.030
32943155356521), (2054, 0.030308363102700166), (3928, 0.030302148725765823), (1443,
0.030218441046223374), (2333, 0.03021084453069152), (741, 0.030197133885458017), (5
2, 0.03018245470107544), (1592, 0.03015243135727702), (2204, 0.03014929092685537),

(315, 0.030148920668698065), (3743, 0.030130962666952157), (4008, 0.0300632984799643
48), (1337, 0.030063073922606327), (1232, 0.03003882024706265), (3858, 0.03003757825
7637323), (276, 0.030035534861912905), (1362, 0.030029774381100886), (1162, 0.030011
3239739253), (3086, 0.029973002426384506), (1857, 0.02994956194695472), (2532, 0.029
909850198565756), (4506, 0.029887571721278913), (4012, 0.029887103619596282), (2247,
0.029882527026796557), (3619, 0.029870418592994934), (2864, 0.029869445351926205),
(3106, 0.02983696085018368), (3142, 0.029832128183312788), (211, 0.02982390826371068
5), (1394, 0.02980985823371892), (1635, 0.02974846766584079), (243, 0.02970175929625
292), (841, 0.029695076657601), (19, 0.02968626711977731), (1331, 0.0296679467969726
97), (386, 0.02965908189547961), (292, 0.029637035409404967), (40, 0.029604400185067
402), (1181, 0.02957607607291457), (973, 0.029574286842999085), (3202, 0.02953841408
185413), (2790, 0.029524464593025945), (4254, 0.029523893148616673), (189, 0.0294837
5476268129), (2763, 0.029478615118976897), (2442, 0.029462613894081127), (4209, 0.02
946035471536927), (1779, 0.029459817144237448), (4022, 0.02945151791400584), (613,
0.029439799478706983), (3908, 0.02940034060581049), (918, 0.02937013579894319), (108
1, 0.029366575006229675), (4554, 0.02936435854975045), (3293, 0.029362170119061767),
(1879, 0.02935651787512665), (388, 0.02934921712825725), (1411, 0.02934714174769422
6), (2742, 0.029318959907519586), (4239, 0.029309778727981246), (2587, 0.02925657372
431103), (197, 0.02924749073264552), (2899, 0.02924126018824487), (254, 0.0292393766
2577296), (4105, 0.029236208145831964), (4164, 0.029219991082797412), (115, 0.029205
437008730902), (1112, 0.02917610781736512), (3203, 0.02916587845601703), (295, 0.029
150304221924946), (3717, 0.02914875131896051), (3244, 0.029144017553061), (4514, 0.0
29122337929354876), (79, 0.029113928626152034), (90, 0.02910820842351401), (1131, 0.
02909138439562463), (2539, 0.02908320534287104), (2999, 0.029063516731651903), (425
2, 0.029057052508361145), (4273, 0.029039967615580354), (1943, 0.02903909398730330
4), (3434, 0.02901340180892563), (3847, 0.028991423943799232), (2446, 0.028949760447
109608), (1207, 0.028933812704153447), (3400, 0.02893376136077479), (3757, 0.0289246
8693879513), (929, 0.028892795434023986), (2961, 0.028844440036226753), (555, 0.0288
35549065099268), (353, 0.028830612918124995), (4386, 0.028816190558475167), (1670,
0.02879239276386738), (2427, 0.028784922682624833), (1412, 0.028779662134377625), (9
41, 0.028759980051565475), (3284, 0.02874741487048716), (3111, 0.02871314588864139
2), (1490, 0.02869088694045427), (2581, 0.028689781019858), (3662, 0.028683516994180
723), (135, 0.02868261916813157), (2020, 0.028654550524500857), (959, 0.028598183747
496796), (2085, 0.028594412014338495), (1256, 0.02857893319033331), (503, 0.02855833
5212354184), (3602, 0.028534327636764778), (3939, 0.0285325923124759), (1015, 0.0285
22202797192253), (4496, 0.02850937208468575), (2829, 0.028501651318413403), (175, 0.
02848895697576779), (799, 0.028485737447099754), (2202, 0.028471410994612575), (244
8, 0.028435360339650977), (3499, 0.02842634049699244), (2769, 0.02841372641299809),
(1644, 0.0284051403211077), (64, 0.028404894035231252), (761, 0.028394240042822164),
(1466, 0.028320021861633106), (2389, 0.02831126068370613), (3906, 0.0283009474357253
55), (3778, 0.028259274724955118), (2756, 0.02824916218904628), (2573, 0.02821981714
658567), (2311, 0.028213272868602368), (2364, 0.028206019882467508), (2260, 0.028198
22542529289), (1546, 0.028197686626645405), (4599, 0.028194480916451325), (3229, 0.0
28189712953565225), (1146, 0.02818822002210506), (1575, 0.02814387701249745), (1078,
0.028131182878539572), (4100, 0.02811374067687117), (2815, 0.02811290499833126), (35
4, 0.028092556012041955), (2280, 0.028090418960616013), (3726, 0.02808293149563382
4), (672, 0.028080516396443968), (2233, 0.028057445099759312), (999, 0.0280464357738
60068), (2210, 0.028023565952007147), (4522, 0.027997420183782068), (1668, 0.0279719
5137612423), (3377, 0.02796278607354603), (81, 0.02794528136793173), (1127, 0.027921
419084988117), (1202, 0.027910207620286995), (2435, 0.027910142510684136), (3868, 0.
027909785585801657), (3945, 0.027900819315858597), (2360, 0.02788330183434064), (394
6, 0.027799019657750428), (2659, 0.027795956225076), (1954, 0.027776657744013528),
(3072, 0.027773070148091546), (3136, 0.02776827173916815), (2704, 0.0277599535134151
86), (4472, 0.027753517605867153), (3464, 0.027750303051607812), (1312, 0.0277335752
34789337), (1109, 0.027724036021638455), (736, 0.027711499197496867), (1375, 0.02766
969073233455), (183, 0.027644775352911852), (3385, 0.0276423993198935), (2425, 0.027

641447857352862), (4519, 0.027641060884158773), (710, 0.027599939819980525), (2820,
0.02757955141152403), (369, 0.027554991866846026), (4704, 0.02755107141247336), (147
3, 0.02752364297483371), (822, 0.02751652787278477), (3898, 0.027514066275856085),
(3678, 0.027500012370676348), (3408, 0.027475376427606172), (4408, 0.027472956222500
587), (316, 0.027439128948452064), (114, 0.02742976146472843), (554, 0.0273364654150
42144), (2048, 0.02733570859398382), (3050, 0.027322690830130215), (131, 0.027274655
4799872), (1250, 0.02727072394112056), (2915, 0.027264879644821115), (3236, 0.027226
49248697842), (3247, 0.02721969734120153), (2615, 0.02721812566439879), (4378, 0.027
21743863977713), (3659, 0.027135351125975853), (1747, 0.027125770893092132), (3294,
0.027082458320694926), (1339, 0.027037333574238043), (4625, 0.027035064284016603),
(714, 0.02702036550624075), (3001, 0.026988659324938612), (1180, 0.0269873243847950
1), (560, 0.026948105474084955), (516, 0.026948099229211347), (3874, 0.0269271034845
4094), (3361, 0.026916611287387024), (2567, 0.026906876494602393), (1967, 0.02689508
947853423), (2736, 0.026887005543794428), (1522, 0.026846566547207188), (3988, 0.026
833284021455805), (4051, 0.026832076226780934), (1734, 0.026818297763628863), (3836,
0.026785282913353692), (923, 0.026737792232875084), (2687, 0.02673082814336546), (88
1, 0.026720355420258073), (4135, 0.026715223763690975), (2931, 0.02667906753486043
4), (2927, 0.026660957131927718), (769, 0.026654856838325223), (858, 0.0266000542739
91522), (4732, 0.02658047073158657), (4167, 0.02657907615836804), (2307, 0.026549044
29489527), (148, 0.02654830260044715), (2755, 0.026544621098285273), (1765, 0.026470
240858585455), (296, 0.026459567913143592), (3643, 0.02643988544473405), (695, 0.026
427326406438738), (4421, 0.026401327295201806), (673, 0.026391236015151927), (2222,
0.026385699481707194), (3088, 0.02636032002322339), (743, 0.026294183527874625), (45
07, 0.02629046389058557), (678, 0.02628436556703755), (2216, 0.026278164821076228),
(1230, 0.02624860488649873), (2007, 0.02623971423777043), (1981, 0.02622956610940177
5), (1467, 0.02621788695346807), (2019, 0.02620975823102622), (1046, 0.0261957688606
64152), (2584, 0.026189207628373564), (449, 0.026179367801091525), (3834, 0.02616116
8557652853), (99, 0.026147609016141292), (279, 0.026139750727758506), (4038, 0.02613
85622310165), (3043, 0.026138123902603193), (2275, 0.026113741510890576), (1187, 0.0
26063767446498107), (280, 0.026059654002839903), (4638, 0.02605735428282043), (852,
0.0260437875929925), (3470, 0.026042314663075636), (3845, 0.025976878807820743), (24
2, 0.02596280025216613), (408, 0.025955338303779565), (3515, 0.025942054980027687),
(3080, 0.02593252628831628), (1687, 0.025927763570016374), (2536, 0.0259099108409198
87), (862, 0.025889571270560203), (2078, 0.025880706040512208), (111, 0.025868460781
820198), (2452, 0.025848180826074527), (2094, 0.02583515634206981), (2759, 0.0257933
1492350329), (35, 0.025792416651174273), (1492, 0.02578392611522216), (1689, 0.025776
40269568751), (1796, 0.025755866711754498), (1153, 0.025722455536786062), (89, 0.025
715410181997155), (2456, 0.02567848145200457), (411, 0.025665699806897826), (95, 0.0
25648894216477 99), (1731, 0.0256420372576307), (1408, 0.025636739337390285), (3724,
0.0256217270399132), (4430, 0.025620425298163683), (1377, 0.025618729675440363), (19
52, 0.025603899893977516), (4291, 0.025602300444489825), (2105, 0.02559898181703091
5), (2403, 0.025548645238506567), (4151, 0.025495013894241345), (1982, 0.02547234732
5829457), (2597, 0.02547058832478942), (3796, 0.025465562245690885), (160, 0.0254561
09332957284), (452, 0.025450151876301928), (1757, 0.02541814734408382), (1595, 0.025
41498779945982), (4629, 0.025384493463633688), (4319, 0.025380193282361255), (627,
0.025374921909953645), (2014, 0.025349207709217652), (1324, 0.02534185104536406), (7
40, 0.025316251525823445), (3865, 0.02531517904992564), (0, 0.02531512269737111), (1
441, 0.025309749409157523), (1921, 0.025281830978181936), (4584, 0.02526883447970260
4), (1889, 0.02524835968971871), (3826, 0.02524187742166132), (20, 0.025210054010122
055), (4120, 0.025206219807988355), (1826, 0.025205395499702174), (1606, 0.025204389
615138215), (2417, 0.025198045508376697), (3859, 0.025167591128221548), (1878, 0.025
147809940367815), (58, 0.025146268236415702), (635, 0.02511799848844787), (1541, 0.0
25113263910305602), (719, 0.025098893938504195), (1091, 0.025051791391380876), (167
5, 0.025041519709610438), (757, 0.02504044763241721), (2002, 0.025039274493230384),
(993, 0.025012133285939552), (1856, 0.02499576497226054), (2086, 0.02498071985849169
3), (934, 0.024963375177119905), (2212, 0.024949799481613175), (4, 0.024929726723526

918), (220, 0.024925248865578585), (2869, 0.024899792820574917), (4284, 0.0248977177
47778072), (23, 0.024870438166594427), (967, 0.02486759973259156), (53, 0.0248527604
67374575), (318, 0.02484370068577551), (3972, 0.02483274145449593), (1074, 0.0248316
80205808962), (3889, 0.024820831202536156), (1969, 0.02481791143280848), (4372, 0.02
479895124378561), (1272, 0.024788436669427188), (647, 0.024758356045488268), (2324,
0.024745694280172124), (536, 0.024709079565036842), (4645, 0.0246919909655625), (187
0, 0.024679406821881085), (1108, 0.02467378377900406), (3382, 0.02466671237286574),
(2148, 0.024618857390416513), (1564, 0.02461725209610294), (250, 0.02461441100501990
4), (3337, 0.024604373831795614), (4774, 0.02459422889255241), (617, 0.0245808348068
46426), (2169, 0.02457124722412299), (2378, 0.024550153498530215), (3510, 0.02453647
5046640752), (3667, 0.024524349708350518), (2163, 0.024502747033782368), (50, 0.0244
73647577403737), (3507, 0.02445344878453156), (1435, 0.0244195061602232), (3672, 0.0
24403838937443217), (2510, 0.02440239443065885), (4618, 0.024359280031183518), (308
3, 0.024349835998103662), (744, 0.024328443396996222), (1025, 0.024326126301632486),
(664, 0.024311299603893725), (433, 0.024284594182904715), (537, 0.0242776819347061
5), (238, 0.02426903109788739), (1030, 0.024246886096932208), (1679, 0.0242445472580
111), (4037, 0.024222956839934705), (208, 0.02419897314958249), (3747, 0.02419246848
1663364), (4663, 0.024163588968421926), (1632, 0.02415532971118427), (1630, 0.024152
976998750332), (3078, 0.024143378199026195), (3251, 0.024094728581014403), (1910, 0.
024075304155847868), (3887, 0.024040298510155534), (2964, 0.02399680854553707), (176
8, 0.023990905648439757), (534, 0.023985316166857368), (1637, 0.023979367336911212),
(3965, 0.023966007445392082), (1320, 0.02394145804020361), (4415, 0.0239183568038424
42), (2178, 0.023917814759109417), (3745, 0.023895805485874087), (2582, 0.0238859260
14894625), (3575, 0.0238854996279511), (2225, 0.023881009027613864), (3224, 0.023863
329128368506), (1075, 0.02386223268207443), (933, 0.023829904876997883), (4047, 0.02
382274621901415), (370, 0.02378016286738358), (2092, 0.023759327990221575), (4106,
0.023752152836940085), (1005, 0.023749380413372243), (1237, 0.023748477729621457),
(1257, 0.023741133133852284), (3020, 0.02373838630851025), (2057, 0.0237080034009229
7), (43, 0.023689369843914104), (2522, 0.02367965149206115), (3291, 0.02367872810724
211), (1619, 0.023674792256187502), (2887, 0.02366854162109285), (2485, 0.0236105809
7952176), (603, 0.023608559652421243), (4325, 0.02360830506485523), (4552, 0.0236020
27507068947), (2614, 0.023595331642463503), (702, 0.023585376006270774), (1132, 0.02
3580888972459302), (4259, 0.023576193859579795), (2783, 0.023570970576071857), (182,
0.02353656978819377), (1663, 0.02353232186537981), (1850, 0.023527216513489886), (15
56, 0.023473703031252284), (1894, 0.02345182848817197), (4261, 0.02337377526735665
7), (126, 0.023353967193491126), (2757, 0.023338152973196875), (294, 0.0233117917522
51575), (4251, 0.023296212848429303), (1244, 0.023291964886609697), (1000, 0.0232648
7032397238), (1714, 0.023259362870364012), (624, 0.023256724026578128), (790, 0.0232
55012007745723), (108, 0.023241981578083354), (212, 0.02324123457380759), (2341, 0.0
2323731807251312), (803, 0.02322558415268294), (3371, 0.02322481068427533), (1147,
0.023191479743118984), (495, 0.023187014523033456), (3754, 0.023185673568415097), (1
669, 0.023147894763598367), (4631, 0.0231445258831369), (3952, 0.02312819732015248
7), (387, 0.023127653059049586), (3940, 0.023120274527044574), (4701, 0.023113170960
263957), (3710, 0.02306355003939454), (2237, 0.02304919835869488), (2583, 0.02302420
727837438), (2367, 0.02302019666880139), (2695, 0.023017457919675435), (275, 0.02301
404884039334), (205, 0.02298215684826604), (4019, 0.022977782980952967), (2399, 0.02
2973571750498186), (1380, 0.02296003928356085), (1996, 0.02294561494723391), (1612,
0.022943313874495386), (31, 0.02290308453698018), (676, 0.022896476422471015), (117
6, 0.022893457942976626), (1419, 0.022886057045970763), (3417, 0.02287145629901430
5), (4438, 0.022870792953686863), (1527, 0.022850734653789555), (871, 0.022845843248
684217), (3632, 0.02284431799979458), (334, 0.022833976332210133), (3953, 0.02282993
2411491248), (2550, 0.022820806370494315), (1118, 0.0228065418199093), (2779, 0.0228
05851491782297), (1519, 0.022800943234727596), (828, 0.022784980087159606), (2197,
0.02274164651064088), (4062, 0.022740775405213007), (927, 0.02274077420912185), (209
0, 0.02273945608357307), (287, 0.02273805394855659), (1800, 0.022727310755579308),
(728, 0.022722227379584077), (1354, 0.022712737635219946), (3668, 0.0226975341435707

02), (1899, 0.022681210769162134), (3489, 0.02267651371268215), (3354, 0.02266755339
048144), (1524, 0.022652613152636603), (2170, 0.022639537781951522), (4493, 0.022635
97819989708), (3848, 0.022635179749546985), (1254, 0.02263332376324078), (3490, 0.02
2631742989093227), (1468, 0.02263157217854836), (3695, 0.022626125255064973), (3771,
0.02262085936625541), (1279, 0.022620557935525098), (3116, 0.022588030355625822), (3
158, 0.022577085131124948), (901, 0.02256589929032795), (1788, 0.02256581224400534
5), (2761, 0.022563717532086677), (1869, 0.022547764366525922), (252, 0.022527942902
764643), (3911, 0.022525889537273178), (2091, 0.022517506395989453), (468, 0.0225162
5813538535), (1073, 0.022495004724559745), (3964, 0.022487871705361817), (3703, 0.02
247095838252873), (3644, 0.02246803151051133), (3869, 0.022456094618819183), (1500,
0.022444863543466864), (3699, 0.022433763024109387), (2296, 0.022423453660892587),
(93, 0.02241215665416583), (2372, 0.022397070739033156), (1707, 0.02239175219501396
8), (3151, 0.022390879571115428), (400, 0.022379891670723233), (1548, 0.022374201721
8123), (456, 0.02234564140242137), (3604, 0.02233947474424601), (917, 0.022329315902
897218), (2407, 0.022324214699764172), (1853, 0.022310864730738632), (4602, 0.022307
021773752184), (873, 0.022294294272584284), (903, 0.02228764760119481), (634, 0.0222
8742829296616), (565, 0.02228676521747925), (3329, 0.022281683124023476), (251, 0.02
227356932657924), (1301, 0.022244056799122433), (421, 0.022237855115106564), (1946,
0.022231660339560783), (2336, 0.02219597296322629), (1770, 0.022181256487692737), (1
305, 0.022180881228978204), (767, 0.022176230603847574), (1194, 0.0221726686406088
7), (2622, 0.022165024393882717), (423, 0.022164506254997107), (3813, 0.022149358102
777045), (2473, 0.022144469585115324), (2726, 0.02213971063872107), (204, 0.02213050
945729524), (982, 0.022129547246681848), (4594, 0.02210554483241709), (2046, 0.02209
4312947349776), (2579, 0.02208403983258324), (407, 0.022082253077502827), (677, 0.02
2081994233331022), (37, 0.022073100012780275), (350, 0.022061634090390726), (75, 0.0
22057119691207824), (3481, 0.022042587180242275), (1122, 0.02202831581670876), (427
9, 0.022024514850632875), (2063, 0.02200818844793445), (1970, 0.021998124176773493),
(955, 0.02198916642979698), (4735, 0.02197488302956456), (1936, 0.0219538289606603
6), (2640, 0.02193353948740738), (500, 0.021931087102158506), (2926, 0.0219225424096
62633), (2747, 0.021913261649682676), (2304, 0.021905402272654223), (2689, 0.0218706
11615505447), (3238, 0.02186863197915005), (4144, 0.02185209805770838), (3406, 0.021
846087600919415), (3501, 0.02183859071095311), (3114, 0.02182121672994405), (2625,
0.021818440733496373), (4391, 0.021806824652795005), (3978, 0.021804884294299585),
(4376, 0.021801046159450426), (2604, 0.02179497168763226), (2681, 0.0217802751133135
3), (3318, 0.021779899180964468), (3704, 0.021777945620316614), (109, 0.021776146296
002157), (2142, 0.02177599767565398), (2215, 0.02176934598333656), (1737, 0.02175420
736919535), (2894, 0.021748636644583534), (3260, 0.02173814978824745), (1470, 0.0216
9221680887711), (2068, 0.02167841015806747), (3312, 0.021677066955678212), (2833, 0.
021656480065505173), (833, 0.021656424197982804), (181, 0.021652893869808187), (184
2, 0.02161818377017973), (3189, 0.02161038523360965), (3352, 0.02160092649701403),
(3669, 0.021598348634490647), (2356, 0.02158623381877749), (618, 0.02157815435944520
5), (1095, 0.02157535030315407), (4326, 0.02157391943089125), (71, 0.021555882140030
593), (2675, 0.0215419178433829), (2136, 0.02153008742220820 2), (1596, 0.02152997496
0177826), (2203, 0.02152236314601559), (4084, 0.021512587003887043), (1543, 0.021511
62704012102), (733, 0.021511382902596524), (3285, 0.021501094123177543), (1780, 0.02
1495432172595943), (527, 0.02149067133338399), (2305, 0.02148815680098999), (2630,
0.02148728643075887), (80, 0.02148467123933299), (1271, 0.0214807964615722), (4094,
0.021480582162419812), (2657, 0.02146595747035468), (1665, 0.021434439428366), (261
3, 0.021428768775321853), (2952, 0.021417697113726212), (3389, 0.02138433327868704
4), (1507, 0.021367076059543886), (1533, 0.021349851527078183), (2022, 0.02133537645
385125), (572, 0.021304857336037212), (3878, 0.021300307628795336), (3591, 0.0212947
5754254638), (3734, 0.021286263111711556), (4098, 0.021286105142155894), (3531, 0.02
1274974632137365), (1270, 0.021272374341004117), (3138, 0.021266897016741428), (383
1, 0.02125017551073208), (1907, 0.021249325849423597), (1721, 0.021242657560316115),
(2978, 0.02124140768447708), (4077, 0.021238396891828812), (1282, 0.0212361194502915
66), (2111, 0.02123574489107766), (1618, 0.02122657151451314), (357, 0.0212095600483

22404), (105, 0.02119994427869696), (2959, 0.021189971123077016), (586, 0.0211455611
8401161), (1572, 0.021142460933170244), (2746, 0.021128667759444206), (2909, 0.02109
899570669337), (245, 0.021091411367858307), (2876, 0.021078169585329223), (2571, 0.0
21070133100436563), (2598, 0.02104956192007803), (3828, 0.02100589059131794), (1678,
0.0210044775606081), (4610, 0.02099808219179862), (2018, 0.020982624753504646), (419
1, 0.02098122572166964), (2192, 0.020966142140828384), (1053, 0.02096379469007401),
(1745, 0.020957586466537226), (1384, 0.020954077350332955), (708, 0.0209469126084152
8), (591, 0.020933909686590944), (4182, 0.020933494176032633), (4017, 0.020932291299
746476), (930, 0.02092726424669351), (2330, 0.020915467859018835), (2788, 0.02089255
347403614), (3209, 0.020878623482585897), (3505, 0.020877538885462288), (154, 0.0208
61732883614623), (2371, 0.020858622646527342), (1370, 0.02085133594949017), (4156,
0.020822643470693958), (4529, 0.020808032182087318), (3733, 0.020802578265744335),
(943, 0.020796079676191383), (1228, 0.020779494190111946), (3897, 0.0207571127482616
8), (4099, 0.02074048248269452), (1848, 0.020732962869425876), (2891, 0.020710165036
60198), (4461, 0.020700562426783264), (884, 0.02069524992863983), (259, 0.0206940829
01925582), (3562, 0.020688125901358635), (3688, 0.020683466548061143), (483, 0.02067
8981721570948), (854, 0.020651091114809082), (2914, 0.02064456738632578), (2879, 0.0
20625305508139076), (533, 0.020607458605063626), (332, 0.020559468783685246), (1103,
0.020550784118254034), (2439, 0.02052714041748429), (2317, 0.02050854134295655), (14
25, 0.020498630997590347), (4569, 0.020495527614960714), (2147, 0.02048563411582767
6), (4380, 0.020482528702467175), (4054, 0.020476113809639852), (270, 0.020470726082
86063), (2902, 0.02046638767894536), (1094, 0.02046026827116788), (3815, 0.020450125
68681413), (380, 0.020445412922726842), (3646, 0.020441997901347927), (374, 0.020433
371520806647), (2375, 0.020430749249052808), (3755, 0.0204259675670689), (2939, 0.02
039978138276963), (1626, 0.020397431679884087), (185, 0.02038304237119583), (2176,
0.020377266779787042), (898, 0.020345862019732805), (3035, 0.020342589588662453), (1
819, 0.020337825225187404), (234, 0.020332927975252314), (931, 0.02032983426503213),
(3934, 0.02032392460108762), (413, 0.020320877783699225), (3308, 0.02031540428006787
7), (3394, 0.02031495832959946), (1022, 0.020314123936655725), (479, 0.0203058754143
62677), (4796, 0.020273113996351778), (2302, 0.02026149813892408), (899, 0.020258539
852295893), (1428, 0.020219277065378273), (1426, 0.02020687227786065), (1538, 0.0202
0610747068459), (2384, 0.020204105912477472), (501, 0.020194433221440527), (3597, 0.
020193550672841114), (3349, 0.020172739288629046), (4482, 0.020168082889197712), (72
5, 0.02015703632859361), (2621, 0.02015357031000619), (1523, 0.02013780616342415),
(1608, 0.020125990512872644), (2848, 0.020125541058772435), (2392, 0.020111496579386
233), (563, 0.020101194795755693), (4181, 0.020100199966455613), (615, 0.02009753383
9594256), (221, 0.020074947827866736), (3183, 0.020047139404423597), (1712, 0.020042
63383622975), (2922, 0.020041619935676415), (4078, 0.020039476478961014), (177, 0.02
0029123423841614), (3503, 0.020021537199765658), (83, 0.019994216200751452), (1366,
0.019993313149369932), (3903, 0.019991996083841806), (381, 0.019973245359529393), (4
42, 0.019972564229175732), (706, 0.0199623864244022), (3220, 0.019953763182146554),
(3186, 0.01992840065415868), (4467, 0.019924712532179627), (4727, 0.0199029347494418
67), (1849, 0.019902901381266977), (1806, 0.019898305087949535), (4773, 0.0198968544
48345436), (2554, 0.019896754083615294), (326, 0.01988992886308015), (3706, 0.019885
823080128526), (2627, 0.019870788652815257), (1322, 0.01986408056030938), (2195, 0.0
1985260475034096), (3456, 0.019849518552316366), (4067, 0.019845494805572288), (136
1, 0.019833133965028015), (4579, 0.01982926007038003), (3954, 0.01979968882488026),
(3651, 0.019769349550181767), (1834, 0.01976863725904065), (2228, 0.0197456543556385
76), (2355, 0.019737413682336506), (1531, 0.01973666357808295), (936, 0.019735801936
74106), (1427, 0.019692892642776887), (760, 0.019689159848395556), (1913, 0.01968621
374115141), (2491, 0.019680944200211176), (200, 0.019672291888541544), (3430, 0.0196
5708350871151), (3073, 0.019653084213322784), (2524, 0.01963042736661027), (905, 0.0
1961555455800855), (2538, 0.019608409148003356), (2414, 0.019601357245267757), (218
2, 0.019590771070673543), (2445, 0.019579106676952403), (642, 0.019577654785027484),
(4297, 0.01957647368273257), (1861, 0.01957254411962333), (4015, 0.01956884575377500
2), (3145, 0.019566763735771934), (2251, 0.019566043091467852), (431, 0.019559063161

72861), (4324, 0.019543054301100492), (2944, 0.0195305334945897), (1931, 0.019528054
525154097), (1422, 0.019508606032306208), (416, 0.01950646882338211), (3192, 0.01949
1481444918187), (463, 0.019485779814410777), (4145, 0.019476251294938643), (2199, 0.
019457325176647743), (1238, 0.019449288579735782), (3177, 0.019419667468474983), (12
83, 0.01941897961500091), (3981, 0.019414843011404684), (2809, 0.01939930786334127),
(3691, 0.019393907601754353), (4131, 0.01938850094274776), (1664, 0.0193807785259902
56), (4471, 0.019378120241405016), (1484, 0.019377355227446973), (2860, 0.0193754244
21008827), (1818, 0.019362801191886492), (2346, 0.019349639936663277), (3744, 0.0193
42345853627447), (4030, 0.019340416223934877), (3611, 0.01933894479238951), (1947,
0.019338506393074367), (441, 0.01933815623254388), (2575, 0.019330292136337187), (13
14, 0.01931834178864059), (1812, 0.019278369520200022), (4427, 0.01927266798882914),
(2859, 0.01925992784410293), (1912, 0.019251555136440564), (3252, 0.0192354613514581
08), (336, 0.0192215086758088), (1529, 0.01922126836497642), (2480, 0.01921415168985
0353), (3237, 0.019203830253771155), (4375, 0.019193832804282808), (1789, 0.01918797
3527433757), (420, 0.01917965793095018), (2568, 0.01916401166234881), (2214, 0.01915
6430265523303), (631, 0.019153498371042826), (718, 0.01914861485789185), (497, 0.019
131469790203838), (1764, 0.019117169709239306), (3455, 0.019107171350900877), (2778,
0.01910496071162926), (4642, 0.01910199601430621), (2531, 0.01908054548694253), (19
1, 0.019075801251320162), (4108, 0.019068693272798268), (172, 0.019057091235472567),
(3055, 0.01904268391790459), (2177, 0.019030595019174895), (4071, 0.0190263238626861
98), (815, 0.019008907792410646), (4232, 0.0190011688590868), (1059, 0.0189991836078
56466), (3362, 0.018964735871980654), (2610, 0.01894034816392277), (308, 0.018923879
12891352), (1597, 0.01891010136816346), (3959, 0.01889228310844543), (3006, 0.018866
939579886197), (3772, 0.018837726693303164), (851, 0.01883109842733956), (3840, 0.01
882685732596795), (693, 0.018825644668711033), (3127, 0.018821130422907575), (2738,
0.018793617078935836), (683, 0.01878342966515282), (1442, 0.018746623576687924), (37
91, 0.01873420829635461), (2765, 0.018715792154415774), (3143, 0.01868994020081840
2), (4053, 0.018685919921167877), (195, 0.01867767015552487), (2098, 0.0186111722855
54166), (3609, 0.01857762198825906), (3504, 0.01857314345513272), (3094, 0.018563964
022973518), (1517, 0.018543716643440747), (3402, 0.018533365215852), (1195, 0.018532
073439352398), (794, 0.018516917064486584), (3855, 0.018501239589191025), (3600, 0.0
18487660289643285), (1134, 0.01848419862286736), (4379, 0.018484048859335078), (263
8, 0.01848058822578971), (1701, 0.018458944281127387), (1064, 0.0184573138577872),
(1055, 0.018456087030565557), (1308, 0.018447048671171393), (4360, 0.018431962768543
402), (488, 0.0184307721862186), (410, 0.01842626297999868), (2986, 0.01842497656229
629), (2168, 0.01842226613972403), (2175, 0.01842159174913272), (3769, 0.01842106388
2608835), (3660, 0.018415786600251196), (1547, 0.01840224832923179), (467, 0.0183811
52902655365), (2281, 0.018357336384289446), (2975, 0.018352783487321334), (2861, 0.0
1834613911499412), (63, 0.01833889030024455), (4175, 0.018330668281005526), (2354,
0.018327080030753545), (2507, 0.01831647856145454), (658, 0.018314827903677736), (16
31, 0.018306258356871125), (1430, 0.01830201281564051), (2184, 0.01828795312730444
4), (1717, 0.01828553955129434), (2889, 0.018268303863125872), (1472, 0.018257957400
084183), (2702, 0.018256206263264237), (2907, 0.01825563560141021), (3280, 0.0182537
32010575077), (4776, 0.018253509722063053), (2003, 0.018252711034050545), (161, 0.01
8248598639931543), (3918, 0.01824817185717945), (3737, 0.018229791574562133), (3896,
0.018216900910687044), (3480, 0.018213793264644982), (1351, 0.018211622978689647),
(750, 0.01820941311825481), (218, 0.01820775128152958), (2102, 0.01820637627225289
7), (4039, 0.01819749655557526), (4288, 0.01816339886505262), (938, 0.01813461236458
6883), (3857, 0.018117900178294826), (3042, 0.018106749290578088), (3495, 0.01810066
761766072), (1795, 0.018098314488782578), (2632, 0.018086621195293473), (3574, 0.018
07640205053272), (3552, 0.018072052315689015), (1535, 0.018070538742894514), (1839,
0.018061614132809926), (2103, 0.01803548528171557), (2488, 0.018034608553201634), (1
413, 0.018031095825940046), (1203, 0.018025072768320487), (4454, 0.0180186070202305
9), (1204, 0.018014847869633022), (2060, 0.018013744482188336), (4201, 0.01801101079
2295928), (759, 0.01799579972299672), (3141, 0.017986041094174607), (3165, 0.0179783
57336314768), (264, 0.01797732495165702), (3559, 0.017965690757809898), (696, 0.0179

64598586746865), (66, 0.01794874160703813), (1119, 0.01794367291124868), (2612, 0.01
793171710312719), (22, 0.01792276874908081), (3593, 0.017922425249429833), (4654, 0.
01792204070545419), (2534, 0.01792085913649957), (2263, 0.017908675281275585), (340,
0.017904003766269105), (2314, 0.017902590655836777), (1156, 0.017898578279374487),
(4119, 0.01789455089260772), (675, 0.01787172110914862), (3281, 0.01785877240315928
2), (3561, 0.01785828459217152), (1404, 0.017851738612026158), (2529, 0.017841390817
484884), (3095, 0.017839072328777694), (4163, 0.017833374959496953), (3469, 0.017827
4960403451), (1170, 0.01782500402095107), (4222, 0.017794374890383248), (4355, 0.017
7893788924605), (2555, 0.017774695279731263), (1706, 0.017769568833652126), (758, 0.
017760025247642412), (2938, 0.017750792658544044), (395, 0.01769894802791771), (165
9, 0.01768280470469291), (3207, 0.017678045944648724), (2377, 0.01767047550685913),
(2249, 0.017668261050699227), (1372, 0.017666804315864287), (3276, 0.017657375993137
867), (1896, 0.017648617559057802), (1299, 0.017645987253240195), (4233, 0.017641359
488570207), (4348, 0.017625826538003934), (1346, 0.017622689430631954), (4357, 0.017
618608497090134), (3511, 0.01760626410453416), (886, 0.017597153024730687), (4650,
0.017596256750848464), (2268, 0.01757890626814134), (2732, 0.017563111403640516), (2
167, 0.017558988362293984), (1225, 0.01755756885679377), (786, 0.01754006475208908
6), (4535, 0.017535581418998837), (485, 0.01752986107581509), (3592, 0.0175142273715
2352), (1300, 0.017510638080635035), (162, 0.017508564372075104), (2308, 0.017505560
722870284), (1932, 0.017504834537351913), (2049, 0.01749051128727583), (3003, 0.0174
48923729866266), (1201, 0.017434336362412362), (384, 0.01743346319543433), (1571, 0.
0174267609054211), (686, 0.01742628185812934), (4649, 0.01740957091634849), (1565,
0.017407385132509796), (3241, 0.017402116829586093), (1900, 0.01739583785738129), (7
84, 0.017392340241155695), (3160, 0.01737349341373925), (1843, 0.01737128670155446),
(107, 0.01737021995321565), (2798, 0.017356878376739125), (2395, 0.01735639481480994
2), (3586, 0.017337549133132416), (3714, 0.01731356656461907), (2563, 0.017304789987
042445), (3802, 0.01729044944538959), (2258, 0.017284678826654527), (4573, 0.0172761
1124121416), (772, 0.01727299383550612), (3528, 0.017259946321743685), (1253, 0.0172
44356378755998), (877, 0.017236054816731523), (405, 0.017228008857348275), (3780, 0.
017224564830784668), (3622, 0.017223651182254985), (1883, 0.01721801449694154), (279
9, 0.017212753272364312), (1957, 0.017206634661685045), (1791, 0.01720580893993258
6), (726, 0.01720298216635047), (3449, 0.017186032329095772), (4442, 0.0171691748063
38575), (4604, 0.01715868957488465), (656, 0.01715026457715706), (85, 0.017148934252
787445), (1407, 0.017147499642535726), (2758, 0.017146272976666894), (1722, 0.017143
48746934268), (3640, 0.01713700037066372), (2654, 0.017122195155005328), (4767, 0.01
7105970967803186), (2455, 0.017103881419806684), (1275, 0.017097624635175872), (305
6, 0.017070664226139093), (3631, 0.017053704667855607), (4320, 0.01704858534733508
6), (1807, 0.01704209471712633), (2405, 0.017037531044987925), (3462, 0.017013234854
92115), (775, 0.01700841750112397), (4225, 0.017003891296486277), (717, 0.0170003214
7721337), (2376, 0.01699964381880189), (3459, 0.016993000139441404), (2802, 0.016970
505659059738), (2700, 0.0169649361873103), (3998, 0.016964648384755317), (1987, 0.01
695800236961683), (322, 0.016956729892305192), (3861, 0.016931957473167437), (3590,
0.01692262346324305), (3383, 0.016919907662983753), (1036, 0.0169158005768121), (96
4, 0.016913919497684303), (1761, 0.016892274395189356), (2797, 0.01689008072224315
6), (2557, 0.016883930202002184), (3730, 0.016857903197132696), (2721, 0.01684175205
5711694), (1974, 0.016841460497583626), (2083, 0.016832288661381645), (1516, 0.01681
427633276236), (3286, 0.01680281277180236), (48, 0.01680183887529584), (600, 0.01678
281217245045), (3979, 0.01677076117193053), (1537, 0.016767059622503695), (4448, 0.0
16752807757106525), (1820, 0.01675024411032386), (3910, 0.016746274440802775), (236
3, 0.01674426403610908), (38, 0.016735251293804827), (1304, 0.016727586039875213),
(3635, 0.01671879606662281), (2546, 0.01671720487864706), (1327, 0.01670631873722516
8), (892, 0.016697122039663805), (3789, 0.016691119781322223), (3947, 0.016687819615
777998), (2993, 0.01667372101197228), (3008, 0.01665085230894172), (3011, 0.01664633
3073262944), (2958, 0.016643740533079855), (1937, 0.016643692779351245), (110, 0.016
63593844713563), (2949, 0.01662354445515998), (4528, 0.016612843255447495), (2339,
0.01658709142666543), (1013, 0.016580937261998246), (398, 0.01657709368003764), (317

3, 0.016577061976447954), (129, 0.01657534659515722), (4595, 0.0165637998240981), (1
102, 0.016543564002150606), (3936, 0.016525879286191848), (4545, 0.0165167634140726
5), (948, 0.01650860552222675), (1160, 0.016500915655068834), (879, 0.01648312745526
8996), (2323, 0.01647336922826021), (330, 0.016471203129028878), (4497, 0.0164697556
5419904), (1781, 0.016462017806447998), (3529, 0.016442639263234096), (2021, 0.01644
214799152987), (3856, 0.01643994732934199), (261, 0.016436043710683648), (2230, 0.01
642259007961301), (3253, 0.01641136858084838), (1433, 0.01640852485747147), (1828,
0.016407214488317088), (4238, 0.016386604363445457), (1748, 0.01637881312634858), (2
921, 0.01635456215161986), (61, 0.01635015002611376), (865, 0.016343451888915356),
(352, 0.01632531881971601), (367, 0.016323992554575013), (2816, 0.01631050942558667
8), (2628, 0.016307518156685394), (2875, 0.016307131246229324), (3765, 0.01627254257
681505), (3049, 0.016267715419819664), (3567, 0.01623587639625799), (768, 0.01623163
9482474778), (1978, 0.016225781808052273), (1498, 0.016217518358364266), (69, 0.0161
8329334393077), (2490, 0.016171779351878784), (874, 0.016149801713601915), (589, 0.0
16147977558358416), (4788, 0.01614538632223229), (4793, 0.016136374840089375), (391
2, 0.016123638025176065), (3401, 0.016119454259424935), (3051, 0.01611719938561220
4), (3441, 0.01611281974764695), (3494, 0.01610171812656732), (1915, 0.0160973198417
17375), (847, 0.01609259766631261), (883, 0.016055388176434853), (2180, 0.0160500117
94654435), (1139, 0.01604538145332697), (2991, 0.016033890831893882), (2390, 0.01602
751841314606), (17, 0.01601669810036477), (2544, 0.016013790702558105), (2940, 0.016
011846071586327), (262, 0.016008532494180345), (290, 0.01599797560237545), (2294, 0.
015996749378376703), (2008, 0.015982728574063187), (3914, 0.015977552271821523), (28
1, 0.01596662468711029), (1338, 0.01596364374262817), (1143, 0.015935447827777442),
(1182, 0.01593509883265183 5), (3636, 0.01592280940912606), (437, 0.01590777991169556
7), (3698, 0.015902367834127244), (165, 0.015898883695491906), (3900, 0.015889597006
78226), (241, 0.015876309856328475), (732, 0.015874927643149427), (668, 0.0158671330
47224584), (1152, 0.015836772148492755), (752, 0.015818182406409552), (3832, 0.01580
1123609065645), (4337, 0.015789086912455935), (1916, 0.015762545444655517), (472, 0.
015751681648498644), (972, 0.015736596850609894), (297, 0.015718856581113386), (306
3, 0.01571609532485843), (425, 0.015709173817807263), (3277, 0.015697726113890545),
(3234, 0.015695954365573153), (328, 0.015678536654377837), (466, 0.01567230832636578
4), (138, 0.015637725129214016), (1020, 0.0156319885403262), (168, 0.015630626662545
388), (3485, 0.015624499474612745), (539, 0.015620166799028124), (486, 0.01560838017
9554501), (652, 0.015607602337934932), (1585, 0.015606251271792982), (247, 0.0156046
01725429042), (1728, 0.015600503534506037), (4352, 0.01559907026918832), (1328, 0.01
559789092562981), (553, 0.015596772165942518), (1266, 0.015580760946057096), (2971,
0.015574242828561253), (3420, 0.01557177233324683), (122, 0.01556615351003 5626), (17
46, 0.015557358009521903), (187, 0.01553413453427916), (1514, 0.015527553951531551),
(592, 0.015503618061031099), (2276, 0.015494388556985081), (956, 0.0154857238503790
9), (3626, 0.015479964272307519), (4231, 0.015456706378099405), (3303, 0.01544992367
1919854), (1264, 0.015429401847627516), (2646, 0.015428457714895445), (1966, 0.01541
6505847956232), (4733, 0.015414064265363212), (3246, 0.01539182250128111 5), (3784,
0.015380935132776236), (1269, 0.015361647906030014), (2096, 0.015349580869854933),
(1033, 0.015346049259993975), (2394, 0.015343954673640987), (3520, 0.015309141745697
401), (257, 0.015269302690415787), (102, 0.015251059855876558), (1660, 0.01524437529
7957365), (4032, 0.015237863014281771), (3533, 0.015232340942432473), (3860, 0.01523
1041516259713), (4666, 0.015218842938926289), (158, 0.015178542794596593), (2985, 0.
01515283592268673), (454, 0.01514420210517302), (4486, 0.015127570162180718), (1783,
0.015125416576253315), (389, 0.015121335493388541), (2300, 0.015118484445708492), (4
150, 0.015115440094260984), (4393, 0.01511439464674318), (679, 0.01510548199846792
3), (206, 0.015104619651285126), (864, 0.015095832941339711), (2527, 0.0150747905787
38399), (2839, 0.015064075176666195), (2703, 0.015063769446971059), (838, 0.01505691
6048470145), (3795, 0.015039873261524506), (839, 0.015031364666698273), (716, 0.0150
2579758103098), (1389, 0.01500270486904054), (2413, 0.014981158100012361), (1114, 0.
014958503157539604), (2478, 0.014947312664113824), (1986, 0.014934726112894302), (36
23, 0.014930920496193798), (3335, 0.014930148564769144), (651, 0.01489109665457804

8), (4179, 0.014882003497418519), (805, 0.01487731647798404), (4726, 0.0148711710229
45306), (232, 0.014866095621163873), (457, 0.01486533530711603), (3616, 0.0148546738
40520144), (4605, 0.014849068302119297), (3273, 0.014848491780132991), (848, 0.01484
8207460403377), (2977, 0.014846762409476362), (937, 0.01482783282084929), (3440, 0.0
14825296522977274), (140, 0.014817222177109362), (345, 0.014815714442063393), (807,
0.014812082647965255), (654, 0.014802947264353612), (2727, 0.01480086453141679), (14
79, 0.01479317528968298), (1628, 0.014772866958500436), (3748, 0.01475853628191695
4), (866, 0.0147545082691226), (229, 0.014714592345297876), (1070, 0.014711569917009
434), (3069, 0.014711134738463055), (657, 0.014698218217661397), (521, 0.01466127530
0078583), (1111, 0.014657067953291992), (2244, 0.01465585131176518), (1569, 0.014651
81323774375), (2082, 0.014648987924051421), (974, 0.014645469648839992), (3414, 0.01
4641821376556615), (1727, 0.014639101750610525), (3983, 0.014630136717243532), (163
3, 0.014610411586649176), (240, 0.01459086841112044), (2288, 0.014575587616796275),
(1990, 0.014558528630020558), (1151, 0.014558045430265763), (1647, 0.014538793312907
415), (1438, 0.01450372255063764), (4146, 0.014471031507595471), (971, 0.01446731767
8648858), (523, 0.014448825923945762), (979, 0.014432562751749999), (2033, 0.0144081
37063061224), (4033, 0.014397628392187483), (28, 0.014389581335300713), (3180, 0.014
386788885050529), (1860, 0.014384186580270561), (984, 0.014374169382007224), (894,
0.014344007397738027), (952, 0.014337502799936806), (2089, 0.014334077242662028), (4
283, 0.014332810491009864), (4003, 0.014330406393226287), (2953, 0.01430249206918555
1), (745, 0.014296427336469887), (4383, 0.014295825600831508), (614, 0.0142949180097
58842), (3957, 0.014291579039763224), (882, 0.01426290745322866), (2154, 0.014261090
045043556), (4688, 0.0142582410038743), (3144, 0.014258042751768613), (3391, 0.01424
465592632181), (1686, 0.014227332954245164), (112, 0.01422619779516028), (3412, 0.01
422419058759691), (2400, 0.014201032790993107), (3227, 0.014188259388254864), (493,
0.014182387449480029), (915, 0.014172443789618369), (2888, 0.014170908630617039), (4
009, 0.014154927279777796), (1505, 0.014117239215771592), (2218, 0.0141144508726716
7), (310, 0.014099823434543982), (3944, 0.014094405511030984), (1133, 0.014090954994
041284), (402, 0.014081291604134655), (88, 0.014079698000279712), (544, 0.0140744727
42619288), (859, 0.014067188995630914), (4574, 0.01406379388613152), (1607, 0.014061
167248561286), (643, 0.014055241400417404), (3445, 0.014053682771515474), (2596, 0.0
14035198521420893), (922, 0.014022847549478943), (1440, 0.014012766584761716), (785,
0.01401254306820612), (4055, 0.014009950626018734), (4679, 0.013999750215399308), (3
61, 0.01399571574610273), (306, 0.013993128616778198), (919, 0.013982166900445563),
(3161, 0.013945103609747201), (3201, 0.01394499404557601), (512, 0.01394099704042405
8), (2461, 0.01393657476210283), (1448, 0.013931988276047544), (1294, 0.013902537012
29965), (4382, 0.013895426737584378), (1814, 0.013890288225782392), (2450, 0.0138884
93092364501), (3379, 0.013879795869627953), (2138, 0.013870745401339092), (379, 0.01
386618467491552), (458, 0.013864825269116102), (453, 0.013863948458613266), (1277,
0.013849161288857003), (3916, 0.013838320156849), (2578, 0.013836301560673045), (333
6, 0.013823729288610524), (3290, 0.013821778560342097), (3093, 0.01382064876571966
6), (1178, 0.013818194169235676), (2350, 0.013811241044855878), (3976, 0.01380281189
024388), (2564, 0.013786292530148845), (1559, 0.013785794831521148), (4230, 0.013776
034714659435), (1719, 0.013773664037395458), (1209, 0.013767757831726598), (2449, 0.
013765745545092837), (1786, 0.013756186572400164), (990, 0.013755877372139326), (384
9, 0.013733865912462931), (4155, 0.013732607977739997), (4327, 0.01372834675904457
6), (988, 0.013691169261071875), (2156, 0.013687613871801152), (356, 0.0136825209175
39964), (2043, 0.013661860798531617), (2250, 0.01363497500537029), (2606, 0.01361832
6126692283), (2619, 0.013613724440774687), (543, 0.013606161618896832), (2, 0.013599
520029326722), (59, 0.013588616309692061), (3709, 0.013570805246580792), (4353, 0.01
355571485898728), (481, 0.013549442607797264), (911, 0.013533002154688875), (812, 0.
013532913444182676), (3316, 0.013527131562123526), (3472, 0.013525842339425298), (58
2, 0.013506406082665956), (3605, 0.013502509610412616), (376, 0.013490994569417328),
(3725, 0.013481878781048548), (1518, 0.013480981994518023), (2161, 0.013470707971755
796), (2808, 0.01345961312079089), (680, 0.013446993996021294), (2253, 0.01343142844
2066351), (383, 0.013425762087526855), (1945, 0.013419865151133454), (3548, 0.013417

205086468337), (4346, 0.01340905478827646), (237, 0.013407097722261084), (1868, 0.01
3386345355129746), (2912, 0.01335291975436016), (123, 0.013352335707811847), (1482,
0.013348013447405924), (747, 0.013342619881024961), (1973, 0.013340100815097008), (2
155, 0.013334535307348962), (3920, 0.013332434573706437), (2412, 0.01331697734788298
4), (2775, 0.013310526377249988), (216, 0.013308904195041413), (84, 0.01330888670622
2196), (804, 0.013295202587327762), (2380, 0.013292623127802008), (2295, 0.013286483
189991219), (412, 0.013282417846038267), (3799, 0.013273167504061956), (1459, 0.0132
70934195827033), (2836, 0.013265241723364888), (13, 0.01326374702786566), (4333, 0.0
13254467584407353), (659, 0.01324817479973928), (1445, 0.013233824822724883), (962,
0.01321831626070123), (1117, 0.013209757980126156), (1671, 0.013203531345148078), (3
463, 0.013187342939057281), (2140, 0.013150997355821741), (125, 0.01314499788544409
7), (577, 0.01313342138818549), (3345, 0.013126294183100427), (531, 0.01312423629572
2512), (27, 0.013106673998149269), (4626, 0.013104790516575593), (2471, 0.0131010863
35393194), (700, 0.01307471779468753), (2766, 0.013065575421491586), (535, 0.0130630
06814410263), (3040, 0.01304201955875002), (844, 0.013035984977621692), (1313, 0.013
019787580348535), (203, 0.013006617597419886), (1486, 0.012986286768971267), (3617,
0.012981790080614633), (1615, 0.01297770565525629), (469, 0.012973204775475542), (43
01, 0.012955881810923449), (1260, 0.012933071831449924), (2602, 0.01291823712106581
6), (724, 0.012904924977953914), (1657, 0.01288095757721532), (1658, 0.0128506534095
462), (1742, 0.012841432896343572), (1563, 0.012828588064996519), (1291, 0.012822320
920255028), (1084, 0.012794674636422042), (2277, 0.012769727612697677), (489, 0.0127
67945945677046), (4728, 0.012756795991799913), (3062, 0.012751866613785096), (1984,
0.012732449525331598), (3584, 0.01270363781105749), (590, 0.012689566244122208), (42
14, 0.012687765261049002), (1750, 0.012683369244718658), (1306, 0.01266841721302412
2), (914, 0.01266440931111332), (1457, 0.012659535700538575), (3929, 0.0126551547439
02157), (3305, 0.012598033238040054), (2196, 0.012586639576815585), (1672, 0.0125436
82946555779), (1708, 0.012537489703183169), (3872, 0.012531591919830479), (1588, 0.0
12520420418828145), (1542, 0.012509475622448047), (3527, 0.012499929262499137), (27
3, 0.012493730297180941), (1753, 0.012468725762600427), (3482, 0.01246814374056671),
(3374, 0.01246799472917649), (309, 0.012447534711356482), (377, 0.01244645179477580
5), (2520, 0.012424463940706334), (1137, 0.012393198676248784), (2388, 0.01239277874
6348584), (3012, 0.012388963363858384), (840, 0.01234700492015121), (2558, 0.0123237
75028799834), (1100, 0.012318282989797822), (1573, 0.012317650873172716), (893, 0.01
2315931792271875), (2050, 0.012307645119260018), (3474, 0.012304645188135344), (162
5, 0.012266285368827462), (1096, 0.012217180267611535), (4087, 0.01221016439492715
3), (4597, 0.012196765764959186), (559, 0.012146856076258003), (2074, 0.012125007913
738632), (1960, 0.012122863594735973), (3460, 0.012100606254904103), (3170, 0.012091
943321009983), (2266, 0.012089392779855757), (475, 0.01208595098449301), (4081, 0.01
2072264272219402), (650, 0.012065445896039924), (78, 0.012046810653425794), (1138,
0.012012382378604771), (3888, 0.012010429915946854), (2209, 0.012005213997481702),
(1450, 0.01200294191627159), (4559, 0.012002694175231901), (3074, 0.0119633964313671
83), (787, 0.011955280484410663), (2652, 0.011940911265721117), (300, 0.011906750449
291664), (2639, 0.011904450281487955), (1930, 0.011866643277531164), (1852, 0.011855
724053823681), (2819, 0.011851542733676244), (2200, 0.011845484541211246), (1508, 0.
011824989606273715), (4215, 0.011808939785869502), (11, 0.011808001891564552), (932,
0.011805561898974404), (2229, 0.011804487037548588), (4152, 0.011796506006603412),
(3777, 0.0117574180190466), (3370, 0.01174969014663075), (3808, 0.01171909505672953
7), (4756, 0.011668526156901298), (1695, 0.011665180329931612), (3800, 0.01166197849
0169197), (1298, 0.011650842289952557), (605, 0.011649660850616457), (754, 0.0116384
60113210685), (1735, 0.011636958444510006), (1583, 0.01163216536219096), (1281, 0.01
160519512710677), (3416, 0.011596915149669994), (1041, 0.011585739784242965), (3259,
0.011582072402653344), (1463, 0.011555368030724987), (950, 0.011552700370816793), (5
85, 0.011504377423919045), (2181, 0.011497940963532812), (106, 0.01148195449626371
1), (776, 0.011474268733619225), (2429, 0.011472623353608848), (1817, 0.011456444443
825528), (764, 0.011441669954161434), (3309, 0.011419985499892791), (1166, 0.0114167
36755473642), (4606, 0.011416225215678366), (3661, 0.011402765944382625), (311, 0.01

1388881674615162), (1423, 0.011379127789045778), (1892, 0.011371550377381502), (92,
0.011341251045199981), (371, 0.01132543208956635), (2624, 0.011290104322949144), (24
16, 0.01128592183734534), (1983, 0.011279617573603204), (3126, 0.01127628619865739),
(4489, 0.01126895407748917), (143, 0.01125008459005959), (1206, 0.01123396548707935
3), (4543, 0.011220923472811117), (1955, 0.011214231219779627), (2016, 0.01121287396
9711067), (711, 0.0112090396051443), (735, 0.011203923491842288), (2423, 0.011200377
995020383), (435, 0.011194417840764729), (2674, 0.011176573504569326), (4494, 0.0111
64944615909845), (3364, 0.011161681231402118), (2438, 0.01115940783588043), (1140,
0.011144195561107666), (2591, 0.011140185255314298), (1003, 0.011136545861787504),
(1603, 0.011132776533034102), (1877, 0.011109943904804038), (953, 0.0111012625553961
26), (1599, 0.011080660620207674), (1694, 0.011067411156849118), (820, 0.01100659961
5256178), (2868, 0.011001105993800987), (1129, 0.010999863723701143), (1402, 0.01098
2823601901455), (928, 0.010956482890816462), (3004, 0.010942476625300828), (1042, 0.
010938409138879545), (3486, 0.01092473494325089), (390, 0.01091824586575766), (3671,
0.010888624412790326), (629, 0.010848660624759119), (2437, 0.010819572876558938), (2
853, 0.010811542554524215), (797, 0.01080633614892827), (1446, 0.01073375076241159),
(3932, 0.010732711442709235), (2693, 0.010720236185077343), (2166, 0.010717541396057
24), (302, 0.010708044933990125), (3613, 0.01070732887552611), (3569, 0.010703892723
267387), (3779, 0.01069818693855462), (258, 0.010689688422080963), (3225, 0.01068822
6271161332), (2141, 0.010677538797630138), (4801, 0.010675801655328677), (3798, 0.01
0669295733820213), (3803, 0.010651264325911588), (3399, 0.010649995796497485), (317
2, 0.010644153982213873), (3915, 0.010635703769647691), (451, 0.010633205817964796),
(1246, 0.010619550176089785), (3282, 0.010618334986523918), (4328, 0.010611956636390
087), (2846, 0.010600044798819248), (2856, 0.010597565902991327), (4764, 0.010580938
890773063), (2990, 0.010560021439008917), (3786, 0.01052345684812332), (2113, 0.0105
13386242058848), (1715, 0.010511793570983457), (1449, 0.01051049246276645), (4125,
0.010501408641802424), (4470, 0.010500723776802872), (2542, 0.010497458272264313),
(2874, 0.010493159812379384), (3554, 0.010488878112705529), (4782, 0.010483456668709
399), (1287, 0.010479910941085361), (2171, 0.01045506002247013), (2837, 0.0104541062
0160236), (3448, 0.010435873127939846), (15, 0.010432758242497996), (289, 0.01043240
827881478), (4367, 0.01042329527717722), (1924, 0.010416920962543878), (4414, 0.0104
15766262004633), (341, 0.010410592276632605), (506, 0.01040546758365313), (4241, 0.0
1039320093067048), (4501, 0.010383346262420592), (3181, 0.010359749063065688), (382
3, 0.010357696787555047), (2499, 0.010337581497730581), (4147, 0.01033455643560661
5), (567, 0.010326604547687785), (2391, 0.010321920352559162), (4330, 0.010308400914
699724), (4613, 0.010294760716985882), (1464, 0.010284505669030521), (349, 0.0102640
5294585673), (2239, 0.010263889731607297), (2088, 0.010247263576363835), (1494, 0.01
0241691794116957), (4525, 0.010239175091362967), (689, 0.010230918975361563), (4048,
0.010229944178983711), (3580, 0.010214933653245375), (3886, 0.010196796300999175),
(118, 0.010185051615491632), (4447, 0.010183020209924933), (4345, 0.0101773587896108
52), (2127, 0.010137987118395897), (2326, 0.010122271011278966), (235, 0.01010320324
7825286), (4097, 0.010097053147082874), (2487, 0.010091099831675353), (1341, 0.01008
5586585247039), (3387, 0.010076597968378795), (338, 0.01007104124077427), (3806, 0.0
1005868194329982), (284, 0.010047441739921288), (1115, 0.010045155534122305), (249,
0.010041233622224933), (3907, 0.010037362991773602), (3157, 0.01003521087711719), (2
973, 0.010002826633673762), (2189, 0.009997577637209175), (2056, 0.009997278744652
6), (792, 0.009988465935533025), (2651, 0.009984038131396575), (1456, 0.009982392031
103189), (2227, 0.009957760386353036), (1975, 0.009956953571886654), (575, 0.0099541
81271995429), (977, 0.009952578882221985), (1865, 0.009947359302382813), (4370, 0.00
994239424526903), (391, 0.009929634552399723), (2017, 0.009922309775577952), (1080,
0.009917722985102422), (57, 0.009917061156672187), (814, 0.009917013842729685), (278
4, 0.009909178075821692), (1381, 0.009900008053882567), (2081, 0.00989865939708576),
(1222, 0.009897545148202538), (2345, 0.009897449029082779), (2071, 0.009879386117380
753), (2504, 0.009877157553128763), (995, 0.009864622919803969), (3013, 0.0098621861
7760741), (626, 0.009848353733926413), (524, 0.009841838523789637), (1821, 0.0098347
91570927145), (1190, 0.00983133783005054), (2037, 0.00982907857108947), (348, 0.0098

26441016833736), (2206, 0.009825180774650782), (2928, 0.009816670186017134), (1481,
0.009800395041701682), (450, 0.009793441405616657), (3556, 0.009791365805198233), (2
745, 0.009789391153187966), (2290, 0.009788448576869925), (2603, 0.00976670966600771
6), (2716, 0.00975131439542477), (4213, 0.009741310905190287), (2231, 0.009738785821
36455), (2548, 0.009737526545616474), (2165, 0.009736520709247952), (443, 0.00972594
4990768265), (1335, 0.0097179420605383), (663, 0.00971745672141671), (1623, 0.009712
381823606215), (2872, 0.009704009857189484), (1549, 0.009703900509859342), (3000, 0.
009698613823861262), (829, 0.009694072031898334), (1179, 0.009675526837040626), (175
9, 0.00967239657720542), (4665, 0.009671658716015748), (1845, 0.00965050855111265),
(2150, 0.009641698502016457), (3553, 0.009632270526504868), (4136, 0.009630817587870
16), (3066, 0.009629825534849208), (2153, 0.009627329387371474), (201, 0.00961287139
4229363), (3664, 0.009611710582400939), (3214, 0.009591132244567663), (1732, 0.00958
640864870957), (1762, 0.009569150570978192), (698, 0.009562708219702466), (3752, 0.0
09556287096574403), (1650, 0.009552085915181458), (722, 0.00955041484262565), (3044,
0.009549390973542593), (1958, 0.009543457590741138), (2262, 0.009539549240166003),
(2027, 0.00953146635438233), (4798, 0.009531042329918646), (3665, 0.0095308661539553
75), (4170, 0.009522611433408934), (771, 0.009522138250825733), (3339, 0.00951984847
0270773), (742, 0.009515795367247296), (4763, 0.009510215661582229), (1008, 0.009480
31654424961), (440, 0.009476661648551307), (477, 0.009475612365586384), (4020, 0.009
472162943483471), (4278, 0.009464212926195429), (4138, 0.009451457533864858), (1648,
0.009446061997133983), (4180, 0.00943023987821648), (1169, 0.009424153020576552), (3
360, 0.009416067423415553), (1154, 0.009414801647695524), (190, 0.0094006087062500
9), (1621, 0.009394793734920446), (4502, 0.00939465681159704), (2328, 0.009389574532
83168), (2508, 0.009385714421556335), (1705, 0.009382041231411158), (576, 0.00937155
5126477136), (3397, 0.00936677968908261), (178, 0.009357441930127653), (4149, 0.0093
5405927823007), (3302, 0.00935185632297031), (2426, 0.009343130391775228), (4647, 0.
00933585139697818), (3330, 0.009331447169828247), (86, 0.009327381057907095), (2842,
0.009301902472823349), (3563, 0.009297030069886943), (1942, 0.009289478917794802),
(368, 0.009286540367566893), (4481, 0.009271047852722926), (3973, 0.0092691982683397
4), (4780, 0.009269068742749641), (3875, 0.009257668796753481), (1767, 0.00925444509
582254), (4746, 0.009250526313751945), (2611, 0.009249984214177734), (4596, 0.009245
402613121971), (471, 0.009238062231059729), (1693, 0.009226751800888274), (2409, 0.0
09217842955270813), (2465, 0.009217588136556633), (179, 0.009209159657810564), (171
8, 0.009201847044412254), (897, 0.009201581335226128), (942, 0.009198503282832091),
(1045, 0.0091868649473202), (2120, 0.00917944002843519), (875, 0.00917457121675842
1), (2967, 0.00917322152449968), (3638, 0.009162758847320866), (2729, 0.009158012202
09005), (1525, 0.009157339065228243), (4509, 0.009152777726504628), (3376, 0.0091403
09631343506), (4052, 0.009137680103957734), (1350, 0.009136598835697613), (3608, 0.0
0913650731444795), (214, 0.009119664204350115), (2158, 0.009103888293947683), (3320,
0.009097950977250642), (4787, 0.009088114682982247), (4282, 0.009081174337117321),
(2117, 0.009081037866644992), (1711, 0.009076622623200492), (2245, 0.009065728286658
787), (1150, 0.00906537582338922), (1570, 0.009061741366227418), (3820, 0.0090582956
97759775), (4217, 0.009058096224510323), (4678, 0.009056769495356205), (2420, 0.0090
54142209635392), (3488, 0.009045747024561446), (3558, 0.009034158580559544), (2998,
0.009033248196177252), (239, 0.009026133708670711), (1098, 0.009026025569565293), (4
049, 0.009017544483032625), (2511, 0.009015120299662442), (1554, 0.00901397622203311
6), (1749, 0.009012775603607633), (1171, 0.009012024986638572), (924, 0.009004915710
337635), (2173, 0.009001634129550014), (4696, 0.00899520611993711), (1591, 0.0089939
17973956286), (857, 0.00898777354115968), (2904, 0.008983219882562277), (697, 0.0089
79715217794244), (4577, 0.008968381969534603), (2649, 0.008965839130655243), (4205,
0.008963532932010123), (1061, 0.008960694446576207), (4264, 0.008958424529064913),
(2919, 0.00895625406654035), (427, 0.008950276812039562), (1744, 0.00894013735304537
1), (3199, 0.008937155236889387), (4371, 0.008933780040529984), (3279, 0.00893119971
888063), (625, 0.008928547813392097), (3767, 0.008924923476056553), (2670, 0.0089226
32722430955), (1584, 0.008922551492771418), (1576, 0.008909604562719449), (3759, 0.0
08906941759391942), (4680, 0.008905737538919345), (1751, 0.008905604206778191), (448

8, 0.008898214591463428), (3885, 0.008897098892136657), (1502, 0.00889347569647855),
(2960, 0.00889299840815021), (2152, 0.008892484123575494), (3176, 0.0088875940528874
66), (3925, 0.00888413575941073), (3016, 0.00888114917044104), (557, 0.0088780207941
58493), (2080, 0.008865215351355408), (1499, 0.008864783686100112), (4321, 0.0088647
08440215053), (256, 0.008859149559609284), (2885, 0.008858374271896388), (1882, 0.00
8850653355231598), (4449, 0.008837224799150863), (3741, 0.008834681349579099), (150
1, 0.008834379100895198), (4224, 0.00882742049201765), (1560, 0.008825888751138114),
(3987, 0.008821327147868901), (720, 0.008818675724175216), (949, 0.00881643779193759
7), (2720, 0.008815825767421964), (3760, 0.00881366114865915), (2719, 0.008813526034
982797), (1274, 0.008809322091417276), (2361, 0.008789661272089458), (3436, 0.008780
033224539123), (4677, 0.00877423435389705), (987, 0.008760570772492102), (2468, 0.00
875983390739302), (3270, 0.008756740049841881), (120, 0.008755244893174786), (4531,
0.008753350112336243), (1926, 0.008734337441492154), (3092, 0.008731537672391303),
(382, 0.008727217872332388), (77, 0.008716823552559781), (2462, 0.00871601316358603
3), (1336, 0.008713114481184604), (1010, 0.008712524730554715), (3750, 0.00870894927
0069152), (1980, 0.008697317750059086), (2831, 0.008697112855002414), (2906, 0.00868
9484878656518), (4479, 0.008684982745146274), (1276, 0.008669542824084029), (811, 0.
008666754761311112), (1240, 0.008663063811007912), (3571, 0.008653717143061106), (17
39, 0.008650953199043365), (4281, 0.008643007914679396), (2116, 0.0086408294588161
7), (3301, 0.0086379864659801), (1057, 0.008637735007822455), (2358, 0.0086370003706
17256), (1161, 0.008632937910957851), (1763, 0.008630437830550624), (1805, 0.0086288
90947105694), (2734, 0.008625460195998715), (4659, 0.008618744250420573), (4475, 0.0
08618324636091938), (2901, 0.00860696417898728), (4109, 0.008605110916871304), (135
2, 0.008601196207514825), (3653, 0.008599111860640826), (1388, 0.00858646367476010
8), (3431, 0.008586161946688004), (2786, 0.008581720572390315), (3824, 0.00858052784
6624716), (3381, 0.008576967414987828), (3682, 0.008574932443123913), (3444, 0.00857
2991281966036), (2220, 0.00857074760528601), (3386, 0.00857055808391046), (1197, 0.0
08569489949539294), (2535, 0.008564713539833924), (910, 0.008562719824375384), (379
7, 0.008559462070017666), (1410, 0.008554984992405472), (3568, 0.00855230648035574
9), (4373, 0.008549387756846594), (4634, 0.008548867495514648), (4272, 0.00854580798
0274156), (1905, 0.008543211535626222), (2791, 0.008538822884785868), (2979, 0.00853
2131421870018), (1149, 0.008531038938045889), (1696, 0.008529304605176043), (2040,
0.008526911385754945), (1318, 0.008526103682848115), (2149, 0.008525481441929175),
(2457, 0.008522131723674714), (1367, 0.00850499609895086), (2024, 0.0085043310108606
76), (3683, 0.008502094100846165), (321, 0.008500313679377706), (4365, 0.00849963016
960993), (3701, 0.008498831447368757), (4600, 0.008492458601566376), (2151, 0.008490
22313407311), (669, 0.008476776832315623), (2800, 0.008468968785837246), (2489, 0.00
8467067925442032), (3433, 0.008463161571889338), (2794, 0.008458868150231419), (350
6, 0.00845778611946717), (3582, 0.00845196790553298), (517, 0.008449304195879297),
(3727, 0.008431439219758243), (1832, 0.00843057883447386), (4720, 0.0084274054960508
28), (4040, 0.008423010439333191), (1798, 0.008421812224210398), (4317, 0.0084188818
31423225), (4483, 0.008411900697427017), (1349, 0.00841102396954277), (3245, 0.00841
0226317822496), (3693, 0.008410083577209045), (2807, 0.008409084626136808), (3930,
0.008403319137724874), (3266, 0.00839810809188871), (3514, 0.00839129849723227), (29
57, 0.008384586631018164), (1562, 0.008376065226063241), (2525, 0.00837364100997175
4), (3627, 0.008367735030674645), (1186, 0.008360515643054322), (4249, 0.00835116343
896571), (2031, 0.008349646169041549), (789, 0.008346563812418268), (3620, 0.0083458
85011077026), (3428, 0.008342685956353733), (4744, 0.008342665834602725), (975, 0.00
8341873263310436), (3147, 0.008337693254790833), (2965, 0.008327123080817546), (301
7, 0.00830934872982811), (3101, 0.00830792328482796), (1627, 0.008299623985191733),
(1429, 0.008297681873716949), (1121, 0.008294242471426677), (1379, 0.008290334565682
565), (2947, 0.008287722643896864), (4076, 0.00828719733309934), (3219, 0.0082862585
84938411), (2752, 0.008286110254383104), (3195, 0.008283658812848118), (4014, 0.0082
83002314604746), (3254, 0.008278851327421167), (2115, 0.008276007050561949), (4197,
0.008263699284628442), (4237, 0.008261552707641099), (1656, 0.008260393599957882),
(2133, 0.008253340076687482), (1185, 0.008248533984264714), (2393, 0.008248472819763

14), (1904, 0.008248325920015233), (1263, 0.00824179209911462), (3549, 0.00823367605
7789657), (4556, 0.00823174020504103), (4592, 0.008226455498323862), (3458, 0.008225
991881888113), (4476, 0.008222961450500511), (2441, 0.008221889632725188), (4740, 0.
008218175296016199), (1859, 0.008217356184935931), (3537, 0.008217329852494376), (13
2, 0.008212602697406698), (1598, 0.008211232625811155), (1641, 0.00820843685938150
8), (1309, 0.008208236632796575), (176, 0.008199725823514694), (1315, 0.008197753361
605725), (298, 0.00819747201126473), (648, 0.008196215056085683), (2408, 0.008194961
163281255), (3500, 0.008192139047744663), (3019, 0.008189699647710073), (795, 0.0081
89520966561724), (3867, 0.008184928190010737), (2269, 0.008177598703837795), (2832,
0.008177060740486486), (3508, 0.00816984741170989), (3085, 0.008146770991792316), (4
718, 0.008141129824638679), (3076, 0.008140345973126186), (671, 0.00813965858314374
6), (2495, 0.008136252616856646), (3657, 0.0081332533465417), (3581, 0.0081305462394
45579), (3313, 0.008127061468231849), (2948, 0.008125166438300345), (3684, 0.0081240
52215293338), (1221, 0.008123758336034967), (4668, 0.00812127649467453), (4524, 0.00
8118910070572088), (3864, 0.008110710132581238), (1545, 0.008106087729578507), (295
0, 0.008104747236162288), (628, 0.008093863499275534), (2186, 0.008081774148035419),
(2238, 0.008078951363265828), (4656, 0.008071877138176855), (3442, 0.008058412563203
481), (3566, 0.008057149668978525), (1993, 0.008051298461780137), (3532, 0.008048703
932781206), (3812, 0.00804176299433157), (4192, 0.008041477453170473), (944, 0.00803
9673761352435), (2318, 0.008028620507667924), (2941, 0.008022430308148016), (2502,
0.008021036122187287), (2463, 0.008000617601678538), (802, 0.007997915371082898), (1
909, 0.007995232460302829), (1099, 0.00798657052156911), (2306, 0.0079800229530522
7), (4080, 0.00797770145596698), (2329, 0.007966809564085051), (2483, 0.007963767298
913118), (2185, 0.007963298199192347), (4572, 0.00796218261301683), (3526, 0.0079616
96799759355), (2348, 0.007959372393462219), (2722, 0.007956938647448092), (4377, 0.0
07950081240470907), (4673, 0.00794071427576975), (1051, 0.00793784732812935), (1736,
0.007935721003861168), (346, 0.007930631777048608), (1340, 0.007924608955632453), (1
24, 0.007921651394546356), (2767, 0.00791974459741216), (2569, 0.00791314249965777),
(4195, 0.007913014458346867), (2110, 0.00790637975131982), (855, 0.00790539500071688
4), (1874, 0.007904344677483348), (3483, 0.00790431840179359), (3087, 0.007901869200
902557), (1021, 0.007900666830026687), (313, 0.007900242606542415), (566, 0.00789815
6343199089), (2335, 0.007889334231240852), (2484, 0.007888845267434868), (2911, 0.00
7887976175178972), (3647, 0.007885521427532032), (3821, 0.007884907756170689), (354
1, 0.007881526207729913), (2994, 0.007881245256192233), (2404, 0.00787627802348136
5), (3785, 0.00787323193390393), (4216, 0.0078703182496836), (4173, 0.00785625073160
7695), (558, 0.007854466424362893), (692, 0.007853698558792655), (4769, 0.0078504315
81962624), (2642, 0.007850132168961967), (2804, 0.007848400842318401), (1609, 0.0078
4647007888458), (1688, 0.007844204355925711), (3970, 0.007839233396686473), (4204,
0.007833181814846973), (2865, 0.007830828833211092), (4343, 0.007823808990535185),
(3346, 0.007812482455297258), (3473, 0.007809192211093686), (2561, 0.007806588735680
5), (3132, 0.00780453779510678), (2669, 0.007800982860591928), (1778, 0.007790188768
082973), (3048, 0.007788254002700383), (2428, 0.007779745653856696), (1135, 0.007778
3141153367945), (515, 0.007771177787962552), (2661, 0.007767515016351138), (2038, 0.
007761358710588314), (667, 0.007758480670631074), (849, 0.0077578229151192305), (75
1, 0.0077557918109426955), (2549, 0.007740637190366689), (3378, 0.007728521599656517
5), (3870, 0.007713664152154154), (1406, 0.007703678119017641), (538, 0.007693854925
330783), (4268, 0.007691007015220353), (1255, 0.007689324296503561), (2739, 0.007688
456357432549), (876, 0.0076837729686955744), (1681, 0.0076795369276497815), (3124,
0.007665080966068965), (2467, 0.007662022201831444), (2795, 0.007649638620855087),
(2932, 0.007644523066420075), (2072, 0.0076435028807289845), (429, 0.007639094433465
508), (4381, 0.007628986310298959), (4292, 0.007597582997653554), (996, 0.0075764986
2289347), (4277, 0.0075743121274934565), (4484, 0.007573511428283609), (630, 0.00757
2650679447448), (3817, 0.007571705029118976), (3790, 0.007567434618230866), (3658,
0.007559612572913113), (414, 0.007558294459328721), (4499, 0.0075518192297254864),
(4390, 0.007548256844989767), (1923, 0.007531840305792245), (3850, 0.007528921749279
049), (2397, 0.0075231036405652195), (4708, 0.007520218607396731), (756, 0.007514058

5682151465), (4024, 0.0075136753791302845), (3639, 0.007511476608410464), (1105, 0.0
07507520950666183), (4134, 0.007507165002268491), (730, 0.007505541392708739), (122
6, 0.007504796955659943), (1551, 0.007491678996788645), (3090, 0.00747830317452246
3), (2828, 0.007477424663806525), (3413, 0.007470593693982962), (2660, 0.00746620997
8199735), (4310, 0.007464709026905589), (49, 0.007462062943836138), (3322, 0.0074531
44952559375), (4799, 0.007435200207794489), (4263, 0.00743221492616897), (2607, 0.00
7425068429264804), (1267, 0.007420163817329874), (2424, 0.007414025241205139), (344
7, 0.007408435854578011), (2130, 0.007401088606339198), (1829, 0.00739650185483419
2), (3255, 0.007383222319211602), (1811, 0.007382294067259588), (3879, 0.00737682896
57509525), (2124, 0.0073684666604114536), (548, 0.007368097391411546), (1962, 0.0073
65899595637163), (3052, 0.007353578328557356), (4202, 0.007351418661532418), (2933,
0.00732790404069939), (2593, 0.007326690338957168), (723, 0.007322232782478983), (35
88, 0.007320581378888757), (3045, 0.007314636474626293), (965, 0.00728799343868342
8), (4331, 0.007280559350666071), (3070, 0.007273009934693388), (3716, 0.00726935043
9607031), (3179, 0.007269336091764261), (3263, 0.007259342935721644), (1622, 0.00724
45371562285575), (2626, 0.0072432969922821035), (3307, 0.007240862847337393), (2283,
0.007223035842208167), (1676, 0.007216555284997412), (1550, 0.007205723586617939),
(2801, 0.007192450337442204), (4648, 0.007192235318862787), (4090, 0.007180944061584
36), (3296, 0.007180469313973745), (3839, 0.00717368263340481), (4302, 0.00716962547
58530035), (2867, 0.007167015455204336), (3776, 0.007162240987433001), (1553, 0.0071
58359547476221), (1288, 0.007155601262555008), (3197, 0.007140969295830423), (2131,
0.0071396846732225), (1948, 0.007135786975592431), (3844, 0.007135615257177114), (12
12, 0.00713363590404558), (3018, 0.007126150418614115), (4161, 0.00712197963492772
7), (3437, 0.007111197451608696), (1581, 0.007108888278063722), (3454, 0.00709438662
9024889), (3576, 0.007092370959654787), (2164, 0.0070879422218198115), (4395, 0.0070
83834483934841), (2282, 0.0070413578597557916), (3304, 0.007032911169183759), (2762,
0.007023964134395655), (3421, 0.007016599330783714), (540, 0.006985241536664359), (4
792, 0.006978860831716811), (519, 0.00697827718224983), (3123, 0.00697689229756327
6), (3257, 0.006974398640630545), (3895, 0.006938996010822264), (3357, 0.00693568445
3099439), (983, 0.0069318798567145285), (1667, 0.006913038735809923), (4340, 0.00691
0398014265468), (3175, 0.006900172394531548), (660, 0.006890886527673485), (4576, 0.
00689000185723599), (1639, 0.00688898481806359), (684, 0.006869323823112695), (3512,
0.006863633391340462), (4394, 0.006836932646240273 5), (3876, 0.006835118095338313),
(3594, 0.006825992736765534), (2774, 0.006823794615014855), (3625, 0.006812849225333
099), (3256, 0.006806007187921874), (2728, 0.006793651759084413), (2440, 0.006773356
509660382), (4265, 0.006771129380892885), (3169, 0.006760329558070126), (4397, 0.006
7470530740054956), (4196, 0.006702064358825231), (3373, 0.006686714282916756), (346
7, 0.006685774288609964), (1971, 0.006677196039465772), (2905, 0.00667488200241066
3), (4752, 0.006673211172831979), (3645, 0.006639063050271446), (514, 0.006629023158
113295), (3205, 0.006597564676605912), (2677, 0.006580398417854229), (283, 0.0065451
38429275421), (4404, 0.006538351925547434), (4287, 0.006529840059985693), (2750, 0.0
06514174750681592), (3418, 0.006497633938246931), (3032, 0.0064729343380845504), (40
96, 0.006465464659595971), (3509, 0.0064644170583374825), (3299, 0.00644231037697476
4), (1357, 0.006434786309166769), (4450, 0.006424147434476346), (3880, 0.00641765151
6985444), (2623, 0.006406800582658762), (1462, 0.006405688609516683), (166, 0.006403
384496392318), (2188, 0.0064030161335636914), (3943, 0.006386398644832711), (2972,
0.006384021715683351), (1769, 0.006365628221953609), (4111, 0.006359807876199569),
(3425, 0.006359371515884716), (4258, 0.0063453053181198255), (3358, 0.00632089941105
5797), (2201, 0.006296412789100745), (2616, 0.006262293613899253), (3618, 0.00625068
23878716155), (4734, 0.006244465355735435), (4226, 0.006240093037830945), (4425, 0.0
06175091089703483), (4143, 0.006172798970472118), (3091, 0.00615466903461918), (56,
0.006136094904267454), (985, 0.006133163108993261), (1666, 0.006121681872805396), (4
223, 0.006098225314203334), (3974, 0.0060957348614268676), (1528, 0.0060949314442038
84), (4187, 0.0060941287842083405), (225, 0.006093588113836095), (773, 0.00609123219
87746185), (4620, 0.006072986416382341), (1752, 0.006062587214095426), (1773, 0.0060
58671554193209), (47, 0.006056967859003828), (3275, 0.006043744372866129), (3154, 0.

0060409568104516816), (1933, 0.006029926885612366), (4557, 0.005992759906969406), (3
021, 0.005979733456284166), (2207, 0.00597956077369272), (602, 0.00597890629576117
7), (2122, 0.005962649978188197), (4460, 0.0059600880958697545), (3128, 0.0059595990
162209875), (4546, 0.005955046389491045), (4124, 0.005948704223365627), (2309, 0.005
93627613231451), (530, 0.005930164405386325), (3882, 0.005927301503942796), (2352,
0.005914269667336131), (4129, 0.0059128410836825004), (3650, 0.005911453517416629),
(1398, 0.005907813640041784), (2980, 0.005897718848800492), (1733, 0.005896909715974
8), (44, 0.005896865705087829), (1920, 0.005884147455237689), (3522, 0.0058839262182
05288), (601, 0.005876017634106431), (362, 0.005874135890534837), (3993, 0.005873306
5436512135), (1777, 0.005865537480103254), (180, 0.005863964523086625), (274, 0.0058
56357216856913), (4004, 0.005830939307813409), (1273, 0.005807999381282511), (4061,
0.005804761630695971), (976, 0.00578993808670283), (409, 0.005787379153185329 5), (65
5, 0.005787722187675423), (1636, 0.005776708803441586), (82, 0.00576827305044206), (1
93, 0.005764152853658987), (1012, 0.005761450271103945), (127, 0.005760090641234576
6), (3830, 0.005759821102857798), (2650, 0.00575734175785355), (55, 0.00575527457017
3456), (4174, 0.005753848393351417), (157, 0.005748857276590655), (1047, 0.005741385
211568459), (29, 0.005723805699010054), (355, 0.005717869525329317), (1953, 0.005717
713486632421), (4072, 0.005713835444529831), (573, 0.005702788084032725), (4093, 0.0
05701108578900813), (3615, 0.005701027398240209), (26, 0.005699120907668437), (1844,
0.005697925012441475), (3816, 0.005668891685470773), (1399, 0.005666944437724892),
(222, 0.005661967356230 4604), (1249, 0.005656863725235057), (3941, 0.005656560443357
7625), (1317, 0.005648048512342407), (1321, 0.005640488690272382), (209, 0.005638065
3548724095), (3089, 0.005634743987543985), (1285, 0.005629935367298657), (3523, 0.00
562881473497 48945), (818, 0.005623551909990202), (609, 0.005622055082906893), (2198,
0.005619890180753037), (1083, 0.005611233786916412), (4565, 0.005610193331200937),
(863, 0.0056050268503635555), (2093, 0.005591394414196888), (104, 0.0055891707934778
82), (3098, 0.00558599820155528), (1284, 0.00558426838838599), (596, 0.0055811982704
8137), (1956, 0.0055754579569536455), (511, 0.005574324286548584), (1872, 0.00556183
1391579429), (3689, 0.005554347904035231), (645, 0.005550054924716862), (4465, 0.005
546866060051285), (1302, 0.005545870742472661), (188, 0.005544672081248338), (3153,
0.005538209502594359), (167, 0.005536191629212126), (266, 0.005529500831600524), (7,
0.005521938648290493), (834, 0.005518430748913458 5), (76, 0.005516189273301243), (18
80, 0.005514445043253252), (1586, 0.005513418879816727), (236, 0.00551225271474633
5), (4487, 0.00551196594659381), (3842, 0.005511321634844242), (622, 0.0055091711613
84699), (4623, 0.005508704247561916), (4549, 0.005505685485488944), (4349, 0.0055013
83169600896), (74, 0.005495141965586583), (3545, 0.005494336140216559 6), (394, 0.005
491572077857567), (286, 0.005489881771352674), (33, 0.005487629188043741), (1395, 0.
0054819744504418 63), (320, 0.00547701853389258), (594, 0.005463707592966443), (277,
0.005460965070966977), (1741, 0.005446897459944766), (508, 0.005435696902570209), (3
35, 0.005426700203861826), (3962, 0.005426610775828206 5), (62, 0.005416634534426565
5), (685, 0.005415570504610253 5), (1510, 0.00539988610208904), (570, 0.0053991269933
04583), (607, 0.005396861321667464), (3343, 0.005396837478246529), (1617, 0.00539648
52496758515), (169, 0.005393553666615686), (260, 0.005392803407541181), (227, 0.0053
90690104302215), (737, 0.005390453411546613), (3077, 0.005389591352749593), (478, 0.
0053886847672649), (823, 0.005378680108069031), (808, 0.005373539761798156), (70, 0.
0053713318229465 04), (1610, 0.005369326219846216), (424, 0.005363999198083402), (65
3, 0.0053614195427355614), (432, 0.005354651274163252), (2565, 0.00534230320877042
4), (1319, 0.005340388672171497), (2785, 0.005337151745043536), (1158, 0.00533293960
2783096), (850, 0.005332021757716764), (2264, 0.005321789201754367), (151, 0.0053210
16206774113), (455, 0.005320094859069555), (263, 0.005318416540504533), (94, 0.00531
8310294658278), (3707, 0.005315193390659952 5), (2293, 0.005314507202980626), (2827,
0.005309759852241237), (507, 0.005304626448524463), (945, 0.005304072052753753), (30
7, 0.005291568896276649), (1836, 0.0052912791828490634), (91, 0.005291140630302694),
(604, 0.005287943847300146), (2644, 0.005279616042468476), (1713, 0.0052788304698636
47), (951, 0.005270744600501487), (2918, 0.005265128699522157), (490, 0.005263014548
414424), (2995, 0.005250568386014171), (888, 0.005247979592649513), (3740, 0.0052455

```
037587596474), (4042, 0.005241060083805741), (16, 0.005235971636780309), (2084, 0.00
5235231116409664), (3606, 0.005230724421820102), (3982, 0.005220591162090804), (207
5, 0.005219967962226397), (2712, 0.005215963147795358), (568, 0.00521079049016164),
(2683, 0.005192921868578381), (1702, 0.005190493622697643 5), (640, 0.005188852949417
256), (2129, 0.00518710618679695), (3942, 0.00518697935251437), (1465, 0.00518523006
2999871), (1815, 0.005178792908771225), (392, 0.005170723532574395 5), (3749, 0.00516
5716221369657), (134, 0.005162531642358688 5), (304, 0.005147368420567504), (228, 0.0
05143517638008635), (687, 0.005139584849282332), (1491, 0.005139455403447934), (286
3, 0.00513609193782345), (2386, 0.005133981393820354), (580, 0.005132187938716139),
(68, 0.005130123069684274), (2996, 0.005124339139704501), (1347, 0.00512063990256799
7), (1093, 0.00511614242192896), (1200, 0.005104806007917166), (153, 0.0051047599326
32073), (4275, 0.005099362409852739), (1825, 0.005092744533513066), (246, 0.00508687
65893401805), (1447, 0.005065863956775308), (1854, 0.005054757724731262), (2678, 0.0
05049462851715121), (2781, 0.005033207383864078), (1645, 0.005031870998304777), (156
8, 0.005030433989265982), (3715, 0.004996217147975443), (1101, 0.00498673369798524
4), (45, 0.004962729367343318), (1908, 0.004957465342509351), (4114, 0.0049336910403
17982), (147, 0.004926143004899927), (989, 0.004924861442996708), (1106, 0.004922963
1964249594), (4351, 0.004915546695148099), (3029, 0.004906763874052582), (1104, 0.00
4905997309755556), (621, 0.004894557097422808), (2327, 0.004888511404914587), (282,
0.004885034999473712), (611, 0.0048829783439722175), (595, 0.00488185450288807), (17
43, 0.004874505123190848), (610, 0.004863843383905942), (529, 0.00484842118779606),
(1616, 0.004847094862544482), (2194, 0.004846742555154652), (1079, 0.004842956999368
6675), (3221, 0.004841356325461938 4), (1290, 0.004834469256724305), (1480, 0.0048336
16221008409), (1044, 0.00483235694450333), (1155, 0.004811608235094448), (1697, 0.00
4806718483768213), (1056, 0.004797228573047399), (312, 0.004786517934094848), (1332,
0.0047864257428798076), (2486, 0.004782269736584759), (1310, 0.004781821804758146),
(1437, 0.004772410004958566), (4316, 0.004742445456975398), (3516, 0.004730667189624
703), (4315, 0.0047238512062288), (491, 0.0047044149113152686), (4402, 0.00470313237
5248852), (2023, 0.00469993409590895), (1193, 0.00469657559411346), (487, 0.00468830
4126411212), (4220, 0.004670831880857795), (1199, 0.004644005471996142), (1136, 0.00
46405633039279534), (244, 0.004606293916228297), (2803, 0.004599393040846562), (21,
0.004591239318101996), (253, 0.004581957148163642), (2641, 0.004567764771982903), (3
44, 0.004527095216613905), (4344, 0.004501719528877708), (51, 0.004490688090626269),
(233, 0.00448643551001502), (2521, 0.004468120464552313), (3274, 0.00444924906572035
9), (3722, 0.004389894690296252), (2013, 0.004356932703326003), (3099, 0.00433962083
7882695), (3231, 0.004334164872270242), (541, 0.0043225778169406195), (4563, 0.00431
8625084217244), (3133, 0.004312512451717949), (230, 0.00429551502513737), (1077, 0.0
04292800055937971), (97, 0.004252359556894674), (3884, 0.0042220292169891285), (300
7, 0.004213605300378606), (2515, 0.004190295521719535), (2806, 0.00415711252616485
3), (3804, 0.00410361069412517), (579, 0.004063687013663969), (337, 0.00357603374597
53606), (6, 0.0), (25, 0.0), (34, 0.0), (60, 0.0), (67, 0.0), (87, 0.0), (128, 0.0),
(130, 0.0), (136, 0.0), (141, 0.0), (142, 0.0), (144, 0.0), (145, 0.0), (146, 0.0),
(155, 0.0), (173, 0.0), (192, 0.0), (202, 0.0), (231, 0.0), (255, 0.0), (288, 0.0),
(301, 0.0), (317, 0.0), (325, 0.0), (333, 0.0), (347, 0.0), (359, 0.0), (360, 0.0),
(363, 0.0), (365, 0.0), (375, 0.0), (378, 0.0), (385, 0.0), (397, 0.0), (399, 0.0),
(401, 0.0), (404, 0.0), (418, 0.0), (430, 0.0), (436, 0.0), (439, 0.0), (445, 0.0),
(459, 0.0), (460, 0.0), (461, 0.0), (462, 0.0), (464, 0.0), (465, 0.0), (474, 0.0),
(482, 0.0), (492, 0.0), (496, 0.0), (498, 0.0), (502, 0.0), (504, 0.0), (509, 0.0),
(520, 0.0), (522, 0.0), (526, 0.0), (532, 0.0), (545, 0.0), (550, 0.0), (551, 0.0),
(552, 0.0), (561, 0.0), (562, 0.0), (564, 0.0), (569, 0.0), (593, 0.0), (598, 0.0),
(599, 0.0), (616, 0.0), (619, 0.0), (623, 0.0), (632, 0.0), (637, 0.0), (639, 0.0),
(641, 0.0), (649, 0.0), (682, 0.0), (688, 0.0), (699, 0.0), (701, 0.0), (703, 0.0),
(704, 0.0), (705, 0.0), (707, 0.0), (709, 0.0), (712, 0.0), (713, 0.0), (727, 0.0),
(729, 0.0), (734, 0.0), (738, 0.0), (749, 0.0), (765, 0.0), (770, 0.0), (778, 0.0),
(780, 0.0), (783, 0.0), (793, 0.0), (796, 0.0), (798, 0.0), (801, 0.0), (816, 0.0),
(817, 0.0), (819, 0.0), (824, 0.0), (835, 0.0), (837, 0.0), (843, 0.0), (845, 0.0),
```

```
(853, 0.0), (860, 0.0), (861, 0.0), (868, 0.0), (872, 0.0), (878, 0.0), (885, 0.0),
(887, 0.0), (889, 0.0), (890, 0.0), (891, 0.0), (895, 0.0), (900, 0.0), (902, 0.0),
(906, 0.0), (907, 0.0), (908, 0.0), (909, 0.0), (912, 0.0), (913, 0.0), (916, 0.0),
(921, 0.0), (925, 0.0), (926, 0.0), (940, 0.0), (954, 0.0), (957, 0.0), (960, 0.0),
(961, 0.0), (966, 0.0), (986, 0.0), (991, 0.0), (997, 0.0), (1004, 0.0), (1007, 0.
0), (1011, 0.0), (1014, 0.0), (1019, 0.0), (1023, 0.0), (1028, 0.0), (1032, 0.0), (1
037, 0.0), (1039, 0.0), (1054, 0.0), (1060, 0.0), (1062, 0.0), (1065, 0.0), (1066,
0.0), (1067, 0.0), (1068, 0.0), (1069, 0.0), (1071, 0.0), (1072, 0.0), (1088, 0.0),
(1090, 0.0), (1107, 0.0), (1116, 0.0), (1123, 0.0), (1126, 0.0), (1128, 0.0), (1130,
0.0), (1142, 0.0), (1145, 0.0), (1157, 0.0), (1164, 0.0), (1167, 0.0), (1168, 0.0),
(1173, 0.0), (1189, 0.0), (1198, 0.0), (1205, 0.0), (1208, 0.0), (1210, 0.0), (1211,
0.0), (1218, 0.0), (1219, 0.0), (1224, 0.0), (1229, 0.0), (1231, 0.0), (1235, 0.0),
(1242, 0.0), (1248, 0.0), (1251, 0.0), (1252, 0.0), (1258, 0.0), (1261, 0.0), (1262,
0.0), (1268, 0.0), (1295, 0.0), (1297, 0.0), (1307, 0.0), (1323, 0.0), (1325, 0.0),
(1326, 0.0), (1329, 0.0), (1330, 0.0), (1333, 0.0), (1344, 0.0), (1345, 0.0), (1353,
0.0), (1355, 0.0), (1356, 0.0), (1360, 0.0), (1365, 0.0), (1371, 0.0), (1376, 0.0),
(1378, 0.0), (1383, 0.0), (1385, 0.0), (1386, 0.0), (1387, 0.0), (1391, 0.0), (1393,
0.0), (1396, 0.0), (1401, 0.0), (1405, 0.0), (1414, 0.0), (1416, 0.0), (1424, 0.0),
(1436, 0.0), (1444, 0.0), (1452, 0.0), (1454, 0.0), (1455, 0.0), (1460, 0.0), (1471,
0.0), (1475, 0.0), (1476, 0.0), (1483, 0.0), (1485, 0.0), (1487, 0.0), (1489, 0.0),
(1493, 0.0), (1496, 0.0), (1509, 0.0), (1511, 0.0), (1520, 0.0), (1521, 0.0), (1526,
0.0), (1530, 0.0), (1532, 0.0), (1534, 0.0), (1536, 0.0), (1539, 0.0), (1540, 0.0),
(1544, 0.0), (1555, 0.0), (1557, 0.0), (1567, 0.0), (1580, 0.0), (1582, 0.0), (1587,
0.0), (1590, 0.0), (1593, 0.0), (1601, 0.0), (1602, 0.0), (1604, 0.0), (1613, 0.0),
(1614, 0.0), (1620, 0.0), (1634, 0.0), (1638, 0.0), (1640, 0.0), (1642, 0.0), (1646,
0.0), (1649, 0.0), (1654, 0.0), (1662, 0.0), (1673, 0.0), (1677, 0.0), (1683, 0.0),
(1684, 0.0), (1690, 0.0), (1691, 0.0), (1698, 0.0), (1700, 0.0), (1703, 0.0), (1704,
0.0), (1709, 0.0), (1710, 0.0), (1716, 0.0), (1723, 0.0), (1724, 0.0), (1725, 0.0),
(1729, 0.0), (1730, 0.0), (1738, 0.0), (1754, 0.0), (1755, 0.0), (1756, 0.0), (1760,
0.0), (1766, 0.0), (1772, 0.0), (1774, 0.0), (1776, 0.0), (1784, 0.0), (1797, 0.0),
(1801, 0.0), (1808, 0.0), (1809, 0.0), (1810, 0.0), (1823, 0.0), (1831, 0.0), (1833,
0.0), (1835, 0.0), (1840, 0.0), (1841, 0.0), (1855, 0.0), (1862, 0.0), (1864, 0.0),
(1866, 0.0), (1867, 0.0), (1871, 0.0), (1875, 0.0), (1876, 0.0), (1885, 0.0), (1887,
0.0), (1888, 0.0), (1891, 0.0), (1895, 0.0), (1898, 0.0), (1901, 0.0), (1903, 0.0),
(1906, 0.0), (1911, 0.0), (1917, 0.0), (1919, 0.0), (1922, 0.0), (1928, 0.0), (1934,
0.0), (1938, 0.0), (1940, 0.0), (1941, 0.0), (1949, 0.0), (1950, 0.0), (1951, 0.0),
(1961, 0.0), (1965, 0.0), (1968, 0.0), (1972, 0.0), (1979, 0.0), (1985, 0.0), (1988,
0.0), (1994, 0.0), (1997, 0.0), (1998, 0.0), (2000, 0.0), (2005, 0.0), (2006, 0.0),
(2009, 0.0), (2010, 0.0), (2011, 0.0), (2012, 0.0), (2028, 0.0), (2030, 0.0), (2032,
0.0), (2039, 0.0), (2041, 0.0), (2044, 0.0), (2045, 0.0), (2051, 0.0), (2053, 0.0),
(2058, 0.0), (2059, 0.0), (2064, 0.0), (2065, 0.0), (2067, 0.0), (2073, 0.0), (2076,
0.0), (2077, 0.0), (2079, 0.0), (2087, 0.0), (2095, 0.0), (2097, 0.0), (2101, 0.0),
(2106, 0.0), (2107, 0.0), (2109, 0.0), (2112, 0.0), (2118, 0.0), (2119, 0.0), (2121,
0.0), (2125, 0.0), (2128, 0.0), (2132, 0.0), (2134, 0.0), (2135, 0.0), (2137, 0.0),
(2139, 0.0), (2143, 0.0), (2144, 0.0), (2146, 0.0), (2162, 0.0), (2172, 0.0), (2174,
0.0), (2183, 0.0), (2190, 0.0), (2208, 0.0), (2211, 0.0), (2213, 0.0), (2217, 0.0),
(2219, 0.0), (2224, 0.0), (2226, 0.0), (2232, 0.0), (2235, 0.0), (2236, 0.0), (2241,
0.0), (2243, 0.0), (2246, 0.0), (2248, 0.0), (2252, 0.0), (2254, 0.0), (2256, 0.0),
(2257, 0.0), (2259, 0.0), (2267, 0.0), (2270, 0.0), (2271, 0.0), (2272, 0.0), (2273,
0.0), (2274, 0.0), (2278, 0.0), (2279, 0.0), (2286, 0.0), (2287, 0.0), (2292, 0.0),
(2297, 0.0), (2298, 0.0), (2299, 0.0), (2312, 0.0), (2315, 0.0), (2316, 0.0), (2319,
0.0), (2325, 0.0), (2332, 0.0), (2337, 0.0), (2344, 0.0), (2347, 0.0), (2353, 0.0),
(2359, 0.0), (2369, 0.0), (2370, 0.0), (2373, 0.0), (2374, 0.0), (2379, 0.0), (2382,
0.0), (2385, 0.0), (2396, 0.0), (2401, 0.0), (2402, 0.0), (2406, 0.0), (2410, 0.0),
(2411, 0.0), (2418, 0.0), (2419, 0.0), (2421, 0.0), (2422, 0.0), (2432, 0.0), (2434,
0.0), (2436, 0.0), (2443, 0.0), (2447, 0.0), (2451, 0.0), (2453, 0.0), (2458, 0.0),
```

```
(2459, 0.0), (2460, 0.0), (2464, 0.0), (2469, 0.0), (2472, 0.0), (2475, 0.0), (2476,
0.0), (2477, 0.0), (2479, 0.0), (2481, 0.0), (2493, 0.0), (2494, 0.0), (2497, 0.0),
(2498, 0.0), (2500, 0.0), (2501, 0.0), (2503, 0.0), (2505, 0.0), (2509, 0.0), (2512,
0.0), (2516, 0.0), (2517, 0.0), (2518, 0.0), (2519, 0.0), (2523, 0.0), (2526, 0.0),
(2528, 0.0), (2533, 0.0), (2537, 0.0), (2541, 0.0), (2547, 0.0), (2551, 0.0), (2552,
0.0), (2559, 0.0), (2566, 0.0), (2570, 0.0), (2576, 0.0), (2588, 0.0), (2589, 0.0),
(2590, 0.0), (2594, 0.0), (2595, 0.0), (2599, 0.0), (2600, 0.0), (2601, 0.0), (2605,
0.0), (2608, 0.0), (2617, 0.0), (2618, 0.0), (2620, 0.0), (2629, 0.0), (2634, 0.0),
(2636, 0.0), (2637, 0.0), (2643, 0.0), (2648, 0.0), (2653, 0.0), (2656, 0.0), (2658,
0.0), (2662, 0.0), (2663, 0.0), (2664, 0.0), (2665, 0.0), (2666, 0.0), (2668, 0.0),
(2671, 0.0), (2672, 0.0), (2673, 0.0), (2679, 0.0), (2680, 0.0), (2682, 0.0), (2684,
0.0), (2686, 0.0), (2690, 0.0), (2691, 0.0), (2694, 0.0), (2696, 0.0), (2697, 0.0),
(2698, 0.0), (2699, 0.0), (2701, 0.0), (2705, 0.0), (2706, 0.0), (2707, 0.0), (2709,
0.0), (2710, 0.0), (2711, 0.0), (2714, 0.0), (2717, 0.0), (2718, 0.0), (2723, 0.0),
(2724, 0.0), (2725, 0.0), (2733, 0.0), (2735, 0.0), (2737, 0.0), (2743, 0.0), (2744,
0.0), (2748, 0.0), (2751, 0.0), (2754, 0.0), (2760, 0.0), (2764, 0.0), (2768, 0.0),
(2770, 0.0), (2772, 0.0), (2776, 0.0), (2782, 0.0), (2787, 0.0), (2810, 0.0), (2811,
0.0), (2817, 0.0), (2821, 0.0), (2825, 0.0), (2834, 0.0), (2838, 0.0), (2840, 0.0),
(2841, 0.0), (2843, 0.0), (2844, 0.0), (2845, 0.0), (2847, 0.0), (2850, 0.0), (2852,
0.0), (2854, 0.0), (2855, 0.0), (2857, 0.0), (2866, 0.0), (2870, 0.0), (2873, 0.0),
(2877, 0.0), (2880, 0.0), (2881, 0.0), (2882, 0.0), (2883, 0.0), (2886, 0.0), (2890,
0.0), (2893, 0.0), (2895, 0.0), (2896, 0.0), (2898, 0.0), (2900, 0.0), (2903, 0.0),
(2908, 0.0), (2910, 0.0), (2913, 0.0), (2916, 0.0), (2917, 0.0), (2920, 0.0), (2923,
0.0), (2924, 0.0), (2925, 0.0), (2930, 0.0), (2934, 0.0), (2935, 0.0), (2936, 0.0),
(2937, 0.0), (2942, 0.0), (2943, 0.0), (2945, 0.0), (2951, 0.0), (2954, 0.0), (2955,
0.0), (2956, 0.0), (2962, 0.0), (2963, 0.0), (2966, 0.0), (2968, 0.0), (2969, 0.0),
(2970, 0.0), (2974, 0.0), (2976, 0.0), (2981, 0.0), (2984, 0.0), (2987, 0.0), (2988,
0.0), (2989, 0.0), (2992, 0.0), (2997, 0.0), (3002, 0.0), (3005, 0.0), (3015, 0.0),
(3022, 0.0), (3023, 0.0), (3024, 0.0), (3025, 0.0), (3026, 0.0), (3027, 0.0), (3028,
0.0), (3034, 0.0), (3036, 0.0), (3037, 0.0), (3038, 0.0), (3039, 0.0), (3041, 0.0),
(3054, 0.0), (3057, 0.0), (3058, 0.0), (3059, 0.0), (3060, 0.0), (3061, 0.0), (3064,
0.0), (3065, 0.0), (3067, 0.0), (3071, 0.0), (3079, 0.0), (3081, 0.0), (3082, 0.0),
(3084, 0.0), (3096, 0.0), (3100, 0.0), (3102, 0.0), (3103, 0.0), (3104, 0.0), (3105,
0.0), (3107, 0.0), (3109, 0.0), (3110, 0.0), (3113, 0.0), (3115, 0.0), (3117, 0.0),
(3118, 0.0), (3119, 0.0), (3121, 0.0), (3122, 0.0), (3129, 0.0), (3130, 0.0), (3131,
0.0), (3134, 0.0), (3135, 0.0), (3137, 0.0), (3139, 0.0), (3140, 0.0), (3149, 0.0),
(3150, 0.0), (3152, 0.0), (3155, 0.0), (3156, 0.0), (3159, 0.0), (3163, 0.0), (3164,
0.0), (3166, 0.0), (3167, 0.0), (3171, 0.0), (3174, 0.0), (3178, 0.0), (3182, 0.0),
(3185, 0.0), (3187, 0.0), (3188, 0.0), (3190, 0.0), (3191, 0.0), (3193, 0.0), (3194,
0.0), (3196, 0.0), (3198, 0.0), (3200, 0.0), (3204, 0.0), (3206, 0.0), (3210, 0.0),
(3212, 0.0), (3213, 0.0), (3215, 0.0), (3216, 0.0), (3217, 0.0), (3222, 0.0), (3223,
0.0), (3226, 0.0), (3228, 0.0), (3230, 0.0), (3233, 0.0), (3235, 0.0), (3240, 0.0),
(3248, 0.0), (3249, 0.0), (3258, 0.0), (3261, 0.0), (3262, 0.0), (3264, 0.0), (3265,
0.0), (3268, 0.0), (3269, 0.0), (3271, 0.0), (3272, 0.0), (3278, 0.0), (3283, 0.0),
(3288, 0.0), (3289, 0.0), (3292, 0.0), (3295, 0.0), (3298, 0.0), (3300, 0.0), (3310,
0.0), (3311, 0.0), (3314, 0.0), (3317, 0.0), (3319, 0.0), (3324, 0.0), (3327, 0.0),
(3328, 0.0), (3331, 0.0), (3340, 0.0), (3341, 0.0), (3342, 0.0), (3344, 0.0), (3353,
0.0), (3356, 0.0), (3363, 0.0), (3365, 0.0), (3366, 0.0), (3367, 0.0), (3369, 0.0),
(3380, 0.0), (3384, 0.0), (3388, 0.0), (3392, 0.0), (3393, 0.0), (3395, 0.0), (3396,
0.0), (3398, 0.0), (3403, 0.0), (3407, 0.0), (3410, 0.0), (3411, 0.0), (3415, 0.0),
(3419, 0.0), (3422, 0.0), (3426, 0.0), (3429, 0.0), (3432, 0.0), (3435, 0.0), (3438,
0.0), (3443, 0.0), (3452, 0.0), (3453, 0.0), (3457, 0.0), (3461, 0.0), (3468, 0.0),
(3471, 0.0), (3476, 0.0), (3478, 0.0), (3484, 0.0), (3487, 0.0), (3491, 0.0), (3493,
0.0), (3496, 0.0), (3497, 0.0), (3498, 0.0), (3502, 0.0), (3513, 0.0), (3517, 0.0),
(3519, 0.0), (3521, 0.0), (3524, 0.0), (3525, 0.0), (3530, 0.0), (3534, 0.0), (3535,
0.0), (3536, 0.0), (3539, 0.0), (3540, 0.0), (3542, 0.0), (3543, 0.0), (3547, 0.0),
```

```
(3551, 0.0), (3555, 0.0), (3557, 0.0), (3560, 0.0), (3565, 0.0), (3570, 0.0), (3572,
0.0), (3573, 0.0), (3577, 0.0), (3578, 0.0), (3583, 0.0), (3585, 0.0), (3587, 0.0),
(3589, 0.0), (3598, 0.0), (3601, 0.0), (3610, 0.0), (3612, 0.0), (3614, 0.0), (3621,
0.0), (3629, 0.0), (3633, 0.0), (3641, 0.0), (3642, 0.0), (3649, 0.0), (3654, 0.0),
(3655, 0.0), (3656, 0.0), (3663, 0.0), (3666, 0.0), (3670, 0.0), (3673, 0.0), (3674,
0.0), (3675, 0.0), (3676, 0.0), (3677, 0.0), (3681, 0.0), (3686, 0.0), (3687, 0.0),
(3690, 0.0), (3692, 0.0), (3696, 0.0), (3697, 0.0), (3702, 0.0), (3705, 0.0), (3708,
0.0), (3712, 0.0), (3713, 0.0), (3718, 0.0), (3720, 0.0), (3721, 0.0), (3723, 0.0),
(3728, 0.0), (3729, 0.0), (3731, 0.0), (3732, 0.0), (3736, 0.0), (3738, 0.0), (3742,
0.0), (3746, 0.0), (3751, 0.0), (3761, 0.0), (3763, 0.0), (3768, 0.0), (3773, 0.0),
(3781, 0.0), (3783, 0.0), (3787, 0.0), (3788, 0.0), (3792, 0.0), (3793, 0.0), (3794,
0.0), (3801, 0.0), (3805, 0.0), (3807, 0.0), (3809, 0.0), (3810, 0.0), (3811, 0.0),
(3814, 0.0), (3818, 0.0), (3822, 0.0), (3825, 0.0), (3827, 0.0), (3835, 0.0), (3838,
0.0), (3843, 0.0), (3846, 0.0), (3852, 0.0), (3853, 0.0), (3862, 0.0), (3863, 0.0),
(3871, 0.0), (3877, 0.0), (3881, 0.0), (3892, 0.0), (3894, 0.0), (3902, 0.0), (3904,
0.0), (3909, 0.0), (3913, 0.0), (3917, 0.0), (3922, 0.0), (3923, 0.0), (3924, 0.0),
(3926, 0.0), (3927, 0.0), (3931, 0.0), (3933, 0.0), (3937, 0.0), (3938, 0.0), (3949,
0.0), (3950, 0.0), (3955, 0.0), (3958, 0.0), (3960, 0.0), (3961, 0.0), (3963, 0.0),
(3967, 0.0), (3968, 0.0), (3969, 0.0), (3977, 0.0), (3980, 0.0), (3985, 0.0), (3989,
0.0), (3990, 0.0), (3992, 0.0), (3994, 0.0), (3995, 0.0), (3996, 0.0), (3997, 0.0),
(3999, 0.0), (4000, 0.0), (4001, 0.0), (4002, 0.0), (4005, 0.0), (4006, 0.0), (4007,
0.0), (4016, 0.0), (4018, 0.0), (4021, 0.0), (4023, 0.0), (4025, 0.0), (4026, 0.0),
(4029, 0.0), (4031, 0.0), (4034, 0.0), (4035, 0.0), (4036, 0.0), (4043, 0.0), (4044,
0.0), (4046, 0.0), (4056, 0.0), (4057, 0.0), (4058, 0.0), (4059, 0.0), (4060, 0.0),
(4064, 0.0), (4066, 0.0), (4068, 0.0), (4069, 0.0), (4070, 0.0), (4074, 0.0), (4075,
0.0), (4082, 0.0), (4083, 0.0), (4085, 0.0), (4086, 0.0), (4089, 0.0), (4091, 0.0),
(4092, 0.0), (4095, 0.0), (4101, 0.0), (4102, 0.0), (4103, 0.0), (4104, 0.0), (4107,
0.0), (4112, 0.0), (4113, 0.0), (4115, 0.0), (4116, 0.0), (4118, 0.0), (4121, 0.0),
(4122, 0.0), (4123, 0.0), (4126, 0.0), (4128, 0.0), (4130, 0.0), (4133, 0.0), (4137,
0.0), (4140, 0.0), (4141, 0.0), (4148, 0.0), (4157, 0.0), (4162, 0.0), (4165, 0.0),
(4166, 0.0), (4168, 0.0), (4169, 0.0), (4171, 0.0), (4178, 0.0), (4185, 0.0), (4186,
0.0), (4188, 0.0), (4190, 0.0), (4194, 0.0), (4198, 0.0), (4199, 0.0), (4206, 0.0),
(4207, 0.0), (4208, 0.0), (4210, 0.0), (4212, 0.0), (4218, 0.0), (4219, 0.0), (4227,
0.0), (4229, 0.0), (4234, 0.0), (4235, 0.0), (4236, 0.0), (4240, 0.0), (4242, 0.0),
(4243, 0.0), (4244, 0.0), (4245, 0.0), (4247, 0.0), (4248, 0.0), (4250, 0.0), (4253,
0.0), (4257, 0.0), (4260, 0.0), (4262, 0.0), (4266, 0.0), (4269, 0.0), (4270, 0.0),
(4271, 0.0), (4274, 0.0), (4280, 0.0), (4289, 0.0), (4290, 0.0), (4293, 0.0), (4294,
0.0), (4296, 0.0), (4298, 0.0), (4299, 0.0), (4303, 0.0), (4304, 0.0), (4305, 0.0),
(4306, 0.0), (4307, 0.0), (4308, 0.0), (4309, 0.0), (4314, 0.0), (4318, 0.0), (4322,
0.0), (4323, 0.0), (4332, 0.0), (4334, 0.0), (4335, 0.0), (4336, 0.0), (4338, 0.0),
(4341, 0.0), (4342, 0.0), (4347, 0.0), (4354, 0.0), (4358, 0.0), (4359, 0.0), (4361,
0.0), (4362, 0.0), (4363, 0.0), (4364, 0.0), (4366, 0.0), (4369, 0.0), (4374, 0.0),
(4384, 0.0), (4385, 0.0), (4387, 0.0), (4389, 0.0), (4396, 0.0), (4398, 0.0), (4400,
0.0), (4403, 0.0), (4405, 0.0), (4406, 0.0), (4407, 0.0), (4409, 0.0), (4410, 0.0),
(4412, 0.0), (4413, 0.0), (4416, 0.0), (4417, 0.0), (4419, 0.0), (4420, 0.0), (4422,
0.0), (4423, 0.0), (4424, 0.0), (4426, 0.0), (4429, 0.0), (4431, 0.0), (4433, 0.0),
(4434, 0.0), (4435, 0.0), (4436, 0.0), (4437, 0.0), (4441, 0.0), (4443, 0.0), (4444,
0.0), (4445, 0.0), (4446, 0.0), (4451, 0.0), (4452, 0.0), (4453, 0.0), (4455, 0.0),
(4456, 0.0), (4457, 0.0), (4458, 0.0), (4459, 0.0), (4462, 0.0), (4463, 0.0), (4464,
0.0), (4466, 0.0), (4468, 0.0), (4469, 0.0), (4474, 0.0), (4478, 0.0), (4480, 0.0),
(4485, 0.0), (4490, 0.0), (4491, 0.0), (4492, 0.0), (4495, 0.0), (4500, 0.0), (4503,
0.0), (4504, 0.0), (4505, 0.0), (4508, 0.0), (4510, 0.0), (4511, 0.0), (4512, 0.0),
(4513, 0.0), (4516, 0.0), (4517, 0.0), (4520, 0.0), (4521, 0.0), (4523, 0.0), (4530,
0.0), (4532, 0.0), (4533, 0.0), (4534, 0.0), (4536, 0.0), (4537, 0.0), (4538, 0.0),
(4540, 0.0), (4541, 0.0), (4542, 0.0), (4544, 0.0), (4547, 0.0), (4548, 0.0), (4550,
0.0), (4551, 0.0), (4553, 0.0), (4560, 0.0), (4561, 0.0), (4562, 0.0), (4564, 0.0),
```

```
(4566, 0.0), (4567, 0.0), (4568, 0.0), (4570, 0.0), (4571, 0.0), (4575, 0.0), (4578,
0.0), (4581, 0.0), (4582, 0.0), (4586, 0.0), (4587, 0.0), (4588, 0.0), (4589, 0.0),
(4590, 0.0), (4591, 0.0), (4593, 0.0), (4598, 0.0), (4603, 0.0), (4607, 0.0), (4609,
0.0), (4611, 0.0), (4612, 0.0), (4614, 0.0), (4615, 0.0), (4617, 0.0), (4619, 0.0),
(4622, 0.0), (4624, 0.0), (4627, 0.0), (4628, 0.0), (4632, 0.0), (4633, 0.0), (4635,
0.0), (4636, 0.0), (4637, 0.0), (4639, 0.0), (4640, 0.0), (4641, 0.0), (4644, 0.0),
(4651, 0.0), (4652, 0.0), (4653, 0.0), (4655, 0.0), (4657, 0.0), (4658, 0.0), (4660,
0.0), (4661, 0.0), (4662, 0.0), (4667, 0.0), (4669, 0.0), (4671, 0.0), (4674, 0.0),
(4676, 0.0), (4681, 0.0), (4682, 0.0), (4683, 0.0), (4684, 0.0), (4685, 0.0), (4686,
0.0), (4687, 0.0), (4689, 0.0), (4690, 0.0), (4691, 0.0), (4692, 0.0), (4693, 0.0),
(4694, 0.0), (4695, 0.0), (4697, 0.0), (4698, 0.0), (4699, 0.0), (4700, 0.0), (4702,
0.0), (4703, 0.0), (4705, 0.0), (4706, 0.0), (4709, 0.0), (4710, 0.0), (4711, 0.0),
(4712, 0.0), (4714, 0.0), (4715, 0.0), (4716, 0.0), (4719, 0.0), (4721, 0.0), (4722,
0.0), (4723, 0.0), (4725, 0.0), (4729, 0.0), (4731, 0.0), (4736, 0.0), (4737, 0.0),
(4738, 0.0), (4741, 0.0), (4743, 0.0), (4745, 0.0), (4747, 0.0), (4748, 0.0), (4750,
0.0), (4751, 0.0), (4753, 0.0), (4754, 0.0), (4755, 0.0), (4758, 0.0), (4761, 0.0),
(4762, 0.0), (4765, 0.0), (4766, 0.0), (4768, 0.0), (4770, 0.0), (4771, 0.0), (4775,
0.0), (4777, 0.0), (4779, 0.0), (4781, 0.0), (4783, 0.0), (4784, 0.0), (4785, 0.0),
(4786, 0.0), (4790, 0.0), (4794, 0.0), (4797, 0.0), (4800, 0.0), (4802, 0.0)]
```

In [26]:

```python
# print the name of similar movies based on the index

print('Movies suggested for you : \n')

i = 1

for movie in sorted_similar_movies:
  index = movie[0]
  title_from_index = movies_data[movies_data.index==index]['title'].values[0]
  if (i<=20):
    print(i, '.',title_from_index)
    i+=1
```

```
Movies suggested for you :

1 . Batman
2 . Batman Returns
3 . Batman & Robin
4 . The Dark Knight Rises
5 . Batman Begins
6 . The Dark Knight
7 . A History of Violence
8 . Superman
9 . Beetlejuice
10 . Bedazzled
11 . Mars Attacks!
12 . The Sentinel
13 . Planet of the Apes
14 . Man of Steel
15 . Suicide Squad
16 . The Mask
17 . Salton Sea
18 . Spider-Man 3
19 . The Postman Always Rings Twice
20 . Hang 'em High
```

In [27]:
```
movie_name = input(' Enter your favourite movie name : ')

list_of_all_titles = movies_data['title'].tolist()

find_close_match = difflib.get_close_matches(movie_name, list_of_all_titles)

close_match = find_close_match[0]

index_of_the_movie = movies_data[movies_data.title == close_match]['index'].values[

similarity_score = list(enumerate(similarity[index_of_the_movie]))

sorted_similar_movies = sorted(similarity_score, key = lambda x:x[1], reverse = Tru

print('Movies suggested for you : \n')

i = 1

for movie in sorted_similar_movies:
  index = movie[0]
  title_from_index = movies_data[movies_data.index==index]['title'].values[0]
  if (i<=20):
    print(i, '.',title_from_index)
    i+=1
```

```
Movies suggested for you :

1 . Spider-Man
2 . Spider-Man 3
3 . Spider-Man 2
4 . The Notebook
5 . Seabiscuit
6 . Clerks II
7 . The Ice Storm
8 . Oz: The Great and Powerful
9 . Horrible Bosses
10 . The Count of Monte Cristo
11 . In Good Company
12 . Finding Nemo
13 . Clear and Present Danger
14 . Brothers
15 . The Good German
16 . Drag Me to Hell
17 . Bambi
18 . The Queen
19 . Charly
20 . Escape from L.A.
```

In [ ]: